Deltares Systems



Delft3D FM Suite 1D2D

Modelling suite for integral water solutions

D-Rainfall Runoff

User Manual

Deltares

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User Manual

Version: 1.00
Revision: 80288

20 June 2026

D-Rainfall Runoff, User Manual

DRAFT

Published and printed by:
Deltares
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The Netherlands

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1 Hydrology

1.1 Hydrology (Rainfall Runoff modules)

1.1.1 Deltares Rural Rainfall-Runoff concept

The Rural concept is the first concept implemented in RR, and focussed on typical polder situations in the Netherlands and abroad. It uses a number of different node types (each with different formulations) describing relevant parts of polders. These node types are:

- ◇ paved node
- ◇ unpaved node
- ◇ greenhouse node
- ◇ open water area with precipitation and evaporation
- ◇ waste water treatment node
- ◇ RR-boundary node
- ◇ RR bifurcation node
- ◇ RR connection node
- ◇ RR industry node

1.1.2 RR paved node concept

The paved node concept available in SOBEK-Rural and RR describes the rainfall-runoff on paved areas. In paved areas water can be stored on the surface (“on-the-street storage”) and in a sewer system. The first one represents the storage on paved areas like roofs and roads. The second one represents the water stored in sewer mains of separated or combined sewer systems. The representation of the rainfall-runoff process in paved areas is shown.

The storage on the street and the sewer storage can be considered to be two reservoirs. The rainfall-runoff module calculates a water balance of these reservoirs. When precipitation occurs on the paved area, first the on-the-street storage reservoir is filled. If this reservoir is full, it starts spilling into the sewer reservoir. The amount of on-the-street storage is reduced by evaporation.

Water can enter the sewer reservoir in two ways: first by spilling from the on-the-street storage, and second by return flow from domestic water use (dry weather flow). Depending on the type of sewer system, the inflow from the surface and the dry weather flow are mixed in one sewer storage reservoir, or put into separate sewer storage reservoirs.

When the sewer storage reservoir contains water, the sewer pumps are switched on, and water is pumped from the sewer to the local open water or to a boundary representing a waste-water treatment plant outside the system. If the sewer is full, it can also spill directly into the open water. Flows from paved to unpaved areas and vice versa are neglected.

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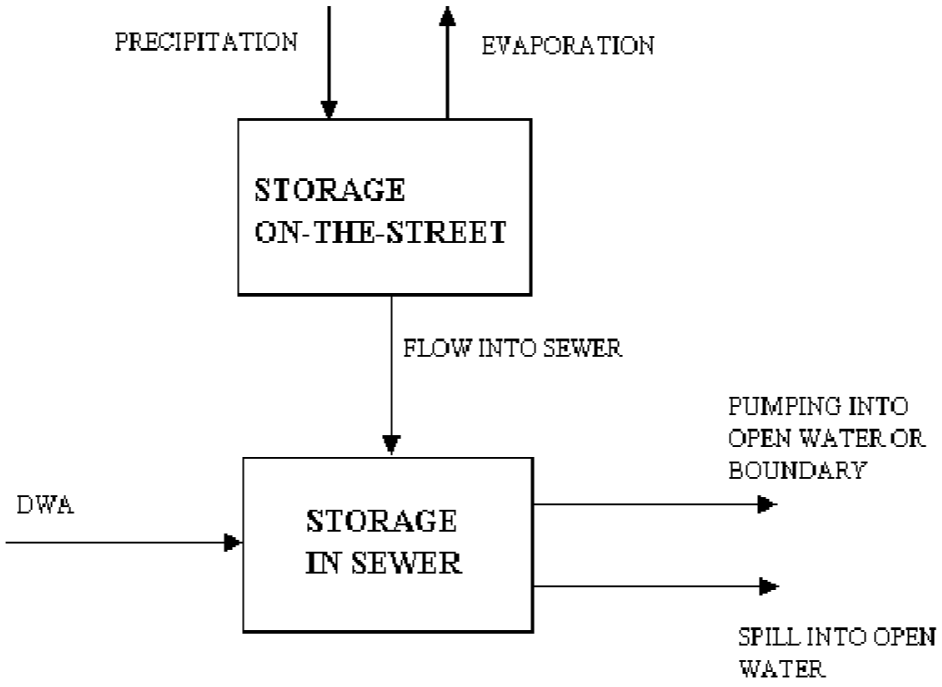


Figure 1.1: *The storage on the street and the sewer storage can be considered as two reservoirs*

Representation of the rainfall-runoff process in paved areas.

You can model multiple types of paved area by dividing the total area into sub-areas with specific characteristics. In this way areas with and without sewers can be modelled. By definition, there is no infiltration to groundwater in paved areas. When modelling urban areas in cities, as a rule of thumb about 50 percent of the urban area can be considered as paved area, and 50 percent can be considered as unpaved area (private gardens and public parks). Detailed land use maps including the foot-print of buildings can give better estimates of the distribution over paved and unpaved area.

1.1.3 RR unpaved node concept

The unpaved node concept is the important concept for Rural areas in polders, since it often covers the largest part of the area. This node type transforms rainfall that falls on a parcel of land into an outflow towards open water, typically modelled in 1D. The unpaved node area receives rainfall, which infiltrates into the soil and/or is stored on the land. The infiltrated water fills up the groundwater, so that the groundwater level will rise. Through different soil layers, each with its own drainage capacity, water will then flow out of the soil towards the connected open water (groundwater outflow to open water, or drainage). Also evaporation, infiltration and percolation can be modelled. If desired, unsaturated zone processes can be included in the calculations.

Because the RR module is a so-called zero-dimensional model, an unpaved node can be seen as a container where water flows in from above (precipitation) or below (seepage from deep groundwater), where water is stored (change in groundwater table & volume of water in the unsaturated zone) and water flows out, (evaporation, groundwater outflow, surface runoff and infiltration). The picture below gives an impression of the various terms which apply to

the unpaved area node.

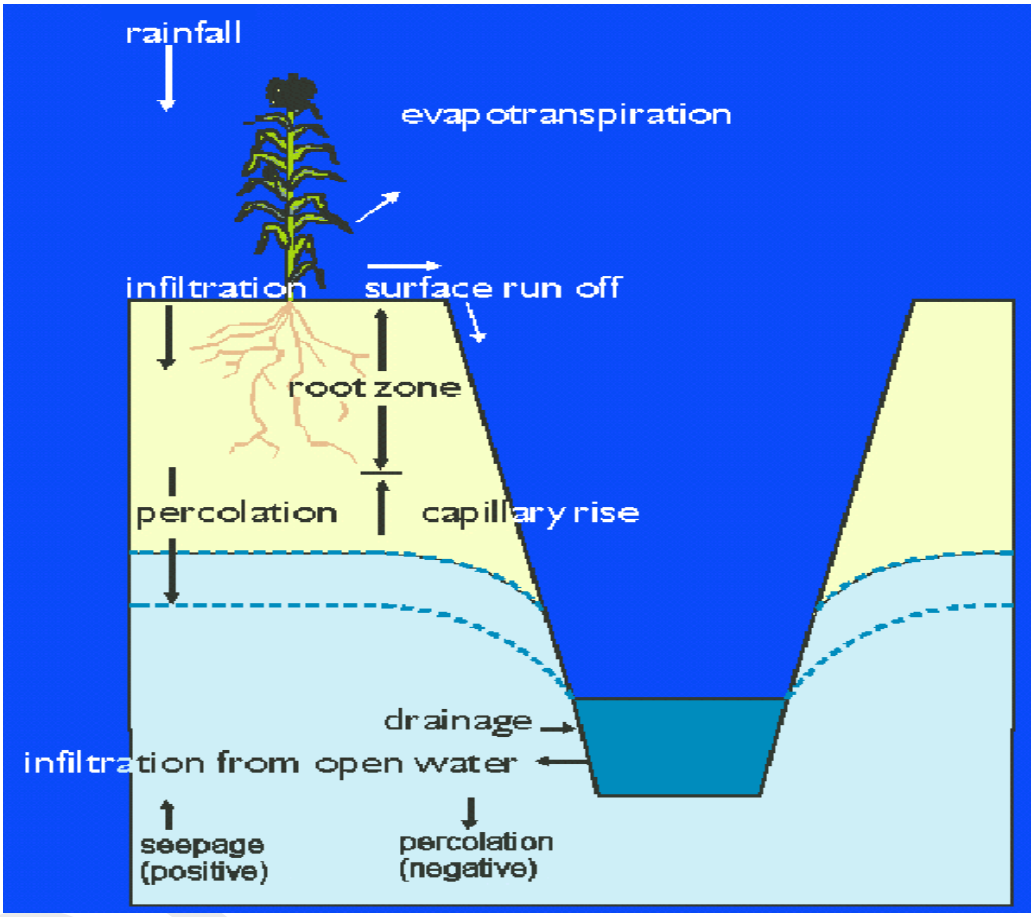


Figure 1.2: *Representation of the rainfall-runoff process in unpaved areas*

In addition, the user can choose whether to use the above described definition of modelling the transport processes in the unpaved area, or to use a more detailed description. In that case the unsaturated zone is modelled by means of root zone reservoir.

The unsaturated zone is modelled as vertically oriented 1-D model using steady-state approach called CAPSIM. The 1-D column model consists of a reservoir representing the root zone and the subsoil. The equilibrium moisture storage in the root zone is defined as the amount of moisture corresponding with a steady-state situation with no-flow conditions to or from the root zone. If the equilibrium moisture storage for the root zone is exceeded, excess water will percolate to the saturated zone. If the moisture storage is less than the equilibrium moisture storage, upward flow from the saturated zone is simulated through capillary rise. The height of the phreatic surface is calculated from the water balance of the subsoil, using a storage coefficient which is dependent on the depth of the groundwater table.

The root zone reservoir is used for calculations of evapotranspiration. The subsoil reservoir is used to calculate the saturated storage coefficient.

Evapotranspiration is determined by the crop and the moisture content in the root zone. For these calculations, recorded values of precipitation and potential evapotranspiration of a reference crop and woodland must be available. The potential evapotranspiration for other crops or vegetation types are derived from the values for the reference crop by conversion.

Some important characteristics of the unsaturated zone, i.e. upward flux, storage coefficient and equilibrium moisture storage of the root zone, are calculated from the soil moisture retention-, and hydraulic conductivity curves using the CAPSIM model which is based on the assumption of steady state soil moisture flow (Wesseling, 1991). For twenty-one different standard Dutch soil types, sets of pre-defined tables are provided with RR.

For the unpaved node, optionally irrigation can be defined based on the conditions of the subsoil.

1.1.4 RR greenhouse node concept

The RR greenhouse node concept is applicable for greenhouse horticulture area, and is developed based on greenhouse practices in the Netherlands. The rainfall-runoff process on greenhouses is described by volume balances in the storage on the greenhouse roofs and storage in greenhouse open rainwater basins or closed silo's (underground or closed basins). Rainfall falls on the greenhouse area on the greenhouse roofs and directly into the greenhouse open storage basins. Water from the roofs is also stored in the open rainwater basins and silo's. Water from the storage basins is used for irrigation of the crops in the greenhouses. Water can evaporate from the greenhouse roofs and storage basins. When the basins are full, excess water is discharged to the connected RR-boundary or corresponding lateral in the 1D model schematisation. This is illustrated in the next figure.

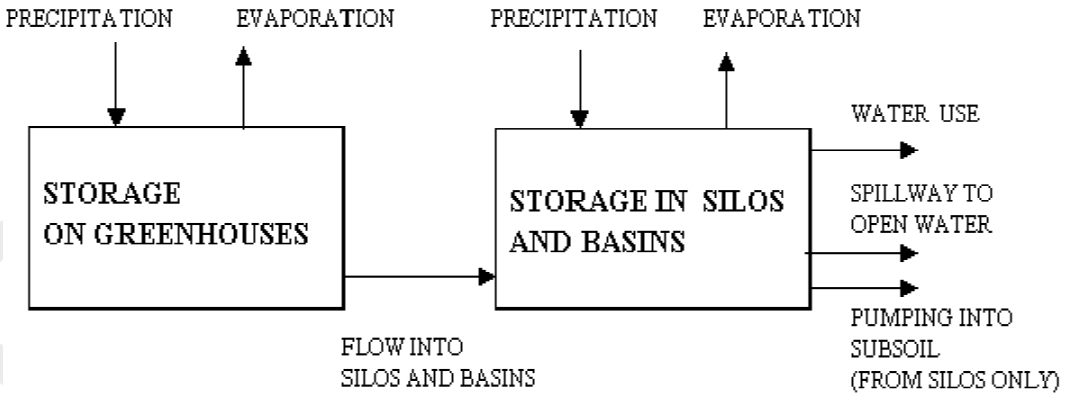


Figure 1.3: *Volume balance in the greenhouse areas*

The rainwater basins have been divided into ten categories, depending on their volume per hectare of draining glass surface. In the RR computations, for each category the lower limit of the storage is used. For example: all basins with a storage between 2500 and 3000 cubic meters per hectare of glass are considered as basins with a capacity of 2500 cubic meters per hectare of glass. The initial conditions are taken from a time table with free storage in the rainwater basins, or from a restart file.

The (remaining) storage present in the basins at the beginning of the computation is an important variable determining whether spilling from the basins will occur or not. Therefore, a table with historical data derived by WUR/Alterra with typical available storage for Dutch conditions is provided as input for the greenhouse nodes. Also time series of typical greenhouse water demands are provided as input. Note that the default values are for Dutch situations. These values can not be added through the GUI, but the data is available in (editable) ASCII input files.

1.1.5 RR open water node precipitation and evaporation concept

The RR open water precipitation and open water evaporation node concepts are used to define a lateral inflow in a 1D model by defining a total area and a precipitation or evaporation time series. This node is not meant as an open water node representing storage. To include the storage and water level variation, this has to be represented in the modelled 1D system.

1.1.6 RR waste water treatment node concept

The RR-WWTP node (WWTP=waste water treatment plant; in Dutch: RWZI) is used to have the possibility for paved nodes to connect to two different outlets, viz. the outlet location of the pumped sewer system of the paved node, and the outlet location of the sewer overflows. A WWTP node is normally used to pass water that flows from paved nodes via the WWTP to an RR-boundary (often also a lateral inflow in the 1D model) assuming that on a timestep basis the inflow and outflow of the WWTP are equal. It is also possible to impose discharge measurument data as outflow from the waste water treatment plant towards the downstream RR-boundary. In the latter case, with differences in inflow and outflow, the WWTP also has a storage change balance term (which should be zero on the long term).

1.1.7 RR boundary node concept

The RR-Boundary node represents a boundary of the RR system and typically acts as outflow point of the RR model. In some cases also inflow to the RR model is possible, e.g. when an RR-unpaved node is connected to an RR-boundary where the boundary water level is higher than the groundwater level at the unpaved node. Often the RR-boundary is acting as a lateral inflow to the 1D model, but an RR-boundary node is not necessarily connected to the 1D model and can be just an outlet of the RR model where water leaves the RR-system (and does not enter the 1D model). At the RR-boundary a water level can be specified as a constant, a predefined time series, or real-time (on-line coupled) from the 1D model.

1.1.8 RR-bifurcation or RR-connection node

The RR-bifurcation and RR-Connection node can be used in combination with RR Rural model concept and other concepts like the RR-Sacramento or RR-HBV concept. Using the RR-bifurcation node, flows can be distributed to 1 or more outlet locations (RR-boundaries), while the RR-connection node is used to combine flows in one outflow link. The RR-Connection node does not require any input data. The *RR-Connection node* functions as a confluence: adding up all flows from incoming links and supplying that to the downstream link. At the *RR-bifurcation node*, the incoming flow can be distributed over multiple links. If no additional data is specified and there are n downstream links of the RR-bifurcation, each downstream links receives the same amount of water (all links get $1/n$ of the inflow. The distribution can also be specified by specifying fractions (adding up to 1.00) for each downstream link in the order of definition of these links in the RR link input file.

1.1.9 RR-industry node

With an *RR - Industry node* you can simulate lateral discharges to and from the RR water system. This node type is typically used to simulate industrial water extraction and -return (e.g. water for process cooling). Demands can be specified as a time series. Discharges can be specified as a time series, or using a return flow percentage of the allocated demand time series (meaning that is the allocation is less than the demand, also the return flow will be reduced). However, when the model is a combined RR-1D model, often the demands and discharges are specified as lateral inflows or outflows in the 1D model.

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1.1.10 RR-links

The RR-nodes are typically connected by RR-links. These links indicate how the nodes are connected within RR. The standard RR-link just indicates the connection. There is no calculation associated with the RR-link.

Besides the standard RR-link there is a type called RR-sewerage link, typically from paved nodes to WWTP nodes indicating the connection of a sewer system to a treatment plant and from the treatment plant to RR-boundaries. There is no specific calculation associated with the RR-sewerage link. The main function of this type of link is for quick selection purposes within the GUI. Another link type is the RR-routing link. This is a link with additional computations, allowing to model delays quickly in RR using the Muskingum method. This is described in a separate section. This RR-routing link is an alternative for modelling delays with a 1D model. Nowadays, with 1D cross-section data abundantly available and fast 1D modelling, most often a 1D modelling approach of water courses with cross-sections, friction, etc. is applied.

1.1.11 RR Rural Technical Reference

This subsection contains more explanation on various aspects of the modelling of the various node types and concepts used within the RR-Rural nodes (paved area, unpaved area, greenhouses, RR-boundaries etc).

1.1.11.1 Alpha reaction factor

The reactionfactor α is an important parameter in the De Zeeuw-Hellinga equation which is used in RR to calculate the different components of the groundwater/subsurface flow:

$$q_t = q_{t-1}e^{-\alpha\Delta t} + (I + S)(1 - e^{-\alpha\Delta t}) \tag{1.1}$$

q_t	specific discharge at time t [m/d]
q_{t-1}	specific discharge at time $t-1$ [m/d]
Δt	time step [d]
α	reaction factor [$1/d$]
I	infiltration [m/d]
S	seepage (percolation) [m/d]

In RR, the De Zeeuw-Hellinga equation is used to calculate the following components of flow:

- ◇ groundwater drainage (towards drainpipes or channels);
- ◇ surface run-off;
- ◇ infiltration from open water.

For each process, the user must define specific reaction factors.

Reaction factor groundwater drainage
In RR, the ground can be divided into different ground layers, each with it's own α reaction factor. The total specific discharge is calculated by applying the De Zeeuw-Hellinga equation to all layers, and summing the result (see Figure 1.4)

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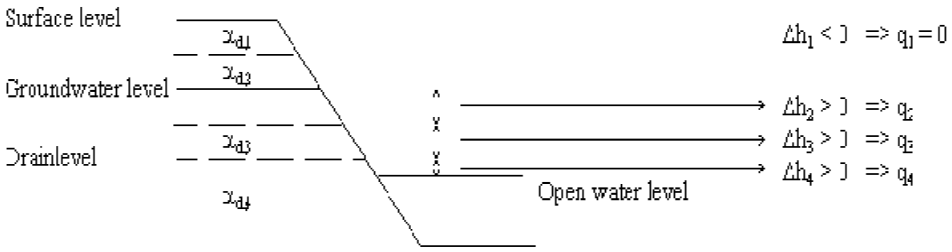


Figure 1.4: *Drainage levels*

Usually, the division between the ground layers is placed at the level of the drains, which are then simulated by giving the above drain level ($\alpha_{d,1}$ to $\alpha_{d,3}$ in the figure) a much higher value than the a below drain level ($\alpha_{d,4}$ in Figure 1.4).

To obtain indications for the values of the reaction factor one can:

- ◇ use the values in the Table below (Vademecum, 1988):

α	Discharge type
100–200	Surface runoff from steep slopes
1–10	Surface runoff from soils with impervious subsoil
0.3–0.7	Drainage discharge from well-drained agricultural soil
0.03–0.07	Discharge from grassland without drainage system

- ◇ Measure the decrease of discharge in time. This only can be done after a period of (heavy)rain, and no additional precipitation, and no (or very little) evaporation.
- ◇ Derive the α value from

$$q = \alpha \quad \mu \quad \Delta h \tag{1.2}$$

and

$$q = \frac{\Delta h}{W} \tag{1.3}$$

and equals

$$\alpha = \frac{1}{\mu W} \tag{1.4}$$

where

q	= specific discharge [m/d]
Dh	= groundwater level above drainage depth or open water level [m]
α	= reaction coefficient [$1/d$]
m	= storage coefficient [$-$]
W	= drainage resistance [d]

Reaction factor surface runoff

When the groundwater level branches the surface level or the precipitation excess exceeds the infiltration capacity, water is stored on land. When the storage on land is filled in an RR-unpaved node, the surface runoff process starts.

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Therefore, the user must define a reaction factor α_s. The above table gives indications for the values of this reaction factor.

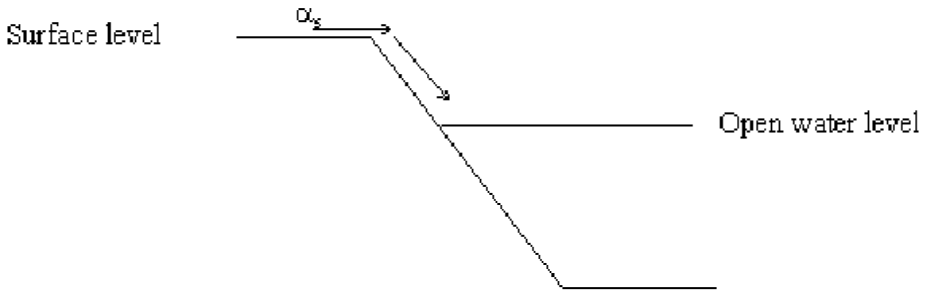


Figure 1.5: *Surface run-off*

Reaction factor infiltration from open water

In dry periods the groundwater level can decrease. When the groundwater level is lower than the open water level, infiltration from the open water occurs. In RR, this process is described by the modified De Zeeuw-Hellinga equation. Again, the user must define a specific reaction factor, α_i When using Ernst resistances, an infiltration resistance γ_i is specified.

1.1.11.2 Capillary rise

Capillary rise describes the unsaturated flow in the subsoil from the groundwater to the root zone. Capillary rise occurs in case of a water deficit in the root zone. Water deficit is defined as a water content less than equilibrium moisture storage. If the water content in the root zone exceeds equilibrium, the excess water percolates to the groundwater. The excess water is the potential root zone volume minus the equilibrium moisture content. To assess whether capillary rise or percolation will occur over a time step, the *potential root zone volume* is determined from the net precipitation and the evapotranspiration, based on the soil moisture storage in the root zone at the previous time step:

$$V'(t) = V(t - \Delta t) + (P_n - E)\Delta t \tag{1.5}$$

where:

$V'(t)$	potential root zone volume at time t [m]
$V(t - \Delta t)$	root zone volume after the previous time step [m]
P_n	net precipitation [m/d]
E	evapotranspiration [m/d]
Δt	time step [d]

In case of a water deficit in CAPSIM, the capillary rise flux is calculated in 2 steps. First the potential capillary rise flux is calculated. The potential capillary rise flux depends on the groundwater level, the soil physical unit and the root zone thickness:

$$q_{pot} = f(s, d_g, d_r) \tag{1.6}$$

with:

q_{pot}	Potential capillary rise [mm/d]
s	Soil physical unit [-]
d_g	Depth groundwater level [m]
d_r	Root zone thickness [m]

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The potential capillary rise flux is calculated with the 1-D, steady state simulation model CAPSEV (Wesseling, 1991), assuming pF=3 in the root zone. The potential capillary rise fluxes are tabulated.

Secondly CAPSIM uses the potential capillary rise flux, the actual moisture storage in the root zone and the equilibrium moisture storage to calculate the actual capillary rise flux, according to:

$$q_{act} = \begin{cases} q_{pot} \frac{v_{eq} - v_{act}}{v_{eq} - v_{pF3}} & v_{act} > v_{pF3} \\ q_{pot} & v_{act} < v_{pF3} \end{cases} \tag{1.7}$$

where:

q_{act}	Actual capillary rise flux [mm/d]
v_{eq}	Equilibrium soil moisture content [m]
v_{act}	Actual soil moisture content [m]

Equilibrium soil moisture content v_{eq}

The moisture storage of the root zone at equilibrium condition is calculated with the function:

$$V_{eq} = f(s, d_g, d_r) \tag{1.8}$$

where:

V_{eq}	the moisture storage of the root zone at equilibrium condition [m]
s	soil physical unit [-]
d_r	thickness of root zone [m]
d_g	depth of the groundwater level [m]

Soil moisture content at pF=3, (V_{pF3})

The moisture storage of the root zone at pF=3 is calculated assuming steady conditions and a groundwater level of 10 m below the soil surface.

1.1.11.3 Crop factors agricultural crops

The fixed input file with crop names and crop factors as a function of time. 1 year of data is enough, since it is assumed that the crop factors are constant over the years; the variation is taken into account in the reference evaporation data and not in the crop factors.

The header of the file contains the number of crops and crop names.

```
*Number of crops
16
*Names
'1 grass '
'2 corn '
'3 potatoes '
'4 sugarbeet '
'5 grain '
'6 miscellaneous '
'7 non-arable land '
'8 greenhouse area '
'9 orchard '
'10 bulbous plants '
'11 foliage forest '
'12 pine forest '
```

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```
'13 nature '
'14 fallow '
'15 vegetables '
'16 flowers '
*Year/Month/Day/Cropfact 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
 0000 1 1 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 2 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 3 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 4 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 5 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 6 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 7 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 8 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 9 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.00
 0000 1 10 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.0
 0000 1 11 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.0
 0000 1 12 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.0
 0000 1 13 0.95 1.00 1.00 1.00 1.00 0.95 1.00 0.00 1.00 0.00 0.90 1.20 0.95 1.00 1.00 0.0
etc.
```

1.1.11.4 Crop factors open water

The fixed input file with open water evaporation factors is a function of time. 1 year of data is enough; it is assumed that the evaporation factors are the same for all years.

```
*Surface water
1
*Names
'0.0 Surface water '
*Year/Month/Day/Cropfact 1
 0000 1 1 0.50
 0000 1 2 0.50
 0000 1 3 0.50
 0000 1 4 0.50
 0000 1 5 0.50
 0000 1 6 0.50
 0000 1 7 0.50
 0000 1 8 0.50
 0000 1 9 0.50
 0000 1 10 0.50
 0000 1 11 0.50
 0000 1 12 0.50
 0000 1 13 0.50
etc.
```

1.1.11.5 De Zeeuw-Hellinga drainage formula

To simulate the flow of groundwater towards the drainage system (i.e. drain pipes and/or channels) the modified equation of De Zeeuw-Hellinga is one of the available options in the Rainfall-Runoff module. The theory behind this equation is treated here.

The head loss of groundwater flowing to a drain can be subdivided in head loss caused by:

- ◇ vertical flow;
- ◇ horizontal flow;
- ◇ radial flow nearby the drain/channel;
- ◇ entrance in the drain/channel.

In the equation of De Zeeuw-Hellinga it is assumed that the head loss is mainly caused by radial flow and entrance loss. The head loss by horizontal flow and vertical flow is neglected.

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This equation is based on a reservoir containing a volume of ground. The reservoir discharges through a capillary tube. The discharge depends on the difference in pressure head. Thus:

$$Q = c \quad \Delta h \tag{1.9}$$

Q	discharge [m^3/d]
c	proportional coefficient [m^2/d]
Δh	difference in pressure head [m]

and

$$q = \frac{Q}{A} = c_2 h = \alpha \mu h \tag{1.10}$$

q	specific discharge [m/d]
Q	discharge [m^3/d]
A	area [m^2]
c_2	proportional coefficient [$1/d$]
h	groundwater level above drainage depth [m]
α	reaction coefficient [$1/d$]
μ	storage coefficient

The water balance for time dt is:

$$(I + S - q)\Delta t = \mu \Delta h \tag{1.11}$$

I	infiltration [m/d]
S	seepage (percolation) [m/d]
t	time [d]

This equals:

$$\Delta t = \frac{\mu}{I + S - \alpha \mu h} \Delta h \tag{1.12}$$

and eliminating the integration constant ($t = 0$, $q = q_0$) they lead to the equation of De Zeeuw-Hellinga:

$$q_t = q_0 e^{-\alpha t} + (I + S)(1 - e^{-\alpha t}) \tag{1.13}$$

q_t	specific discharge at time t [m/d]
q_0	specific discharge at time 0 [m/d]

With this equation the groundwater flow to a drain can be calculated for a constant $I + S$. When $I + S$ is not constant, the $I + S$ series can be seen as a series of constant $I + S$, positive and negative. For each constant $I + S$ in the series the q_t can be calculated, with t depending on the start of the component. Because the De Zeeuw-Hellinga equation is linear, superposition may be applied. It is easier though to calculate the q_t for several short successive periods wherein $I + S$ is constant. This results in the following modified equation:

$$q_t = q_{t-1} e^{-\alpha \Delta t} + (I + S)(1 - e^{-\alpha \Delta t}) \tag{1.14}$$

q_{t-1}	specific discharge at time $t - 1$ [m/d]
Δt	time step [d]

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For each time step infiltration and seepage must be constant.

The total volume of water flowing to the drain can be calculated by integrating the equation of De Zeeuw-Hellinga and multiplying it with the area:

$$V_t = A \int_0^{\Delta t} q_t \, dt \tag{1.15}$$

V_t Volume of groundwater flow to drain at time t $[m^3]$

V_t becomes:

$$V_t = A \int_0^{\Delta t} q_t \, dt \tag{1.16}$$

$$= A \left[-\frac{1}{\alpha} q_0 e^{\alpha t} + (I+S)(t + \frac{1}{\alpha} e^{-\alpha t}) \right]_0^{\Delta t} \tag{1.17}$$

$$= A \left[-\frac{1}{\alpha} q_0 e^{\alpha \Delta t} + (I+S)(\Delta t + \frac{1}{\alpha} e^{-\alpha \Delta t}) + \frac{1}{\alpha} q_0 - \frac{1}{\alpha}(I+S) \right] \tag{1.18}$$

$$= A \left[\frac{q_0 - (I+S)}{\alpha} (1 - e^{-\alpha \Delta t}) + (I+S)\Delta t \right] \tag{1.19}$$

and

$$q_0 = \alpha \mu \Delta h_0 \tag{1.20}$$

Substituting gives:

$$V_t = \frac{A\alpha\mu\Delta h - A(I+S)}{\alpha} (1 - e^{-\alpha \Delta t}) + A(I+S)\Delta t \tag{1.21}$$

The average discharge $Q_{t,average}$ $[m^3/d]$ is computed by

$$Q_{t,average} = \frac{V_t}{\Delta t} \tag{1.22}$$

Thus the average discharge in a time step to the drain as is used by the Rainfall-Runoff module is:

$$Q_{t,average} = \frac{A\alpha\mu\Delta h - A(I+S)}{\alpha \Delta t} (1 - e^{-\alpha \Delta t}) + A(I+S) \tag{1.23}$$

This is the so-called modified equation of De Zeeuw-Hellinga. Two important differences from the 'normal' equation of De Zeeuw-Hellinga:

- ◇ the storage coefficient μ is explicit, and
- ◇ the pressure head Δh is in the Rainfall-Runoff module calculated per time step as the difference between the groundwater level and the open water level.

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1.1.11.6 DrainageDeltaH option

```
DrainageDeltaH=-1 ! -1=parallell systems (default), 0=stacked system;
```

Given the definition of the drainage levels, and the Hellinga-de Zeeuw alfa reaction factors α or the Ernst drainage resistances γ, there are two options for computing the relevant heads on which the α factors or Ernst drainage resistances are applied to. These options are is a 'parallell' option (default), and a 'stacked' option. These options are illustrated in the next graphs.

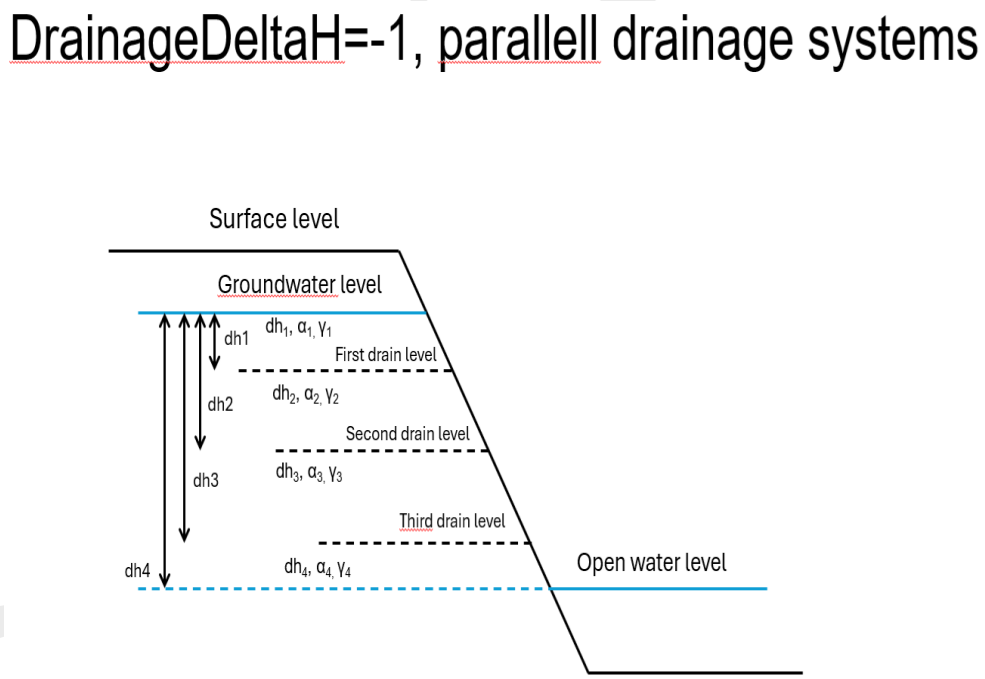


Figure 1.6: *Parallell drainage systems (default:* `DrainageDeltaH=-1`*)*

DrainageDeltaH=0, stacked drainage systems

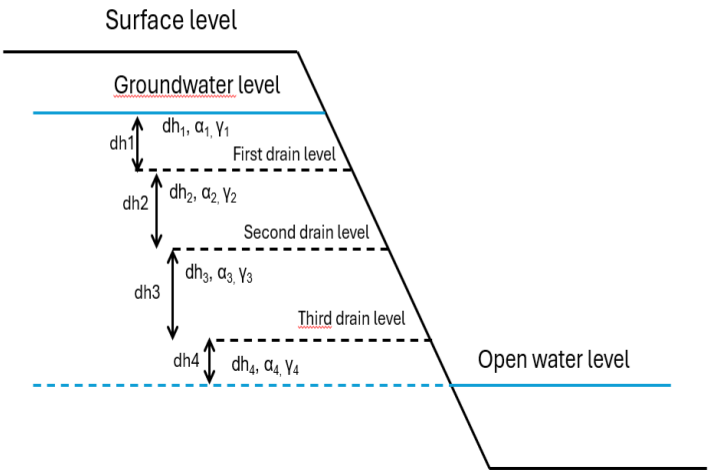


Figure 1.7: *Stacked drainage systems (*`DrainageDeltaH=0`*)*

See also: <Delft_3B.Ini> as described in RR input file formats.

1.1.11.7 Dry Weather Flow (DWF)

The DWF or Dry Weather Flow can be taken into account at the RR-paved node. Also the DWF is part of the formulation in the RR-NWRW node or RR-Urban concept, based on the Nationale Werkgroep Riolering en Waterkwaliteit (NLingenieurs, 1978).

The dry weather flow represents return flow from domestic use, like showers, washing, flushing the toilets, etc.

The dry weather flow can be specified in one of the following ways:

- ◇ as a constant flow [m^3/s] during the whole day
- ◇ as a variable flow [m^3/s] for 24 hours; 1 value for each hour.
- ◇ as a constant flow [m^3/s] per day per person, multiplied with the number of persons
- ◇ as a variable flow [m^3/s] for each hour per person, multiplied with the number of persons

For paved area, the dry weather flow is discharged into the DWF sewer in case of a separated or improved separated system. In case of a mixed system it is discharged into the mixed sewer system, where it is mixed with sewer inflow due to rainfall.

For a NWRW node the dry weather flow is added to the computed sewer inflow with the NWRW rainfall-runoff model.

1.1.11.8 Ernst drainage formula

One of the options of modelling the drainage from unpaved area towards open water is the Ernst formulation (Ernst, 1978). Other options available are the De Zeeuw-Hellinga formulation and the Krayenhoff van de Leur formulation.

The following figure illustrates the principles of the Ernst method.

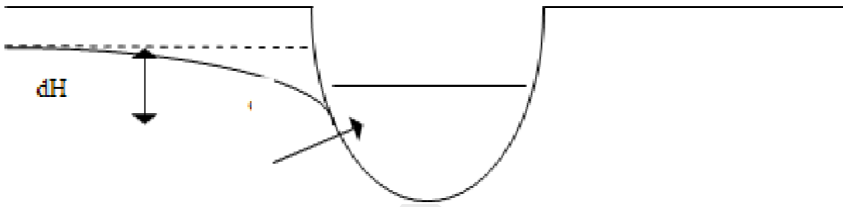


Figure 1.8: *Principles of Ernst method*

The equation reads:

$$q = \frac{dH}{\gamma f} \tag{1.24}$$

where:

q	drainage $[m/d]$
dH	difference between groundwater level and drainage basis $[m]$
γ	drainage resistance $[d]$
f	factor (0.65 - 0.85 (Ernst, 1978)), depending on the shape of the groundwatertable $[-]$

The f factor can not be entered via the User Interface of SOBEK and Delft3D FM Suite 1D2D. Instead, the user has to take this factor into account in the drainage resistance.

1.1.11.9 Evaporation (when using CAPSIM)

The actual evapotranspiration ET_{act} is calculated as:

$$ET_{act} = \alpha_E \quad E_r \tag{1.25}$$

with

$$\alpha_E = f\left(\frac{V}{V_{eq}}\right) \tag{1.26}$$

where:

α_E	relative evapotranspiration factor $[-]$
V	actual soil moisture storage in the root zone $[m]$
V_{eq}	equilibrium soil moisture storage in the root zone $[m]$
E_r	potential evapotranspiration $[m]$

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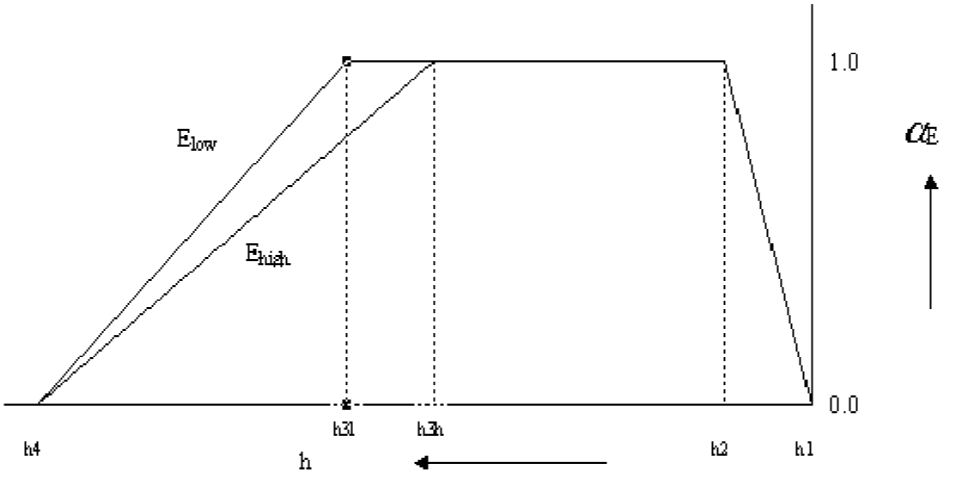


Figure 1.9: *Reduction coefficient for root water uptake, α_E, as a function of soil water pressure head h (Feddes et al., 1978).*

Water uptake by roots is zero at soil water pressure h_4 which is assumed to be the wilting point, see Figure 1.9. Soil water pressure h_3 is called the reduction point. In between the wilting point and the reduction point the potential evapotranspiration rate is linearly reduced. The location of the reduction point depends on the potential evapotranspiration. If the potential evapotranspiration is less than or equal to 1 mm/d then the curve E_{low} is used. If the potential evapotranspiration is equal to or greater than 5 mm/d then the curve E_{high} is used. In between CAPSIM linearly interpolates between the reduction curves E_{low} and E_{high}.

Between soil water pressures h_2 and h_3 the evapotranspiration is at maximum ('potential'). As a result of oxygen deficiency in the root zone, water uptake is hampered for some crops between soil water pressures h_2 and h_1. CAPSIM does not take this reduction into account ($\alpha_E = 1$ between soil water pressures h_2 and h_1).

CAPSIM doesn't compute the soil waterpressure head directly, but uses the relative root zone storage V/V_{eq}. The relative root zone storages for h_4, h_{3l}, h_{3h}, h_2 and h_1 are calculated by CAPSEV and tabulated in CAPSIM. Then Figure 1.9 can be better explained by:

$$\alpha_E = 0 \quad \text{when} \quad 0 \leq V/V_{eq} < V/V_{eq}(h_4) \tag{1.27}$$

$$0 \leq \alpha_E \leq 1 \quad \text{when} \quad V/V_{eq}(h_4) \leq V/V_{eq} < V/V_{eq}(h_{3l} \text{ or } h_{3h}) \tag{1.28}$$

$$\alpha_E = 1 \quad \text{when} \quad V/V_{eq} \geq V/V_{eq}(h_{3l} \text{ or } h_{3h}) \tag{1.29}$$

1.1.11.10 Evapo(transpi)ration

Open water

For open water, related to Makkink evapotranspiration, the following 'crop' factors are used. These values are based on Hooghart and Lablans (1988).

Decade[1)]	**Value**	**Decade**	**Value**	**Decade**	**Value**
jan-01	0.50	may-01	1.30	sep-01	1.17

jan-02	0.50	may-02	1.30	sep-02	1.17
jan-03	0.70	may-03	1.30	sep-03	1.17
feb-01	0.80	jun-01	1.31	oct-01	1.00
feb-02	1.00	jun-02	1.31	oct-02	0.90
feb-03	1.00	jun-03	1.31	oct-03	0.80
mar-01	1.20	jul-01	1.29	nov-01	0.80
mar-02	1.30	jul-02	1.27	nov-02	0.70
mar-03	1.30	jul-03	1.24	nov-03	0.60
apr-01	1.30	aug-01	1.21	dec-01	0.50
apr-02	1.30	aug-02	1.19	dec-02	0.50
apr-03	1.30	aug-03	1.18	dec-03	0.50

[1)] Decades are parts of a month, defined as: first 10 days, second 10 days and rest of the month.

Related input file:
crop factor open water

Unpaved areas

The potential evapotranspiration in unpaved areas depends on the vegetation. The standard way to determine the potential evapotranspiration is to multiply the reference evaporation (p.e. Makkink) by a so called crop factor. The crop factor can differ per vegetation, in time and per location on earth. The default crop factors are defined for the Dutch situation for a limited number of crops, listed in the table below.

crop nr.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
decade	**grass**	**corn**	**potatoes**	**sugarbeet**	**grain**	**miscellaneous**	**non-arable land**	**greenhouse area**	**orchard**	**bulbous plants**	**foliage forest**	**pine forest**	**nature**	**fallow**	**vegetables**	**flowers**
jan-01	0.95	1.00	1.00	1.00	1.00	0.95	1.00	0.00	1.00	1.00	0.90	1.20	0.95	1.00	1.00	0.00
jan-02	0.95	1.00	1.00	1.00	1.00	0.95	1.00	0.00	1.00	1.00	0.90	1.20	0.95	1.00	1.00	0.00
jan-03	0.95	0.71	0.71	0.71	0.71	0.95	0.71	0.00	0.71	0.71	0.90	1.20	0.95	0.71	0.71	0.00
feb-01	0.95	0.63	0.63	0.63	0.63	0.95	0.63	0.00	0.63	0.63	0.90	1.20	0.95	0.63	0.63	0.00
feb-02	0.95	0.50	0.50	0.50	0.50	0.95	0.50	0.00	0.50	0.50	0.90	1.20	0.95	0.50	0.50	0.00
feb-03	0.95	0.40	0.40	0.40	0.40	0.95	0.40	0.00	0.40	0.40	0.90	1.20	0.95	0.40	0.40	0.00
mar-01	0.95	0.33	0.33	0.33	0.33	0.95	0.33	0.00	0.33	0.33	1.00	1.20	0.95	0.33	0.33	0.00
mar-02	0.95	0.23	0.23	0.23	0.23	0.95	0.23	0.00	0.23	0.23	1.00	1.20	0.95	0.23	0.23	0.00
mar-03	0.95	0.23	0.23	0.23	0.23	0.95	0.23	0.00	0.23	0.78	1.00	1.20	0.95	0.23	0.23	0.00
apr-01	1.00	0.23	0.23	0.23	0.65	1.00	0.23	0.00	1.04	0.91	1.05	1.20	1.00	0.23	0.23	0.00
apr-02	1.00	0.23	0.23	0.23	0.78	1.00	0.23	0.00	1.04	0.91	1.05	1.20	1.00	0.23	0.52	0.00
apr-03	1.00	0.23	0.23	0.23	0.91	1.00	0.23	0.00	1.04	0.91	1.05	1.20	1.00	0.23	0.65	0.00
may-01	1.00	0.52	0.15	0.52	1.04	1.00	0.15	0.00	1.43	1.04	1.15	1.20	1.00	0.15	0.78	0.00
may-02	1.00	0.52	0.65	0.52	1.04	1.00	0.15	0.00	1.43	1.04	1.15	1.20	1.00	0.15	0.91	0.00
may-03	1.00	0.52	0.91	0.52	1.04	1.00	0.15	0.00	1.43	1.04	1.15	1.20	1.00	0.15	1.04	0.00
jun-01	1.00	0.79	1.05	0.79	1.18	1.00	0.15	0.00	1.57	1.05	1.20	1.20	1.00	0.15	1.18	0.00
jun-02	1.00	1.05	1.05	1.05	1.18	1.00	0.15	0.00	1.57	1.05	1.20	1.20	1.00	0.15	1.18	0.00

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crop nr.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
decade	grass	corn	potatoes	sugarbeet	grain	miscellaneous	non-arable land	greenhouse area	orchard	bulbous plants	foliage forest	pine forest	nature	fallow	vegetables	flowers
jun-03	1.00	1.18	1.18	1.18	1.18	1.00	0.15	0.00	1.57	0.92	1.20	1.20	1.00	0.15	1.18	0.00
jul-01	1.00	1.29	1.16	1.16	1.03	1.00	0.16	0.00	1.68	0.77	1.25	1.20	1.00	0.16	1.03	0.00
jul-02	1.00	1.27	1.14	1.14	0.89	1.00	0.16	0.00	1.65	0.64	1.25	1.20	1.00	0.16	0.76	0.00
jul-03	1.00	1.24	1.12	1.12	0.74	1.00	0.16	0.00	1.61	0.50	1.25	1.20	1.00	0.16	0.16	0.00
aug-01	1.00	1.21	1.09	1.09	0.61	1.00	0.17	0.00	1.33	0.17	1.10	1.20	1.00	0.17	0.17	0.00
aug-02	1.00	1.19	0.83	1.07	0.17	1.00	0.17	0.00	1.31	0.17	1.10	1.20	1.00	0.17	0.17	0.00
aug-03	0.90	1.18	0.83	1.06	0.25	0.90	0.25	0.00	1.18	0.25	1.10	1.20	0.90	0.25	0.25	0.00
sep-01	0.90	1.17	0.70	1.05	0.26	0.90	0.26	0.00	1.17	0.26	1.05	1.20	0.90	0.26	0.26	0.00
sep-02	0.90	1.17	0.26	1.05	0.26	0.90	0.26	0.00	1.17	0.26	1.05	1.20	0.90	0.26	0.26	0.00
sep-03	0.90	1.17	0.26	1.05	0.26	0.90	0.26	0.00	1.17	0.26	1.05	1.20	0.90	0.26	0.26	0.00
oct-01	0.90	0.40	0.30	0.40	0.30	0.90	0.30	0.00	0.30	0.30	1.00	1.20	0.90	0.30	0.30	0.00
oct-02	0.95	0.45	0.44	0.44	0.44	0.95	0.44	0.00	0.44	0.44	1.00	1.20	0.95	0.44	0.44	0.00
oct-03	0.95	0.50	0.50	0.50	0.50	0.95	0.50	0.00	0.50	0.50	1.00	1.20	0.95	0.50	0.50	0.00
nov-01	0.95	0.50	0.50	0.50	0.50	0.95	0.50	0.00	0.50	0.50	0.95	1.20	0.95	0.50	0.50	0.00
nov-02	0.95	0.71	0.71	0.71	0.71	0.95	0.71	0.00	0.71	0.71	0.95	1.20	0.95	0.71	0.71	0.00
nov-03	0.95	0.83	0.83	0.83	0.83	0.95	0.83	0.00	0.83	0.83	0.95	1.20	0.95	0.83	0.83	0.00
dec-01	0.95	1.00	1.00	1.00	1.00	0.95	1.00	0.00	1.00	1.00	0.90	1.20	0.95	1.00	1.00	0.00
dec-02	0.95	1.00	1.00	1.00	1.00	0.95	1.00	0.00	1.00	1.00	0.90	1.20	0.95	1.00	1.00	0.00
dec-03	0.95	1.00	1.00	1.00	1.00	0.95	1.00	0.00	1.00	1.00	0.90	1.20	0.95	1.00	1.00	0.00

Related input file:
crop factor agricultural crops

Remarks:

- ◇ The crop factor file is specified for a limited number of crops and suitable for the Dutch situation or for simular conditions only. In other situations (or other crops) please define your own crop factors. This can be done by changing the input file <../../fixed/cropfact>. We recommend saving the default crop factor file first. We also recommend to save the changed crop factor file at a different location as well, because when installing a new release of SOBEK the crop factor file could be replaced by a new default crop factor file.
- ◇ The values in the table above are based on the Makkink evapotranspiration. The used information comes from the Dutch Cultuurtechnisch Vademecum (3rd edition), Hooghart, Spieksma et. al, Lysimeter results and PAWN (DEMGEN) documentation.
- ◇ The growing season of the specified crops is typically somewhere between April and October. For instance, corn is typically grown from May-01 until October-01. Bulbous plants are early crops, from March-03 until July-03. Outside the growing season the crop factor values are the same as the values for fallow (bare soil).
- ◇ Non-arable land is assumed to be the same as fallow.
- ◇ Miscellaneous is assumed to be the same as grass.
- ◇ Nature depends entirely on the kind of nature. Therefore it is assumed to be the same as grass.
- ◇ For greenhouse crops (greenhouse and greenhouse-flowers), the values are set to zero. For proper use, please use the greenhouse node. Other flowers are assumed to be bulbous plants.
- ◇ For vegetables peas and beans are assumed.
- ◇ For both open water as well as unpaved areas a fatal error will occur when a crop factor

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is used with a value lower than 0 or higher than 2.5, and RR will stop.
- ◇ When a crop factor with value 0 is used, a warning will be given in the log file (<SOBEK_3B.log>). The program continues.

The actual evapotranspiration can be determined by using the potential evapotranspiration (default) or by taking into account the reduction for root water uptake as a function of soil water pressure head (using CAPSIM).

1.1.11.11 Hydrologic Cycle

Water circulates through the hydrosphere through a maze of paths, in a cycle without a beginning or an end. Water evaporates from the oceans, is transported to other parts of the atmosphere, and precipitates on the land or the oceans. From the land, it will eventually branch to the seas via different processes known as infiltration, percolation, groundwater flow and river flow. This cycle of processes is known as the *hydrologic cycle*.

Basically, the rainfall-runoff module focuses on the following transport processes:

- ◇ precipitation
- ◇ evapo(transpi)ration
- ◇ surface runoff
- ◇ infiltration
- ◇ (Drainage) outflow
- ◇ Seepage
- ◇ Percolation

In the RR rural concept, the user can choose from many different ways to model the flows of water over and under the ground surface, depending on the area type. The following area types are available:

- ◇ paved area
- ◇ unpaved area
- ◇ greenhouse area
- ◇ open water (precipitation and evaporation only)
- ◇ industrial area
- ◇ waste water treatment plants

Please refer to the corresponding sections for more information about the way the water transport and storage processes are modelled.

1.1.11.12 Improved separated sewer

The sewer type of a paved area node can be one of the following types:

- ◇ mixed sewer
- ◇ separated sewer
- ◇ improved separated sewer

A separated sewer system has separate sewer systems for rainfall and dry weather flow. The improved separated system is a small modification of a separated system, designed to further reduce the sewer overflows of a separated system. Although overflows from the rainfall sewer system do not contain waste loads originating from dry weather flows (DWF), they do contain waste loads from street surfaces, and thus can have negative impacts on water quality of the

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receiving water (the so-called ‘first flush’).

The improved separated sewer system has a connection between the rainfall sewer and the DWF sewer. This connection is used to store overflows from the rainfall sewer in the DWF sewer, if storage in the DWF sewer is available and no overflow of the DWF sewer is caused. If storage in the DWF sewer is not enough to store the overflow of the rainfall sewer system, the rainfall sewer will still spill into open water (but less than in case of a normal separated system).

1.1.11.13 Infiltration

Infiltration is the process by which water infiltrates from the surface of the ground into the *root zone* (subsoil), and is thus a part of the *Hydrologic Cycle*. The infiltration capacity is influenced by many factors, like the condition of the soil surface and its vegetative cover, the soil properties and the current moisture content of the soil.

In RR, the infiltration capacity of the unpaved areas is considered to have a constant value in time, and can be entered in either mm/hour or mm/day.

1.1.11.14 Infiltration from open water

In dry periods the groundwater level can decrease. When the groundwater level is lower than the open water level, infiltration from the open water occurs. The infiltration rate depends on 1 the difference between open water level and groundwater level and 2 the resistance of the channel

In RR, this process is described by the modified De Zeeuw-Hellinga equation, the Ernst approeach or the de Krayenhoff van de Leur approach. When the user selects the De Zeeuw-Hellinga approach, the user must define a specific reaction factor, α; in case of the Ernst approach, a resistance has to be specified (see separate sections on these approaches). The difference between the open water level and groundwater level is computed by RR.

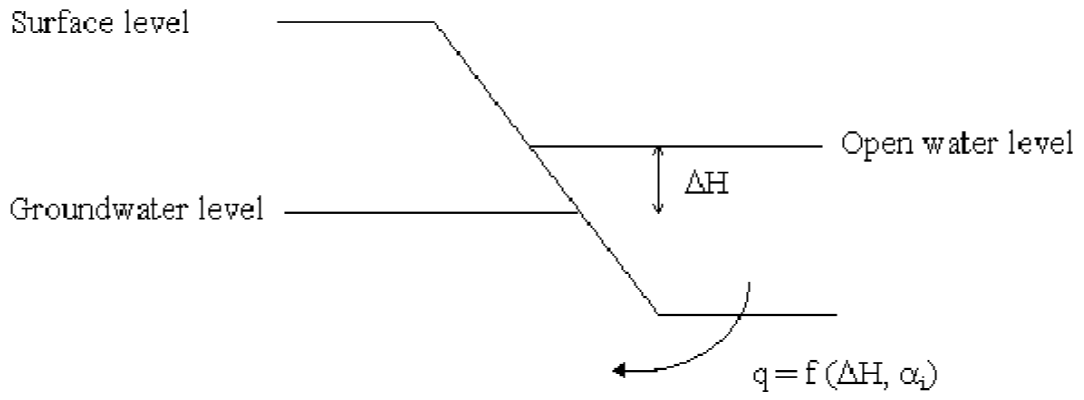


Figure 1.10: *Infiltration from open water*

1.1.11.15 Irrigation unpaved area

For the unpaved area, default simulation is without irrigation. However, for some applications it can be important to add irrigation. These options are not yet supported by the GUI, but can be easily added in the unpaved inputfile following the file descriptions of this manual (see: <unpaved.3B.Ini>).

Typically the irrigation is triggered when the soil moisture content of the unsaturated zone is low (e.g. corresponding with dry equilibrium condition of pF=2.7) Irrgation can be specified from local groundwater within the unpaved node, from local surface water (RR open water or boundary node connected with 1D system) or from external sources outside the modelled domain (which could be deep groundwater or water supplied from outside the area). Usually when irrigation is used in the model, the unsatured zone is also modelled using CAPSIM. However, this is not necessary. Note that in case of modelling without CAPSIM, evaporation reduction is not modelled, and irrigation from unpaved groundwater makes no sense since the irrigated water is taken from groundwater, added to the rain, used for evaporation and infiltration, and infiltration is added to the groundwater again. With CAPSIM, evaporation reduction is modelled ((See also section 1.1.11.9).

Irrigation is modelled following common Dutch practices. It is used in the irrigation season (in the Netherlands: between April 1st and October 1st) and triggered when the soil moisture content is low (typically pF=2.7). When reaching this critical level, farmers start irrigating until their targed soil moisture content (say pF=2.4) is reached. For the first irrigation in the season, an initial irrigation amount is supplied (e.g. 12.0 mm/day) with an initial duration (e.g. 2 days). For next irrigation supplies, a maximum gift in mm/day is specified (e.g. 4.0 mm/day) and a minimum waiting period (e.g. 5 days) after the previous irrigation supply.

Irrigation losses are taken into account by specifying an irrigation efficiency factor (typically 1.2), meaning that the actual sprinkling gift is to be multiplied with this factor to get the actual abstraction(from surface water, groundwater or external sources). The losses are assumed to be evaporation losses. The irrigation period within the day is typically the whole day, but could be specified as a subset of hours during the day in which the specified initial or or maximum gift is supplied.

Considering a small example for grass on soiltype Capsim soil type 1 (peat soil with earthed topsoil), where grass has a root zone depth of 40cm. Following the CAPSIM tables, the following relation between pF, groundwater level and equilibrium soil moisture content is derived. Note that when using CAPSIM the unsaturated content will drop below the equilibrium soil moisture content due to crop evapotranspiration, and the dry conditions of pF=2.4 or pF=2.7 are reached at higher groundwater levels. In case of not using CAPSIM, the groundwater levels from the relation are used which may be a reason to specify lower critical and target pF values than in case of using CAPSIM.

With an example unpaved area with grass, rootzone depth 40 cm, on CAPSIM soil type 1, using a critical pF value of 2.7, a target pF value of 2.4, initial supply 2.5 or 6 mm/day and duration 5 days, and next supplies with 4 or 12 mm/day, irrigation from groundwater, surface water or external supply, or no irrigation, the following graphs show some output results.

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pF	GWL	equilibrium SMC
1.0	0.1	249.4
2.0	1.0	219.4
2.3	2.0	199.2
2.4	2.52	192.5
2.5	3.16	186.
2.6	3.98	179.
2.7	5.01	172.7
3.0	10.0	154.7

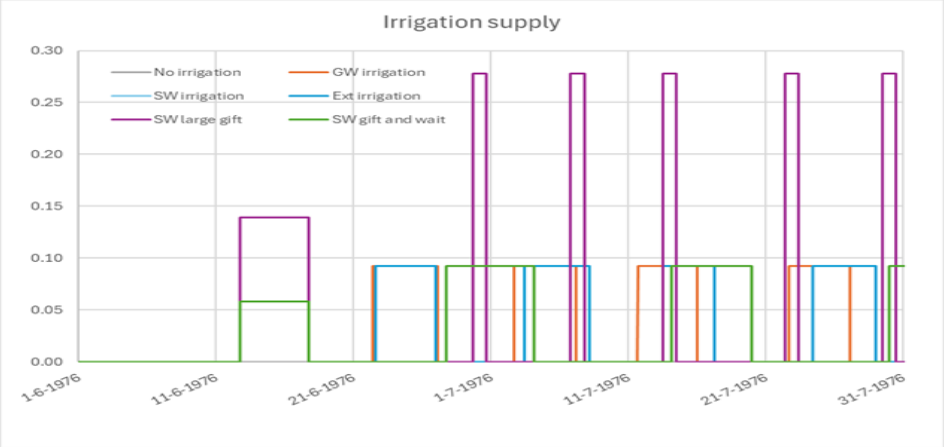


Figure 1.11: *Irrigation supply for different cases*

The irrigation supply is zero for the case without irrigation (grey line). Another distinguishable line is the line with irrigation from surface water with larger gifts (purple line). The other lines, which have the same parameters for irrigation but only irrigate from different sources, show similar results, only with a shift in time. The irrigation from local phreatic groundwater leads quicker to a new irrigation supply, whereas the irrigation from surface water and external supply have equal timing. When a minimum period between successive irrigation supplies is specified, this influences the timing of the next irrigation supply since both the soil moisture needs to be below the critical pF value, and the minimum number of days after the previous irrigation supply has to be taken into account.

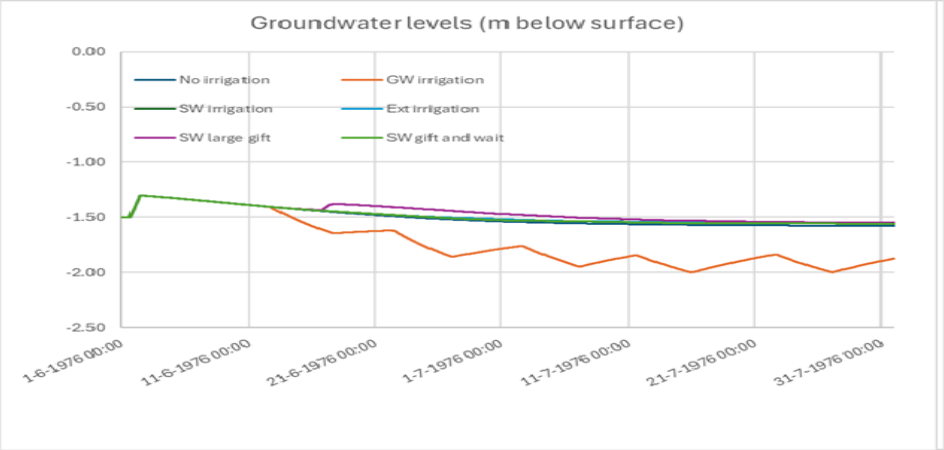


Figure 1.12: *Groundwater level for different cases*

The groundwater levels for the case with irrigation from local groundwater drop the fastest, while the irrigation with large gifts from surface water or external sources leads to a slightly higher groundwater level than the other options. The groundwater levels for cases with irrigation from surface water or external water, or no irrigation at all are very similar (with the level being the lowest for the case without irrigation).

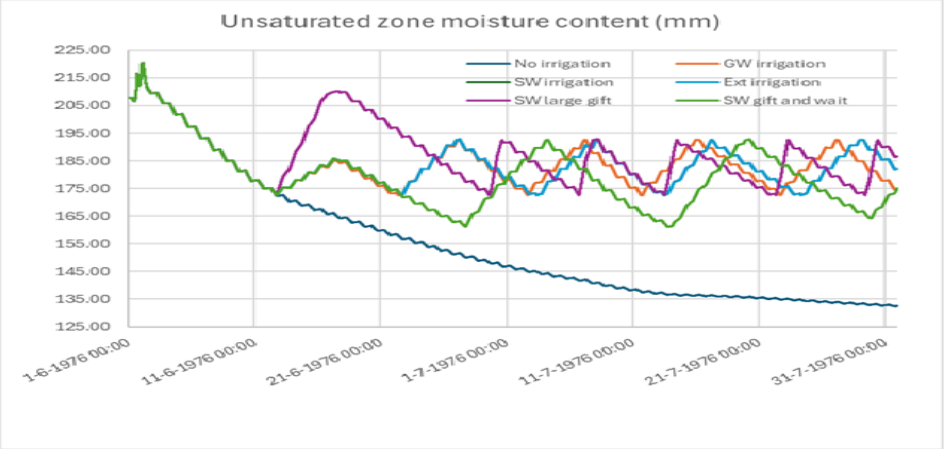


Figure 1.13: *Unsaturated Zone Soil Moisture Content for different cases*

The unsaturated zone content is dropping the most for the case without irrigation (dark blue line), while it rises highest for the case with the large gift. This is because of the specified initial duration, which causes the surface water irrigation to continue above the target pF value 2,4 (unsatured zone content 192.5 mm). The case with a longer minimum period between irrigation supplies (light green line) shows that due to this higher setting, irrigation starts later and causes dropping of the unsaturated zone content below 172.7 mm (pF=2.7)

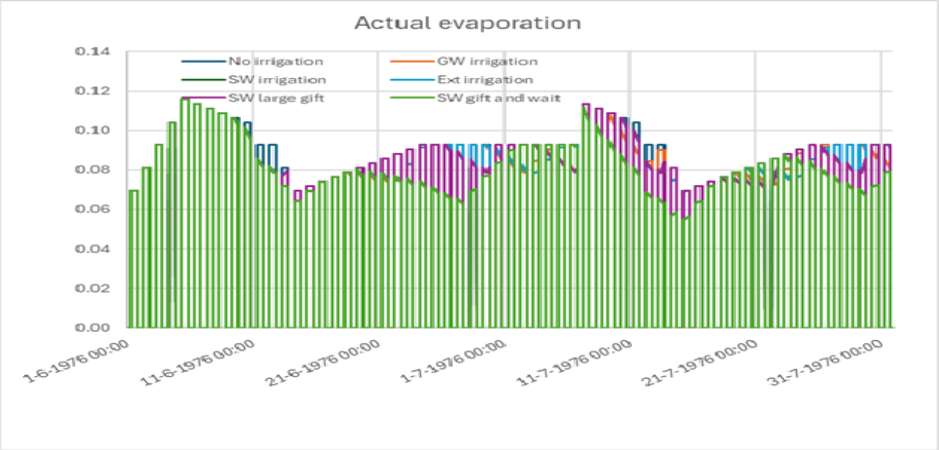


Figure 1.14: *Actual evaporation for different cases*

The actual evaporation shows the daily pattern which is imposed in this calculation (only evaporation between 7 in the morning and 7 in the evening) The potential evaporation in the 2-month calculation is different for each day, but it can be derived from the figure that evaporation reduction is also different due to the differences in timing of the irrigation and the actual irrigation supplies and soil moisture content. Without irrigation, the actual evaporation at the end of July has dropped to 0.025 m3/s, while with irrigation it is between 0.07 and 0.09 m3/s depending on the selected irrigation parameter values.

1.1.11.16 Krayenhoff van de Leur drainage formula

One of the options of modelling the runoff for unpaved area is the Krayenhoff van de Leur formulation. Another available option is the de Hellinga-de Zeeuw formulation.

The following figure illustrates the principles of the Krayenhoff van de Leur method.

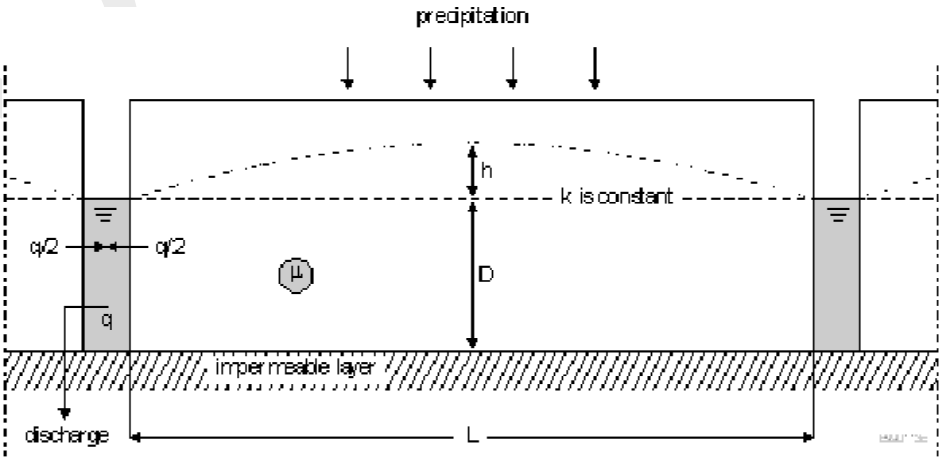


Figure 1.15: *Drainage according to Krayenhoff van de Leur*

The reservoir coefficient j is either specified directly, or calculated as:

$$j = \frac{\mu L^2}{\pi^2 k_D} \tag{1.30}$$

The (cumulative) drainage flow q is computed as:

$$q = \frac{8P}{\pi^2} \sum_{n=1,3,5,\ldots}^{n\to\infty} \frac{1}{n^2}(1 - e^{-n^2 \frac{t}{j}}) \tag{1.31}$$

and the (cumulative) highest groundwater level between the ditches as computed as:

$$h = \frac{\pi^2 P}{8\mu} j \left(1 - \frac{32}{\pi^3} \sum_{n=1,3,5,\ldots}^{n\to\infty} \frac{1}{n^3} e^{-n^2 \frac{t}{j}}\right) \tag{1.32}$$

where:

j	reservoir coefficient [d]
m	storage coefficient (specific yield) [$-$]
L	distance between drainage ditches [m]
k_D	soil transmissibility (transmissivity) [m^2/d]

k_D can be derived from:

k	soil permeability [m/d]
D	thickness of permeable layer [m]
P	precipitation [m]
q	(cumulative) drainage flow [m/d]
h	difference between highest groundwater level and ditch level [m]
t	time [d]

Given the area A in [m^2], precipitation and drainage flow can be converted to [m^3/s]

1.1.11.17 Minimum filling percentage for greenhouse storage basin

When the water level in the greenhouse storage basins becomes equal/lower than this minimum filling percentage, the withdrawal of water out of the basins will be stopped. Default this minimum filling percentage of the storage basins is set at 10 % in RR.

1.1.11.18 Mixed sewer

The sewer type of a paved area node can be one of the following types:

- ◇ mixed sewer
- ◇ separated sewer
- ◇ improved separated sewer

A mixed sewer system means that there is only one system for the discharge of both rainfall water and dry weather flow (return flow from domestic water use, e.g. showers, toilets, etc.)

Spilling from the sewer system after heavy rainfall thus contains a mix of rainfall water and domestic return flow. Spilling from a mixed sewer system therefore in general has a negative effect on the water quality of the receiving open water.

1.1.11.19 Paved area node

General

The paved area node is used to simulate the rainfall-runoff process on paved or impervious areas.

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A paved area is characterized by:

- ◇ area [ha]
- ◇ maximum storage on street [mm]
- ◇ maximum storage in sewer [mm]
- ◇ sewer pump capacity [m^3/s]
- ◇ sewer type (mixed, separated, or improved separated system)
- ◇ DWF (dry weather flow)

Important processes are surface runoff or sewer overflow, occurring when the storage on land or storage in the sewer exceeds its maximum and rapid runoff occurs.

1.1.11.20 Paved node surface runoff

When the amount of water on the RR-paved surface exceeds the maximum storage, surface runoff will occur. The surface runoff can be implemented using either

- ◇ no delay,
- ◇ using a runoff delay coefficient, or
- ◇ using a QH-relation.

The first option, no delay, means all surface runoff branches the outflow point in the same timestep. Surface runoff can be blocked however if the open water level at the connecting node exceeds the surface level of the paved node. But when using the 'no delay' option the surface storage will immediately runoff as soon as the water level drops below the paved surface level. This may lead to very spiky discharges from the paved node to the RR-open water or RR-Boundary node.

The second option, using a runoff delay coefficient, is similar to the RR-Urban runoff model. Also see the description of delay of runoff in the RR-Urban documentation, 1.1.18.2. This is formulating the surface runoff using the rational method ($q = ch$). The runoff delay coefficient c is specified as a number between zero and one (1/min). A coefficient of one means all runoff occurs in 1 minute, while a coefficient of 0.1 means only 10 % of the excess volume will branch the open water in 1 minute.

The impact of the runoff delay coefficient on the runoff pattern is shown in the following graph. The graph shows the rainfall time-series used, and the runoff using runoff delay coefficients of 0.1, 0.2 and 0.5. It is assumed that there is no infiltration or sewer, and that the maximum storage on the street is zero.



Note: The example Figure 1.16 is actually taken from a simulation of the RR-Urban inflow model, but the behaviour for the RR-Paved node is the same.

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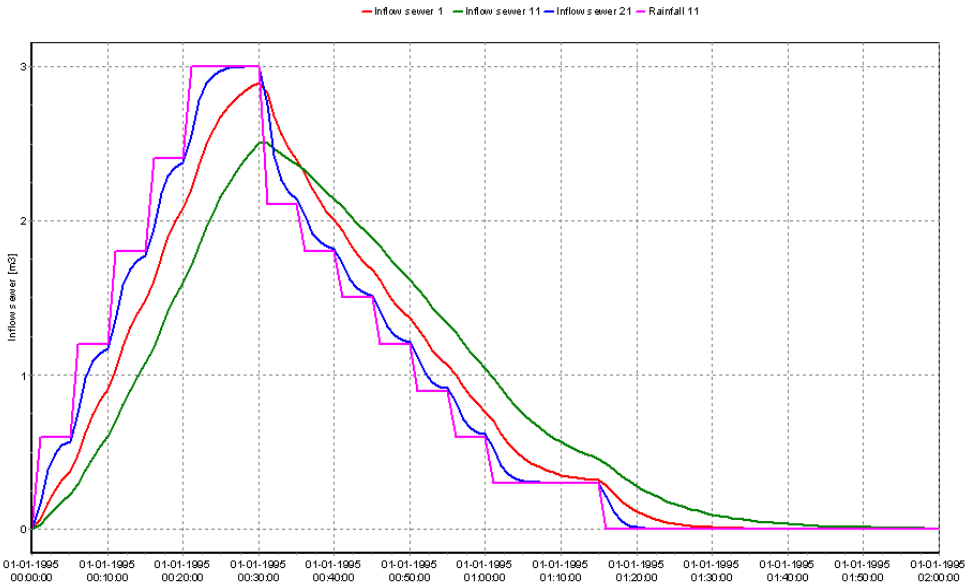


Figure 1.16: *Impact of delay coefficient on runoff pattern*

The third option, using a QH-relation, allows limiting of the Q based on the water level h. Based on the interpolation table defined by the user a maximum Q is determined with which the overflowing discharge is limited.

Note: The QH relation option is not yet available in the GUI of Delft3D FM Suite 1D2D.

The used h is the water level of the open water node or RR-CF-connection node downstream of the paved node, where the overflowing discharge of the RR-paved node is discharged to. See the following two examples:

1 Figure 1.17 shows the result of water overflowing from two paved nodes to open water nodes without delay of discharge from the paved nodes.
2 Figure 1.18 shows the discharges after using a QH-relation.

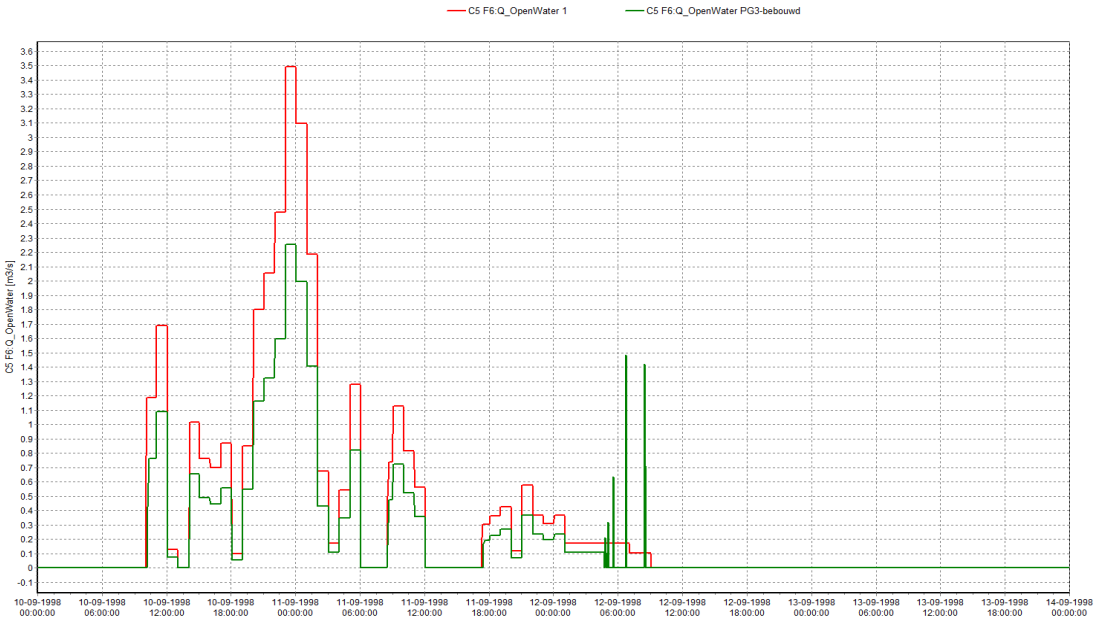


Figure 1.17: *Discharge from Paved to Open Water without delay*

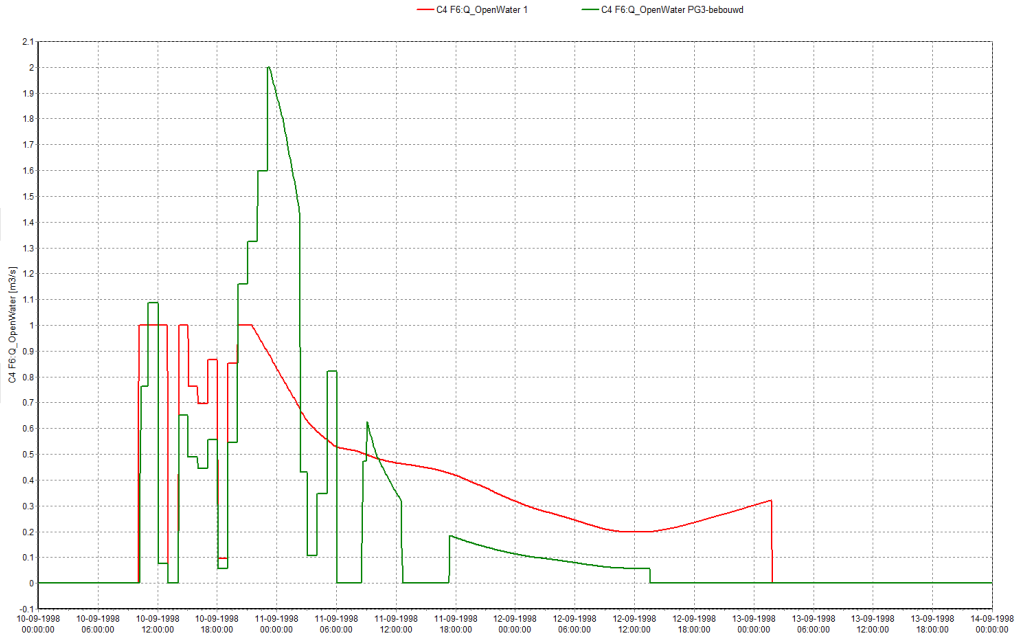


Figure 1.18: *Discharge from Paved to Open Water using QH-relation*

1.1.11.21 Percolation

If the moisture storage in the root zone exceeds the equilibrium content, percolation occurs from root zone to groundwater:

$$q_p = \frac{V_{eq} - (V(t - \Delta t) + \Delta V)}{\Delta t} \tag{1.33}$$

q_p Percolation $[m/d]$
V Soil moisture storage in the root zone $[m]$

V_{eq} Equilibrium moisture storage in the root zone [m]
Δt Time step [d]

If percolation occurs, it is assumed that the percolating water branches the groundwater table within the current time step of the groundwater.

The equilibrium moisture storage in the root zone is defined as the amount of moisture corresponding with a steady-state simulation with no-flow conditions to or from the root zone.

1.1.11.22 Root zone

The root zone for which the water balance is considered has a thickness which is a function of the land use type and the soil physical unit:

$$d_r = f(j, s) \tag{1.34}$$

where:

d_r root zone thickness
j land use type (1,. . . ,16)
s soil physical unit (1,. . . ,21)

The thickness of the root zone is assumed to be constant, regardless of season and year. The moisture storage of the root zone at equilibrium condition is calculated with the function:

$$V_{eq} = f(s, d_r, dg) \tag{1.35}$$

where:

V_{eq} the moisture storage of the root zone at equilibrium condition [m]
s soil physical unit [$-$]
d_r thickness of root zone [m]
d_g depth of the groundwater level [m]

Initial storage in the root zone

The initial storage in the root zone can be defined by one of the three following options:

- ◇ the equilibrium root zone storage for the given crop-soil combination and specified initial groundwater level;
- ◇ the root zone storage at pF=2;
- ◇ the root zone storage at pF=3.

Root zone thickness defining more than one crop

When the user defines more than one crop per unpaved node, RR calculates the overall root zone by weighted averaging. So, the area of the individual crops is also taken into account.

For calculating the evaporation, RR uses the crop with the largest sub-area.

1.1.11.23 RR routing link

The standard RR-links are links indicating connections between e.g. an unpaved node and an open water node. The link flow in that case represents the flow from the unpaved area to the open water. There is no difference between link inflow and link outflow.

Within SOBEK or Delft3D FM Suite 1D2D, the hydrodynamic flow aspects involving time delay between link inflow and outflows are typically handled in the 1D model. However, in some

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cases a simple approach may be convenient. For instance, if only flow measurements at an upstream location and a downstream location (without additional inflows in between) are known, but no cross-section information is available to set-up a 1D model. In such situations a simple routing approach can be used. An example of such a simple approach is the well-known Muskingum method. This method is available in DRR by using the so-called RR-Routing link.

Standard Muskingum method

The standard Muskingum method is using a layered routing approach. The inflow hydrograph is divided into a number of layers, where each layer has its own routing coefficient.

According to the standard Muskingum method (McCarthy, 1938) the outflow Q at time step $i+1$ is computed using outflow Q at time step i and inflows I at time step i and $i+1$:

$$Q_{i+1} = c_1 I_i + c_2 I_{i+1} + c_3 Q_i \tag{1.36}$$

where:

$$c_1 = \frac{\Delta T + 2Kx}{2K(1-x) + \Delta T} \tag{1.37}$$

$$c_2 = \frac{\Delta T - 2Kx}{2K(1-x) + \Delta T} \tag{1.38}$$

$$c_3 = \frac{2K(1-x) - \Delta T}{2K(1-x) + \Delta T} \tag{1.39}$$

The parameter K is a proportionality factor with the dimension of time, it is dependent on the time step you select. K is the travel time of a flood wave through the branch. The parameter x is a dimensionless weighting factor with $0 \leq x \leq 0.5$. In natural streams values of x ranging from 0 to 0.3 are often found. Great accuracy in determining x may not be necessary as the final result is relatively insensitive to the value of x.

It is easily verified that $c_1 + c_2 + c_3 = 1$ and further that ΔT and K should have the same time units. The use of the method requires choices on ΔT, K and x. The routing interval ΔT is the calculation time step size in RR. The chosen time step size should be less than K, as otherwise peaks will be missed at the downstream side. It is advised to take the RR computation time step size ΔT as $1/2$ to $1/4$ of K. If, however, a ΔT is selected which is much smaller than K, then c_2 becomes negative. This will lead to a dip in outflows, or even to negative outflows when the inflow starts rising. To avoid negative coefficients, the routing interval should be within the following range: $2Kx \leq \Delta T \leq K$. For values of x close to 0.5 this requirement leaves little freedom in the selection of ΔT. If one is mostly interested in the higher part of the flood wave the above condition can be relaxed somewhat by accepting slightly negative values for c_2.

The Muskingum routing method allows the routing to be split in multiple layers. This can be of great value when the river contains a minor bed and a floodplain. In the floodplain the Muskingum parameters will be completely different from the parameters of the minor bed.

An example of this computation is given below. Suppose that you use 2-layers for routing the flood wave:
layer 1: $x = 0.4 \quad K = 3.5 \quad Q_{max} = 100\,\mathrm{m^3\,s^{-1}}$
layer 2: $x = 0.1 \quad K = 9$

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RR will then route all discharges up to $Q = 100\,\mathrm{m}^3\,\mathrm{s}^{-1}$ with the parameters of the first layer, all discharges above $Q = 100\,\mathrm{m}^3\,\mathrm{s}^{-1}$ will be routed with the parameters of the second layer

(and using $Q_{layer2} = Q_i - 100[\mathrm{m}^3/\mathrm{s}]$).

Network validation rules for RR-routing link and RR-connection nodes

The following rules apply to the use of the RR-routing link and RR-connection nodes:

- ◇ The RR-routing link can be used to connect:
  - ▫ a Sacramento node to an RR-connection node;
  - ▫ a HBV node to an RR-connection node;
  - ▫ an RR-connection node to an RR-connection node;
  - ▫ an RR-connection node to a RR-boundary node;
- ◇ At RR-connection nodes, multiple upstream links are possible, but only 1 downstream (routing) link is allowed.

1.1.11.24 Separated sewer

The sewer type of a paved area node can be one of the following types:

- ◇ mixed sewer
- ◇ separated sewer
- ◇ improved separated sewer

A separated sewer has separate sewer systems for rainfall and dry weather flow. This system is designed to reduce water quality impacts of sewer overflows on the receiving open water. In a mixed sewer system spilling contains DWF water (domestic return flows). In a separated sewer system however, sewer overflows usually only originate from the rainfall sewer system and not from the DWF sewer system. Therefore, the sewer overflows do not contain the waste loads from the domestic return flows.

1.1.11.25 Silo capacity/Pump capacity

Apart from storage in rainfall basins in some greenhouse areas there is also storage of water in silo's. These silo's, usual capacity approximately 200 $m^3\,ha^{-1}$, function as a temporary storage before pumping the water into the soil (subsoil storage). The pump capacity for subsoil storage usually is about 15 $m^3\,h^{-1}$.

1.1.11.26 Soil surface level

There are two options to define the soil surface level:

Surface level = constant

The user can define any constant surface level he wants, usually the (almost) lowest surface level in the unpaved area.

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Storage on land

- ◇ groundwater level rise: when the groundwater table branches the lowest surface level, surface runoff occurs only when the storage on land is filled.
- ◇ high precipitation rate: when the net precipitation rate exceeds the infiltration rate, water will be stored on land. Surface runoff occurs only when the storage on land is filled.

Surface level = not constant

When this option is chosen, the user can specify the soil surface level in a more detailed way.

RR does take into account up to a maximum of 100 sub-areas (default). The user can define these areas in a table, see Figure 1.19 taken from the SOBEK2 GUI:

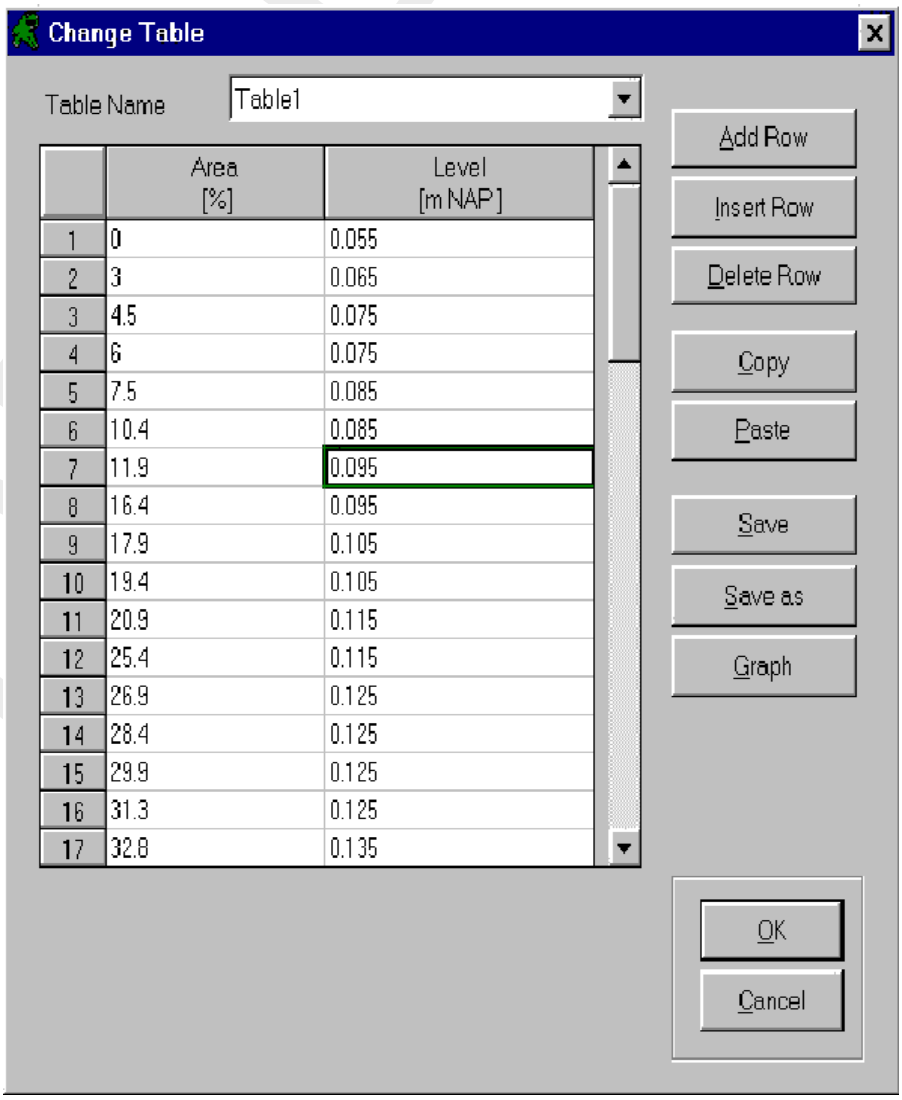


Figure 1.19: *Surface Level As a Table in the SOBEK2 GUI*

The user should make sure that 0 and 100 % are present in this table. In between the user can define as many rows as he wants. Above the 100 % level it is assumed that the area does not increase.

Note: By showing the graph of the S-curve you can check whether you entered the S-curve data correctly.

Initial groundwater level

When the surface level is not constant, it is assumed that the initial groundwater level is defined w.r.t. the lowest surface level. For example, in Figure 1.19 the initial groundwater level is defined w.r.t. the level of 0.055 m NAP.

Reaction factor α

In RR, it is possible to schematise the drainage process by defining four different layers. For each layer a different reaction factor α is defined. When the surface level is not constant, the reaction factor layers are defined relative to the surface level according to the table, see Figure 1.20.

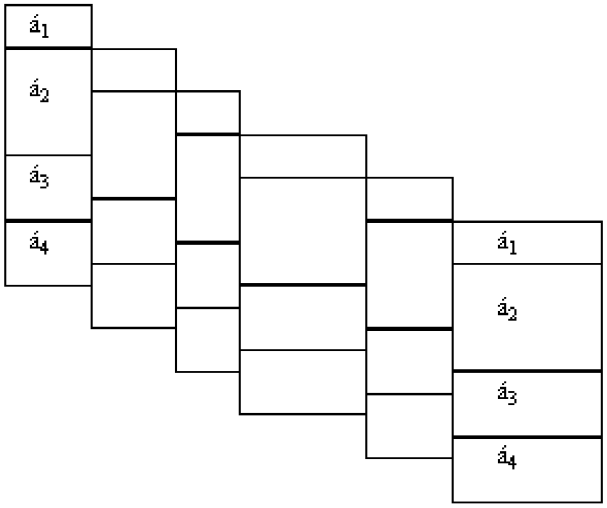


Figure 1.20: *Example of definition of reaction factor layers when the surface level is not constant (in this case the number of sub-areas defined in the table is equal to 6)*

Actually, the above figure is a bit more complex, because default 100 sub-areas are considered. The surface level of each sub-area is determined by interpolation of the surface level table defined by the user.

Storage on land

When the surface level is not constant:

- ◇ groundwater level rise: when the groundwater table reaches the lowest surface level defined in the surface level and area table, immediately surface runoff occurs. Storage on land is not considered.
- ◇ high precipitation rate: when the net precipitation rate exceeds the infiltration rate, water will be stored on land. Surface runoff occurs only when the storage on land is filled.

Storage coefficient i

Surface level = not constant AND NOT using CAPSIM for unsaturated zone

The storage coefficient used to calculate the groundwater level fluctuation, is determined as the average of the initial storage coefficients for each sub-area. The initial storage coefficient is determined by the soil type and the distance between the soil surface and initial groundwater level.

When the initial groundwater level exceeds part of the surface level (storage on land) then the storage coefficients of this part of the surface are excluded from the averaging procedure.

Surface level = not constant AND using CAPSIM for unsaturated zone

CAPSIM computes the unsaturated zone water balance and needs, among others, the root zone depth and initial groundwater level. If the surface level is not constant, the root zone depth and initial groundwater level passed to CAPSIM are defined relative to the surface level (Figure 1.21). The root zone depth is equal for each sub-area, because the root zone depth only depends on the soil type and the crop.

For each sub-area, separate actual evaporation, percolation and capillary rise, and volume of water in the unsaturated zone are computed.

The groundwater level is determined using the total percolation/capillary rise flow for all sub-areas. The storage coefficient used to calculate the groundwater level fluctuation, is determined as the average of the actual storage coefficients for each sub-area. When the groundwater level exceeds part of the surface level (storage on land) then the storage coefficients of this part of the surface are excluded from the averaging procedure.

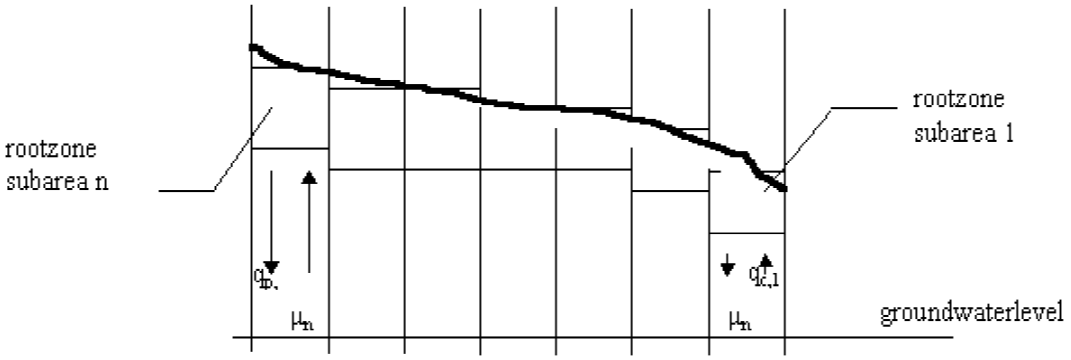


Figure 1.21: *Rootzone sub areas*

1.1.11.27 Storage coefficient

The storage coefficient m represents the percentage of soil-volume which is available for storage of water. Once the storage coefficient is known, the total storage capacity can be calculated:

$$V = \mu \quad d \tag{1.40}$$

V	storage capacity [mm]
μ	storage coefficient [m/m]
d	depth to groundwater table [mm]

With the storage coefficient, the change of the groundwater level can be calculated.

Option 'Unsaturated zone = none'

Calculating groundwater level

For calculating the groundwater level the average storage coefficient is used during the simulation period (Table 1.3). This average storage coefficient depends on the drainage basis, i.e. the distance between the surface level and the initial groundwater level. These coefficients are provided by the former Winand Staring Centre-DLO, now part of WUR (Wageningen University and Research). .

The soil types are:

1 loamy, humous fine sand,
2 peat,
3 heavy clay,
4 humous clay and peat,
5 light loamy sand, medium coarse sand,
6 loamy silt,
7 humous clay and peat with silty top layer,
8 clay and light clay,
9 loamless, medium coarse and coarse sand,
10 silt,
11 very light clay,
12 sand with a silty top layer.

Table 1.3: *Average storage coefficients depending on soil type and drainage basis.*

Drainage basis [m]	**Soil type**											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
0.1	0.0074	0.0077	0.0094	0.0063	0.0046	0.0049	0.0049	0.0048	0.0027	0.0029	0.0025	0.0013
0.2	0.0183	0.0153	0.0166	0.0134	0.0124	0.0103	0.0118	0.0092	0.0098	0.0066	0.0052	0.0032
0.3	0.0302	0.0226	0.0226	0.0206	0.0220	0.0158	0.0181	0.0132	0.0162	0.0105	0.0079	0.0052
0.4	0.0419	0.0305	0.0278	0.0279	0.0323	0.0211	0.0237	0.0171	0.0228	0.0145	0.0107	0.0074
0.5	0.0532	0.0387	0.0323	0.0350	0.0427	0.0262	0.0289	0.0206	0.0294	0.0186	0.0134	0.0096
0.6	0.0664	0.0467	0.0363	0.0420	0.0528	0.0310	0.0339	0.0240	0.0359	0.0226	0.0160	0.0120
0.7	0.0805	0.0545	0.0399	0.0486	0.0625	0.0364	0.0386	0.0271	0.0422	0.0265	0.0185	0.0143
0.8	0.0938	0.0621	0.0431	0.0551	0.0715	0.0416	0.0430	0.0300	0.0484	0.0303	0.0210	0.0167
0.9	0.1061	0.0702	0.0461	0.0613	0.0801	0.0466	0.0472	0.0328	0.0542	0.0341	0.0235	0.0191
1.0	0.1173	0.0784	0.0489	0.0673	0.0880	0.0514	0.0512	0.0355	0.0598	0.0377	0.0258	0.0214
1.2	0.1372	0.0939	0.0538	0.0786	0.1024	0.0605	0.0586	0.0404	0.0704	0.0446	0.0303	0.0261
1.5	0.1614	0.1158	0.0600	0.0941	0.1208	0.0729	0.0685	0.0470	0.0845	0.0541	0.0366	0.0329

Calculating drainage flux

For calculating the drainage flux according to De Zeeuw- Hellinga a constant storage coefficient is used. This storage coefficient, which is the same as for calculation the changes in groundwater level, is the storage coefficient corresponding with the initial groundwater level.

Option 'Unsaturated zone = CAPSIM'

Groundwater level

For calculating the groundwater level a storage coefficient is used depending on the actual groundwater level during the simulation period.

Drainage flux

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The storage coefficient is calculated by averaging all coefficients within in the domain of the groundwater level and the open water level. Since the groundwater level and the open water level may change during the simulation period, the average storage coefficient may change.

The coefficients for calculating the groundwater level and the drainage flux are tabulated in RR. These storage coefficients depend on the soil type, the root zone thickness end the groundwater level. Below the storage coefficients are depicted for a root zone thickness of 20 cm. RR also takes into account storage coefficients for root zone thicknesses of 10 cm, 50 cm, 100 cm and 200 cm.

All coefficients are provided by Alterra (part of WUR - Wageningen University and Research).

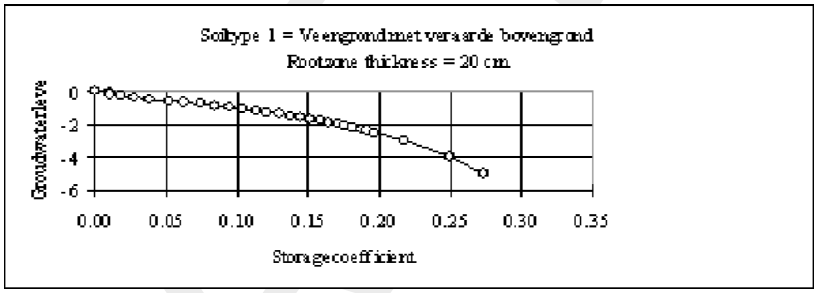


Figure 1.22: *Relation between storage coefficient and groundwaterlevel for soil type 1 (peat soil with earthed topsoil), rootzone 20 cm*

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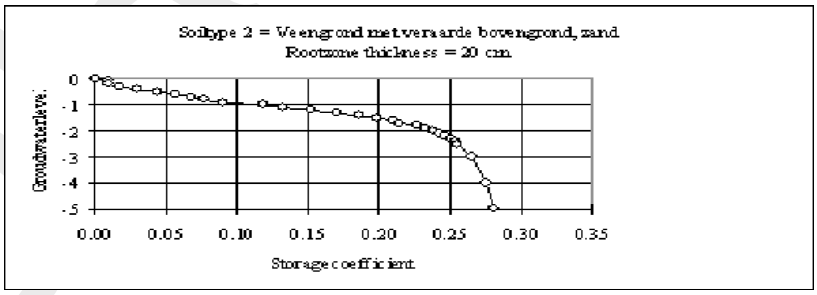


Figure 1.23: *Relation between storage coefficient and groundwaterlevel for soil type 2 (peat soil with earthed topsoil, sand) rootzone 20 cm*

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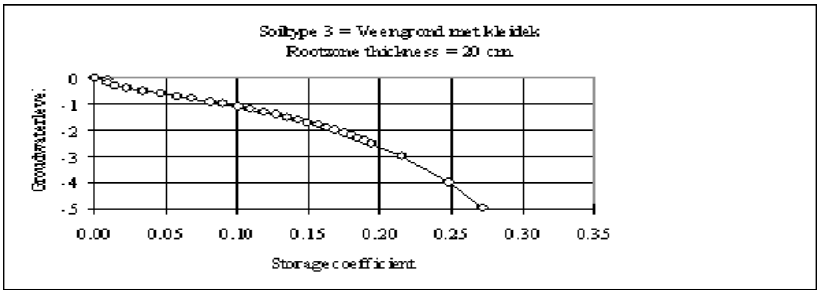


Figure 1.24: *Relation between storage coefficient and groundwaterlevel for soil type 3 (peat soil with clay cover), rootzone 20 cm*

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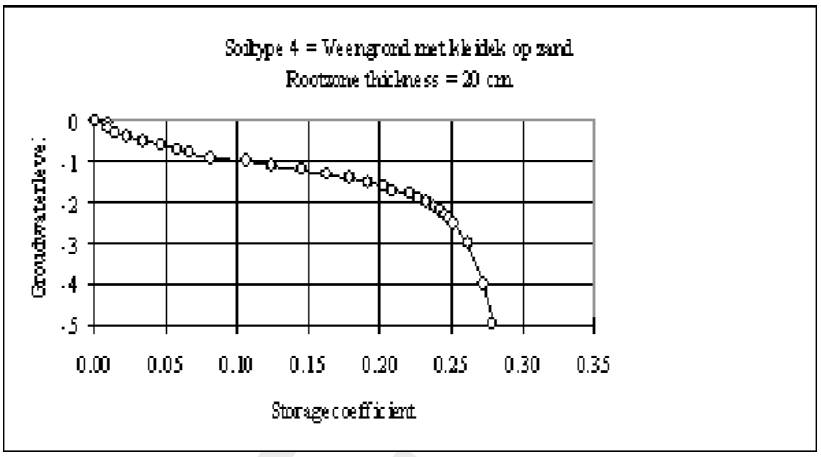


Figure 1.25: *Relation between storage coefficient and groundwaterlevel for soil type 4 (peat soil with clay cover on sand), rootzone 20 cm*

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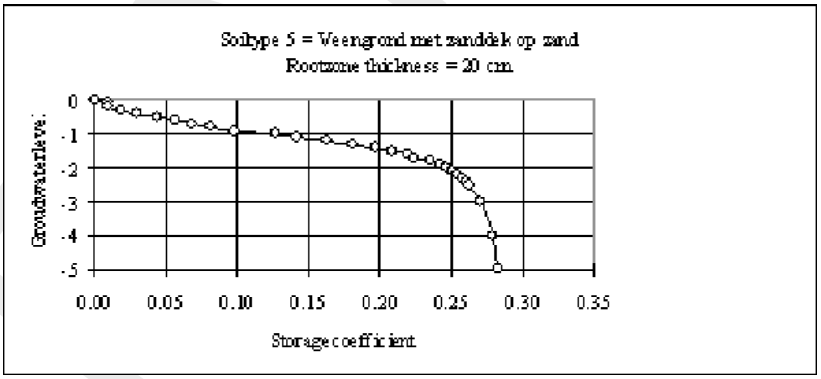


Figure 1.26: *Relation between storage coefficient and groundwaterlevel for soil type 5 (peat soil with sand cover on sand), rootzone 20 cm*

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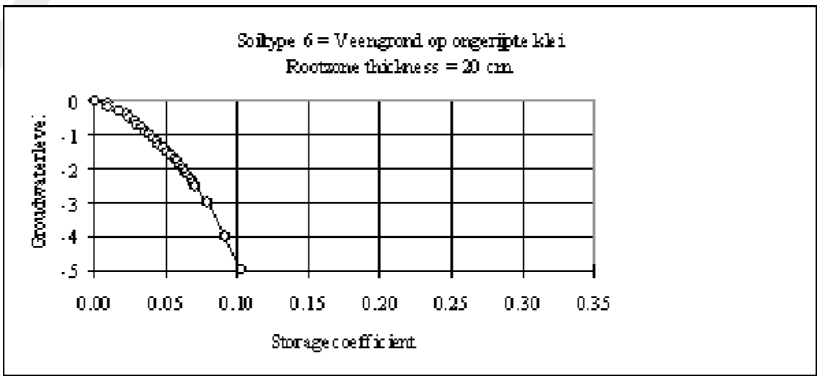


Figure 1.27: *Relation between storage coefficient and groundwaterlevel for soil type 6 (peat soil on unripened clay), rootzone 20 cm*

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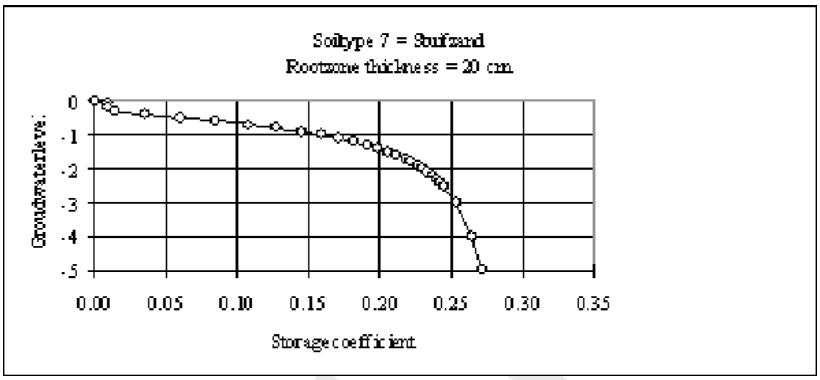


Figure 1.28: *Relation between storage coefficient and groundwaterlevel for soil type 7 (blown sand), rootzone 20 cm*

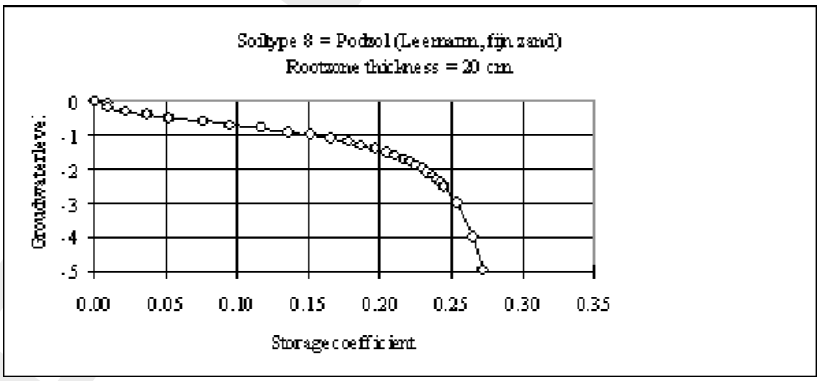


Figure 1.29: *Relation between storage coefficient and groundwaterlevel for soil type 8 (podzol; poor loam, fine sand), rootzone 20 cm*

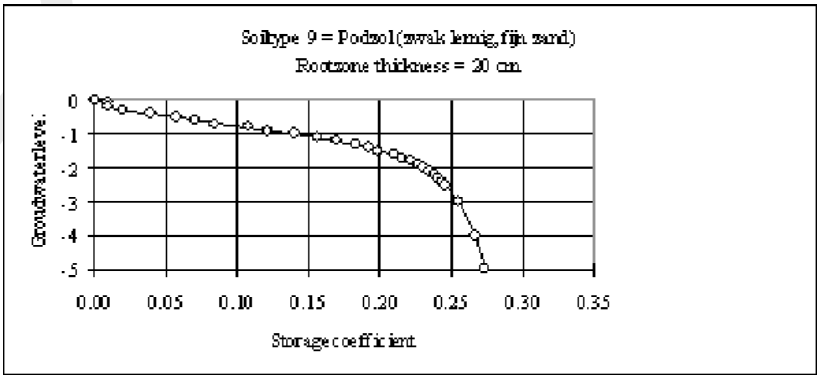


Figure 1.30: *Relation between storage coefficient and groundwaterlevel for soil type 9 (podzol; weak loam, fine sand) rootzone 20 cm*

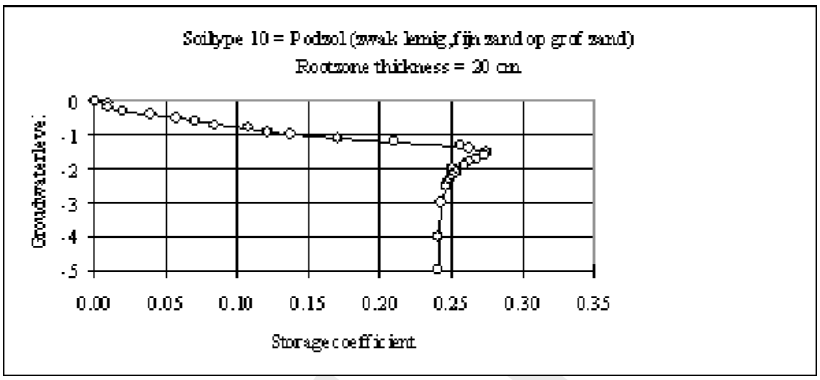


Figure 1.31: *Relation between storage coefficient and groundwaterlevel for soil type 10 (podzol, wek loam, fine sand on coarse sand), rootzone 20 cm*

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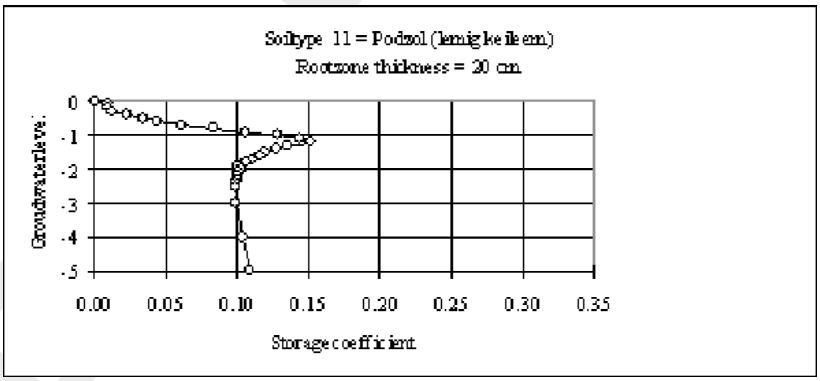


Figure 1.32: *Relation between storage coefficient and groundwaterlevel for soil type 11 (podzol, loamy boulder clay), rootzone 20 cm*

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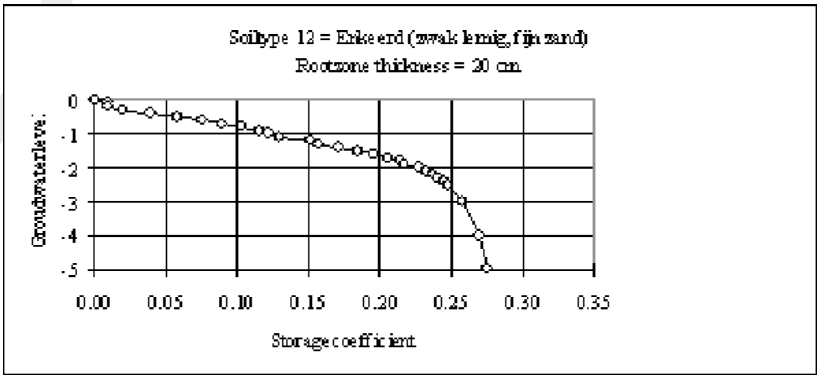


Figure 1.33: *Relation between storage coefficient and groundwaterlevel for soil type 12 (enkeerd, weak loam, fine sand), rootzone 20 cm*

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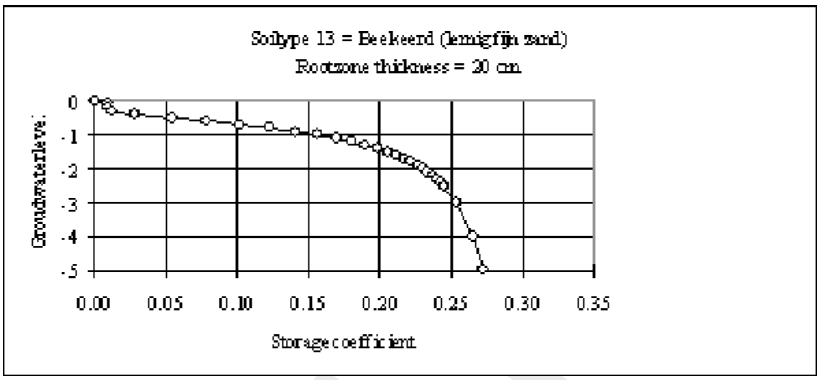


Figure 1.34: *Relation between storage coefficient and groundwaterlevel for soil type 13 (beekeerd, loamy fine sand), rootzone 20 cm*

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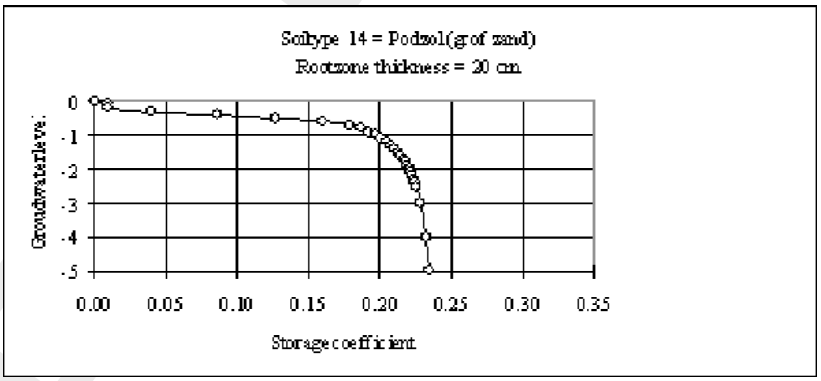


Figure 1.35: *Relation between storage coefficient and groundwaterlevel for soil type 14 (podzol, coarse sand), rootzone 20 cm*

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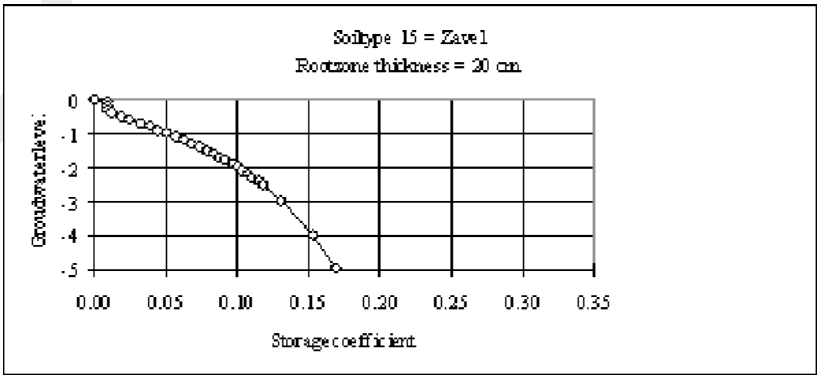


Figure 1.36: *Relation between storage coefficient and groundwaterlevel for soil type 15 (loamy sand), rootzone 20 cm*

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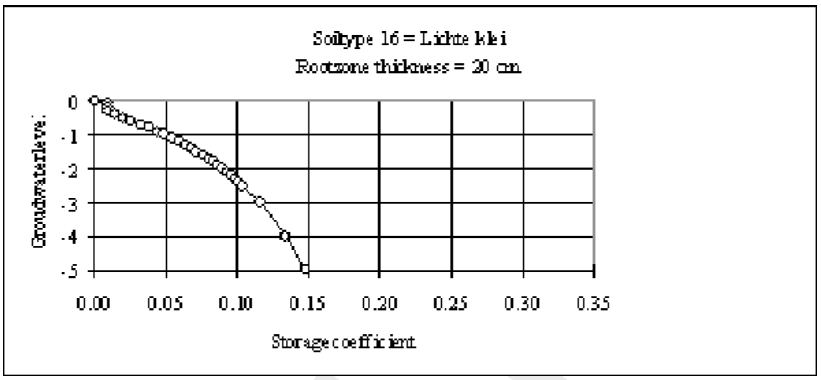


Figure 1.37: *Relation between storage coefficient and groundwaterlevel for soil type 16 (light clay), rootzone 20 cm*

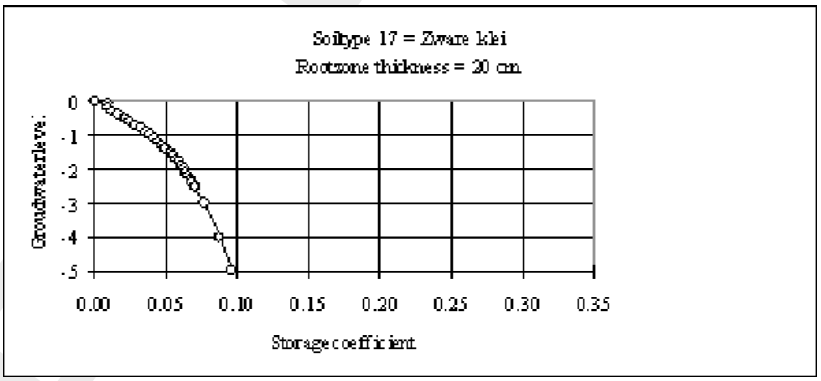


Figure 1.38: *Relation between storage coefficient and groundwaterlevel for soil type 17 (heavy clay), rootzone 20 cm*

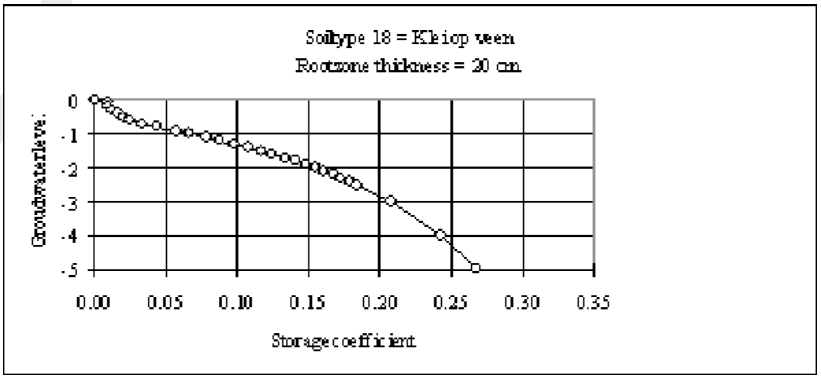


Figure 1.39: *Relation between storage coefficient and groundwaterlevel for soil type 18 (clay on peat), rootzone 20 cm*

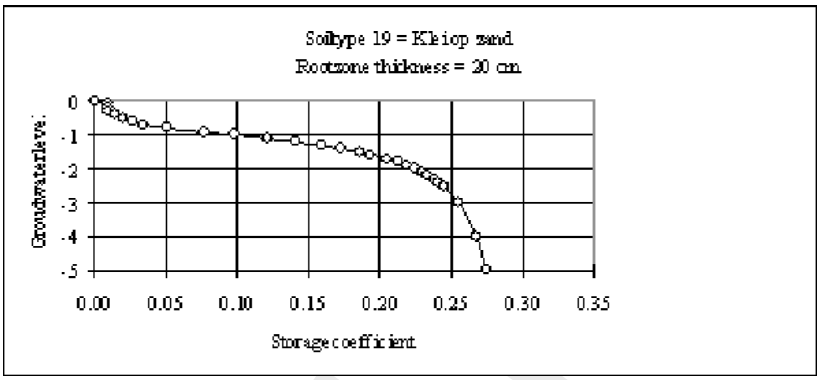


Figure 1.40: *Relation between storage coefficient and groundwaterlevel for soil type 19 (clay on sand), rootzone 20 cm*

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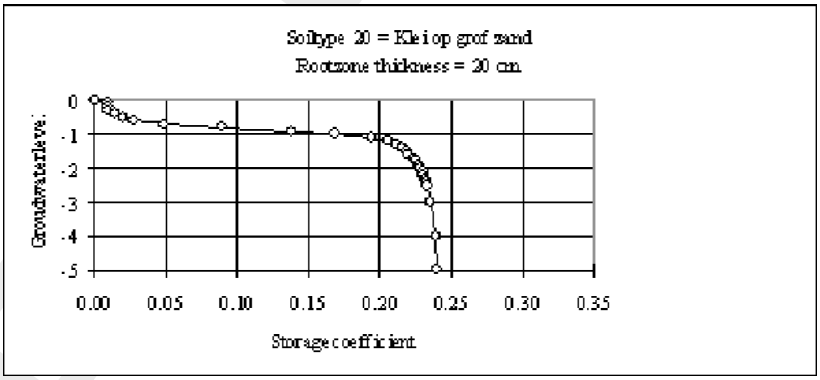


Figure 1.41: *Relation between storage coefficient and groundwaterlevel for soil type 20 (clay on coarse sand), rootzone 20 cm*

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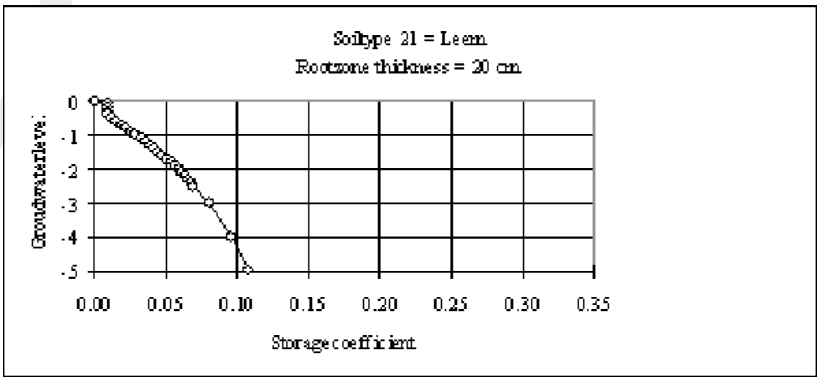


Figure 1.42: *Relation between storage coefficient and groundwaterlevel for soil type 21 (loam), rootzone 20 cm*

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1.1.11.28 Surface runoff

Surface runoff in paved areas

In paved areas surface runoff occurs when the ‘storage on street’ reservoir is filled above its maximum capacity.

Surface runoff in unpaved areas

In unpaved nodes surface runoff to open water can occur in two cases:

1 when the ‘storage on land’ reservoir is filled by the precipitation (minus evaporation and infiltration into the soil).
2 when the groundwater level has reached the soil surface level. In this case, the storage on land has not been taken into account, so surface runoff will immediately take place when the groundwater level has reached the surface level.

In both cases the runoff process is simulated by means of the de Zeeuw-Hellinga equation (requiring a specific surface runoff reaction factor) or the Ernst equation (requiring drainage resistance for the surface layer).

Notice that when the soil surface is defined as a constant level, the total area defined in the unpaved node is part of the surface runoff process. Often this causes very large discharges to the open water.

When the soil surface is defined as a variable level, only the inundated part of the soil surface is part of the surface runoff process.

1.1.11.29 Time step output

By default the time step of the results is equal to the time step of the computation. Especially when a small computational time step is chosen, this results in large output files. The user can choose another option in RR the Settings task in order to reduce the number of output values.

Output reduction options in Settings (SOBEK2 GUI):

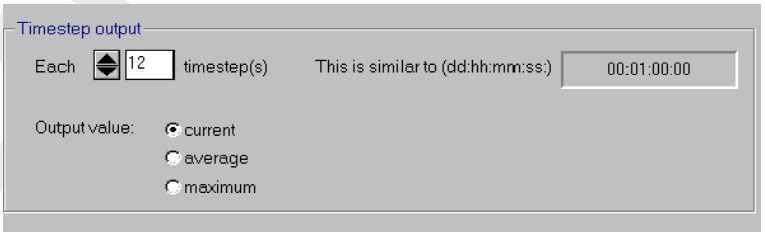


Figure 1.43: *Output reductions in Settings SOBEK2 GUI*

In this case each 12th computed value is shown in the results graphs. The user can define any interval he or she wants.

Other options;

- ◇ Current: each nth computed value is written to the output file;
- ◇ Average: the average value of n computed values is written to the output file;
- ◇ Maximum: the maximum value of n computed values is written to the output file.

Note: the user should realise that the chosen options are of great importance to the shape of the output graph, see this example:

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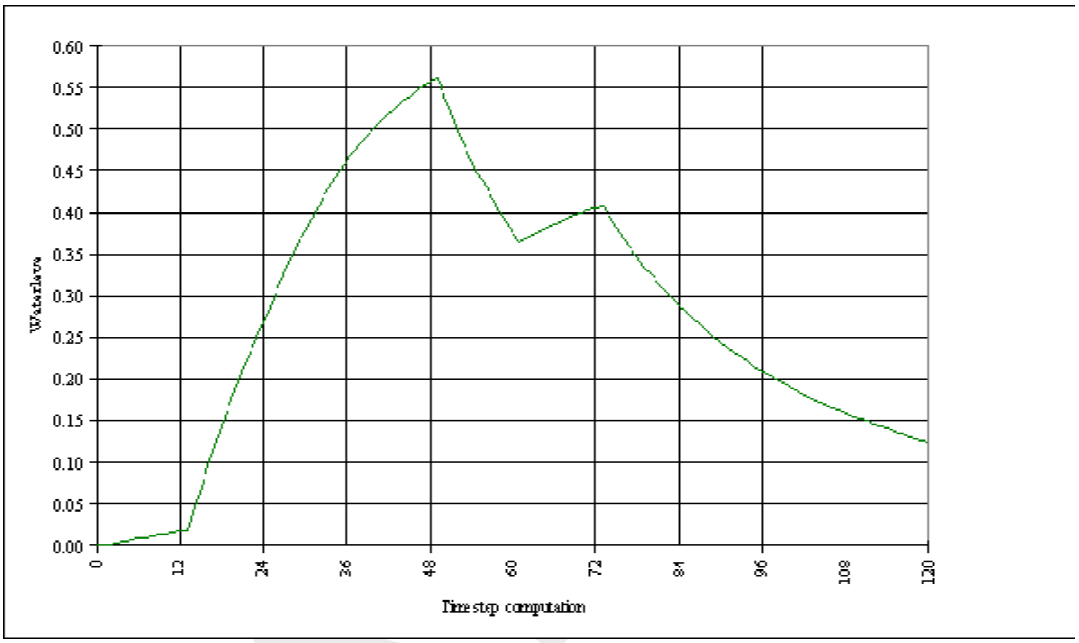


Figure 1.44: *Time step computation*

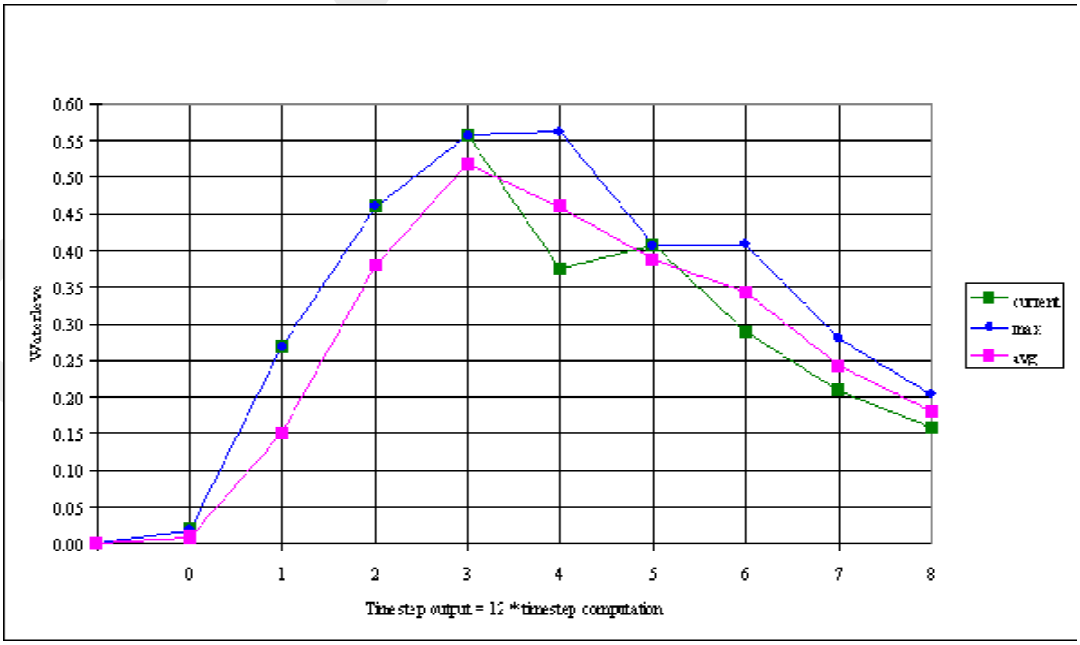


Figure 1.45: *Time step output = 12 ∗ time step computation*

When RR is used to compute series of precipitation events, then the maximum realised level is written to the output file, based on all computed values. The options current, average and maximum are not relevant anymore. Note that this series computation is supported within SOBEK2, but not yet within Delft3D FM Suite 1D2D.

1.1.11.30 Unpaved area node

General

The unpaved area node is used to simulate the rainfall-runoff process on unpaved areas.

An unpaved area is characterized by:

- ◇ total unpaved area (sum of crop areas)
- ◇ groundwater area
- ◇ area per crop
- ◇ soil surface level
- ◇ soil type
- ◇ storage coefficient
- ◇ root zone
- ◇ saturated zone
- ◇ evaporation
- ◇ capillary rise/percolation from root zone to groundwater
- ◇ storage on land
- ◇ infiltration capacity
- ◇ drainage resistance value / reaction factor (dependent on the chosen method)
- ◇ seepage/percolation (constant, variable in time or calculated from a defined groundwater head in the lower confined aquifer and the resistance value of the confining layer)
- ◇ surface runoff

The unpaved area is modeled using boxes representing storage on land, storage in the unsaturated zone, and the saturated zone. The unsaturated zone is optional. The surface area is divided into area for different crops. The sum of the crop area is the total unpaved area. This area is used for the surface storage and unsaturated zone computations. The saturated zone computations use the groundwater area. This area is by default equal to the total unpaved area, but can be defined separately by the user.

1.1.12 Sacramento Rainfall-Runoff model

1.1.12.1 Sacramento, the Segment module: implemented in DRR

The segment module simulates the rainfall-runoff process in part of the catchment, where the attention is on the land-phase of the rainfall-runoff process. It is assumed that the open water system in the segments contributes little to the shaping of the hydrograph. The conceptualisation of the processes as described in the segment module is presented in Figure 1.46 and Burnash *et al.* (1973).

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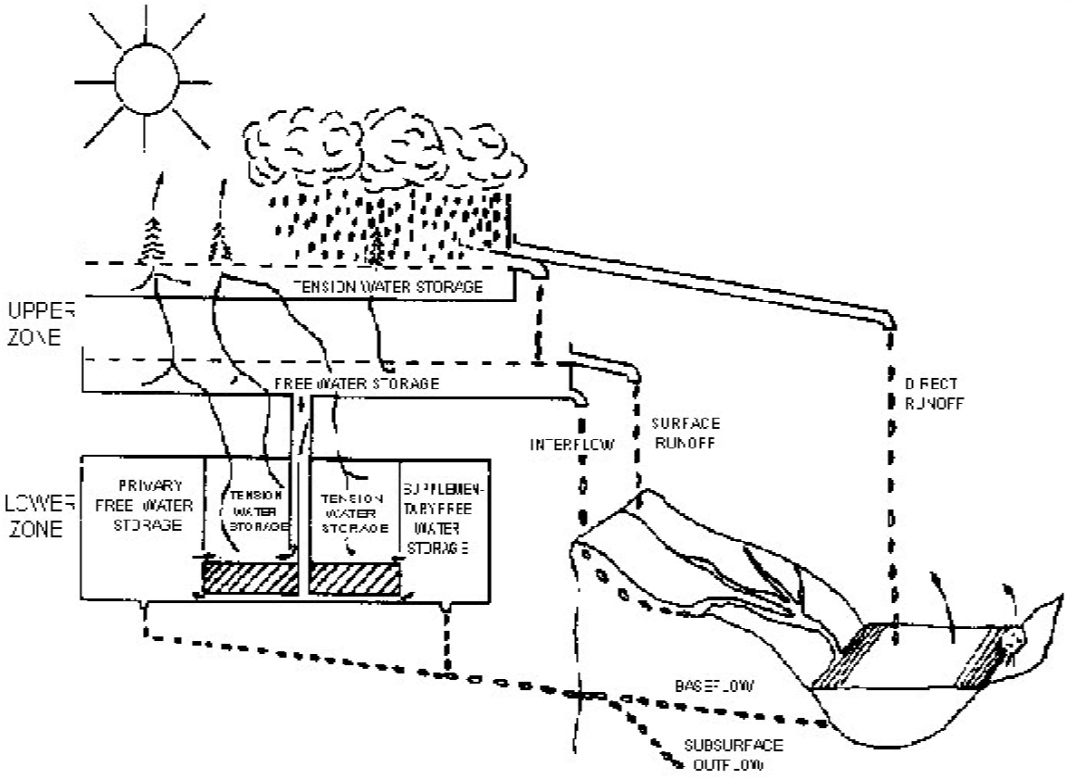


Figure 1.46: *Conceptualisation of the rainfall-runoff process in a segment.*

The segment module is divided into the following components, (see also Figure 1.47):

- ◇ Impervious area with transfers to direct runoff
- ◇ Pervious area
  - ▫ Upper zone
    - ○ Tension storage with transfers to evaporation, free water storage
    - ○ Free water storage with transfers to evaporation, percolation,
    - ○ surface runoff and interflow
  - ▫ Lower zone
    - ○ Tension storage with transfers to evaporation. free water storage
    - ○ Free water storage with transfers to base flow

From the *impervious* areas, precipitation immediately discharges to the channel. However, impervious areas, which drain to a pervious part before branching the channel, are not considered impervious. Both zones have a tension and a free water storage element. Tension water is considered as the water closely bound to soil particles. Generally first the tension water requirements are fulfilled before water enters the free water storage.

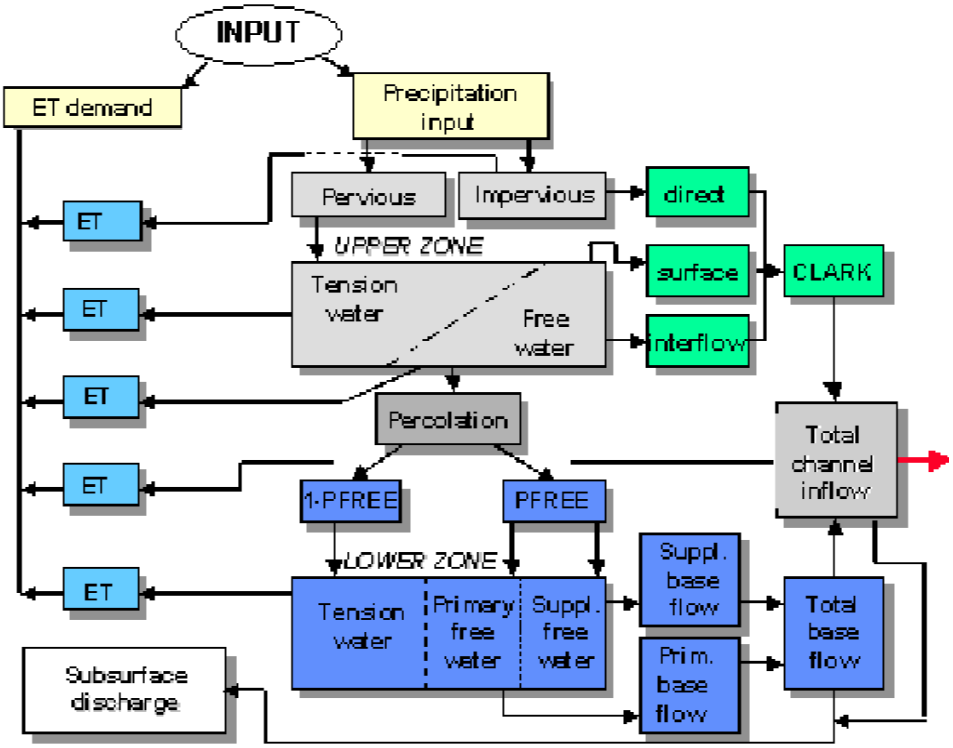


Figure 1.47: *Schematisation of the rainfall-runoff process in a segment*

In the following sub-sections the various components are described in detail.

1.1.12.2 Upper zone storage

The upper zone tension storage represents that precipitation volume required under dry conditions:

- ◇ to meet all interception requirements, and
- ◇ to provide sufficient moisture to the upper soil so that percolation can begin.

If the maximum storage capacity of the upper-zone tension storage is exceeded, water becomes available for the upper zone free water storage, a temporary storage from which water percolates to the lower zone system and from which water discharges to the channel via the interflow component. The preferred flow direction from the upper zone is the vertical direction, i.e. percolation to the lower zone system.

Interflow occurs only when the precipitation rate exceeds the percolation rate. The upper zone is treated as a linear storage element which is emptied exponentially: discharge = storage $*$ storage depletion coefficient. The upper zone free water storage depletion coefficient is denoted by UZK and the upper zone free water content by *UZFWC* then the interflow takes place at a rate:

$$Q_{\text{interflow}} = UZFWC * UZK \tag{1.41}$$

When the precipitation intensity exceeds the percolation intensity and the maximum interflow drainage capacity, then the upper zone free water capacity ($UZFWM$) is completely filled and the excess precipitation causes surface runoff.

1.1.12.3 Lower zone storage

The lower zone consists of the

- ◇ tension water storage, i.e. the depth of water held by the lower zone soil after wetting and drainage (storage up to field capacity) and
- ◇ two free water storages: the primary and supplemental storage elements representing the storages leading to a slow and a fast groundwater flow component, respectively. The introduction of two free lower zone storages is made for greater flexibility in reproducing observed recession curves caused by groundwater flow.

1.1.12.4 Percolation from upper to lower zones

The percolation rate from the upper zone to the lower zone depends on the one hand on the lower zone demand, i.e. requirements determined by the lower zone water content relative to its capacity and on the other hand on the upper zone free water content relative to its capacity.

The lower zone percolation demand is denoted by $PERC_{act.dem}$. The upper zone free water content relative to its capacity is $UZFWC/UZFWM$. Hence, the actual percolation intensity then reads:

$$PERC = PERC_{act.dem} * UZFWC/UZFWM \tag{1.42}$$

The lower zone percolation demand has a lower and an upper limit:

- ◇ the minimum lower zone percolation demand, and
- ◇ the maximum lower zone percolation demand.

The minimum lower zone percolation demand occurs when all three lower zone storages are completely filled. Then by continuity the percolation rate equals the groundwater flow rate from full primary and supplemental reservoirs. Denoting the minimum demand by *PBASE* then it follows:

$$PERC_{min.dem} = PBASE = LZFPM * LZPK + LZFSM * LZSK \tag{1.43}$$

where:

$LZFPM$	lower zone primary free water storage capacity
$LZFSM$	lower zone supplemental free water storage capacity
$LZPK$	drainage factor of primary storage
$LZSK$	drainage factor of supplemental storage

The maximum lower zone percolation demand takes place if the lower zone is completely dried out i.e. if its content = 0. Then the maximum percolation rate is expressed as a function of *PBASE*:

$$PERC_{max.dem} = PBASE \quad (1 + ZPERC) \tag{1.44}$$

with: $ZPERC \gg 1$ usually.

The actual lower zone percolation demand depends on the lower zone content relative to its capacity. Computationally it means that *ZPERC* has to be multiplied by a function G of the relative lower zone water content such that this function:

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- ◇ equals 1 in case of a completely dry lower zone
- ◇ equals 0 in case of a completely saturated lower zone
- ◇ represents an approximate exponential decay of the percolation rate in case of a continuous recharge.

In the Sacramento model this function has the following form:

$$G = \left(\frac{\sum(\text{lower zone capacities} - \text{lower zone content})}{\sum(\text{lower zone capacities})} \right)^{REXP} \tag{1.45}$$

and the *actual percolation demand* is given by (see Figure 1.48):

$$PERC_{act.dem} = PBASE * (1 + ZPERC * G) \tag{1.46}$$

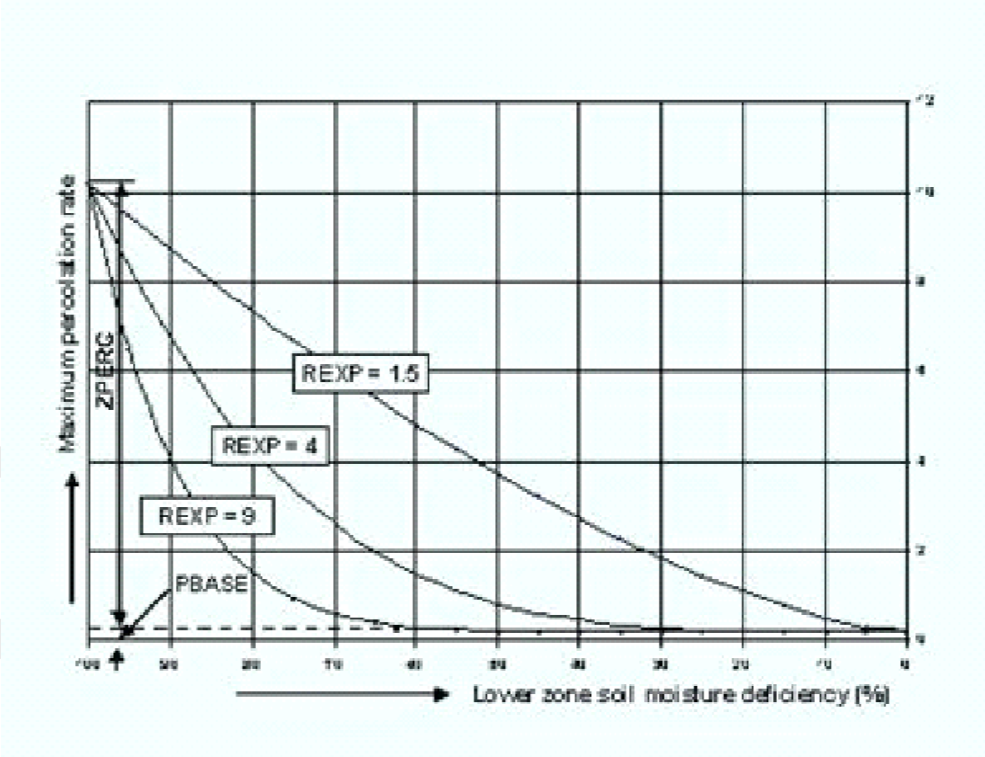


Figure 1.48: *Actual percolation demand representation*

1.1.12.5 Distribution of percolated water from upper zone

The percolated water drains to three reservoirs, one tension and two free water reservoirs. Based on the preceding comments one would expect that the lower zone tension storage is filled first before percolation to the lower zone free water storages takes place. However, variations in soil conditions and in precipitation amounts over the catchment cause deviations from the average conditions. This implies that percolation to the free water reservoirs and hence groundwater flow takes place before the tension water reservoir is completely filled. The model allows for this to let a fraction of the infiltrated water percolate to the two free water storages. When the tension water reservoir is full, all percolated water drains to the primary and supplemental free water storage in a ratio corresponding to their relative deficiencies.

1.1.12.6 Groundwater flow

Base flow to the river from groundwater depends on the contents of the two lower zone free water storages and two drainage constants expressed in fractions of the content per day. If the actual contents of the primary and supplemental free water zones are denoted by *LZFPC* and *LZFSC* respectively then the total base flow QBASE becomes, in accordance with the linear reservoir theory:

$$QBASE = LZFPC * LZPK + LZFSC * LZSK \tag{1.47}$$

The drainage factors $LZPK$ and $LZSK$ can be determined from the recession part of the hydrograph by plotting that part of the hydrograph on semi-logarithmic paper (Fig. 6).

In the lowest part of the recession curve only the slow base flow component is acting while in the higher stages both base flow components contribute.

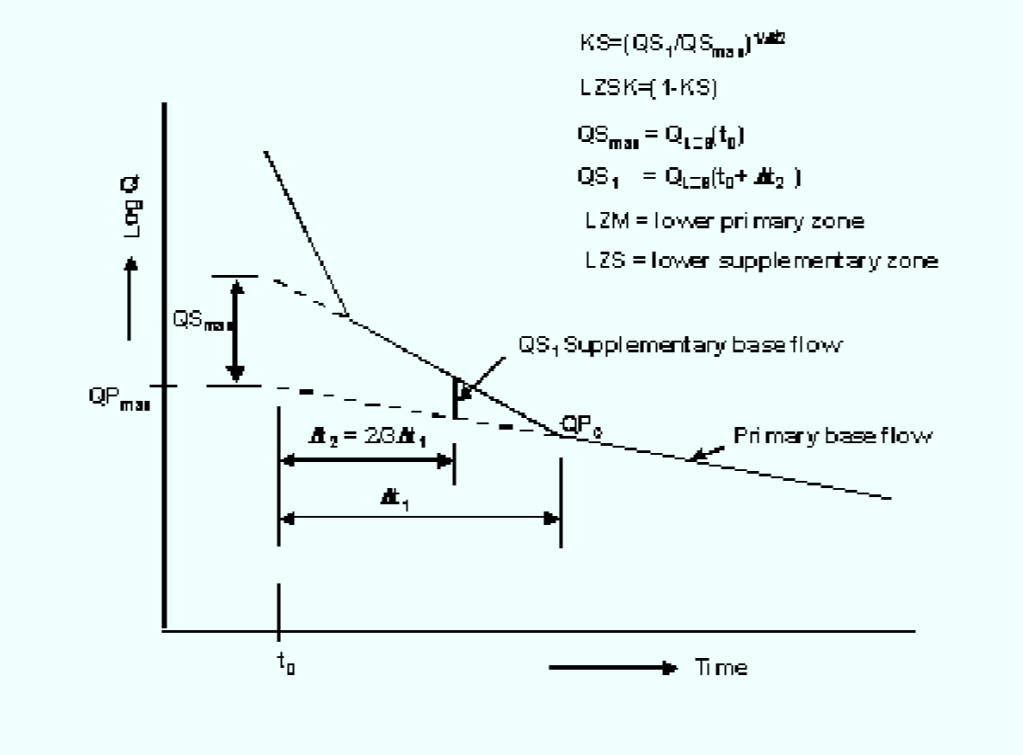


Figure 1.49: *Principle of computation of lower zone recession coefficient*

The drainage factor $LZPK$ follows from:

$$K = \left(\frac{QP_{t0+\Delta t}}{QP_{t0}} \right)^{1/\Delta t} \tag{1.48}$$

and

$$LZPK = 1 - K \tag{1.49}$$

where:

K	recession coefficient of primary base flow for the time unit used
Δt	number of time units, generally days

$QP_{t0+\Delta t}$ a discharge when recession is occurring at the primary base flow rate
QP_{t0} the discharge t time units later

If QP_{max} represents the maximum value of the primary base flow, then the maximum water content of the lower zone becomes:

$$LZFPM = \frac{QP_{max}}{LZPK} \tag{1.50}$$

and similarly the supplemental lower zone free water capacity is determined; at least this procedure provides first estimates of the lower zone free water capacities (Figure 1.49).

The total base flow contributes completely or in part to the channel flow. A complete contribution occurs if subsurface discharge (i.e. discharge from the segment, which is not measured at the outlet) is absent. Otherwise a fraction of the total base flow represents the subsurface flow.

1.1.12.7 Actual evapotranspiration

Evaporation at a potential rate occurs from that fraction of the basin covered by streams, lakes and riparian vegetation. Evapotranspiration from the remaining part of the catchment is determined by the relative water contents of the tension water zones. If *ED* is the potential evapotranspiration, then the actual evapotranspiration from the upper zone reads:

$$E_1 = ED * \frac{UZTWC}{UZTWM} \tag{1.51}$$

i.e. the actual rate is a linear function of the relative upper zone water content. Where $E_1 < ED$ water is subtracted from the lower zone as a function of the lower zone tension water content relative to the tension water capacity:

$$E_2 = (ED - E_1)\frac{LZTWC}{UZTWM + LZTWM} \tag{1.52}$$

If the evapotranspiration should occur at such a rate that the ratio of content to capacity of the free water reservoirs exceeds the relative tension reservoir content, then water is transferred from free water to tension water such that the relative loadings balance. This correction is made for the upper and lower zone separately. However, a fraction $RSERV$ of the lower zone free water storage is unavailable for transpiration purposes.

1.1.12.8 Impervious and temporary impervious areas

Besides runoff from the pervious area, the channel may be filled by rainwater from the impervious area. With respect to the size of the impervious area it is noted that in the Sacramento model a distinction is made between **permanent and temporary impervious areas** where temporary impervious areas are created when all tension water requirements are met, i.e. an increasing fraction of the catchment assumes impervious characteristics.

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1.1.12.9 Routing of surface runoff

Before the runoff from the impervious areas, the overland- and interflow branch the channel, they may be transformed according to a unit hydrograph leading to an adapted time distribution of these flow rates.

Use can be made here of the Clark method, which is a combined time-area and storage routing method. The model requires the construction of a time-area diagram. For this isochrones are constructed representing points of equal travel time to the segment outlet, see Figure 1.50 The areas between successive isochrones is determined and subsequently properly scaled by the time of concentration T_c. The latter is defined as the time required to have the effect of rainfall fallen in the most remote part felt at the segment outlet. The time-area diagram can be thought of as the outflow from the segment if only translation and no deformation takes place of an instantaneous unit supply of rain over the entire segment. Subsequently, the time area diagram flow is routed through a linear reservoir, which characterises the effect of storage in the open drainage system of the segment. This reservoir is represented by the second parameter: the recession coefficient k. It is noted that the output from the reservoir represents the instantaneous unit hydrograph (IUH). This has to be transformed into say a 1-hour unit hydrograph, dependent on the chosen routing interval

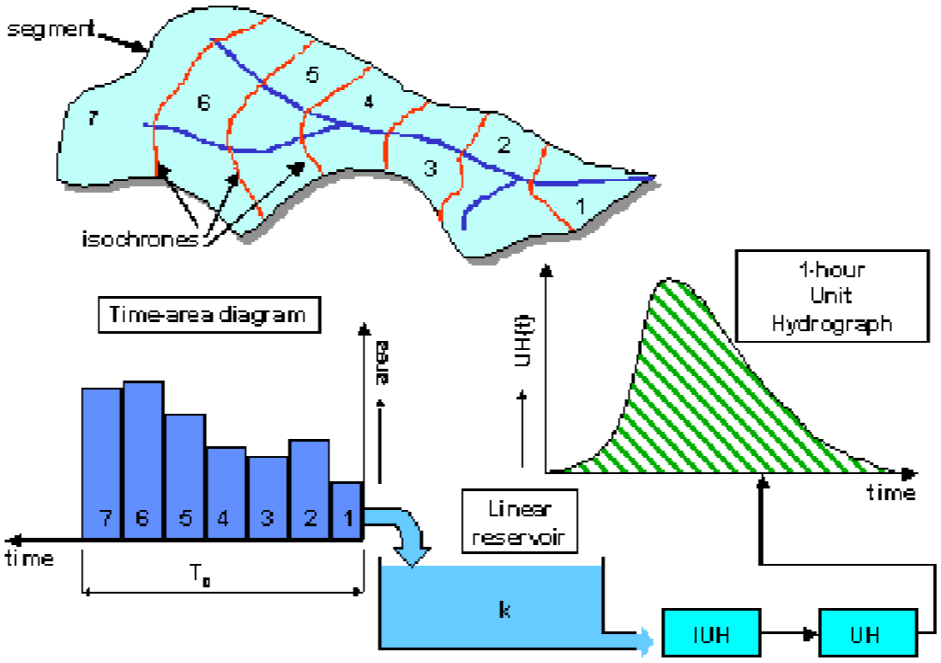


Figure 1.50: *Principles of the Clark method for simulating surface runoff and interflow.*

The two parameters T_c and k can be obtained from observed rainfall and discharge hydrographs. The time of concentration is equal to the time interval between cessation of rainfall and the time the hydrograph has receded to its inflection point. Alternatively it is determined from physical features of the segment as length and slope. A large number of empirical formulas are available which relate the time of concentration to topographical features of the basin. It is noted, though, that these formulas have generally only local validity. The best is to estimate the celerity from the flow velocities in the drainage system taking account of the following characteristics of celerity:

- ◇ If the stream remains inbank, the celerity is about 1.5 to 1.7 times the cross-sectional flow velocity
- ◇ If the flow becomes overbank the above celerity has to be multiplied with the ratio of the drain width and the total width of the flow at the water surface (i.e. inclusive of the floodplain)

To the time required to travel through the drainage system one has to add the overland flow time.

The recession coefficient k is determined from the slope of recession part of the surface runoff hydrograph, similar to the procedure for groundwater.

1.1.12.10 Sacramento - Estimation of segment parameters

Overview of parameters

The following groups of parameters can be distinguished for a particular segment:

- ◇ Segment

  [-] Segment area [km^2]
- ◇ Direct runoff

  PCTIM Permanently impervious fraction of segment contiguous with stream channels
  ADIMP Additional impervious fraction when all tension water requirements are met
  SARVA Fraction of segment covered by streams, lakes and riparian vegetation
- ◇ Upper soil moisture zone

  UZTWM Capacity of upper tension water zone [mm]
  UZFWM Capacity of upper free water zone [mm]
  UZK Upper zone lateral drainage rate (fraction of contents per day)
- ◇ Percolation

  ZPERC Proportional increase in percolation from saturated to dry conditions in lower zone
  REXP Exponent in percolation equation, determining the rate at which percolation demand changes from dry to wet conditions
- ◇ Lower zone

  LZTWM Capacity of lower zone tension water storage [mm]
  LZFPM Capacity of lower zone primary free water storage [mm]
  LZFSM Capacity of lower zone supplemental free water storage [mm]
  LZPK Drainage rate of lower zone primary free water storage (fraction of contents per day)
  LZSK Drainage rate of lower zone supplemental free water storage (fraction of contents per day)
  PFREE Fraction of percolated water, which drains directly to lower zone free water storages
  RSERV Fraction of lower zone free water storages which is unavailable for transpiration purposes
  SIDE Ratio of unobserved to observed base flow
  SSOUT Fixed rate of discharge lost from the total channel flow [$mm/\Delta t$]

  Note that the unit of SSOUT is sensitive for the timestep size (so SSOUT should be adjusted when the DRR computation timestep is adjusted).
- ◇ Surface runoff

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[-] Unit hydrograph ordinates

◇ Internal routing interval

PM	Time interval increment parameter
PT1	Lower rainfall threshold
PT2	Upper rainfall threshold

Basically two procedures are available to get first estimates for the majority of the segment parameters:

◇ from observed rainfall and runoff records: this method is usually applied and works well provided that the model concepts are applicable and that reliable records are available for some time covering the majority of the range of flows
◇ from soil characteristics: this method is particularly suitable if no runoff records are available, i.e. for ungauged catchments.

With respect to gauged catchments the following grouping of parameters according to the method of estimation can be made:

◇ Parameters computed and estimated from basin map solely:
segment area and SARVA
◇ Parameters estimated from observed rainfall and runoff records:
readily: LZFPM, LZPK, LZFSM, LZSK, PCTIM
approximately: UZTWM, UZFWM, UZK, LZTWM, SSOUT and PFREE
◇ Parameters estimated from topographic maps and rainfall and runoff records:
unit hydrograph ordinates obtained from Clark method
◇ Parameters to be obtained through trial runs:
ZPERC, REXP, SIDE, ADIMP, RSERV
◇ Internal routing parameters, as per requirement:
PM, PT1, PT2
In the next sub-sections guidelines are given for the determination and estimation of the segment parameters for gauged catchments.

1.1.12.11 Segment parameter estimation for gauged catchments.

The estimation of the segment parameters is presented according to their order of appearance in the previous sub-section. The sequence in which the estimation is done in practice is different from this order, for which reference is made to the end of the sub-section.

Segment

Segment area

To allow a good comparison between the observed and simulated runoff from the basin, the segment area (km^2) should refer to the total segment area draining upstream of the gauging station. Any difference between total segment area up to the main stream and the area upstream of the gauging station can be accommodated for in the channel routing part.

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Direct runoff

PCTIM

Permanently impervious fraction of the basin contiguous with stream channels. It can be determined from small storms after a significant period of dry weather. Then the volume of direct runoff (= observed runoff - base flow) divided by the volume of rain gives the percentage impervious fraction of the basin. *PCTIM* should not be close to 1!

An example is given below.

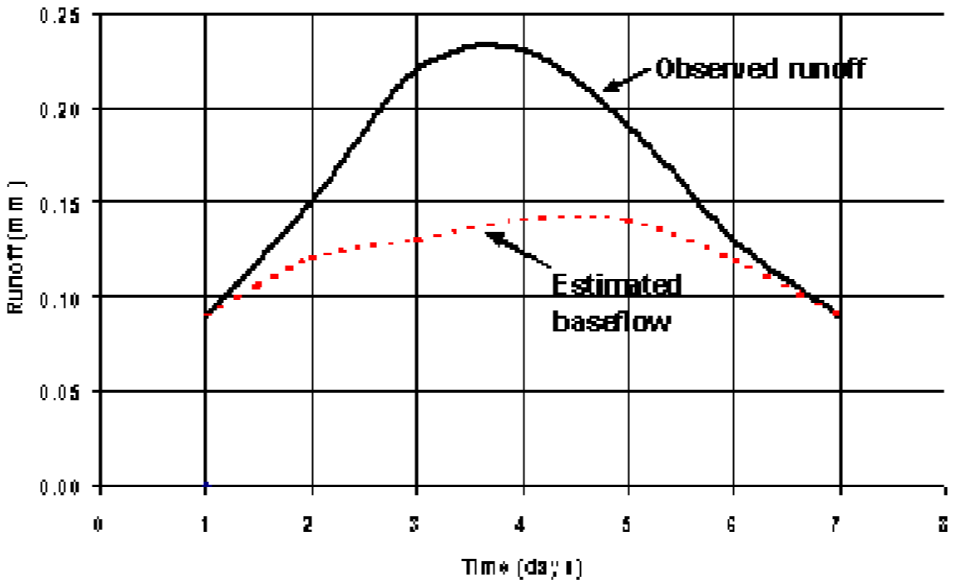


Figure 1.51: *Calculation of PCTIM*

ADIMP

Fraction of the basin, which becomes impervious as all tension water requirements are met. It can be estimated from small storms after a very wet period. As before, the volume of direct runoff divided by the volume of rain gives the total percentage of impervious area. The estimate for $ADIMP$ follows from:

$$ADIMP = \text{Total Percentage Impervious} - PCTIM \tag{1.53}$$

SARVA

Fraction of the basin covered by streams, lakes, and riparian vegetation, under normal circumstances. The *SARVA* area is considered to be the same as or less than *PCTIM* (see below). Detailed maps may be referred to in order to estimate the extent of paved areas, which drain directly to the streams so that differences between *PCTIM* and *SARVA* can be approximated. Generally, *SARVA* appears to range between 40 % and 100 % of the *PCTIM* value.

Upper soil moisture zone

UZTWM - the upper tension storage capacity

The depth of water, which must be filled over non-impervious areas before any water becomes available for free water storage. Since upper zone tension water must be filled before any stream flow in excess of the impervious response can occur, its capacity can be approximated from hydrograph analysis. Following a dry period when evapotranspiration has depleted the upper soil moisture, the capacity of upper zone tension water can be estimated. That volume of rainfall, which is retained before runoff from the pervious fraction is visible, is identified as *UZTWM*. To that rainfall volume the losses to evaporation during the considered period should be added. All periods of rain following a dry period should be checked for estimation of this parameter. Generally the capacity of the upper zone tension will vary between 25 and 175 mm, depending on the soil type.

Following the logic of the Curve Number method (see SCS model concept description in this DRR-manual), where the initial abstraction before rainfall becomes effective is estimated as 20 % of the potential maximum retention, the *UZTWM* becomes:

$$UZTWM = 50.8 \left(\frac{100}{CN} - 1 \right) \; [mm] \tag{1.54}$$

CN-values range from 30 to about 90 for rural areas and are a function of:

- ◇ soil type (soil texture and infiltration rate); hydrological soil groups A–D are distinguished
- ◇ land use, type of land cover, treatment and hydrologic or drainage condition

It is also a function of antecedent moisture condition, for which the condition “dry” should be taken in view of the meaning of $UZTWM$. Based on this assumption $UZTWM$ would vary between 120 and 6 mm, values which are in the range of those given above, particularly if one realises that the 20 % of the potential maximum as initial abstraction is an average value. Reference is made to the SCS method description in this manual and to standard textbooks on hydrology for CN-values.

UZFWM - the upper free water storage capacity

Upper zone free water represents that depth of water, which must be filled over the non-impervious portion of the basin in excess of $UZTWM$ in order to maintain a wetting front at maximum potential. This volume provides the head function in the percolation equation and also establishes that volume of water, which is subject to interflow drainage. Generally its magnitude ranges from 10–100 mm. It is not generally feasible to derive the magnitude of the upper zone free water from direct observations, and successive computer runs are required in order to establish a valid depth.

However, if a rough estimate of UZK is available (see below), then a rough value of $UZFWM$ can be obtained from the hydrograph at the time of the highest interflow, by reducing the flow value with primary and supplemental base flow.

UZK - the upper zone lateral drainage rate

The upper zone lateral drainage rate is expressed as the ratio of the daily withdrawal to the available contents. Its range is roughly 0.18 to 1.0, with 0.40 generally serving as an effective initial estimate. Though basically, this factor is not capable of direct observation and must be determined by successive computer runs, Peck (1978) suggests the following approximate procedure. UZK is roughly related to the amount of time that interflow occurs following a

period with major direct and surface runoff. A long period of interflow results in a small value for UZK. Assuming that interflow is observed during N consecutive days and that interflow becomes insignificant when it is reduced to less than 10 % of its maximum value it follows:

$$(1 - UZK)^N = 0.10 \quad \text{or} \quad UZK = 1 - 0.1^{1/N} \tag{1.55}$$

Values for UZK as a function of N can be read from Figure 1.52.

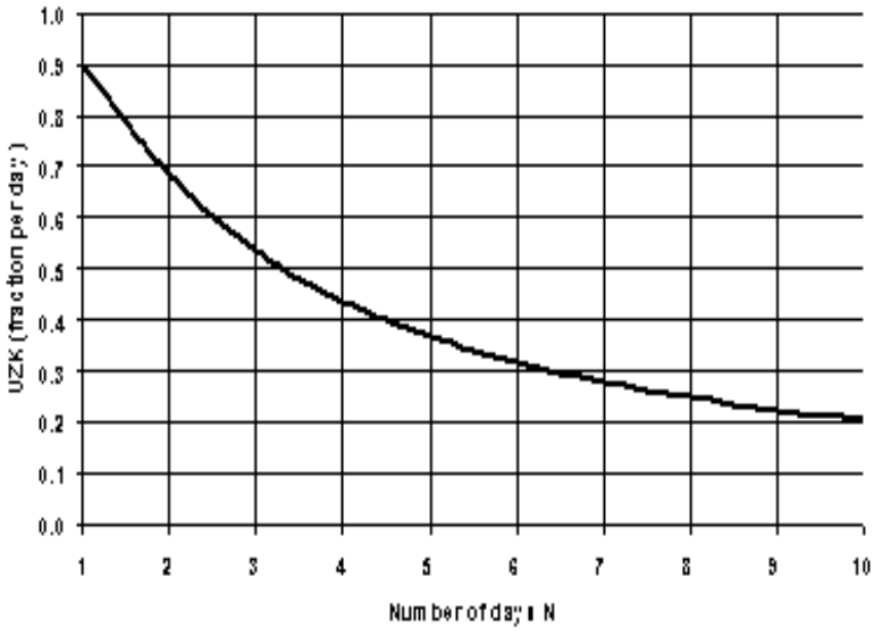


Figure 1.52: *UZK as function of number of days with significant interflow*

Percolation

ZPERC

The proportional increase in percolation from saturated to dry condition is expressed by the term *ZPERC*. The value of *ZPERC* is best determined through computer trials. The initial estimate can be derived by sequentially running one or two months containing significant hydrograph response following a dry period. The value of *ZPERC* should be initially established so that a reasonable determination of the initial run-off conditions is possible.

Amstrong (1978) provides a procedure to derive *ZPERC* from the lower zone tension and free water reservoir capacities and drainage rates, using Equation (1.43) and Equation (1.44). The maximum percolation takes place when the upper zones are full and the lower zones are empty. Assuming that the maximum daily percolation will be the maximum contents of the lower zones, from equation Equation (1.44) it follows for *ZPERC*:

$$ZPERC = \frac{LZTWM + LZFPM + LZFSM - PBASE}{PBASE} \tag{1.56}$$

If data would be available on maximum percolation rates *ZPERC* can be estimated using equation Equation (1.44). Values for *ZPERC* ranging from 5 to 80 have been used.

REXP

The exponent in the percolation equation which determines the rate at which percolation demand changes from the dry condition, $(ZPERC + 1)PBASE$, to the wet condition, *PBASE*. Figure 1.48 illustrates how different values of the exponent affect the infiltration rate. It is recommended that an initial estimate of this exponent is made from the same record which is used in determining an initial estimate of *ZPERC*. The interaction between *PBASE*, *ZPERC* and *REXP* may require a shift of all three terms whenever it becomes clear that a single term should be changed. Visualising the percolation curve generated by these three terms helps to ascertain the necessary changes. The observed range of *REXP* is usually between 1.0 and 3.0. Generally a value of about 1.8 is an effective starting condition. Values for *REXP* for different soils are given by Amstrong (1978) and are presented in Table 1.4.

Table 1.4: *Percolation exponent REXP for different soil types*

Soil classification	**REXP**
Sand	1.0
Sandy loam	1.5
Loam	2.0
Silty loam	3.0
Clay, silt	4.0

Lower zone

LZTWM - lower zone tension water capacity [mm]

This volume is one of the most difficult values to determine effectively. In as much as carryover moisture in this storage may exist for a period of many years, its total capacity may not be readily discernible from available records. If a drought condition during the period of record in the basin or in the area being studied has been sufficient to seriously affect the transpiration process of deep rooted plants, then the period of record is usually sufficient to determine the maximum storage value of lower zone tension water. Often, however, field data is not adequate for this purpose. As a result, unless great care is taken, the depth of lower zone tension water storage may inadvertently be set near the maximum deficit experienced during the period of record rather than the actual capacity of the zone. It has been noted that the plant growth of an area is a relatively effective indicator of the capacity of the lower zone tension water zone. In heavily forested regions of deep-rooted conifers, this zone may be approximately 600 mm in magnitude. In areas of deep-rooted perennial grasses this depth is more likely to be close to 150 mm. Where vegetation is composed primarily of relatively shallow-rooted trees and grasses, this depth may be as little as 75 mm. It should be realised that this zone represents that volume of water, which will be tapped by existing plants during dry periods.

An approximate procedure to estimate *LZTWM* from a water balance analysis is presented by Peck (1978). For this a period is selected with direct and/or surface runoff following an extended dry spell. The selected period is bounded by the times t_1 and t_2. At both times t_1 and t_2 only base flow occurs. The start t_1 is selected immediately prior to the occurrence of direct/surface runoff and t_2 immediately following a period of interflow. The times t_1 and t_2 can best be selected from a semi-log plot of the runoff, see Figure 1.53.

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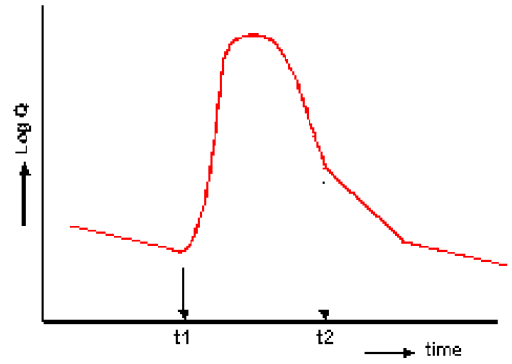


Figure 1.53: *Selection of period for LZTWM estimation*

Assuming that *UZTW* is full and *UZFWC* is empty at times t_1 and t_2 the water balance for the period $t_1 - t_2$ then reads:

$$P - R - E - DLZFPC - DLZFSC = DLZTWC \tag{1.57}$$

where:

P precipitation from t_1 to t_2 $[mm]$
R total runoff from t_1 to t_2 $[mm]$
E segment evaporation $[mm]$; this amount would small for most wet period and may be neglected
$DLZFPC$ change in storage in LZ primary free water reservoir from t_1 to t_2 $[mm]$
$DLZFSC$ change in storage in LZ supplemental free water reservoir from t_1 to t_2 $[mm]$
$DLZTWC$ change in the lower zone tension water $[mm]$

DLZTWC is a lower limit of *LZTWM* since:

1 The lower zone tension water reservoir may not have been fully empty at t_1
2 The lower zone tension water reservoir may not have been completely filled at t_2

Hence some 10 to 20 % (or more) may be added to the value obtained through Equation (1.57). If such ideal cases as assumed above cannot be found, water balances for periods of 3 to 4 months may be considered.

In Equation (1.57) *LZFPC* and *DLZFSC* are computed as follows:

$$DLZFPC = LZFPC(t_2) - LZFPC(t_1) \quad \text{where} \quad LZFPC(t) = QP(t)/LZPK \tag{1.58}$$

$$DLZFSC = LZFSC(t_2) - LZFSC(t_1) \quad \text{where} \quad LZFSC(t) = QS(t)/LZSK \tag{1.59}$$

The primary baseflows QP at times t_1 and t_2 are estimated by extrapolation from other periods. Let the discharges at t_1 and t_2 be denoted by $Q1$ and $Q2$, then the supplemental baseflows follow from:

$$QS(t_1) = Q1 - QP(t_1) \quad \text{and} \quad QS(t_2) = Q2 - QP(t_2) \tag{1.60}$$

LZFPM - lower zone primary free water storage

The maximum capacity of the primary lower zone free water, which is subject to drainage at the rate expressed by $LZPK$. The value of the lower zone primary free water maximum can be approximated from hydrograph analysis. For this the primary base flow, obtained from a semi-log plot of the lower end of the recession curve, is extended backward to the occurrence of a peak flow. Assuming that the primary free water reservoir is completely filled then, so that it outflow is at maximum ($QPmax$), its value is determined from Equation (1.50). The effectiveness of this computation in determining the maximum capacity is dependent upon the degree to which the observed hydrograph provides a representation of the maximum primary base flow. If only a portion of the groundwater discharge is observable in the stream channel, the estimated capacity based upon surface flows must be increased to include the non-channel components by applying the term *SIDE* (See below).

LZFSM - lower zone supplemental free water storage

The maximum capacity of the lower zone supplemental free water reservoir, which is subject to drainage at the rate expressed by *LZSK*. A lower limit of the lower zone free water supplemental maximum can be approximated from hydrograph analysis. Figure 1.49 illustrates the computation of the lower zone free water supplemental maximum. Note that first the primary base flow has to be identified and corrected for, see also Equation (1.60). The effectiveness of this computation in determining the maximum capacity is dependent upon the degree to which the observed hydrograph provides a representation of the maximum base flow capability of the basin. If only a portion of the groundwater discharge is observable in the stream channel, the estimated capacity based upon surface flows must be increased to include the non-channel components by applying the term *SIDE* (See below).

LZPK - lateral drainage of the lower zone primary free water reservoir.

Lateral drainage rate of the lower zone primary free water reservoir expressed as a fraction of the contents per day. The coefficient is determined from the primary base flow recession curve. Selecting flow values from this curve at some time interval Dt apart provides with the help of Equation (1.48) and Equation (1.49) the required estimate, see also Figure 1.49.

LZSK - lateral drainage of the lower zone supplemental free water reservoir

Lateral drainage rate of the lower zone supplemental free water reservoir, expressed as a fraction of the contents per day. Its computation is outlined in Figure 1.49. The procedure is similar to that of *LZPK*, with the exception that the flow values have to be corrected for the primary base flow.

PFREE

The fraction of the percolated water, which is transmitted directly to the lower zone free water aquifers. Its magnitude cannot generally be determined from hydrograph analysis. An initial value of 0.20 is suggested. Values will range between 0 and 0.50. The analysis of early season base flow allows an effective determination of *PFREE*. The relative importance of *PFREE* can be determined from storms following long dry spells that produce runoff (*UZTW* completely filled). If the hydrograph returns to approximately the same base flow as before then little filling of the lower zone free water reservoirs did take place and hence the *PFREE*-value can be rated small, 0 to 0.2. If, on the contrary, the base flow has increased significantly a *PFREE*-value as high as 0.5 may be applicable.

RSERV

Fraction of the lower zone free water, which is unavailable for transpiration purposes. Generally this value is between zero and 0.40 with 0.30 being the most common value. This factor has very low sensitivity.

SIDE

Represents that portion of base flow, which is not observed in the stream channel. When the soil is saturated, if percolation takes place at a rate, which is greater than the observable base flow, the need for additional soil moisture drainage becomes manifest. *SIDE* is the ratio of the unobserved to the observed portion of base flow. When the saturated soils do not drain to the surface channel, *SIDE* allows the correct definition of *PBASE*, in order that the saturated percolation rate may be achieved. In an area where all drainage from base flow aquifers branches surface channels, *SIDE* will be zero. Zero or near zero values occur in a large proportion of basins. However, in areas subject to extreme subsurface drainage losses, *SIDE* may be as high as 5.0. It is conceivable that in some areas the value of *SIDE* may be even higher.

SSOUT

The sub-surface outflow along the stream channel, which must be provided by the stream before water is available for surface discharge. This volume expressed in mm/time interval is generally near zero. It is recommended that the value of zero be utilised, and *SSOUT* is applied only if the ${}^{10}\log Q$ vs. time plot requires a constant addition in order to achieve a valid recession characteristic. If constant volumes of flow are added to observed stream flow, the slope of the discharge plot will be altered. That value, which is required to linearize the primary recession, is the appropriate value of *SSOUT*. It should be realised that where *SSOUT* is required, an effective determination of lower zone free water storages and discharge rates will require inclusion of the *SSOUT* value (mm/Δt)

Surface runoff

Unit hydrograph ordinates for the routing of flow from the impervious and pervious surfaces as well as interflow towards the segment outlet can be obtained through standard unit hydrograph procedures. It requires the selection of rainfall events (corrected for losses) with their resulting flood hydrographs (corrected for base flow). Note that for each event the net rainfall amount should match with the surface runoff and interflow amount. Various procedures are available to arrive at a unit hydrograph. If the rainfall intensity during the storm varies, multiple linear regression and discrete convolution techniques may be applied. The regression technique is readily available in spreadsheet software. The resulting unit hydrographs generally will show some irregularities and hence requires some smoothing afterwards. Unit hydrographs from various storms may appropriately be averaged to arrive at a representative unit hydrograph for the segment.

Another option is to use the Clark method. The principle of the Clark method was dealt with in Sub-section 2.2.8. First requirement is the derivation of a time-area diagram. If a Digital Elevation Model (DEM) is available from a catchment with appropriate software automatic calculation of the time-area diagram is possible. In the absence from a DEM the time-area diagram is derived from a basin map. By estimating travel times to the basin outlet (from river and terrain slopes, assumed roughness and flow depth) isochrones can be determined. The areas between successive isochrones is determined leading to a first estimate of the time-area diagram. The total time base of the time-area diagram should be the concentration time T_c, but due to inaccurate assessment of celerities in the basin it may differ from that. Therefore, the time base of the time-area diagram is scaled by a more appropriate estimate of T_c. An estimate for T_c may be obtained as the time lapse between the cessation of rainfall and

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the occurrence of recession on the falling limb of the hydrograph of surface runoff. The time base of the time-area diagram is scaled by this time lapse. Alternatively, the concentration time is estimated from an empirical formula applicable to the region. E.g. for a number of small catchments in the Indus basin the following equation applies:

$$T_c = \frac{1}{119}\frac{L}{\sqrt{S}} \tag{1.61}$$

where:

T_c	concentration time [h]
L	length of river [km]
S	slope of main river

The units of the time-area diagram (km^2) are converted into m^3/s by multiplication with $0.278/\Delta t$, with Δt in hours. Subsequently, the time-area diagram is routed through a linear reservoir, with reservoir coefficient k, estimated from the slope of the recession curve of the surface water hydrograph. The routing is carried out by the following equation:

$$O_{i+1} = c_1 I_{av} + c_2 O_i \tag{1.62}$$

$$I_{av} = \frac{1}{2}(I_{i+1} + I_i) \tag{1.63}$$

$$c_1 = \frac{\Delta t}{k + \Delta t/2} \tag{1.64}$$

$$c_2 = \frac{k - \Delta t/2}{k + \Delta t/2} \tag{1.65}$$

$$c_1 + c_2 = 1 \tag{1.66}$$

where:

I_{av}	average inflow during Δt (input is in form of histogram)
O	outflow from the linear reservoir

The outflow from the reservoir is the Instantaneous Unit Hydrograph (IUH) for the basin, which has to be transformed by averaging or S-curve technique into the Unit Hydrograph resulting from a rainfall of duration equal to the routing interval.

Internal routing interval

PM	Time interval increment parameter
Pt_1	Lower rainfall threshold
Pt_2	Upper rainfall threshold

In case the time step used in the model is larger than 1 hour, the model simulates the redistribution of water between the various reservoirs with a time step, which is smaller than the time interval of the basic data. Particularly for the infiltration process this effect could be important. Also the rainfall will be lumped to that smaller interval. The number of increments in the time interval is derived from:

$$N_{\Delta t} = 1 + PM * (UZFWC * F + P_{\mathit{eff}}) \tag{1.67}$$

where:

$$F = 1 \quad \text{for} \quad P_{\mathit{eff}} < Pt_1 \tag{1.68}$$

$$F = 1/2 P_{\mathit{eff}}/Pt_2 \quad \text{for} \quad Pt_1 \leq P_{\mathit{eff}} \leq Pt_2 \tag{1.69}$$

$$F = 1 - 1/2 Pt_2/P_{\mathit{eff}} \quad \text{for} \quad P_{\mathit{eff}} > Pt_2 \tag{1.70}$$

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The most important parameter is seen to be PM. Taking a very small value for PM (say $PM = 0.01$), then $N_{\Delta t}$ remains approximately1. If e.g. $PM = 0.1$ then $N_{\Delta t}$ becomes substantially larger than 1. To limit the increase of $N_{\Delta t}$ a low value for Pt_1 is to be chosen in combination with a large value of Pt_2, which will reduce the value of F.

Sequence of parameter estimation

From the presentation above it will be clear that certain parameters should be estimated before other can be assessed. The following sequence is recommended of which the first three steps are mandatory:

1 Segment area
2 Lower zone primary free water parameters LZPK and LZFPM
3 Lower zone supplemental free water parameters LZSK and LZFSM
4 Impervious fraction PCTIM
5 Upper zone parameters UZTWM, UZK and UZFWM
6 Lower zone tension capacity LZTWM
7 Percolation parameters ZPERC and REXP
8 Remaining parameters

Linear reservoirs

An essential feature of the Sacramento model is that the free water reservoirs are considered as linear reservoirs, i.e. there is a linear relation between the reservoir storage *S* and the outflow Q:

$$S = kQ \tag{1.71}$$

If the recharge is indicated by I, the continuity equation for the linear reservoir reads:

$$\frac{dS}{dt} = I - Q \tag{1.72}$$

Eliminating S from above equations results in a linear first order differential equation in Q:

$$\frac{dQ}{dt} + \frac{1}{k}Q - \frac{1}{k}I = 0 \tag{1.73}$$

With I constant and at $t = t_0$, $Q_t = Q_{t_0}$ the solution to Equation (1.73) reads:

$$Q_t = I\left(1 - \exp\left(-\frac{t - t_0}{k}\right)\right) + Q_{t_0}\exp\left(-\frac{t - t_0}{k}\right) \quad \text{for} \quad t \geq t_0 \tag{1.74}$$

When there is no recharge to the reservoir ($I = 0$) Equation (1.74) reduces to:

$$Q_t = Q_{t_0}\exp\left(-\frac{t - t_0}{k}\right) \quad \text{for} \quad t \geq t_0 \tag{1.75}$$

This equation can be compared with Equation (1.48), using the same notation:

$$Q_t = Q_{t_0}K^{(t-t_0)} \tag{1.76}$$

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Hence:

$$K = \exp\left(-\frac{1}{k}\right) \quad \text{or} \quad k = -\frac{1}{\ln K} \tag{1.77}$$

Expressing time in days, then the amount of water released from the reservoir in 1 day amounts according to equation Equation (1.71):

$$S_0 - S_1 = kQ_0 - kQ_1 = kQ_0\left(1 - \exp\left(-\frac{1}{k}\right)\right) = S_0(1-K) \tag{1.78}$$

This is seen to match with e.g. the equations for the lower zone primary free water reservoir, where:

$$S_0 = LZFPC \quad \text{and} \quad 1 - K = LZPK \tag{1.79}$$

Equation (1.77) provides a means to express the lower zone free water parameters in terms of dimensions and physical properties of aquifers. Consider the phreatic aquifer shown in Figure 1.54, which has the following dimensions and properties:

1 The width of the aquifer perpendicular to the channel is L
2 The water table at the divides is h_0 above the drainage base
3 The specific aquifer yield is m
4 The aquifer transmissivity is T.

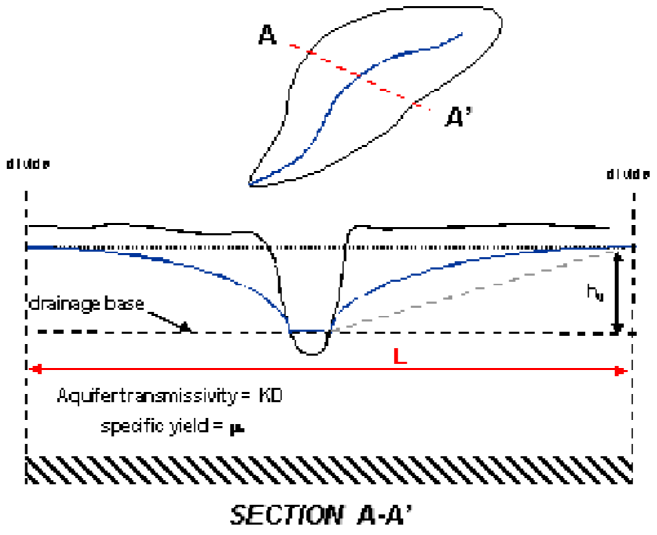


Figure 1.54: *Schematic cross-section through basin aquifer*

The amount of water stored above the drainage base per unit length of channel available for drainage is:

$$S = mc_1Lh_0 \quad \text{with} \quad \frac{1}{2} < c_1 < 1 \tag{1.80}$$

The discharge to the channel per unit length of channel according to Darcy with the Dupuit assumption

$$Q = -2T\frac{dh}{dx} = \frac{2Tc_2h_0}{(L/2)} \quad \text{with} \quad c_2 > 1 \tag{1.81}$$

Combining the above two equations by eliminating h_0 and bringing it in the form of the linear storage discharge relation Equation (1.71) :

$$S = \frac{\mu L^2}{4cT}Q \quad \text{with} \quad c = \frac{c_2}{c_1} > 1 \tag{1.82}$$

Hence for the reservoir coefficient k in Equation (1.71) it follows:

$$k = \frac{\mu L^2}{4cT} \tag{1.83}$$

The reservoir coefficient k is seen to be proportional to the square of the aquifer width and inversely proportional to T, which is logical as k is a measure for the reside-time of the percolated water in the groundwater zone. The value of c varies between 2 and 2.5 dependent on the shape of the water table. For the parameters K and $LZPK$ for the lower zone primary free water storage it then follows:

$$K = \exp\left(-\frac{4cT}{\mu L^2}\right) \quad \text{and} \quad LZPK = 1 - \exp\left(-\frac{4cT}{\mu L^2}\right) \tag{1.84}$$

A similar story applies for the lower zone free supplemental reservoir, which can be viewed as representing the drainage from the shallower based denser network of the smaller channels, see Figure 1.55. Since its main difference is with the aquifer width L, which is much smaller than for the deeper based primary channel network, its reservoir coefficient will be smaller than that of the primary free water storage and consequently $LZSK \gg LZPK$.

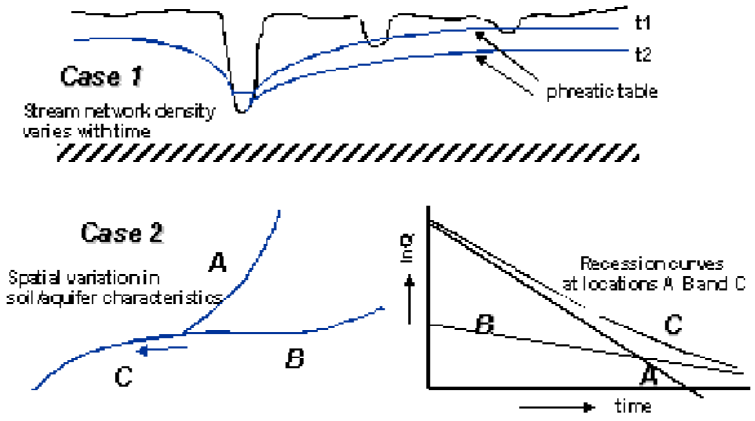


Figure 1.55: *Cases of multiple exponential decay of recession curve*

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Note that similar differences in a basin between fast and slow draining aquifers if different soils are present leading to different transmissivities.

1.1.13 DRR-HBV Rainfall Runoff concept

1.1.13.1 HBV Rainfall Runoff concept

The Hydrologiska Byrans Vattenbalansavdelning (HBV) model was introduced back in 1972 by the Swedisch Meteorological and Hydrological Institute (SMHI). It has started as a simple lumped model for river basins, but also distributed model versions (Lindström *et al.*, 1997) are available. The HBV model concept is are also available in D-HYDRO. The version available is based on the description of Killingtveit and Sælthun (Killingtveit and Sælthun, 1995). The model concept is illustrated in Figure 1.56.

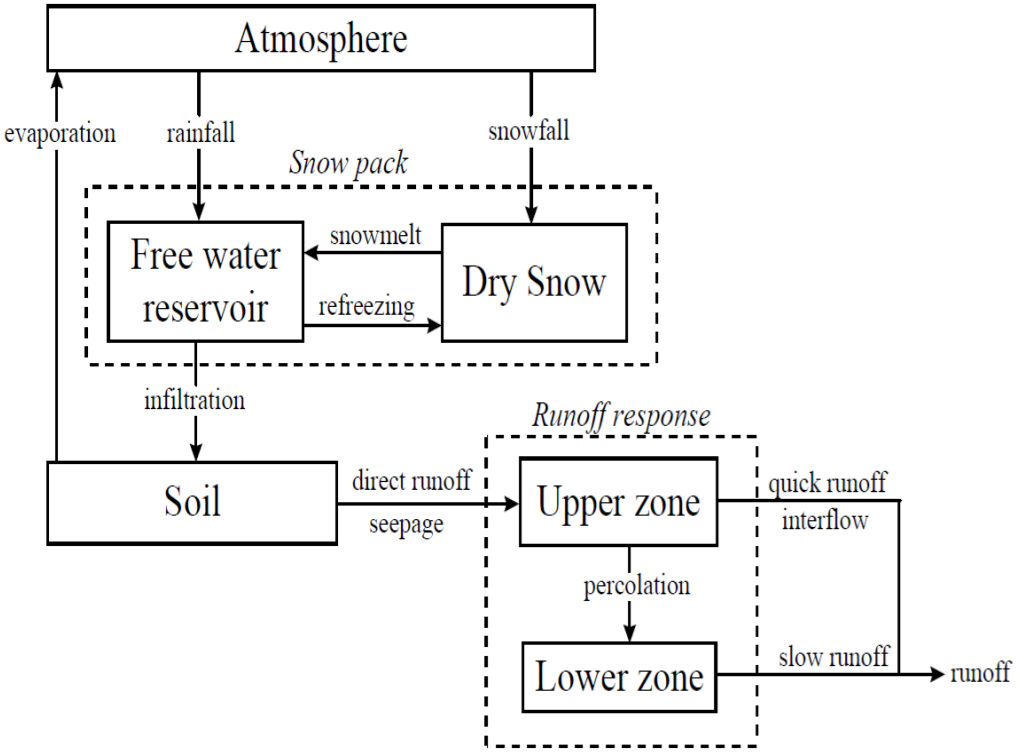


Figure 1.56: *Schematic view of HBV model (Killingtveit and Sælthun, 1995*

The HBV model contains three components to represent the land-phase of the hydrological cylde, i.e. a snow routine, a soil routine and a runoff response routine. These are described below.

1.1.13.2 The snow routine

Precipitation enters the model via the snow routine. If the air temperature, T_a, is below a user-defined threshold T_x (approximately 0 degrees Celcius), precipitation occurs as snowfall, whereas it occurs as rainfall if $(T_a >= T_x)$. If precipitation occurs as snowfall, it is added to the dry snow component within the snow pack. Otherwise it ends up in the free water reservoir, which represents the liquid water content of the snow pack. Between the two components of the snow pack, interactions take place, either through snow melt (if temperatures are above a threshold T_s or through snow refreezing (if temperatures are below threshold T_s). The

respective rates of snow melt and refreezing are:

$$F_m = C_x(T_a - T_s) \quad \text{for} \quad T_a > T_s \tag{1.85}$$

$$F_r = C_x C_r(T_s - T_a) \quad \text{for} \quad T_a < T_s \tag{1.86}$$

where

F_m	snow melt [mm]
F_r	refreezing [mm]
C_x	snow melt and refreezing rate constant [mm/day/degree]
C_r	freezing efficiency model parameter [-]
T_a	air temperature (daily average) [degrees Celcius]
T_s	threshold temperature for snow melt/refreezing [degrees Celcius]

The air temperature, T_a, is related to measured daily average temperatures, with additional compensation for vertical gradients (a constant temperature decrease of T_L degrees Celsius per 1000 m is assumed to exist; a typical value for T_L, temperature altitude constant, is 6 degrees per km). In the original HBV-concept, elevation differences within the catchment are represented through a distribution function (i.e. a hypsographic curve) which makes the snow module semi-distributed. In the modified version that is applied here, the temperature, T_a, is represented in a fully distributed manner, which means for each RR-runoff node the temperature is related to the specified node elevation. Note that this requires that the original temperature time series used as input for the model is a temperature time series at the reference level (0 meters above mean sea level) The fraction of liquid water in the snow pack (free water) is at most equal to a user defined fraction, C_p, of the water equivalent of the dry snow content. If the liquid water concentration exceeds C_p, either through snow melt or incoming rainfall, the abundant water becomes available for infiltration into the soil:

$$S_{in} = max((SW - C_p * SD); 0) \tag{1.87}$$

where

S_{in}	volume of water added to the soil module [mm]
SW	free water content of the snow pack [mm]
SD	dry snow content of the snow pack [mm]

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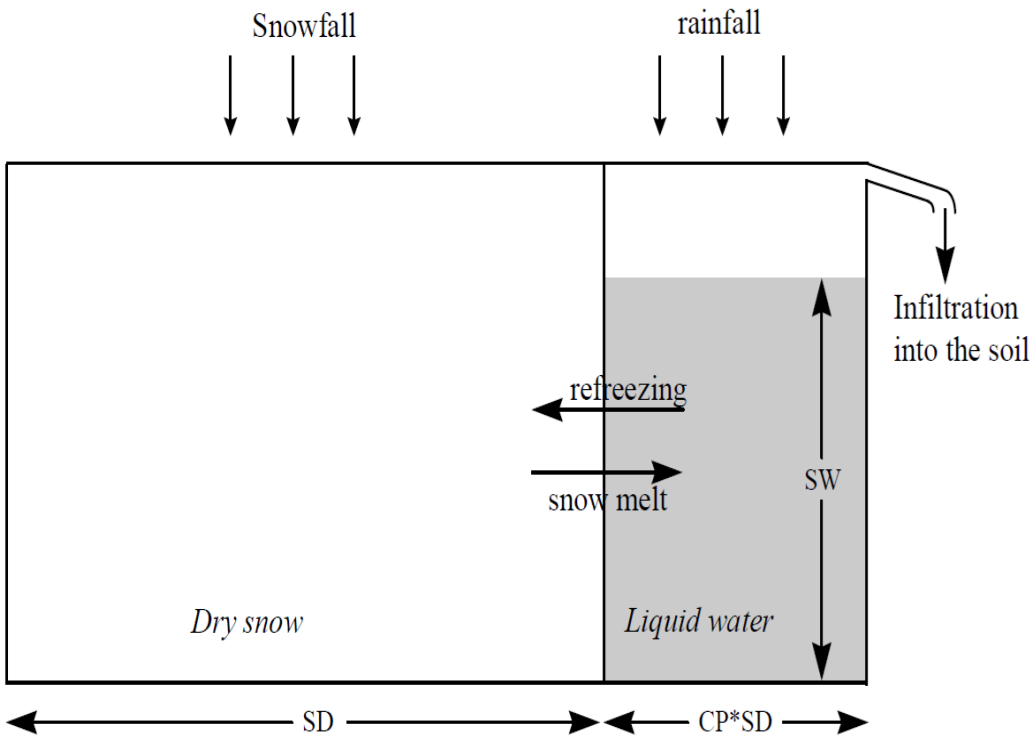


Figure 1.57: *Schematic view of the HBV snow routine*

1.1.13.3 The soil routine

The incoming water from the snow routine, S_{in}, is available for infiltration in the soil routine. The soil layer has a limited capacity, F_c, to hold soil water, which means if F_c is exceeded the abundant water cannot infiltrate and, consequently, becomes directly available for runoff.

$$S_{dr} = max((SM + S_{in} - F_c); 0) \tag{1.88}$$

where

S_{dr}	abundant soil water or direct runoff [mm]
SM	soil moisture content [mm]
F_c	field capacity to hold soil water [mm]
I_{net}	net amount of water infiltrating into the soil [mm]

The net amount of water that infiltrates into the soil, I_{net}, equals:

$$I_{net} = S_{in} - S_{dr} \tag{1.89}$$

Part of the infiltrating water, I_{net}, will runoff through the soil layer (seepage). This runoff volume, SP, is related to the soil moisture content, SM, through the following power relation:

$$SP = \left(\frac{SM}{F_c}\right)^{\beta} * I_{net} \tag{1.90}$$

where β is an empirically based dimensionless parameter. Application of Equation (1.90) implies that the amount of seepage water increases with increasing soil moisture content. The fraction of the infiltrating water which doesn't runoff, $I_{net} - SP$, is added to the available amount of soil moisture, SM.

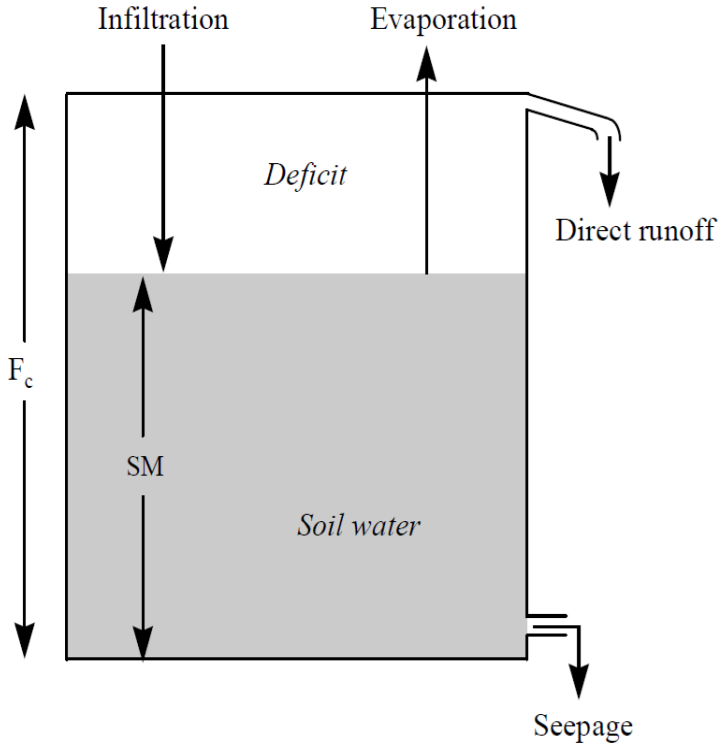


Figure 1.58: *Schematic view of the HBV soil routine*

A percentage of the soil moisture will evaporate. This percentage is related to the measured potential evaporation and the available amount of soil moisture:

$$E_a = \frac{SM}{T_m} * E_p \quad \text{for} \quad SM < T_m \tag{1.91}$$

$$E_a = E_p \quad \text{for} \quad SM >= T_m \tag{1.92}$$

where E_a is the actual evaporation, E_p is the potential evaporation and $T_m (<= F_c)$ is a user defined threshold, above which the actual evaporation equals the potential evaporation.

E_a	actual evaporation [mm]
E_p	potential evaporation [mm]
T_m	soil moisture threshold above which actual evaporation equals potential evaporation [mm] defined as a fraction of field capacity
F_c	field capacity to hold soil water [mm]

1.1.13.4 The runoff response routine

The volume of water which becomes available for runoff, $S_{dr} + SP$, is transferred to the runoff response routine. In this routine the runoff delay is simulated through the use of a number of linear reservoirs. Three types of runoff are distinguished:

- ◇ Quick runoff.
- ◇ Interflow
- ◇ Slow runoff (baseflow)

Two linear reservoirs are defined to simulate these three different processes: the upper zone (generating quick runoff and interflow) and the lower zone (generating slow runoff). The available runoff water from the soil routine (i.e. direct runoff, S_{dr}, and seepage, SP) in principle ends up in the lower zone, unless the percolation threshold, P_m, is exceeded, in which case the redundant water ends up in the upper zone:

$$\Delta V_{LZ} = min(P_m; (S_{dr} + SP)) \tag{1.93}$$

$$\Delta V_{UZ} = max(0; (S_{dr} + SP - P_m)) \tag{1.94}$$

where is the content of the upper zone, V_{LZ} is the content of the lower zone and Δ means 'increase of'.

V_{UZ}	content of upper zone [mm]
V_{LZ}	content of lower zone [mm]
Δ	increase of [mm]
P_m	maximum percolation [mm/day]
SP	runoff volume [mm]
SM	soil moisture content [mm]
S_{dr}	direct runoff [mm]

The lower zone is a linear reservoir, which means the rate of slow runoff, Q_{LZ}, which leaves this zone during one time step equals:

$$Q_{LZ} = K_{LZ} * V_{LZ} \tag{1.95}$$

where K_{LZ} is the reservoir constant.

The upper zone is also a linear reservoir, but it is slightly more complicated than the lower zone because it is divided into two zones: A lower part in which interflow is generated and an upper part in which quick flow is generated (see Figure 1.59):

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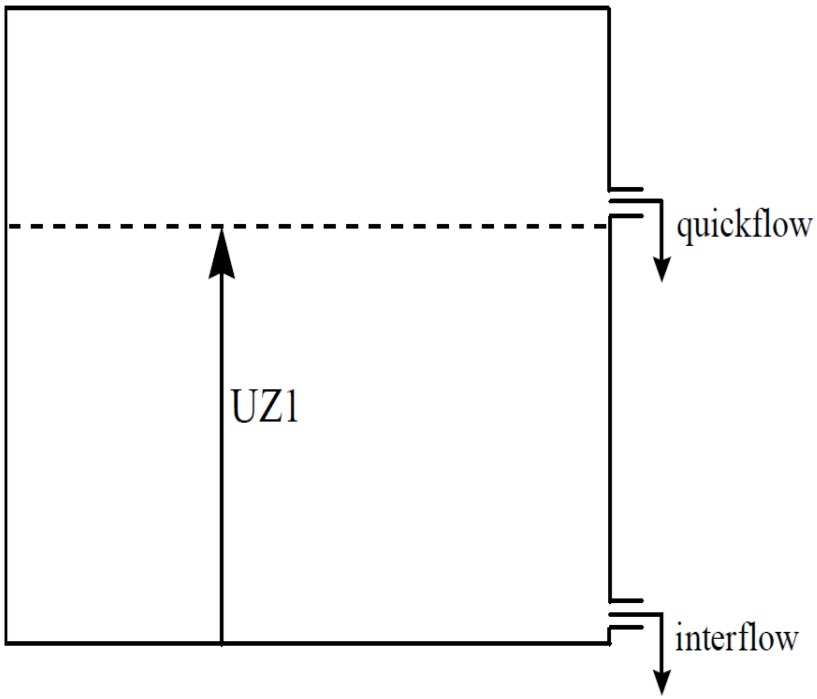


Figure 1.59: *Schematic view of the Upper Zone*

If the total water content of the upper zone, V_{UZ}, is lower than a threshold $UZ1$, the upper zone only generates interflow. On the other hand, if V_{UZ} exceeds $UZ1$, part of the upper zone water will runoff as quick flow:

$$Q_i = K_i * min(UZ1; V_{UZ}) \tag{1.96}$$

$$Q_q = K_q * max((V_{UZ} - UZ1); 0) \tag{1.97}$$

Where Q_i is the amount of generated interflow in one time step, Q_q is the amount of generated quick flow in one time step and K_i and K_q are reservoir constants for interflow and quick flow respectively.

The total runoff rate, Q, is equal to the sum of the three different runoff components:

$$Q = Q_{LZ} + Q_i + Q_q \tag{1.98}$$

The runoff behaviour in the runoff response routine is controlled by two threshold values P_m and $UZ1$ in combination with three reservoir parameters, K_{LZ}, K_i and K_q. In order to represent the differences in delay times between the three runoff components, the reservoir constants have to meet the following requirement:

$$K_{LZ} < K_i < K_q \tag{1.99}$$

1.1.13.5 HBV parameters

The HBV model as described in the previous sections has a number of parameters. These parameters are summarized in the next table. The table also contains a typical range of the parameters and proposed default value.

Table 1.5: *HBV model parameters*

Name	Description	Typical Range	Suggested	Unit
T_x	Snowfall temperature	-1.0 .. +2.0	0.0	^oC
T_s	Snowmelt temperature	-1.0 .. +2.0	0.0	^oC
T_L	Temperature altitude constant	4.0 .. 10.0	6.0	$^oCkm^{-1}$
C_x	Snow melting constant	3.0 .. 6.0	4.0	$mm/day/^oC$
C_r	Freezing efficiency	0.0 .. 0.01	.005	-
F_c	Field capacity	75 .. 300	300	mm
C_p	Free water fraction		0.1	-
T_m	Threshold	70-100% of F_c	80% of F_c	mm
β	Beta	1.0 .. 4.0	3.0	-
P_m	Maximum percolation	0.5 .. 1.0	0.6	mm/day
$UZ1$	Upper zone threshold	10 .. 40	20	mm/day
K_{LZ}	Base flow reservoir constant	.0005 .. .01	0.01	1/day
K_i	Interflow reservoir constant	0.05 .. 0.15	0.1	1/day
K_q	Quickflow reservoir constant	0.1 .. 0.5	0.3	1/day

1.1.14 SCS Rainfall Runoff concept

The SCS Curve Number Concept is a simple, widely used and efficient method for determining the amount of runoff from a rainfall event in a river basin. It is also known as the Curve Number method. This method is typically designed for a single storm event and this concept is available in SOBEK2 and D-HYDRO.

Note: the SCS implementation is not yet available in the D-HYDRO GUI, but is available in the computational core via the ASCII inputfiles.

1.1.14.1 SCS Curve Number method

The SCS model was developed by the US Department of Agriculture, Soil Conservation Service (see technical note SCS-TP-149 (Kent, 1973). An updated short description is available from the US Department of Agriculture (technical note TR55: (USDA, 1986). The method is suitable for estimating runoff from small catchments for short events (typically one day). Precipiation excess (also called direct runoff) is estimated as a function of cumulative precipitation, soil cover, land use, and antecedent soil moisure, based on the following equation:

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$$P_e = \frac{(P - I_a)^2}{P - I_a + S} \quad \text{for} \quad P > I_a \tag{1.100}$$

$$P_e = 0 \quad \text{for} \quad P <= I_a \tag{1.101}$$

where

P_e	excess rainfall [mm]
P	accumlated rainfall depth at time t [mm]
I_a	initial abstraction, also called initial loss [mm]
S	potential maximum retention [mm]

Until the accumulated rainfall P exceeds the initial abstraction, I_a, the precipitation excess, P_e, and hence the runoff, will be zero. From analysis of results from many small experimental watersheds, the Soil Conservation Service developed an empirical relationship between I_a and S:

$$I_a = 0.2 * S \tag{1.102}$$

The initial abstraction losses represent interception, initial infiltration, surface depression storage, evapotranspiration and other factors. The empirical relation of equation 1.102 was derived for agricultural watersheds.

Consequently, the cumulative precipitation excess P_e at time t is (for $P > I_a$):

$$P_e = \frac{(P - 0.2S)^2}{P + 0.8S} \tag{1.103}$$

Incremental excess for a time interval [t, t+1] is computed as the difference between the accumulated excess, P_e (t+1) at the end of the period and the accumulated excess, P_e (t), at the beginning of the period. The maximum retention S and watershed characteristics are related through an intermediate parameter, the curve number (commonly abbreviated CN) through:

$$S = \frac{25400 - 254CN}{CN} \tag{1.104}$$

The CN Curve number refers to CN_{II}, i.e. the Curve Number value for average antecedent moisture conditions (AMC). CN-values range from 100 (for water bodies) to approximately 30 for permeable soils with high infiltration rates. The CN for a watershed can be estimated as a function of land use, soil type, and antecedent watershed moisture, using tables published by the SCS and given below.

The USDA publications TR-149 (1976) and TR55 (1999 update) provide tables relating the cure number CN to hydrologic soil groups and land cover. The hydrologic soil groups distinguishes soil groups A,B,C,D and follows the US classification of soil groups depending on minimum infiltration rate. Group A soils have low runoff potential and high infiltration rates (even when thoroughly wetted). Typically these are well- drained soils, sandy or gravel, with high vertical water tranmissivity (sand, loamy sand . Group B soils have moderate infiltration rates when wetted. The soils are moderately to well drained with moderately fine to moderately coarse textures, Group C soils have low infiltration rates when wetted, and mainluy

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consists of soil with layers with moderately fine to fine texture, with low vertical water transmissivity Group D soils have high runoff potential and very low infiltration rates when wetted. Typically these are clay soils with high swelling potential, soils with a permanent high water table or nearly impervious layers close to the surface, with very low infiltration rates.

Table 1.6: *Runoff Curve Numbers Cultivated Agricultural land*

Cover type	**Treatment**	**Hydrologic Condition**	**A**	**B**	**C**	**D**
Fallow	Bare soil	-	77	86	91	94
	Crop residue cover (CR)	Poor	76	85	90	93
		Good	74	83	88	90
Row crops	Straight row (SR)	Poor	72	81	88	91
		Good	67	78	85	89
	SR+CR	Poor	71	80	87	90
		Good	64	75	82	85
	Contoured (C)	Poor	70	79	84	88
		Good	65	75	82	86
	C + CR	Poor	69	78	83	87
		Good	64	74	81	85
	Contoured terraced (CT)	Poor	66	74	80	82
		Good	62	71	78	81
	CT + CR	Poor	65	73	79	81
		Good	61	70	77	80
Small grain	SR	Poor	65	76	84	88
		Good	63	75	83	87
	SR + CR	Poor	64	75	83	86
		Good	60	72	80	84
	C	Poor	63	74	82	85
		Good	61	73	81	84
	C + CR	Poor	62	73	81	84
		Good	60	72	80	83
	CT	Poor	61	72	79	82
		Good	59	70	78	81
	CT + CR	Poor	60	71	78	81
		Good	58	69	77	80
Close seeded	SR	Poor	66	77	85	89
or broadcast		Good	58	72	81	85
legumes or	C	Poor	64	75	83	85
rotation		Good	55	69	78	83
meadow	CT	Poor	63	73	80	83
		Good	51	67	76	80

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Note: Crop residue cover (CR) only applies is the residue is on at least 5**Note:** Hydraulic condition poor means that factors influences infiltration and runoff (e.g. density and canopy of vegetative areas, year-round cover, percent of residue cover, surface roughness) are such that infiltration is impaired, and runoff is higher. **Note:** Hydraulic condition good means that infiltration is better than average, and runoff is lower.

Table 1.7: *Runoff Curve Numbers Other Agricultural land*

Cover type	**Treatment**	**Hydrologic Condition**	**A**	**B**	**C**	**D**
Pasture grass land	or range - continuous	Poor	68	79	86	89
	forage for grazing	Fair	49	69	79	84
		Good	39	61	74	80
Meadow	grass, protected from grazing and generally moved for hay	-	30	58	71	78
Brush-weed-grass	mixture with brush as	Poor	48	67	77	83
	main element	Fair	35	56	70	77
		Good	30	48	65	73
Woods - grass combination	(orchard or tree farm)	Poor	57	73	82	86
		Fair	43	65	76	82
		Good	32	58	72	79
Woods		Poor	45	66	77	83
		Fair	36	60	73	79
		Good	30	55	70	77
Farmsteads	buildings, lanes, driveways, surrounding	-	59	74	82	86

Table 1.8: *Runoff Curve Numbers Urban areas*

Cover type	**Cover and Hydrologic Condition**	**Average percent impervi-ous**	**A**	**B**	**C**	**D**
Open space	(lawns, parks, golf courses, cemeteries etc					
	Poor condition (grass cover <50 percent)		68	79	86	89
	Fair condition (grass cover 50-75 percent)		49	69	79	84

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	Good condition (grass cover > 75 percent)		39	61	74	80
Impervious areas	Parking lots, roofs, driveways		98	98	98	98
	Streets and roads					
	...paved; curbs and storm sewers		98	98	98	98
	...paved; open ditches (incl. right of way)		83	89	92	93
	...gravel (including right of way)		76	85	89	91
	...dirt (including right of way)		72	82	87	89
Western desert urban areas	Natural desert, pervious areas		63	77	85	88
	Artifucial desert, impervious area, shrub		96	96	96	96
Urban district						
	Commercial and business	85	89	92	94	95
	Industrial	72	81	88	91	93
Residential	(by average lot size)					
	.. 1/8 acre or less (town)	65	77	85	90	92
	.. 1/4 acre	38	61	77	83	87
	.. 1/3 acre	30	57	72	81	86
	.. 1/2 acre	25	54	70	80	85
	...1 acre	20	51	68	79	84
	.. 2 acres	12	46	65	77	82
Developing urban areas						
	Newly graded areas (pervious, no vegetation)		77	86	91	94
	Idle lands (use agricultural land, other)					

The above tables with Curve Numbers are applicable for average antecedent moisture conditions. Depending on the antecedent moisture conditions (AMC), curve number CN_I (dry AMC), CN_{II} (medium AMC) or CN_{III} (wet AMC) are distinguished. The dry and wet CN can be derived from CN_II as follows:

$$CN_I = \frac{4.2 * CN_{II}}{10 - 0.058 * CN_{II}} \tag{1.105}$$

$$CN_{III} = \frac{23 * CN_{II}}{10 + 0.13 * CN_{II}} \tag{1.106}$$

In the input of the SCS model concept, the user specifies the CN_{II} value and the initial antecedent moisture conditions (dry, average or wet). Using the above formula the CN value

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is adjusted for the specified AMC conditions. Figure 1.60 shows the precipitation excess or direct runoff as a function of Curve Number CN and cumulative precipitation P. The figure is taken from USDA TR55 and therefore is using standard US units (inches). The figure shows that for CN=100 the direct runoff is equal to the cumulative precipitation and follows the y=x 45 degree line in the graph, while for lower CN values the direct runoff starts later (due to initial losses 0.2 S) and slowly rises (due to the remaining losses) and finally will also reach the point where any additional rainfall is direct runoff (the line in the graph ultimately go parallell to the 45 degrees y=x line)

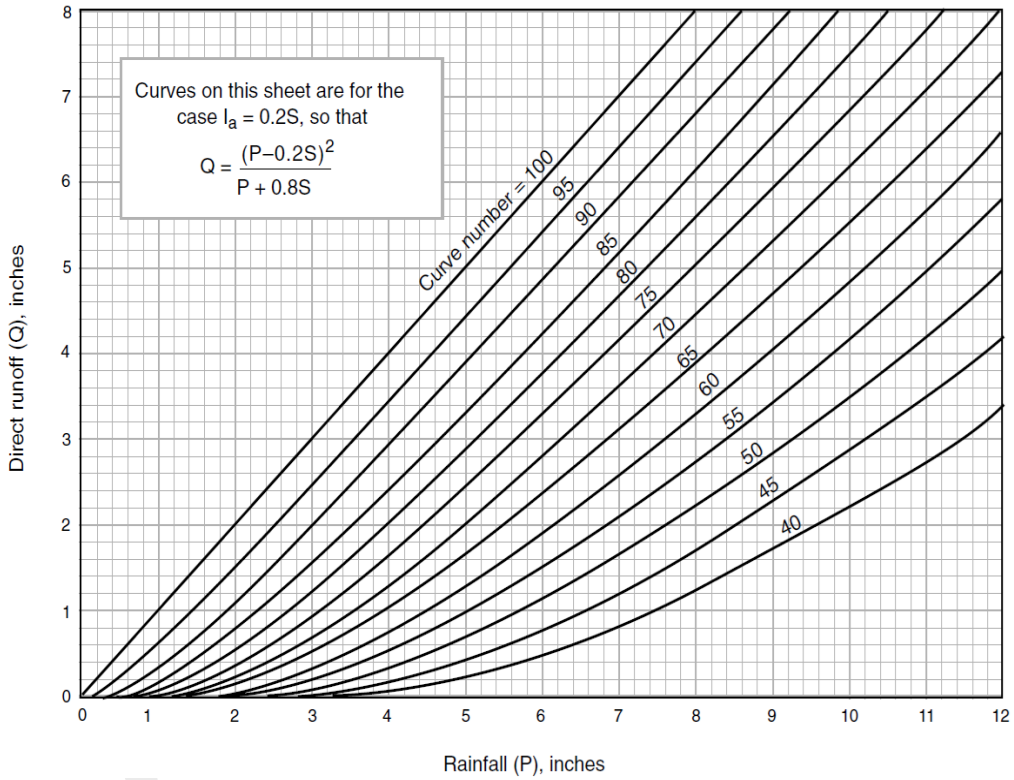


Figure 1.60: *Direct runoff (or precipitation excess) versus cumulative precipitation for different CN values*

In order to translate the excess precipitation P_e to a hydrograph, the standard SCS method uses the SCS unit hydrograph and SCS time lag formula. Other unit hydrograph approaches are also available in DRR. These are the HEC-HMS unit hydrograph or the Snyder unit hydrograph.

1.1.14.2 SCS Unit Hydrograph

The SCS time lag formula relates the time lag between rainfall peak and runoff peak to the flow path length (to the outlet point of the catchment), the Curve Number and the average basin slope.

$$T_{lag} = 60 * \left(\frac{L^{0.8}(2540 - 22.86CN)^{0.7}}{14104CN^{0.7}Y^{0.5}} \right) \tag{1.107}$$

where

T_{lag} Time lag [minutes]

L	flow path length [m]
CN	SCS Curve Number
Y	average sub basin slope [m/m]

Optionally, the time lag can be defined by the user.

The SCS unit hydrograph can be a dimensionless unit hydrograph presented in the table/figure below, or a triangular unit hydrograph of duration t_p + 1.67 t_p. Both require the estimation of t_p (time to peak) and q_p (peak flow value per mm excess rainfall).

$$t_p = \frac{D}{2} + t_{lag} \tag{1.108}$$

$$t_{lag} = 0.6 t_c \tag{1.109}$$

$$q_p = 0.208 \frac{A}{t_p} \tag{1.110}$$

where

t_p	time to peak [hours]
D	rainfall duration in computational procedure [hours]
t_{lag}	Time lag [hours]
t_c	Time of concentration [hours]
A	catchment area [km^2]
q_p	peak flow value per mm excess rain [m^3/s]

Table 1.9: *Ordinates of SCS dimensionless unit hydrograph*

t/t_p, **time ratios**	q/q_p, **discharge ratio**	Q/Q_a, **mass curve ratio outflow volumes**
0.0	0.0	0.0
0.1	0.03	0.001
0.2	0.10	0.006
0.3	0.19	0.012
0.4	0.31	0.035
0.5	0.47	0.065
0.6	0.66	0.107
0.7	0.82	0.163
0.8	0.93	0.228
0.9	0.99	0.300
1.0	1.00	0.375

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1.1	0.99	0.450
1.2	0.93	0.522
1.3	0.86	0.589
1.4	0.78	0.650
1.5	0.68	0.700
1.6	0.56	0.751
1.7	0.46	0.790
1.8	0.39	0.822
1.9	0.33	0.849
2.0	0.28	0.871
2.2	0.207	0.908
2.4	0.147	0.934
2.6	0.107	0.953
2.8	0.077	0.967
3.0	0.055	0.977
3.2	0.040	0.984
3.4	0.029	0.989
3.6	0.021	0.993
3.8	0.015	0.995
4.0	0.011	0.997
4.5	0.005	0.999
5.0	0.000	1.00

1.1.14.3 HEC-HMS Unit Hydrograph

The HEC-HMS method basically follows the Clark method (as used in Sacramento as well) to translate the excess rainfall by a time-area diagram with duration t_c and routing of this translated excess rainfall through a lineair reservoir. Only a time lag would delay but not attenuate the runoff peaks, whereas in reality attenuation definitely takes place. The unit hydrograph approcht is an effective and well-known procedure to model this process of attenuation. The unit hydrograph approach consists of N ordinals I_1, I_2 ... I_n, for timesteps 1 to n, which are allo >=0 and for which the sum is equal to 1. The method is relatively straigthforward: of all water that becomes available for runoff at a certain timestep, a fraction I_1 reaches the catchment outlet within one timestep, a fraction I_1 + I_2 1 reaches the catchment outlet within two timesteps, etc. The values and number of ordinals depends on the catchment size and slope. In large, flat catchments the number of ordinals N will be relatively large (as it takes some time for all runoff water reaches the outlet), while for small, steep slope catchments N will be small. The HEC-HMS approach defines the time of concentration t_c, i.e. the maximum time it takes for runoff water to reach the catchment outlet as follows:

$$T_c = \frac{T_{lag}}{0.6} \tag{1.111}$$

where

T_c	time of concentration [hours]
T_{lag}	Time lag either, from SCS time lag formula or user defined [hours]

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The ordinals of the time-area diagram are determined as follows (following HEC-HMS). This time-area diagram represents the translation (delay) from rainfall to runoff.

$$I_t = 1.414 \left(\frac{t}{T_c} \right)^{1.5} \quad \text{for} \quad t <= 0.5T_c \tag{1.112}$$

$$I_t = 1 - 1.414 * \left(1 - \frac{t}{T_c} \right)^{1.5} \quad \text{for} \quad 0.5T_c <= t <= T_c \tag{1.113}$$

$$I_t = 0 \quad \text{for} \quad t > T_c \tag{1.114}$$

Basically the HEC-HMS formula for the time-area diagram describes a symmetrical function, in which the the incremental area contributing to discharge at the outlet is highest around $t = 0.5 * T_c$

The Clark method uses a linear storage reservoir at the outlet point. Through the linear reservoir the attenuation of the discharge peak (and extension of the discharge period) is modelled. For the storage equation of the reservoir, we write:

$$dS/dt = In_t - Q_t \tag{1.115}$$

with

dS/dt	storage change
In_t	inflow at time t
Q_t	outflow at time t

For a linear reseroir, the storage and outflow are related by a linear equation

$$S_t = R \;\; Q_t \tag{1.116}$$

with

S_t	storage at time t
R	reservoir coefficient in [hours]
Q_t	outflow at time t

In differential form, this results in the equations

$$Q_t = C_1 In_t + C_2 Q_{t-1} \tag{1.117}$$

with routing constants C_1 and C_2

$$C_1 = \frac{\Delta t}{R + 0.5\Delta t} \tag{1.118}$$

$$C_2 = 1 - C_1 \tag{1.119}$$

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Note: if the reservoir constant R is not specified in the input, the linear reservoir is assumed to be non-existing, meaning outflow is equal to the computed inflow from the time-area diagram.

The distinction between the translation and attenuation is also referred to as convection vs diffusion in (Ponce and Nuccitelli, 2013). The next figure shows the typical effect of the linear reservoir: the runoff peak is lowered and runoff period is extended.

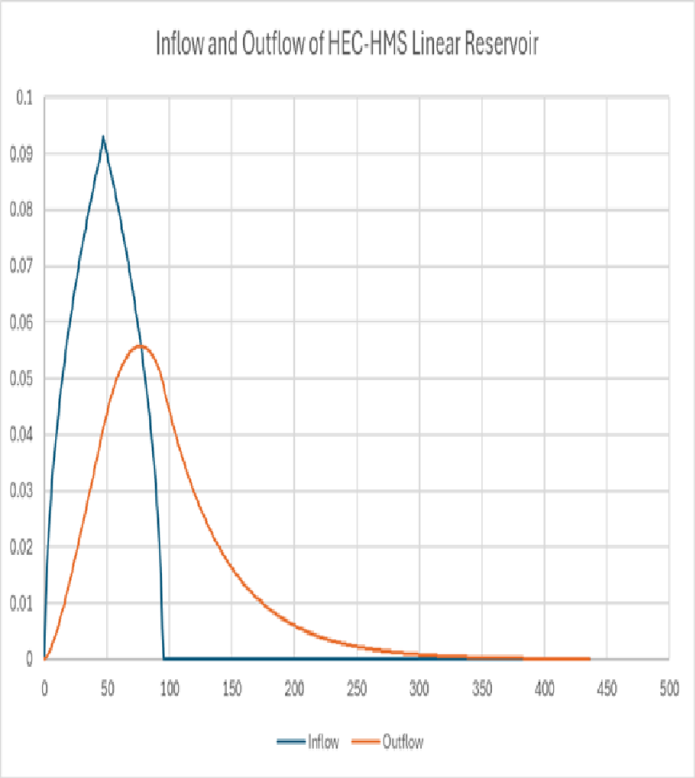


Figure 1.61: *Inflow from the time-area diagram and outflow after routing through a linear reservoir*

In order to translate the excess precipitation P_e to a hydrograph, the standard SCS method uses the SCS unit hydrograph and SCS time lag formula. Other unit hydrograph approaches are also available in DRR. These are the HEC-HMS unit hydrograph or the Snyder unit hydrograph.

1.1.14.4 Snyder Unit Hydrograph

In addition to the Clark Unit Hydrograph (in the Sacramento RR concept) and the SCS Dimensionless Unit hydrograph and HEC-HMS Unit hydrograph within the SCS RR concept, it is also possible to use the Snyder Unit Hydrograph for SCS nodes. The Snyder Unit Hydrograph is a parametric unit hydrograph originally developed for analysis of ungauged watersheds in the Appalachian Highlands in the US. The Snyder Unit Hydrograph is parameterized by two parameters, C_p, the peaking constant which is region dependent, and t_l, the basin lag time. The basin lag time is the duration between the middle of a rainfall event and the moment of maximum runoff, and correspondingly the peaking time t_P is half the duration of the rainfall event plus the basin lag time.
The Snyder Unit hydrograph is defined by 7 support points, the start, peak and end time, and four more defined by the values at which the hydrograph reaches 50% and 75% of peak value, respectively.

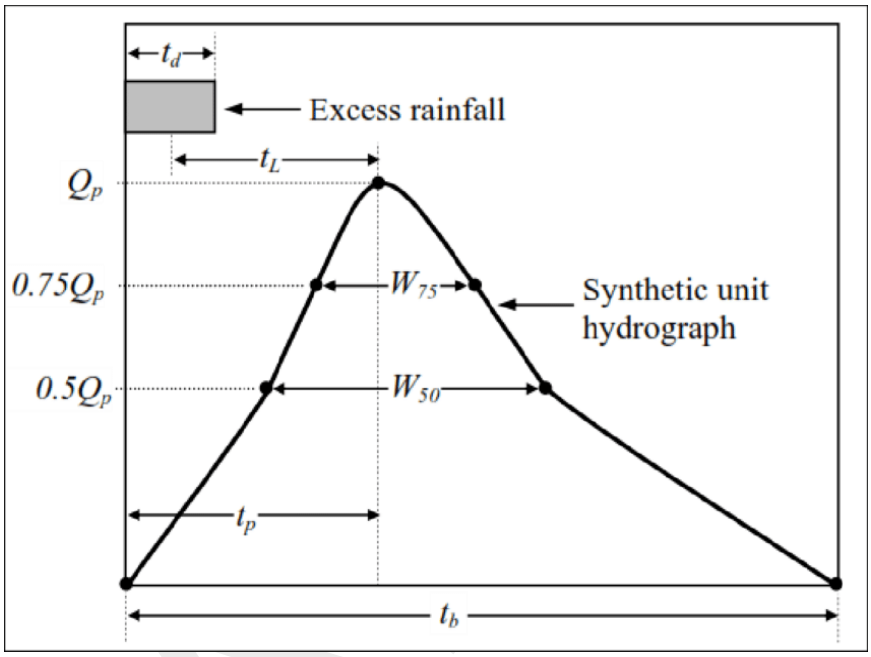


Figure 1.62: *Illustration of a Snyder hydrograph*

With:

$$Q_P = \frac{2.75\, C_p\, A}{t_P} \qquad W_{50} = \frac{2.14}{\frac{Q_P}{A}^{1.08}}$$
$$W_{75} = \frac{1.22}{\frac{Q_P}{A}^{1.08}} \qquad t_b = 11.11\frac{A}{Q_P} - 1.5\, W_{50} - W_{75} \tag{1.120}$$

Where A is the runoff node area in km^2.

Intermediate values of the hydrograph are obtained by interpolation. Between the first 6 support points linear interpolation is used. At any point between the last two support points the graph can be set to decay exponentially.

Additionally, a base runoff flow can be added to the Snyder unit hydrograph, starting at the beginning of the simulation with a user specified value. It can either remain constant over the course of the simulation or decay in either linear or exponential fashion.

A description of the input parameters for the Snyder Unit Hydrograph can be found in section: 1.2.11

1.1.14.5 Green-Ampt Infiltration

Besides the standard SCS implementation as described before, and the different options for unit hydrographs (being the SCS unit hydrograph, the HEC-HMS unit hydrograph or the Snyder unit hydrograph), the Green-Ampt infiltration has been added as a seperate option. The Green-Ampt formulation is optional. In case it is switched on, the excess rainfall is adjusted with the computed infiltration according to the Green-Ampt formulation (Green and Ampt, 1911). The Green-Ampt infiltration model is implemented in many model packages like e.g. HEC-HMS (Hydrologic Engineering Center, 2023) and EPA-SWMM (Rossman and Huber,

2015). The implementation of the infiltration formulation in DRR follows the description of the Storm Water Management Model EPA-SWMM (Rossman and Huber, 2015), pages 106-119.

The Green-Ampt infiltration model can be considered a simplification of the Richards' equation of unsteady water flow in soil. The implementation in DRR requires the following input data:

K_{sat}	Saturated hydrulic conductivity in [mm/hour]
ψ	Suction head at wetting front in [[]mm]
θ_{dmax}	Maximum soil mosture deficit, also assumed as the initial soil moisture deficit [-]

The Green-Ampt method assumes a sharp wetting front in the soil column, where infiltrated water moved vertically downward as a saturated layer, starting at the surface. In the wetted zone the soil moisture content θ is saturated, while in the unwetted zone it is at some initial level. Through the wetted zone, the water flows downward as a function of the saturated hydraulic conductivity K_{sat}, the capillary suction head along the wetting front ψ, the depth of ponded water at the surface d and the depth of the saturated layer below the surface L_s. This is illustrated in the next figure.

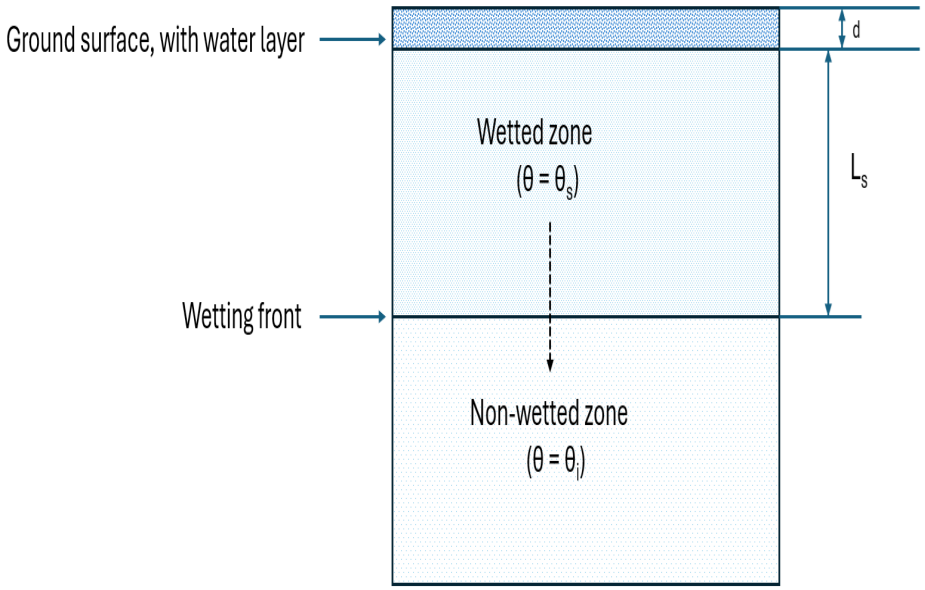


Figure 1.63: *Representation of Green-Ampt infiltration model*

$$f_p = K_{sat}\left(\frac{d + L_s + \psi}{L_s}\right) \tag{1.121}$$

where

f_p	infiltration [mm/hour]
K_{sat}	Saturated hydrulic conductivity in [mm/hour]
ψ	Suction head at wetting front in [mm]

d	depth of water on surface [mm]
L_s	depth of saturated layer below the ground surface [mm]

With

F	cumulative infiltration [mm]
θ_d	soil moisture deficit to be filled below the wetting front [mm]

and using

$$L_s = \left(\frac{F}{\theta_d}\right) \tag{1.122}$$

and substituting this in equation 1.121 and assuming d is small compared to L_s gives

$$f_p = K_{sat}\left(1 + \frac{\psi\theta_d}{F}\right) \tag{1.123}$$

which applies after a saturated layer has developed on the surface.

Before a saturated layer has developed on the surface, the infiltration (capacity) f_p will be equal to the rainfall intensity i: $f_p = i$

Solving equation 1.123 for F and denoting the result when reaching saturation with F_s, and substituting $f_p = i$, gives

$$F_s = \frac{K_{sat}\psi\theta_d}{i - K_{sat}} \tag{1.124}$$

Note that when rainfall intensity i is smaller than K_{sat}, F_s will not be calculated and only the cumulative infiltration F is updated.

An example with K_{sat} = 6.35 mm/hr (0.25 inches/hour), ψ = 165.1 mm (6.5 inch), and θ_d = 0.20, is illustrated in the next figures. Figure 1.64 shows the specified rainfall and computed Green-Ampt infiltration by DRR. The used computational timestep is 15 minutes, with rainfall 5.08 mm/timestep, meaning 20.32 mm/hour (0.8 inch/hour). Figure 1.65 shows the infiltration f (in mm/hour) versus the cumulative infiltration (in mm) for this example. The results match with the example shown in figure 4-5 of the SWMM reference manual.

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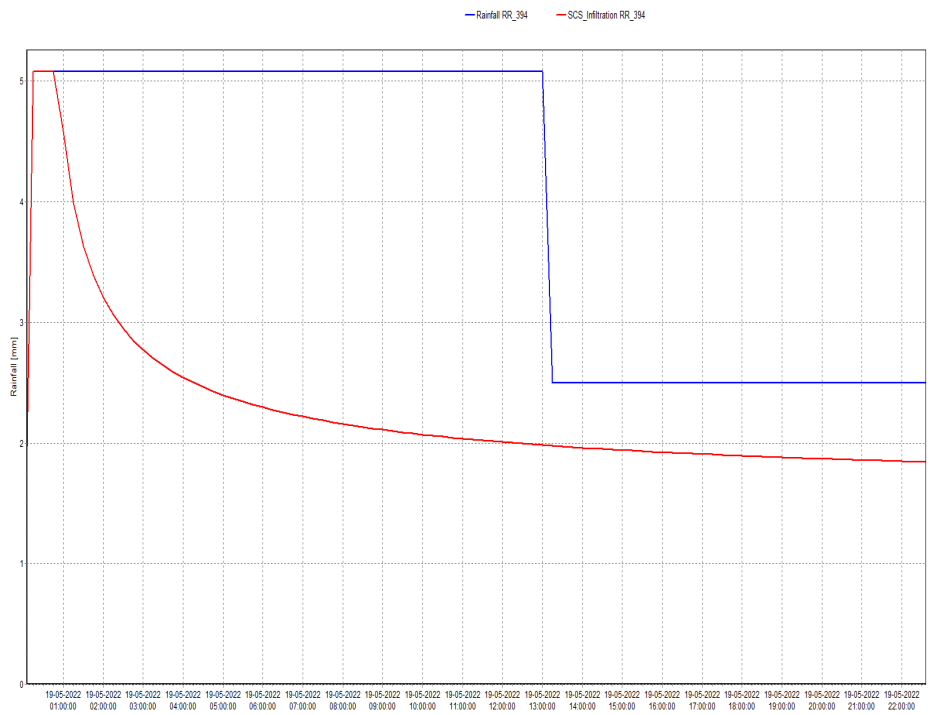


Figure 1.64: *GreenAmpt Infiltration Example, Specified rainfall (blue line) and computed infiltration f (red line)*

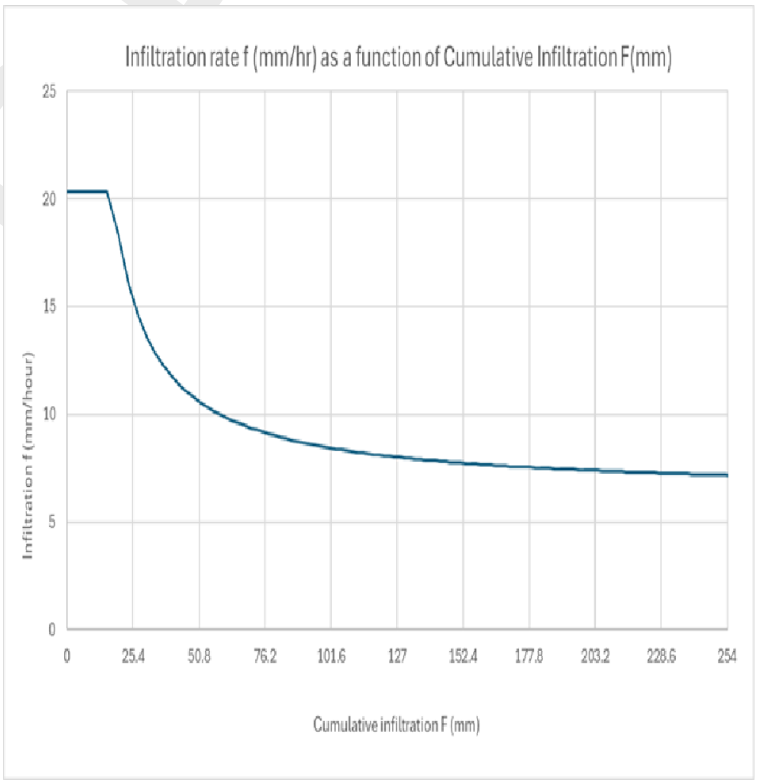


Figure 1.65: *GreenAmpt Infiltration f as a function of cumulative infiltration F*

The initial, flat portion of the curve corresponds to f = i meaning infiltration equals the rainfall intensity, up to the point where F = F_s (Equation 1.124). The remainder of the curve corresponds to the potential rate computed with Equation 1.123. Note that the infiltration rate slowly approaches to the saturated conductivity K_{sat} value of 6.35 mm/hour (0.25 inch/hour)

From equation 1.123 is follows that the infiltration capacity after surface saturation depends on the infiltrated volume, which in turn depends on the infiltration rates in previous time steps. To avoid numerical errors over long time steps, the integrated form of the Green-Ampt equation is more suitable. That is, f_p is replaced by dF/dt and integrated to obtain:

$$F = K_{sat}t + \psi\theta_d \ln\left(1 + \frac{F}{\psi\theta_d}\right) \tag{1.125}$$

With F_1 being the known cumulative infiltration at the start of the time step and F_2 the unknown cumulative infiltration at the end of the time step then one can write:

$$F_2 = C + \psi\theta_d \ln(F_2 + \psi\theta_d) \tag{1.126}$$

where

$$C = K_{sat}\Delta t + F_1 - \psi\theta_d \ln(F_1 + \psi\theta_d) \tag{1.127}$$

is a known constant.

Equation 1.126 can be solved numerically for F_2. The average infiltration capacity f_p over the time step Δt can then be computed as (F_2 - F_1)/Δ t.

Recovery of infiltration capacity

Infiltration is usually dominated by conditions in the uppermost layer of the soil. The thickness of this layer depends on the soil type; for a sandy soil it could be several centimeters, for heavy clay it would be less. The equation used to determine the thickness of the layer L_u is:

$$L_u = 4 * \sqrt{K_{sat}} * \sqrt{25.4} \tag{1.128}$$

where L_u has units of mm and K_{sat} is expressed in mm/hour. Since in DRR Green-Ampt computations the variables are in mm, mm/hour etc., the conversion factor of 25.4 from inches to mm has to be taken into account. Thus for a high K_{sat} of 12.7 mm/hour (0.5 inch/hour) the thickness computed by Equation 1.128 is 71.8 mm (2.83 inches). For a soil with a low hydraulic conductivity, say K_{sat} = 2.54 mm/hour (0.1 inch/hour), the computed thickness is 32.1 mm (1.26 inches). This constant thickness is different from the saturated zone thickness L_s shown in Figure 1.63, which grows over time as infiltration proceeds.

In the Green-Ampt model, the initial soil moisture deficit at the start of a rainfall event determines how much infiltration capacity is available during the event itself. Recall that the moisture deficit θ_d is the difference between the saturated moisture content θ_s and the initial moisture content θ_i. During a dry period the moisture deficit in the upper soil zone, θ_{du}, is regenerated, i.e., its value is increased. The Green-Ampt model keeps continuous track of

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this quantity. At the start of a simulation, θ_{du} is set equal to the user-supplied initial value of θ_{dmax}. During a wet period when infiltration occurs at a rate f over a time step of Δt, θ_{du} is decreased according to: equation Equation (1.129) down to a possible limiting value of 0.

$$\theta_{du} <- \quad \theta_{du} - \frac{f\Delta t}{L_u} \tag{1.129}$$

During a dry period it increases as follows:

$$\theta_{du} <- \quad \theta_{du} + k_r \theta_{dmax} \Delta t \tag{1.130}$$

up to a maximum possible value of θ_{dmax}, where k_r is a recovery constant (1/hour).

Assuming the recovery constant to be depending on K_{sat}, such that tight, clay soils with low K_{sat} take longer to recover than loose, sandy soils with high K_{sat}, the following relationship is used for k_r:

$$k_r = \frac{\sqrt{K_{sat}}}{75 * \sqrt{25.4}} \tag{1.131}$$

where the constant 75 has units of $(in/hr)^{0.5}$. Since in DRR GreenAmpt computations the variables are in mm, mm/hour etc., the conversion factor of 25.4 from inches to mm has to be taken into account.

Note that the time it would take a fully saturated soil to recovery to its maximum capacity is simply 1/k_r.

To complete the recovery process it is necessary to define the minimum amount of time that a soil must remain in recovery before any further rainfall would be considered as an independent event. This time T_r (hour) is computed as:

$$T_r = \frac{0.06}{k_r} = \frac{4.50 * \sqrt{25.4}}{\sqrt{K_{sat}}} \tag{1.132}$$

Thus when a new period of rainfall occurs after a recovery interval of at least T_r hours, the two-stage Green-Ampt infiltration process is re-started with θ_d = θ_{du} and F = 0. Figures 1.66 and 1.67 summarize the functional dependence of the three internally computed recovery parameters L_u, k_r, and T_r on the saturated hydraulic conductivity K_{sat}.

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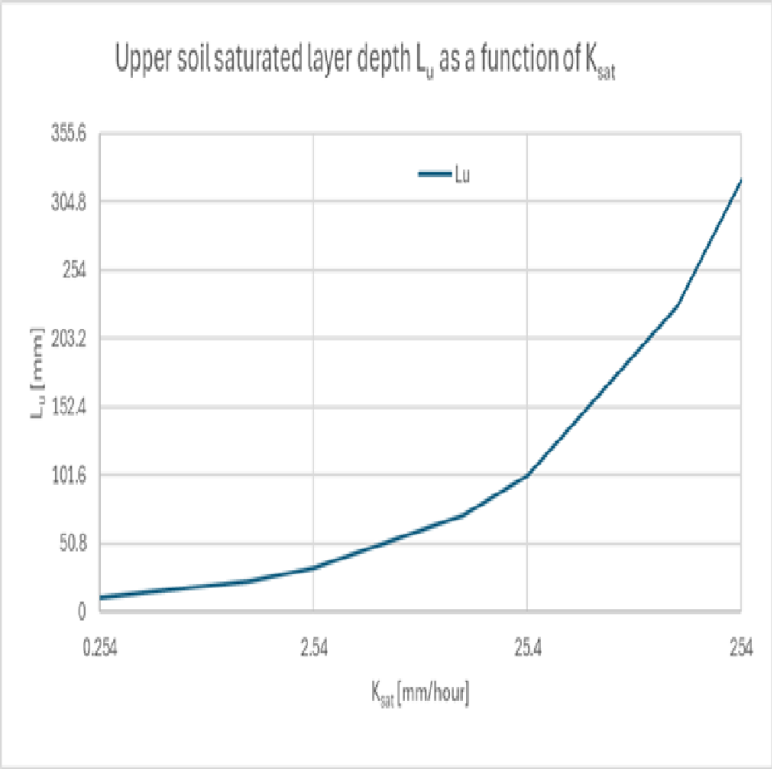


Figure 1.66: *GreenAmpt L_u parameter as a function of saturated hydraulic conductivity*

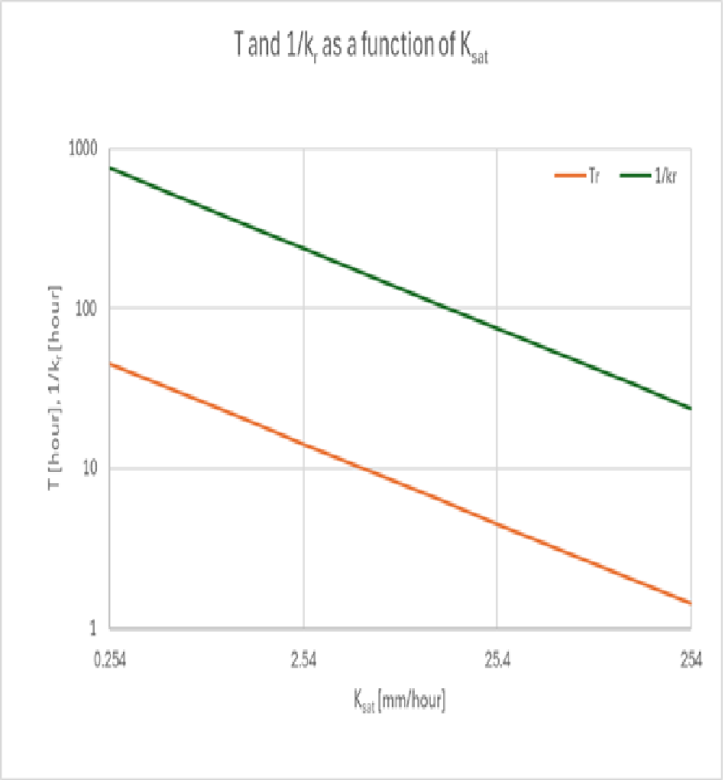


Figure 1.67: *GreenAmpt T and k_r as a function of saturated hydraulic conductivity*

1.1.14.6 GreenAmpt parameter estimation

The GreenAmpt input parameters K_{sat}, ψ and θ_{dmax}, denoting respectivly the saturated hydraulic conductivity in mm/hour, the suction head at wetting front in mm, and the maximum soil moisture deficit, can be estimated based on soil charactics tables (taken from the SWMM reference manual and converted to mm and mm/hour where needed.

The following table (Rawls *et al.*, 1983) gives estimates for different soil classes for the porosity ϕ, effective porosity ϕ_e, wetting front suction head ψ and saturated hydraulic conductivity K_{sat}. Note that the original table uses inches and inches/hour as units for the suction head at the wetting front and the saturated hydraulic conductivity, but in this manual we use mm and mm/hour.

Table 1.10: *Green Ampt parameters for different soil classes*

Soil Class	**Porosity** ϕ **-**	**Effective Porosity** ϕ_e **-**	**Wetting front suction head** ψ **mm**	**Saturated Hydraulic** K_{Sat} **mm/hour**
Sand	0.437 (0.374-0.500)	0.417 (0.354-0.490)	49.5 (9.7-253.5)	120.4
Loamy sand	0.437 (0.363-0.506)	0.401 (0.329-0.473)	61.2 (13.5-279.4)	30.0
Sandy Loam	0.453 (0.351-0.555)	0.412 (0.283-0.541)	110.0 (26.7-454.7)	10.9
Loam	0.463 (0.375-0.551)	0.434 (0.334-0.534)	88.9 (13.2-593.9)	3.3
Silt Loam	0.501 (0.420-0.582)	0.486 (0.394-0.578)	166.9 (29.2-954.0)	6.6
Sandy clay loam	0.398 (0.332-0.464)	0.330 (0.235-0.425)	218.4 (44.2-1080.)	1.5
Clay loam	0.464 (0.409-0.519)	0.309 (0.279-0.501)	208.8 (48.0-911.1)	1.0
Silty clay loam	0.471 (0.418-0.524)	0.432 (0.347-0.517)	273.1 (56.6-1315.)	1.0
Sandy Clay	0.430 (0.370-0.490)	0.321 (0.207-0.435)	239.0 (40.9-1402.)	0.5
Silty Clay	0.479 (0.425-0.533)	0.423 (0.334-0.512)	292.1 (61.2-1394.)	0.5
Clay	0.475 (0.427-0.523)	0.385 (0.269-0.501)	316.2 (64.0-1564.)	0.3

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Note: The effective porosity is defined as the difference between the porosity and the residual moisture content θ_r that remains when a saturated soil is allowed to drain thoroughly.

Note: The numbers in parenthesis in columns 2 to 4 in the above table are approximately plus or minus one standard deviation from the parameter value shown.

Note: As can be concluded from the above table, the (effective) porosity can be very similar for different soil types. The wetting front suction head ψ and (especially) the saturated hydraulic conductivity are more discriminatory. The ranges of parameter values for (effective) porosity and suction head can be largely overlapping for different soils.

Table 1.11: *Typical values for θ_{dmax} for various soils*

Soil Texture	**Typical $\theta_d max$ at soil wilting point**
Sand	0.34
Sandy Loam	0.33
Silt Loam	0.32
Loam	0.31
Sandy clay loam	0.26
Clay loam	0.24
Clay	0.21

1.1.15 Walrus Rainfall Runoff concept

The Wageningen Lowland Runoff Simulator (in short: Walrus) is a lumped rainfall-runoff model concept developed by Wageningen University and Research. Walrus is the successor of the earlier Wageningen model (WagMod) that was implemented in SOBEK2 versions. The Walrus model concept is described in detail in publications ((Brauer *et al.*, 2014a) and (Brauer *et al.*, 2014b)) and is summarized here.

Note: the Walrus implementation is not yet available in the GUI, but is available in the computational core via the ASCII inputfiles.

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1.1.15.1 Walrus

Like other lumped conceptual models like Sacramento, HBV, and NAM, the Walrus model concept distinghuishes a number of connected reservoirs which compute the response of a modelled catchment on forcings as rainfall, potential evapotranspiration, and external supplied water. The model structure is illustrated in Figure 1.68.

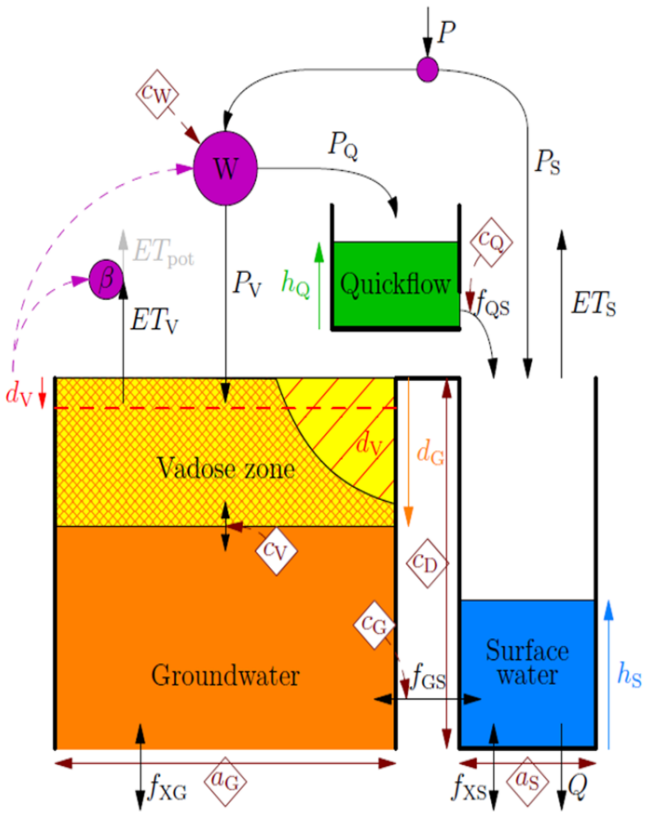


Figure 1.68: *Walrus Model Structure*

The model structures distinguishes the following elements:

- ◇ land surface (purple in Figure 1.68);
- ◇ the vadose zone (also called unsaturated zone) within the soil reservoir (yellow-red hatched);
- ◇ groundwater zone within the soil reservoir (orange);
- ◇ quickflow reservoir (green); and
- ◇ surface water reservoir (blue).;

The fluxes between the reservoirs are black arrows. Model parameters are in brown diamonds. Model states (reservoir contents or shortages) are in the colour of the reservoir they belong to. The variables (parameters) are described in the next paragraph table and text.

Walrus distinghuishes three reservoirs within the five compartments, i.e.

- ◇ the soil water reservoir, comsisting of the vadose zone and groundwater zone;
- ◇ the quickflow reservoir; and
- ◇ the surface water reservoir.

The precipitation P falls on the land surface. A fixed fraction P_s corresponding with the surface

water fraction (a_s) is lead to the surface water reservoir. The remaining rainfall falls on the land. Depending on the soil wetness index W an amount P_V is infiltrating and percolating into the soil, and an amount P_Q is flowing into the quickflow reservoir. Evapotranspiration is removing water from the vadose zone (ET_V) and from the surface water reservoir (ET_S).

The vadose zone is the upper part of the soil reservoir and extends from the soil surface to the groundwater table (d_G). The dryness of soil is indicated by the storage deficit d_V, representing the effective volume of empty pores per unit area. The storage deficit determines the soil wetness index (W) and the evaporation reduction (β). The groundwater zone is the saturated lower part of the soil reservoir. The state is indicated by the groundwater depth d_G (in mm below surface). Walrus assumes that the groundwater depth can drop below the depth of the drainage channels (c_D) in dry periods. The groundwater depth together with the surface water level (h_S) determines the groundwater drainage or infiltration from surface water (f_{GS})

The quickflow reservoir captures precipitation which does not flow through the soil, but flows via the quickflow reservoir to the surface water (f_{QS}). This flow thus represents macropore flow through drainpipes, animal burrows and soil cracks, but also local overland flow. The variable h_Q represents the level of the quickflow reservoir (in mm).

The surface water reservoir determines the outflow Q based on the actual water level h_S and a stage-discharge relation. The parameter c_D is the level of the channel bed (mm below surface). External fluxes adding water to or removing water from the soil water reservoir by seepage (f_{XG}) or adding or removing water to/from the surface water reservoir (f_{XS}) can be imposed by the user as timeseries.

All Walrus input, output and computations are done in mm or mm/hour. DRR allows to specify the forcings (rainfall, potential evapotranspiration, external supplies or extractions) at different timesteps; these are converted when supplied to Walrus.

1.1.15.2 Walrus Parameters

Table 1.12: *Walrus model parameters*

Parameter	Parameter name	Formula or default value	Unit
	States		
d_V	Storage deficit		mm
d_G	Groundwater depth		mm
h_Q	level quickflow reservoir		mm
h_S	Surface water level		mm
	Dependent variables		
W	Wetness index	Function of d_V	-
β	Evapotranspiration reduction factor	Function of d_V	-
dV_{eq}	Equilibrium storage deficit	Function of d_G	mm
	External fluxes		
	input		
P	Precipitation		mm/hour

ET_{pot}	Potential evapo(transpi)ration		mm/hour
f_{XG}	Groundwater seepage or extraction		mm/hour
f_{XS}	Surface water supply or extraction		mm/hour
	External fluxes		mm/hour
	output		mm/hour
ET_{act}	Actual evapo(transpi)ration	$= ET_V + ET_S$	mm/hour
Q	Discharge	Function of h_S	mm/hour
	Internal fluxes		
P_S	Precipitation into surface water	Pa_S	mm/hour
P_V	Precipitation into vadose zone	$P(1-W)a_G$	mm/hour
P_Q	Precipitation into quickflow reservoir	PWa_G	mm/hour
ET_V	Actual evapo(transpi)ration vadose zone	$ET_{pot}\beta a_G$	mm/hour
ET_S	Actual evaporation surface water	$ET_{pot}a_S$	mm/hour
f_{GS}	Grondwater drainage and/or infiltration	$(c_D - d_G - h_S)*$ $max((c_D - d_G, h_S)a_G/c_G$	mm/hour
f_{QS}	Quickflow	$(h_Q/c_Q)a_G$	mm/hour
	Model input parameters		
c_W	Wetness index parameters	200	mm
c_V	Vadose zone relaxation time	4	hour
c_G	Groundwater reservoir constant	5000000	mm.hour
c_Q	Quickflow reservoir constant	10.	hour
	Supplied parameters		
a_S	Surface water area fraction	0.1	-
a_G	Groundwater reservoir area fraction	$1 - a_S$	-
c_D	Channel depth		mm
	User defined or analytical function		

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$W(d_V)$	Wetness index	Function	-
$\beta(d_V)$	Evapotranspiration reduction factor	Function	-
$d_{Veq}(d_G)$	Equilibrium storage deficit	Function	mm
$Q(h_S)$	Stage-discharge relation	Function	mm/hour
	Parameters for analytical functions		
$\zeta 1$	Curvature ET reduction function	0.02	-
$\zeta 2$	Translation ET reduction function	400.	mm
θ_s	Soil moisture content at saturation	Set based on soil type	-
b	Pore size distribution parameter	Set based on soil type	-
ψ_{ae}	Air entru pressure	Set based on soil type	mm
c_S	Bankfull Q	4	mm/hour
x_S	Stage-discharge exponent	1.5	-
h_{Smin}	Surface water level when Q=0.0	0.0	mm

The available Walrus soil types and corresponding parameter values are given in table Table 1.13 Soil type 21 to 31 are standard Dutch soil types, type 32 and 33 are soil types representing the calibrated catchments of Hupsel and Cabauw in the Netherlands. Soil type 34 is a custom soil type for which the user can specify the parameter values for the pore-size distribution b, air entry pressure ψ_{ae} and soil moisture content at saturation θ_s. The given parameters are the parameters of the Brooks-Corey equilibrium soil moisture profile (Brooks and Corey, 1964) with the values for the first 11 standard soil types taken from (Clapp and Hornberger, 1978). The relation describes the soil moisture content θ as a function of height above the groundwater table h [mm], saturated soil moisture content θ_s and the air entry pressure ψ_{ae}, following (Clapp and Hornberger, 1978)

$$\theta = \theta_s \left(\frac{h}{\psi_{ae}} \right)^{-1/b} \tag{1.133}$$

Table 1.13: *Walrus soil types and related Brooks-Corey parameter values*

Soil type nr	**Soil type name**	$b(-)$	$\psi_a e(mm)$	$\theta_s(-)$
21	sand	4.05	121	0.395
22	loamy sand	4.38	90	0.410
23	sandy loam	4.9	218	0.435

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24	silt loam	5.3	786	0.485
25	loam	5.39	478	0.451
26	sandy clay loam	7.12	299	0.42
27	silt clay loam	7.75	356	0.477
28	clay loam	8.52	630	0.476
29	sandy clay	10.4	153	0.426
30	silty clay	10.4	490	0.492
31	clay	11.4	405	0.482
32	cal Hupsel	2.63	90	0.418
33	cal Cabauw	16.77	9	0.639
34	custom	User defined	User defined	User defined

Default analytical functions for the wetness index, evaporation reduction, equilibrium storage deficit and stage-discharge relation are available within Walrus. The user has to supply some parameter values for these functions. As an alternative, there is the option to use a user-defined interpolation table for these functions. In typical practical applications so far, the default Walrus analytical functions have been used.

The following default analytical functions are implemented within Walrus:

For the wetness index W:

$$W = 0.5 + 0.5cos(max(min(d_V, c_W), 0) * 3.1415/c_W) \tag{1.134}$$

For the evaporation reduction β:

$$\beta = 0.5 + 0.5 * \frac{(1 - exp^{\zeta_1(d_V - \zeta_2)})}{(1 + exp^{\zeta_1 * (d_V - \zeta_2)})} \tag{1.135}$$

For the Equilibrium storage deficit d_{Veq}:

$$d_{Veq} = \theta_s * (d_G - \frac{dG^{1-1/b}}{(1-1/b)\psi_{ae}^{-1/b}} - \frac{\psi_{ae}}{1-b} \quad \text{for} \quad d_G > \psi_{ae} \tag{1.136}$$

$$d_{Veq} = d_G \quad \text{for} \quad d_G < 0 \tag{1.137}$$

$$d_{Veq} = 0 \quad \text{otherwise} \tag{1.138}$$

For the stage-discharge relation:

$$Q = c_S * \left(\frac{h_S - h_{Smin}}{(c_D - h_{Smin})} \right)^{x_S} \tag{1.139}$$

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The next figures illustrate the typical pattern of these default analytical functions.

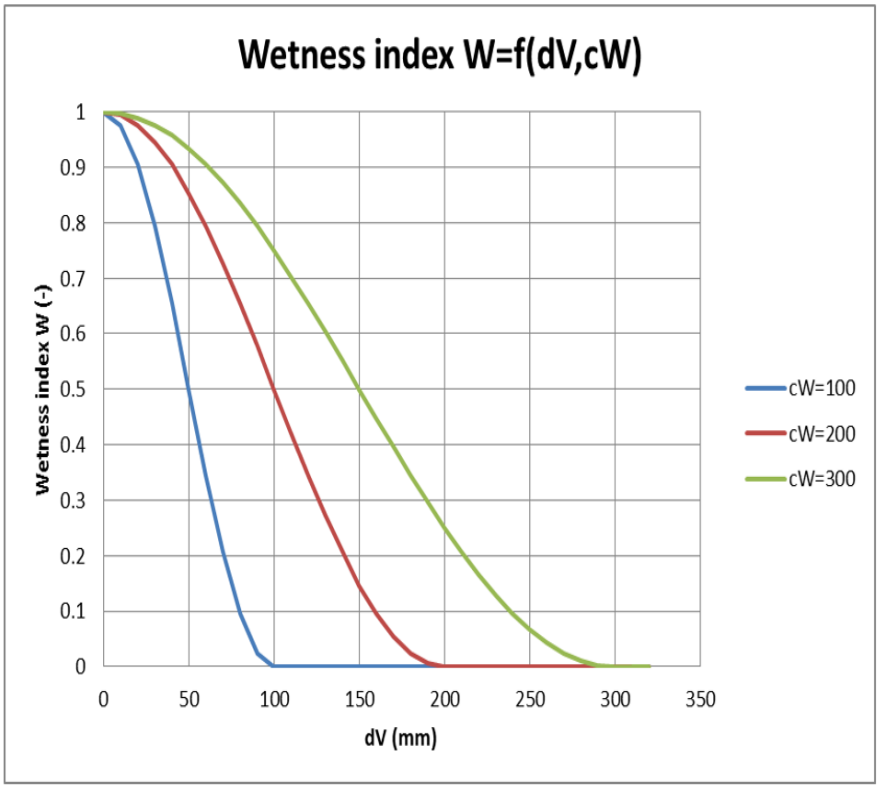


Figure 1.69: *Walrus Wetness Index Function*

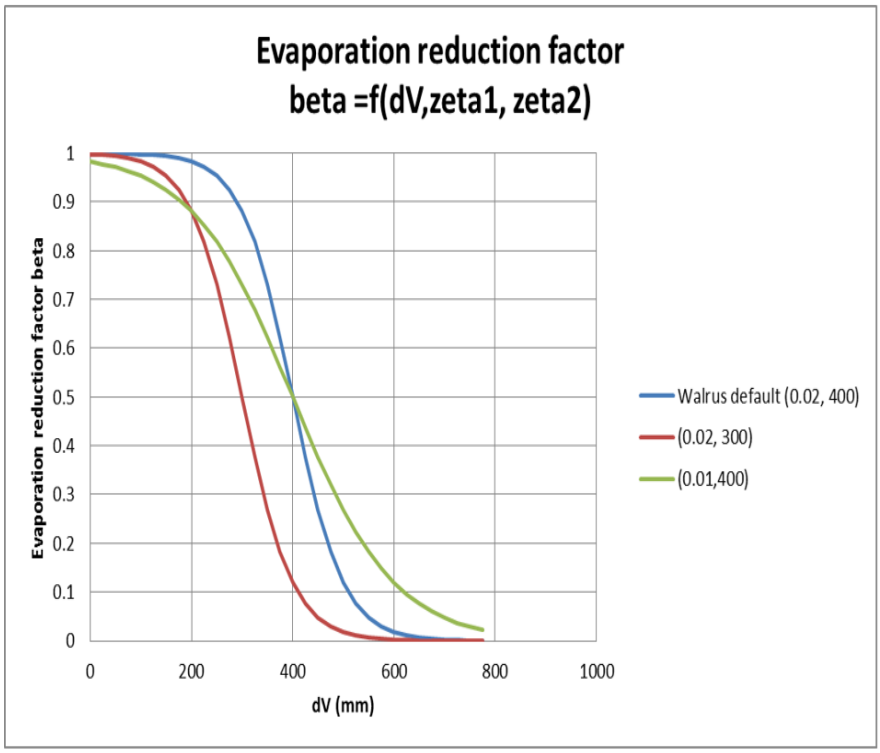


Figure 1.70: *Walrus Evaporation Reduction Function*

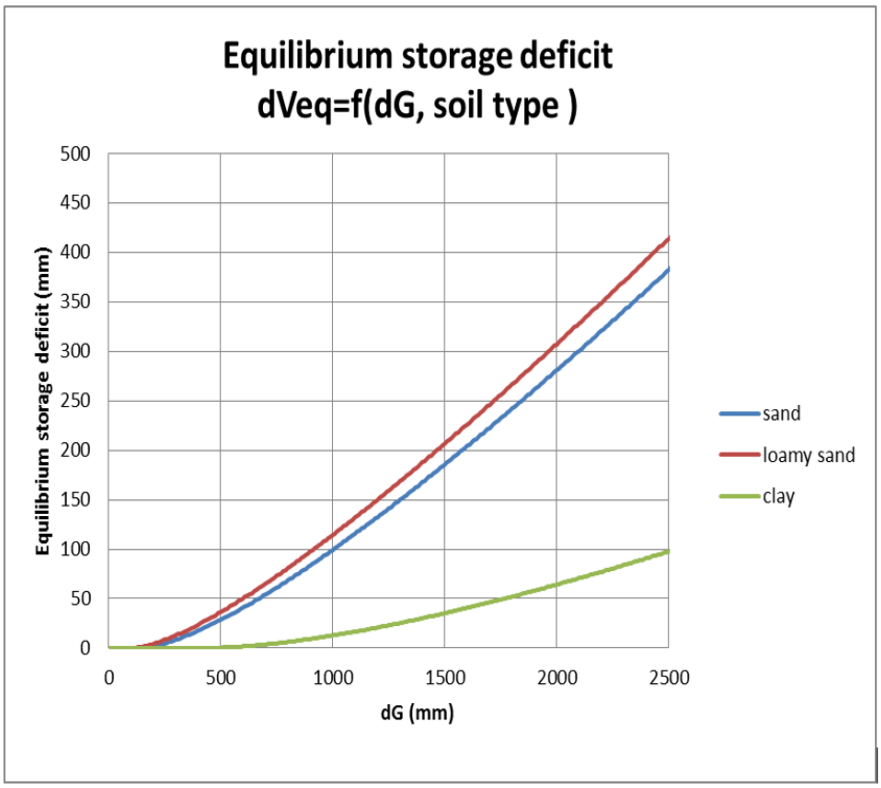


Figure 1.71: *Walrus Equilibrium Storage Deficit Function*

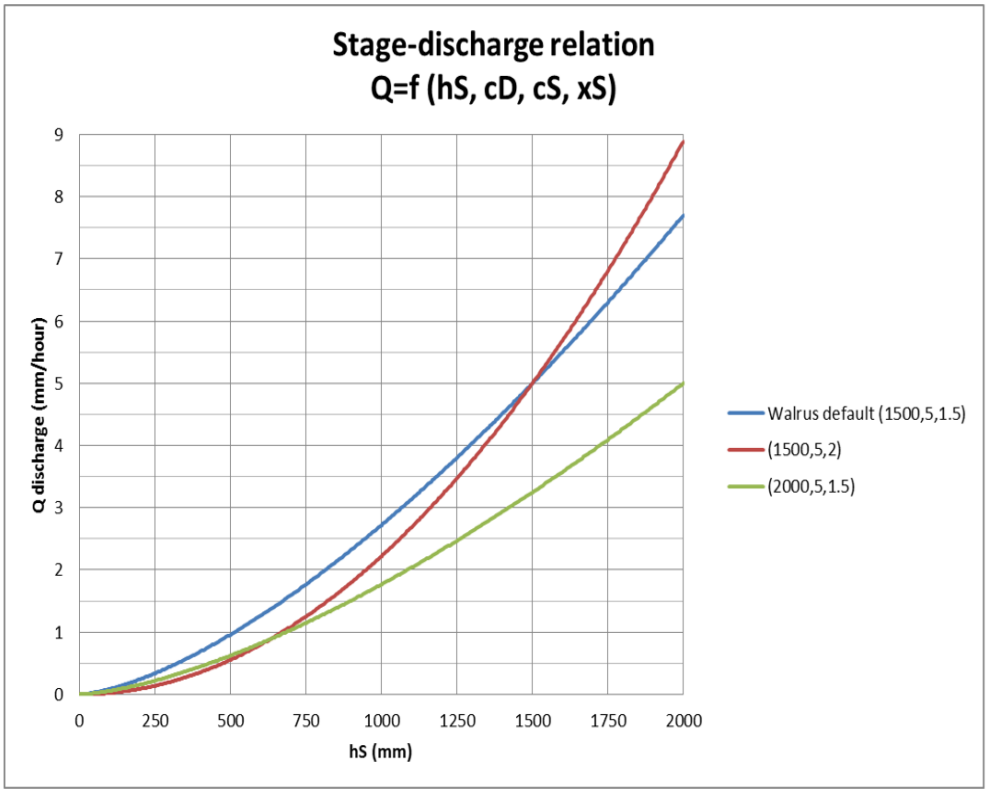


Figure 1.72: *Walrus Stage Discharge Function*

Quickflow is computed in Walrus following a linear reservoir approach. Using the water level in the quickflow reservoir h_Q (in mm), the quickflow reservoir constant c_Q (in hours) and the fraction of the soil area ($a_G = 1 - a_S$), quickflow is computed as

$$f_{QS} = (h_Q/c_Q) * a_G \tag{1.140}$$

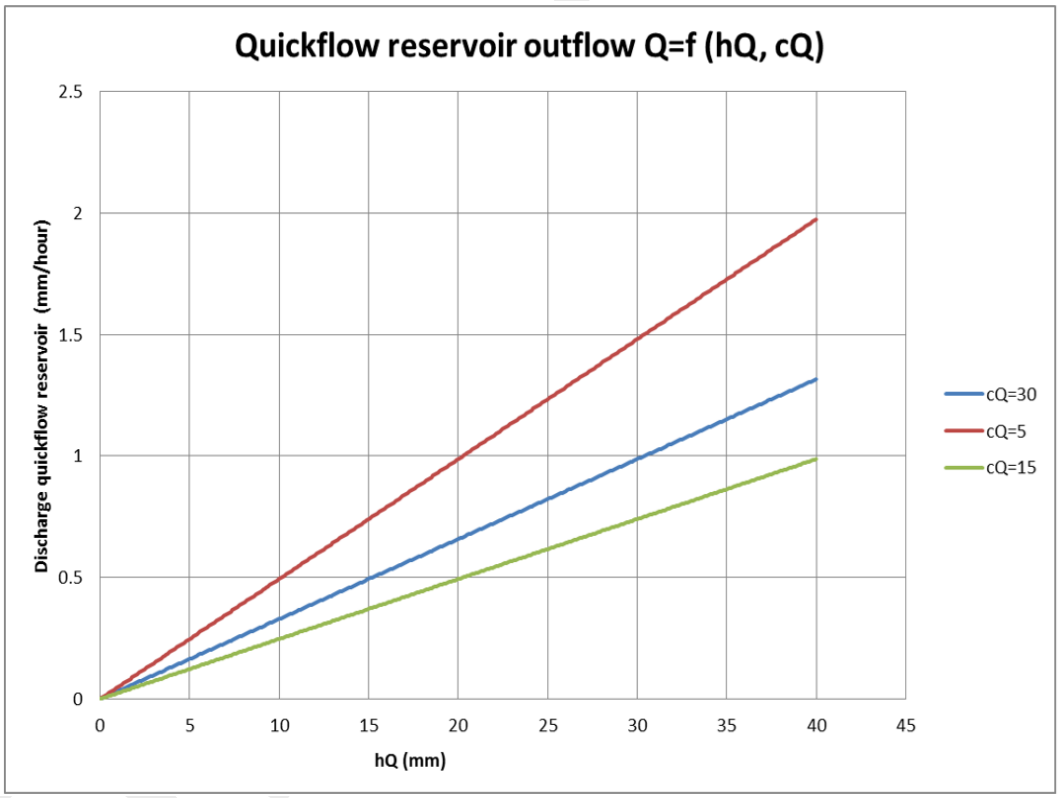


Figure 1.73: *Walrus Quickflow*

The groundwater drainage flow to surface water (or infiltration from surface water into groundwater) is computed based on the groundwater depth (dG), the surface water level (hS) and channel depth (cD), and the groundwater reservoir constant (cG). Of course also the fraction of surface water or groundwater (aS or aG, with aS = 1 - aG) is used. The drainage flow is from groundwater to surface water is computed with the formula:

$$f_{GS} = (c_D - d_G - h_S) * max(c_D - d_G, h_s) * a_G/c_G \tag{1.141}$$

Walrus has an internal timestep reduction mechanism which makes sure that Walrus uses a smaller internal timestep in case of rapid changes. Typically this is in case of heavy rainfall. The internal Walrus timestep reduction is governed by Walrus numerical parameters limiting the amount of precipitation in a computation timestep ($max - Pstep$=10 mm), limiting the water level change ($max - h - change$=10 mm). The following Walrus output variables are available within DRR for each output timestep:

- ◇ Precipitation (mm)
- ◇ Potential evapotranspiration (mm)
- ◇ Actual evapotranspiration (mm)
- ◇ Runoff (mm)
- ◇ Outflow (m3/s)

- ◇ Walrus storage deficit (mm) (d_V)
- ◇ Walrus Groundwater depth (mm) (d_G)
- ◇ Walrus level quickflow reservoir (mm) (h_Q)
- ◇ Walrus surface water level (mm) (h_S)
- ◇ Walrus external supply to surface water F_{XS} (mm)
- ◇ Walrus external supply to groundwater F_{XG} (mm)
- ◇ Walrus precipitation on quickflow reservoir (mm)
- ◇ Walrus precipitation on vadose zone reservoir (mm)
- ◇ Walrus actual evapotranspiration from land (mm)
- ◇ Walrus actual evapotranspiration from surface water (mm)
- ◇ Walrus equilibrium storage deficit (mm) ($d_V eq$)
- ◇ Walrus quickflow from quickflow reservoir to surface water (mm)
- ◇ Walrus baseflow from groundwater reservoir to surface water (mm)
- ◇ Walrus wetness index (-)
- ◇ Walrus Beta evaporation reduction factor (-)

All output in mm is relative to the total Walrus catchment area.

Walrus catchments are included in the DRR balance output files with balances for each node individually, and in the water balance for the whole DRR schematization. In the whole DRR water balance the Walrus flows are included in the following balance terms:

- ◇ Rainfall on the Walrus catchments is included in the balance term 'Rainfall'
- ◇ Evapotranspiration from the Walrus catchments is included in the balance term 'Evaporation RRRunoff-HBV-Wagmod-Walrus'
- ◇ External supply or extraction is included in the balance term 'NetLoss Sacr/LGSI/Wagmod/Walrus'
- ◇ Outflow of the Walrus catchment is, just like outflow of other catchment models in DRR, flowing to RR-boundaries (which may be 1D-Laterals in the Flow model) and is thus included in the 'boundaries out' balance term.
- ◇ Storage change of Walrus catchments is included in the 'Storage change RRRunoff' balance term.

1.1.16 DRR LGSI Rainfall Runoff concept

1.1.16.1 LGSI Rainfall-Runoff concept

The Lowland Groundwater Surface Water Interaction (LGSI) model was introduced in 2009 by the work of vd Velde et.al as described in (Van der Velde *et al.*, 2009) and (Wanders *et al.*, 2011). The version available within DRR is based on the description of (Kuijper *et al.*, 2014). which is based on mentioned earlier publications and (Kuijper *et al.*, 2012). The idea is that the LGSI model is based on available groundwater models (e.g. using Modflow) which are used to derive LGSI model input parameters and functions. In that sense LGSI is a fast running lumped rainfall-runoff model based on detailed computationally intensive groundwater modelling. The LGSI model concept has been applied on the Hupsel brook catchment, the Drentsche Aa catchment, the area of regional water authority Aa en Maas, and on the Dinkel and Vecht area for regional water authority Vechtstromen, all areas in the Netherlands. The model is suitable for flat and hilly areas with sedimentary subsoil, and not yet suitable for non-permeable rocky soils. The model concept is at the moment (2025) used in the operational system FEWS-Vecht where it is applied within a SOBEK RR-1D model for the Dinkel catchment.

Note: the LGSI implementation is not yet available in the D-HYDRO and Delft3D FM 1D2D GUI, but it is available in the computational core via the ASCII inputfiles.

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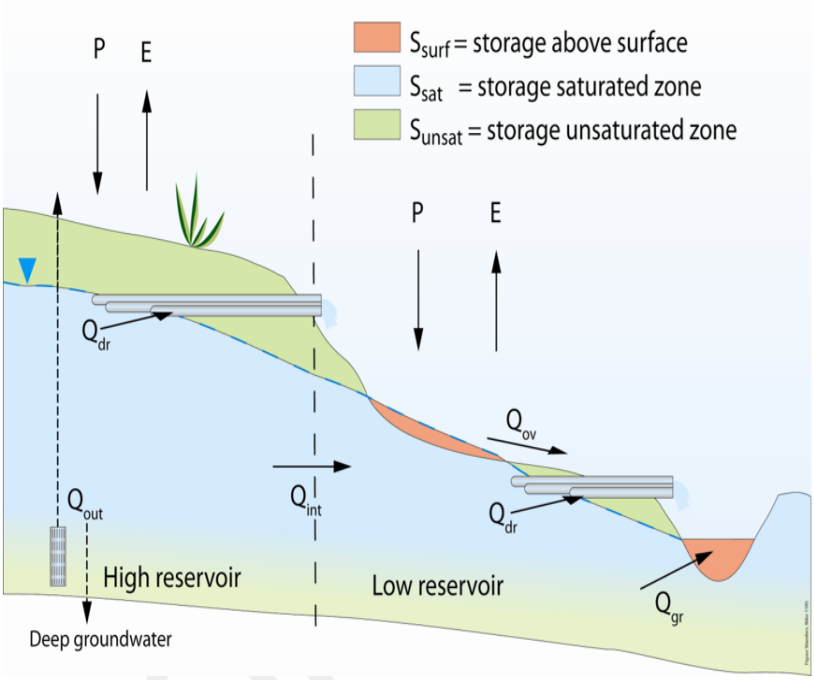


Figure 1.74: *Model concept of LGSI after (Kuijper et. al., 2014)*

The model concept is illustrated in Figure 1.74. In this figure the higher, infiltrating subarea is shown on the left, and the draining low subarea is shown on the right. Used symbols relate to:

P	precipitation
E	evapo(transpi)ration
Q_{dr}	outflow via tube drainage
Q_{out}	seepage to deep groundwater layer, or groundwater abstraction
Q_{int}	discharge from high infiltrating area to low draining area
Q_{ov}	surface discharge from zones with groundwater above surface level
Q_{gr}	groundwater discharge to surface water ditches
Q_h	fast runoff during storm events
S_{surf}	surface storage (above surface level)
S_{sat}	storage in the saturated zone
S_{unsat}	storage in the unsaturated zone

All storages are in mm and fluxes in mm/timestep unless indicated otherwise.

The infiltrating area is the higher part of the catchment where no surface water discharge is occuring (either via water courses or over land surface). Water is infiltrating in the soil, and from this higher area a groundwater flow towards the lower area is occurring. The groundwater depth (relative to surface level) is described with a normal (Gaussian) probability distribution. Zones which are sometimes draining to surface water and sometimes not, are included at the lower, wetter, draining area. They can fall dry during simulations with the LGSI concept. The draining area is the lower, wetter are of the catchment where surface water discharges via water courses or over the surface is taking place. This lower area receives rainfall (just like the higher part of the catchment) and also receives groundwater drainage from the higher area. The groundwater depth is described using a gamma probability distribution. The distinction

of these 2 subcatchment area is made based on information from the detailed groundwater model. It is therefore relevant that the computational period of the groundwater model includes both extreme wet and extreme dry conditions, in order to have data for the whole range of possible groundwater depths.

1.1.16.2 Computational Procedure

The details of the computational procedure are described in (Van der Velde *et al.*, 2009) and (Wanders *et al.*, 2011). Here, the less-detailed description (Kuijper *et al.*, 2014) is followed. The computational procedure is illustrated in Figure 1.75.

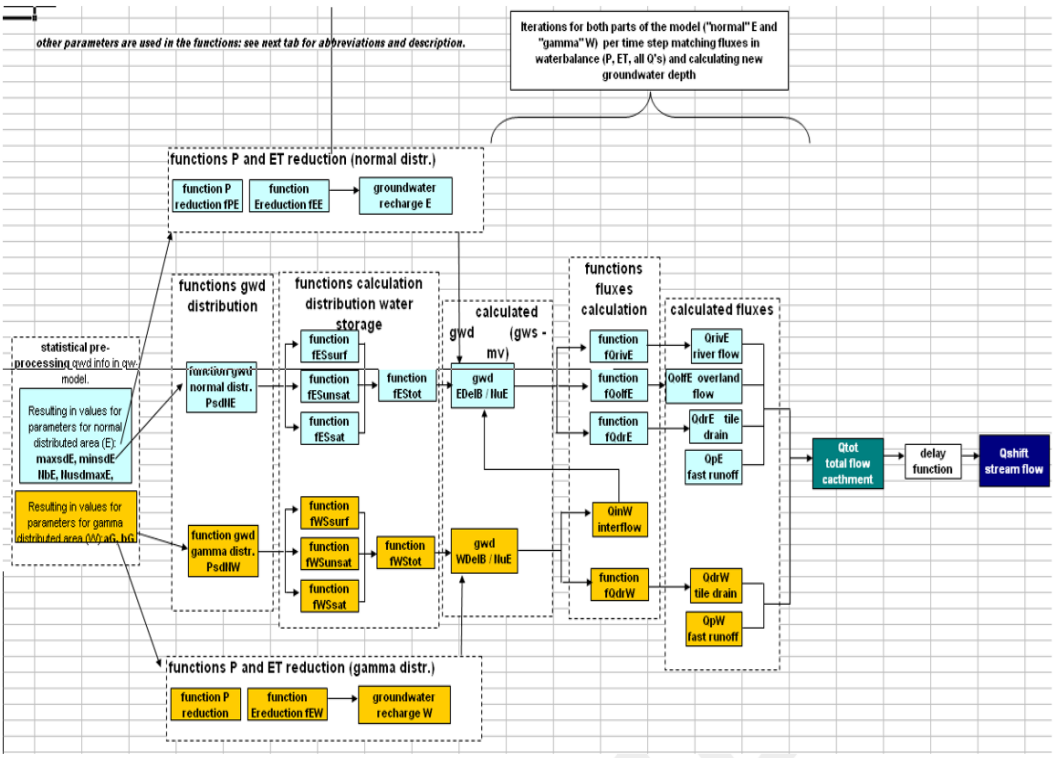


Figure 1.75: *LGSI Computational Procedure*

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The driving principle of the LGSI computations is the groundwater depth distribution and the water storage distribution over different compartments (saturated zone, unsaturated zone, and surface water which generates runoff). During the model simulations, groundwater depth and spatial variations for both the infiltration subarea and draining subarea are computed. The storage S for saturated zone (S_{sat}), unsaturated zone (S_{unsat}) and surface water storage above drain levels (S_{surf}) is determined by groundwater depth and soil characteristics. The computational procedure of Figure 1.75 shows some more details, but in summary, the computation of the discharge for each timestep (typically one hour or one day) consists of 5 steps:

- ◇ 1. Computation of groundwater recharge by precipitation and evaporation
- ◇ 2. Computation of groundwater depth, based on initial groundwater depth and change in storage
- ◇ 3. Computation of storage above surface (drain level) and above tube drainage levels, based on groundwater depths
- ◇ 4. Computation of discharge, based on the storages and flow resistances

◇ 5. Application of discharge routing

These steps are elaborated upon in the next sections.

1.1.16.3 Step 1 LGSI computational procedure

The change in storage is based on the water balance equation:

$$dS_{tot} = P - ET_{act} - Q_{tot} \tag{1.142}$$

where

dS_{tot}	storage change [mm]
P	precipitation [mm]
ET_{act}	actual evapo(transpi)ration [mm]
Q_{tot}	total dischage of previous timestep [mm]

Actual evapotranspiration is compiuted based on reference evapotranspiration (in the Netherlands typically Makkink), seperately for the infiltrating dry subarea and the draining wet area. This computation takes into acount possible evaporation reduction, actual groundwater depth and soil characteristics. Per timestep, the precipitation is distributed of both subcatchments, and storage contents for the saturated, unsaturated and surface water above drain level (the latter only for the draining subarea). For the unsaturated zone, the computations use the van Genuchten equation and parameters (Van Genuchten, 1980).

1.1.16.4 Step 2 LGSI computational procedure

Due to the change in storage computed in step 1, the groundwater depth in the area is changing (increasing or decreasing). Based on the information from the non-stationary distributed groundwater model, for both the high (infiltrating) as the low (draining) subarea the distribution of groundwater depths (and storage) is computed. This is based on a formula for each subarea seperately.

For the infiltrating, higher subarea the formula consists of a gamma-distribution. For the draining, lower area the formula is a normal distribution. Therefore the draining, lower area is also called the N-area, and the infiltrating, higher area the G-area.

Deriving these normal- and gamma distributions can be made clear by showing the groundwater depths in the spatially distributed groundwater model on a certain day in a histogram. The next figure (Figure 1.76) shows an example of these histograms on a wet day (left figure) and a dry day (right figure) The distribution of the groundwater depths on both days is described as the sum (blue line) of a normal distribution (red line) and a gamma distribution (green line) More information can be found in (Van der Velde *et al.*, 2009) and (Wanders *et al.*, 2011).

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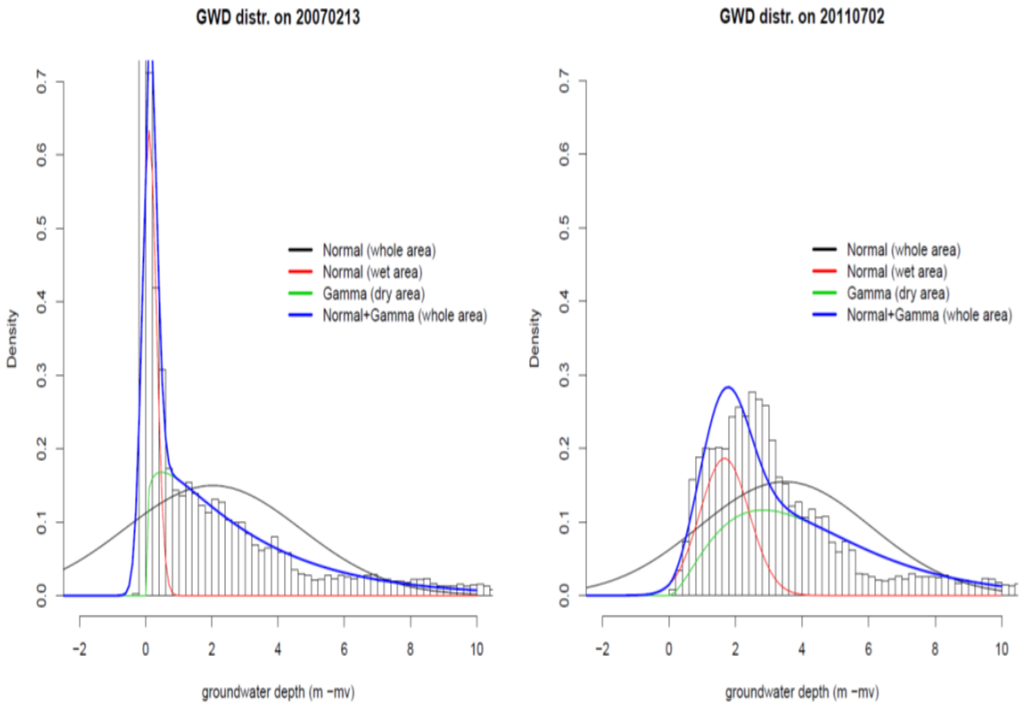


Figure 1.76: *LGSI Example of Spatial Distribution Groundwater Depths*

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The figure shows an example for the Snoeyinksbeek in the Netherlands, with February 13 2007 as a wet day, and July 2, 2011 as a dry day. The histogram shows the spatial distribution of the groundwater depths (in m below surface) from the distributed groundwater model, and the blue line as the resulting approximation as the sum of a normal and a gamma distribution. The area with groundwater depths less than zero represents the storage above the surface level (and is surface water that will come to discharge).

1.1.16.5 Step 3 and 4 of the LGSI computational procedure

Water above the drain level will come to discharge, and there is a groundwater flow from the infiltrating to the draining subarea.

In the **infiltrating subarea**, there are no surface water ditches and no surface water storage above local surface level, meaning that no surface discharge is generated. There may be tube drainage in this higher infiltrating area, which leads to a discharges Q_{dr} to the lower, draining area. In general, this drainage will be small. As a consequence of difference in groundwater levels, a groundwater flow from the high area to the low area will take place, which is denoted with Q_{int}. This flow depends on the drainage resistance of the soil, and the difference in groundwater level of the two sub-areas. From the groundwater in the higher infiltrating subarea also percolation to deep groundwater layers or groundwater abstractions for e.g. drinking water are included in the model (Q_{out}).

In the **draining subarea**, discharge is generated from the storage above surface level (surface distrhct Q_{ov}) pr by groundwater drainage above the bed level of the water courses (Q_{gr}). The flow depends on the available storage above surface level and water course, i.e. the groundwater depths above surface at that actual moment. Furthermore, a quick runoff Q_h is

taking place during periods of intensive rainfall in areas where (ground)water level is above surface. This quick runoff thus depends largely on the actual groundwater depths or levels. And a tube drainage discharge Q_{dr} may occur if tubes are present, when the groundwater level is above the tube drainage level. The discharge of water above surface, bed level of water courses and tube drainage levels may be delayed due to flow or drainage resistances. These resistances influence the actual runoff, and thus the water balance since water remains longer in the compartment 'storage above drainage level' (S_{surf}).

1.1.16.6 Step 5 of the LGSI computational procedure

The discharge computed by LGSI is not instantenously at the outflow point of the catchment. Therefore, there is an option to delay the instantaneous discharges using a simple delay function. The delay function is specified by the user indicating discharge fraction per time step. for every LGSI catchment a delay function can be specified. The next figure (Figure 1.77) shows an example of a delay function for a period of 5 days. The left figure illustrates the continous line actual discharge and associated daily delay steps, both as a fraction of the discharge. The right figure shows the difference between 'direct runoff' or impulse discharge (black line) and 'routed discharge' (red line) using the delay function.

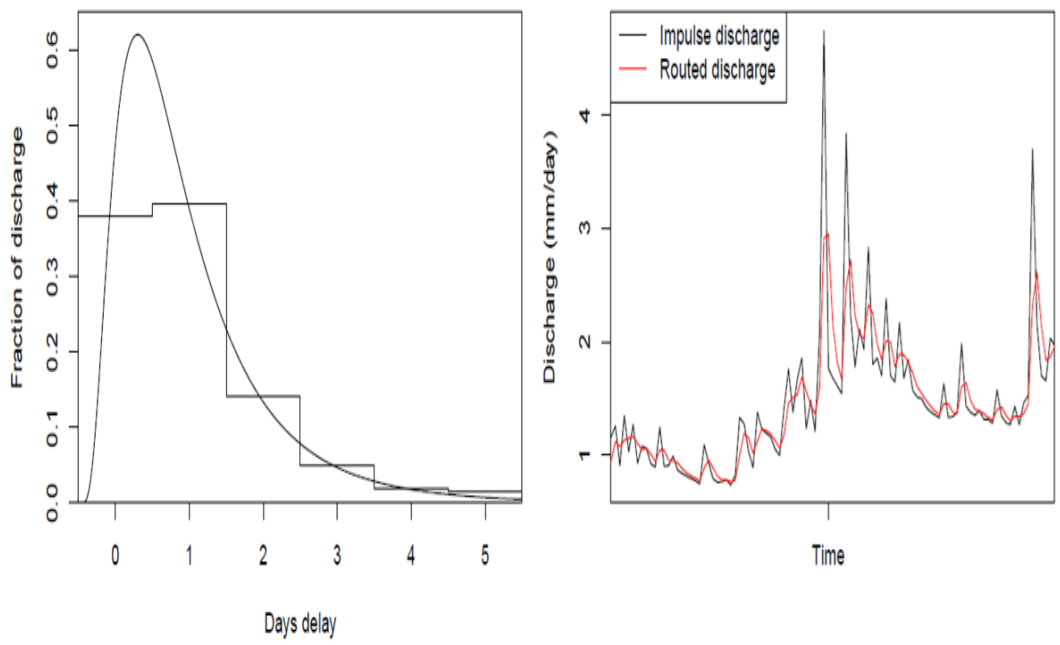


Figure 1.77: *LGSI Example Delay Function With Daily Timestep*

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1.1.16.7 LGSI parameters

The list of model parameters in the LGSI model is described in the next table. This concerns parameters as mentioned in the previous sections and some parameters used for describing the subarea characteristics in terms of groundwater depth distribution, groundwater fluxes between the 2 subareas, soil characteristics, flow and drainage resistance, tube drainage, groundwater abstractions or losses to the deep groundwater. The sub areas are denoted with as the G-area (for the higher, infiltrating area described with a Gamma-distribution function), and the N-area (the lower, draining area described with a Normal distribution function).

Table 1.14: *LGSI model parameters*

Category	Abbreviation	Description	Unit
Subcatchment areas	AN	lower subcatchment area	[m^2]
	AG	higher subcatchment area	[m^2]
groundwaterdepth subcatchment N (draining)	maxsdN	max. stdev gwdepth (N)	[m]
	minsdN	min. stdev gwdepth (N)	[m]
	NbN	shape parameter gwdepth distribution (N)	[-]
	NusdmaxN	area average gwd at maxsdN	[m]
groundwaterdepth subcatchment G (infiltrating)	aG	shape parameter gwd distribution G	[-]
	bG	scale parameter gwd distribution G	[-]
groundwater flow from G to N area	hdif	difference of groundwaterlevel subareas G and N	[m]
	cc	resistance groundwater flow G to N	[day]
Soil Characteristics	Tets	soil porosity (fraction)	[-]
	alp	Van Genuchten parameter 1	[1/m]
	n	Van Genuchten parameter 2	[-]
	ErdN	evaporation reduction depth subarea N	[m]
	EsdN	standard deviation evap.reduction depth subarea N	[m]
	ErdG	evaporation reduction depth subarea G	[m]
	EsdG	standard deviation evap.reduction depth subarea G	[m]
Resistances and quick runoff parameters	Rex	flow resistance water courses subarea N	[day]
	Rov	flow resistence surface N subarea	[day]
	rdr	drainage resistance tube drains	[day]
	m	fraction quick runoff from (gwd < 0) in subarea N	[]
	r	fraction surface water courses, inundated area in subarea N	[]

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Tube drainage in N and G subareas	DdrN DdrG fAsdN fAsdG	tube drainage depth in subarea N tube drainage depth in subarea G fraction non-tube-drained subarea N fraction non-tube-drained subarea G	[m] [m] [-] [-]
Percolation deep Abstraction	QWout	percolation to deep groundwater or groundwater abstraction	 [mm/day]
Initial gw depth for N and G subareas	Ngw1 Ggw1	initial gw depth in N subarea initial gw depth in G subarea	[m] [m]

1.1.16.8 LGSI output parameters

The output variables available from the DRR output for areas modelled with LGSI are described in the next tables. The output can be given for every DRR computation step, of for a selected subset. Where input data is typically in mm of mm/day, output fluxes are typically in mm/timestep, so depending on the computational timestep size. Typically the text says mm, but for fluxes the interpretation is than mm/timestep. Since the LGSI output is combined with output from HBV, SCS and D-NAM rainfall-runoff concepts, the actual order in the output file may be different, but the name will match.

Table 1.15: *LGSI output variables*

Nr	Output variable	Description
1	Rainfall [mm]	Precipitation average for both subareas
2	PotEvap [mm]	Potential evaporation average for both subareas
3	ActEvap [mm]	Actual evaporation average for both subareas
4	Runoff [mm]	Runoff (total for both subareas)
5	Outflow [m3/s]	Total delayed outflow in m3/s of the LGSI node
6	Rainfall1 [mm]	Precipitation subarea 1 (N)
7	Rainfall2 [mm]	Precipitation subarea 2 (G)
8	PotEvap1 [mm]	Potential evaporation subarea 1 (N)
9	PotEvap2 [mm]	Potential evaporation subarea 2 (G)
10	ActEvap1 [mm]	Actual evaporation subarea 1 (N)
11	ActEvap2 [mm]	Actual evaporation subarea 2 (G)
12	Recharge1 [mm]	Groundwater recharge subarea 1 (N)
13	Recharge2 [mm]	Groundwater recharge subarea 2 (G)
14	DrainFlow1 [mm]	Drainage flow subarea 1 (N)
15	Drainflow2 [mm]	Drainage flow subarea 2 (G)
16	Interflow [mm]	Groundwater flow from subarea 2 (G) to subarea 1 (N)
17	Overlandflow1 [mm]	Surface runoff area 1 (N)
18	Overlandflow2 [mm]	Surface runoff area 2 (G)

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19	Qpdirect1 [mm]	Direct runoff subarea 1 (N)
20	Qpdirect2 [mm]	Direct runoff subarea 2 (G)
21	Riverflow1 [mm]	River flow from subarea 1 (N)
22	Riverflow2 [mm]	River flow from subarea 2 (G)
23	OverlndSto1[mm]	Overland storage subarea 1 (N)
24	OverlndSto2[mm]	Overland storage subarea 2 (G)
25	GwStor1 [mm]	Groundwater storage subarea 1 (N)
26	GwStor2 [mm]	Groundwater storage subarea 2 (G)
27	NewVolume1 [m]	Total volume subarea 1 (N), sum of storage on surface, saturated, unsaturated) Note that the volume is described in m3 per m2, so in m
28	NewVolume2 [m]	Total volume subarea 2 (G), sum of storage on surface, saturated, unsaturated) Note that the volume is described in m3 per m2, so in m
29	GWTableDepth1 [m-surf]	Groundwater depth in m below surface, subarea N
30	GwTableDepth2 [m-surf]	Groundwater depth in m below surface, subarea G
31	RunoffDelay	do not use / wrong description
32	Qtot [mm]	Total outflow before applying delay function
33	Qdelayed [mm]	Total outflow after applying delay function
34	GWLevel1 [m]	Groundwater level subarea N (as 30, with - sign)
35	GWLevel2 [m]	Groundwater level subarea G (as 31, with - sign)

1.1.17 Description of the D-NAM rainfall-runoff model

The "**D-NAM**" rainfall-runoff model is developed at Deltares. It concerns an improvement and expansion of the NAM rainfall-runoff model, developed at the Institute of Hydrodynamics and Hydraulic Engineering of the Technical University of Denmark (see Nielsen and Hansen (1973)). NAM is the abbreviaton of the Danish "Nedbør-Afstrømmings-Model" (in Dutch "Neerslag-Afvoer Model"), meaning rainfall-runoff model.

The D-NAM rainfall-runoff model is a deterministic, lumped and conceptual model, that treats each catchment as a single unit. Model parameters and variables, therefore, represent average values for the entire catchment area.

A comparison of the D-NAM model and the NAM model is made in section 1.1.17.18. The D-NAM model concept is only supported in the SOBEK2 GUI, but not in the D-HYDRO and Delft3D FM 1D2D user interface. However, it is also supported in the computational core under D-HYDRO and Delft3D FM 1D2D. So, when working and running outside the GUI, DRR models including the D-NAM concept can be made and run via DIMR.

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1.1.17.1 External forces acting on a D-NAM model

External forces (or boundary conditions) acting on a D-NAM model comprise of:

1 Rainfall (P), defined in the (SOBEK2) "Meteo" Task block.
2 Potential evapotranspiration (EP), defined in the (SOBEK2) "Meteo" Task block.
3 Groundwater pump discharge ($GWPump$), defined on the "GWPump" tab of the D-NAM Node in the (SOBEK2) "Schematisation" Task block.
4 External water level (h). Either defined at a RR boundary Node (rainfall-runoff modelling only) or computed by a 1D Flow model (combined rainfall-runoff and 1D flow modelling).

1.1.17.2 D-NAM storages and their water-storage capacity

The D-NAM model comprises of three different reservoirs:

1 The surface storage, that is situated on the surface of the catchment area and has an **unlimited** water-storage capacity.
2 The lower zone storage, having a limited water-storage capacity only and representing the field capacity of the root zone layer (s_{fc} [-]). In other words the part of the soil moisture content that can be retained against gravity.
3 The groundwater storage, having a limited water-storage capacity only and representing the soil moisture content that can not be retained against gravity.

The lower zone storage and the groundwater storage jointly represent the water-storage-capacity of the modelled soil layer thickness (see Figure 1.78). The soil beneath the surface level (SL [m AD]) is divided into a **root zone layer** with bed level $RZBL$ [m AD] and specific yield $s_{y,rz}$ [-] and a **subsoil layer** with bed level $GWSBL$ [m AD] and specific yield $s_{y,ss}$ [-]. For water-storage-capacities yield, see 1.78:

◇ The maximum water depth in the lower zone storage (L_{max} [mm]) amounts to $s_{fc}(SL - RZBL)$.
◇ The maximum water depth in the root zone part of the groundwater storage ($GWSD_{rz,max}$ [mm]) amounts to $(s_{y,rz} - s_{fc})(SL - RZBL)$.
◇ The maximum water depth in the subsoil part of the groundwater storage ($GWSD_{ss,max}$ [mm]) amounts to $s_{y,ss}(RZBL - GWSBL)$.
◇ The maximum water depth in the groundwater storage ($GWSD_{max}$ [mm]) amounts to $GWSD_{rz,max} + GWSD_{ss,max}$.

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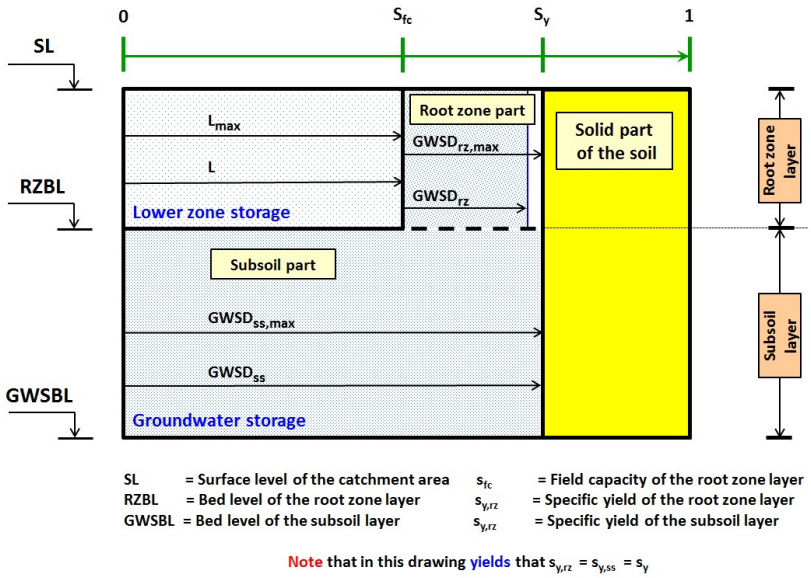


Figure 1.78: *Maximum water depths in lower zone storage and groundwater storage. The solid part of the soil (coloured yellow) can not contain any groundwater.*

1.1.17.3 D-NAM external and internal fluxes

The D-NAM model discerns nine external fluxes (see Figure 1.79):

1. Rainfall into the surface storage (P).
2. Evaporation from the surface storage ($E1$).
3. Interflow out of the surface storage (IF).
4. Overland flow out of the surface storage (OF).
5. Transpiration from the root zone layer ($E2$); being the summation of the transpiration from the lower zone storage ($E2_{LZS}$) and the transpiration from the root zone part of the groundwater storage ($E2_{GWS,rz}$).
6. Groundwater Pump. Either the abstraction of water contained in the groundwater storage ($GW_{ABS,Act}$) or the supply of external water ($GW_{SUP,Act}$); being the summation of the supply of external water into the lower zone storage (DL_{GWPump}) and the supply of external water into the groundwater storage (GW_{GWPump}).
7. Fast base flow component out of the groundwater storage ($FastBF$).
8. Slow base flow component out of the groundwater storage ($SlowBF$).
9. Inflow of external (ground)water ($GWInflow$); being the summation of the inflow of external (ground)water into the lower zone storage (DL_{Ext}) and the inflow of external (ground)water into the groundwater storage (GW_{Ext}).

The D-NAM model discerns four internal fluxes (see Figure 1.79):

1. Infiltrated water (INF) or water in the surface storage that infiltrates into the soil.
2. Infiltration into the lower zone storage (DL).
3. Percolation into the groundwater storage (G).
4. Capillary rise of water (CR), contained in the subsoil part of the groundwater storage.

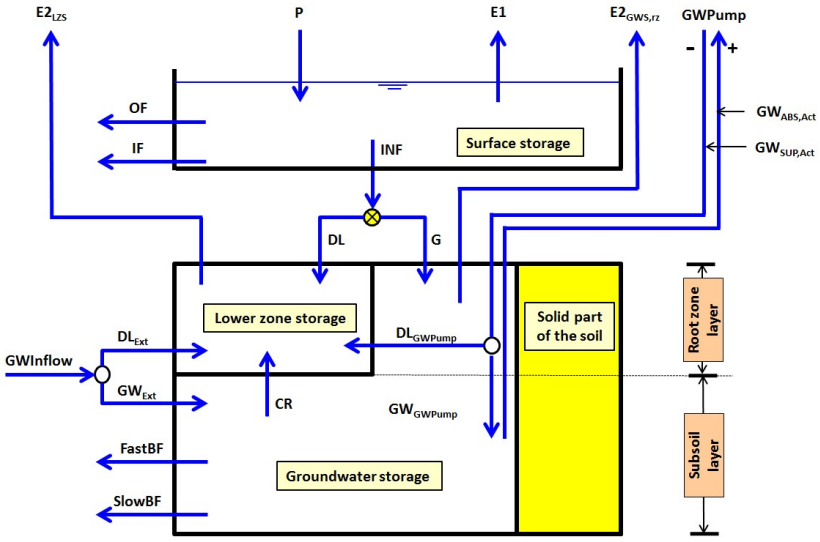


Figure 1.79: *Schematic representation of fluxes in the D-NAM rainfall runoff model.*

1.1.17.4 Computing water depths in the surface flow storage

Water depths in the surface storage (U [mm]) are computed accounting for fluxes in the following sequence (see Figure 1.80):

1 Rainfall P [mm] is added.
2 Evaporation $E1$ [mm] (see section 1.1.17.6) is subtracted.
3 Interflow IF [mm] (see 1.1.17.7) is subtracted.
4 Infiltrated water INF [mm] (see section 1.1.17.8) is subtracted.
5 Overland flow OF [mm] (see section 1.1.17.9) is subtracted.

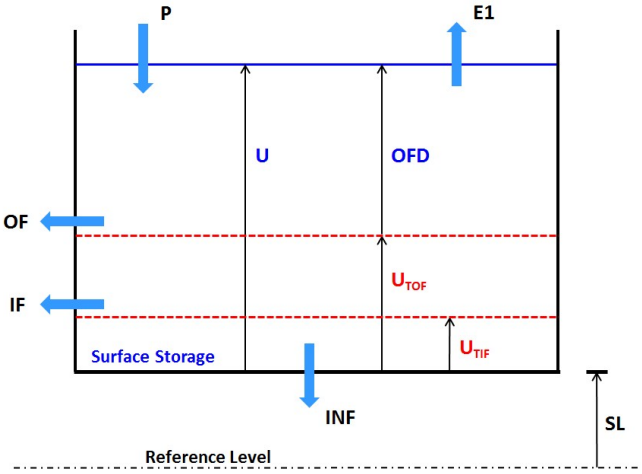


Figure 1.80: *Water depths, thresholds and fluxes related to the surface storage*

1.1.17.5 Computing water depths in the lower zone storage and overland flow storage

Water depths in the lower zone storage (L [mm]) and the groundwater storage ($GWSD$ [mm]) are computed accounting for fluxes (see Figure 1.81) in the following sequence:

1 Infiltration DL [mm] (see section 1.1.17.10) is added to the lower zone storage.
2 Percolation G [mm] (see section 1.1.17.10) is added to the groundwater storage.

3 Transpiration $E2_{LZS}$ [mm] (see section 1.1.17.11) is subtracted from the lower zone storage.
4 Transpiration $E2_{GWS_{rz}}$ [mm] (see section 1.1.17.11) is subtracted from the groundwater storage.
5 Capillary rise CR [mm] (see section 1.1.17.12) is subtracted from the groundwater storage and added to the lower zone storage.
6 Fast base flow $FastBF$ [mm] (see section 1.1.17.13) is subtracted from the groundwater storage.
7 Slow base flow $SlowBF$ [mm] (see section 1.1.17.13) is subtracted from the groundwater storage.
8 External (ground)water DL_{Ext} [mm] (see section 1.1.17.14) is added to the lower zone storage.
9 External (ground)water GW_{Ext} [mm] (see section 1.1.17.14) is added to the groundwater storage.
10 Abstraction by the groundwater pump $GW_{ABS,Act}$ [mm] (see section 1.1.17.15) is subtracted from the groundwater storage.
11 Supply by the groundwater pump DL_{GWPump} [mm] (see section 1.1.17.16) is added to the lower zone storage.
12 Supply by the groundwater pump GW_{GWPump} [mm] (see section 1.1.17.16) is added to the groundwater storage.

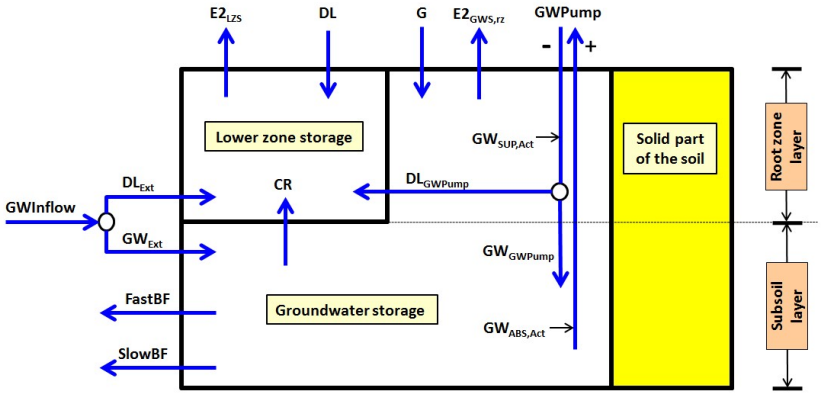


Figure 1.81: *Fluxes related to the lower zone storage and groundwater storage. The solid part of the soil (coloured yellow) can not contain any groundwater.*

1.1.17.6 Evaporation from the surface storage

For the evaporation ($E1$ [mm]) from the surface storage yields:

$$E1 = min(EP, U)$$

where:

EP	User-defined potential evapotranspiration (see section 1.1.17.1) [mm].
U	Water depth in the surface storage [mm].

1.1.17.7 Interflow out of the surface storage

For interflow (IF [mm]) out of the surface storage yields:

$$L_\alpha = (L - L_{TIF})/(L_{max} - L_{TIF}), \text{ where } L_\alpha = 0 \text{ if } L \leq L_{TIF}$$

$$CKIF_{\Delta t} = 1/(1 - (1 - 1/CKIF)^{\Delta t/86400})$$

$$IF = (L_\alpha/CKIF_{\Delta t})\ max(0, U - U_{TIF})$$

Note: No interflow occurs if $U \leq U_{TIF}$ or $L \leq L_{TIF}$

where:

$CKIF$	User-defined reservoir coefficient applied in routing interflow [day].
$CKIF_{\Delta t}$	Reservoir coefficient used in each time-step Δt [s] for routing interflow [Δt].
L	Water depth in lower zone storage [mm].
L_{max}	Maximum water depth in lower zone storage (see section 1.1.17.2 and Figure 1.78) [mm].
L_{TIF}	User-defined lower-zone-storage threshold for interflow (Figure 1.80) [mm].
U	Water depth in the surface storage [mm].
U_{TIF}	User-defined surface-storage threshold for interflow (Figure 1.80) [mm].
Δt	User-defined time-step [s].

1.1.17.8 Infiltrated water into the soil

For infiltrated water (INF [mm]) from the surface storage into the soil yields:

$$INF_{cap,\Delta t} = INFCap\ (\Delta t/3600)$$

$$DL_{max} = L_{max} - L.$$

$$G_{max} = min(GWSD_{max} - GWSD\ ,\ G_{pot,max}\ (\Delta t/86400)\ )$$

$$INF = min(U\ ,\ INF_{cap,\Delta t}\ ,\ DL_{max} + G_{max})$$

Note: Infiltrated water (INF) equals zero if $INFCap = 0$

where:

DL_{max}	Available water-storage-depth in the lower zone storage [mm].
G_{max}	Water depth that can potentially be stored in the groundwater storage [mm].
$G_{pot,max}$	User-defined maximum rate of percolation into the groundwater storage [mm/day].
$GWSD$	Water depth in the groundwater storage [mm].
$GWSD_{max}$	Maximum water depth in the groundwater storage (see section 1.1.17.2 and Figure 1.78) [mm].
$INFCap$	User-defined maximum infiltration rate for water contained in the surface storage [mm/hr].
$INFCap_{\Delta t}$	Amount of water contained in the surface storage that infiltrates into the soil in time-step Δt [mm/hr].
L	Water depth in lower zone storage [mm].
L_{max}	Maximum water depth in lower zone storage (see section 1.1.17.2 and Figure 1.78) [mm].

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U	Water depth in the surface storage [mm].
Δt	User-defined time-step [s].

1.1.17.9 Overland flow out of the surface storage

Overland flow (OF [mm]) is determined using an approximate analytical formula that:

1 Describes overland flow with the Manning roughness formula for open channel flow.
2 Assumes that the catchment area is a rectangular container with drainage length CL [m] and width B [m], having a slope S [-] towards its overland outflow point.
3 Assumes that overland flow occurs in such way, that at each point-in-time yields that water depths in the surface storage are constant in space.

This approximate analytical formula is given below:

- $OFD = max(0\ ,\ U - U_{TOF})$
- $OF = OFD - 1000 \left((OFD/1000)^{-2/3} + (2\ \Delta t\ \sqrt{S})/(3\ n\ CL) \right)^{-3/2}$

Note: No overland flow occurs if $U \leq U_{TOF}$.

where:

OFD	Overland flow depth (see Figure 1.80), defined as the water depth in the surface storage above U_{TOF} [mm].
U	Water depth in the surface storage [mm].
U_{TOF}	Surface-storage threshold for overland flow [mm].
n	Manning value for the surface roughness of the catchment area [$s/m^{1/3}$].
Δt	User-defined time-step [s].

1.1.17.10 Infiltration into the lower zone storage and percolation into the groundwater storage

Infiltrated water (INF [mm], see section 1.1.17.8) is distributed as infiltration in the lower zone storage (DL) [mm] and percolation in the groundwater storage (G) [mm] (see Figure 1.82). This distribution is done as follows:

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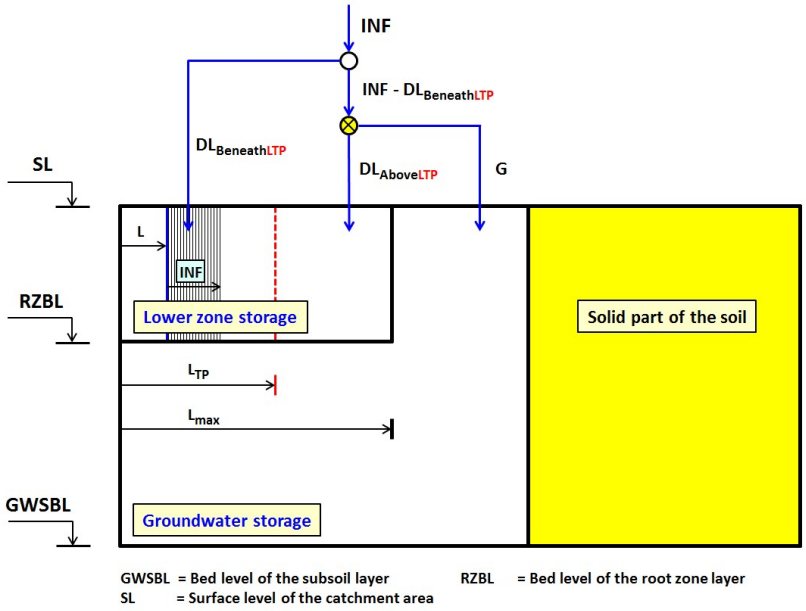


Figure 1.82: *Distributing infiltrated water (INF) over the lower zone storage and the groundwater storage. The solid part of the soil (coloured yellow) can not contain any groundwater.*

1 $DL_{BeneathL_{TP}}$ [mm] is the part of the infiltrated water (INF) that is stored in the lower zone storage beneath the lower-zone-storage threshold for percolation (L_{TP} [mm], see Figure 1.82).

$$DL_{BeneathL_{TP}} = max(0\ ,\ min(L_{TP} - L\ ,\ INF)).$$

where:

L	Water depth in the lower zone storage [mm].
L_{TP}	Lower-zone-storage threshold for percolation [mm].

The value for L_{TP} determines the degree in which infiltrated water (INF) is distributed over the lower zone storage and the groundwater storage as proposed by Nielsen and Hansen (1973). More precisely:

- ◇ If $L_{TP} = L_{max}$ the distribution proposed by Nielsen and Hansen (1973) is overruled. Since, first the lower zone storage is completely filled up before infiltrated water can percolate into the groundwater storage.
- ◇ If $0 \leq L_{TP} < L_{max}$ the distribution proposed by Nielsen and Hansen (1973) becomes applicable as soon as the lower zone storage is filled up to L_{TP}, meaning that part of the infiltrated water (INF) can percolate into the groundwater storage (or $G > 0$) before the lower zone storage is at field capacity (or $L < L_{max}$).

2 $DL_{NH,AboveL_{TP}}$ [mm] (see Figure 1.82) is the part of the infiltrated water (INF) that, in line with the Nielsen and Hansen (1973) distribution, infiltrates into the lower zone storage above L_{TP}.

$$DL_{NH,AboveL_{TP}} = min(AD_{AboveL_{TP}}\ ,(1-(L/L_{max}))(INF - DL_{BeneathL_{TP}}))$$

where:

$AD_{AboveL_{TP}}$	Available water-storage-depth in the lower zone storage above L_{TP} [mm].
L_{max}	Maximum water depth in the lower zone storage (see section 1.1.17.2 and Figure 1.78) [mm].

3 G_{NH} [mm] (see Figure 1.82) is the part of the infiltrated water (INF) that, in line with the Nielsen and Hansen (1973) distribution, percolates into the groundwater storage.

$$G_{NH} = INF - DL_{BeneathL_{TP}} - DL_{NH,AboveL_{TP}}$$

4 The actual percolation (G [mm]) follows from limiting G_{NH} to the user-defined maximum percolation rate $G_{pot,max}$ [mm/day].

$$G = max(G_{NH}\ ,\ G_{pot,max}\ (\Delta t/86400))$$

where:

$G_{pot,max}$	User-defined maximum rate of percolation into the groundwater storage [mm/day].
Δt	User-defined time-step [s].

5 For the actual infiltration into the lower zone storage (DL [mm], see Figure 1.82) yields.

$$DL = INF - G$$

1.1.17.11 Transpiration from the root zone layer

Transpiration from the root zone layer ($E2$ [mm], see Figure 1.81) respectively comprises of transpiration from the root zone part of the groundwater storage ($E2_{GWS,rz}$ [mm]) and transpiration from the lower zone storage ($E2_{LZS}$ [mm]). $E2_{GWS,rz}$ and $E2_{LZS}$ are computed as follows:

$$GWSD_{rz} = max(0,\ GWSD - GWSD_{ss,max})$$

$$E2 = min(L + GWSD_{rz}\ ,\ (EP - E1)(L + GWSD_{rz})/(L_{max} + GWSD_{rz}))$$

$$E2_{GWS,rz} = min(E2\ ,\ GWSD_{rz}).$$

$$E2_{LZS} = E2 - E2_{GWS,rz}$$

where:

$E1$	Evaporation from the surface storage (see section 1.1.17.6) [mm].
EP	User-defined potential evapotranspiration (see section 1.1.17.1) [mm].
$GWSD$	Water depth in the groundwater storage [mm].
$GWSD_{rz}$	Water depth in the root zone part of the groundwater storage [mm].
$GWSD_{ss,max}$	Maximum water depth in the root zone part of the groundwater storage (see section 1.1.17.2 and Figure 1.78) [mm].
L	Water depth in the lower zone storage [mm].
L_{max}	Maximum water depth in the lower zone storage (see section 1.1.17.2 and Figure 1.78) [mm].

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1.1.17.12 Capillary rise

For capillary rise (CR [mm], see Figure 1.81) yields:

$$CR = \min(L_{max} - L\ ,\ GWSD\ ,\ (\Delta t/86400)\ CR_{pot,max}\ )$$

where:

$CR_{pot,max}$	User-defined maximum capillary rise [mm/day].
$GWSD$	Water depth in the groundwater storage [mm].
L	Water depth in the lower zone storage [mm].
L_{max}	Maximum water depth in the lower zone storage (see section 1.1.17.2 and Figure 1.78) [mm].
Δt	User-defined time-step [s].

Note: The surface level (SL [m AD]) and the bed level of the groundwater storage ($GWSBL$ [m AD]) determine the thickness of the soil layer, that is considered in a D-NAM model. So the bed level of the groundwater storage is to be considered as watertight. Therefore, $CR_{pot,max}$ is set equal to zero if $GWSD = 0$, this irrespective of the user-defined value for $CR_{pot,max}$.

1.1.17.13 Fast and slow base flow component

Base flow occurs only if the external water level (h [m AD]) is below the groundwater level (GWL [m AD]). Base flow equals zero if $h \geq GWL$ (see Figure 1.83). Water contained in the lower zone storage can not flow out as base flow, since this water can be retained against gravity.

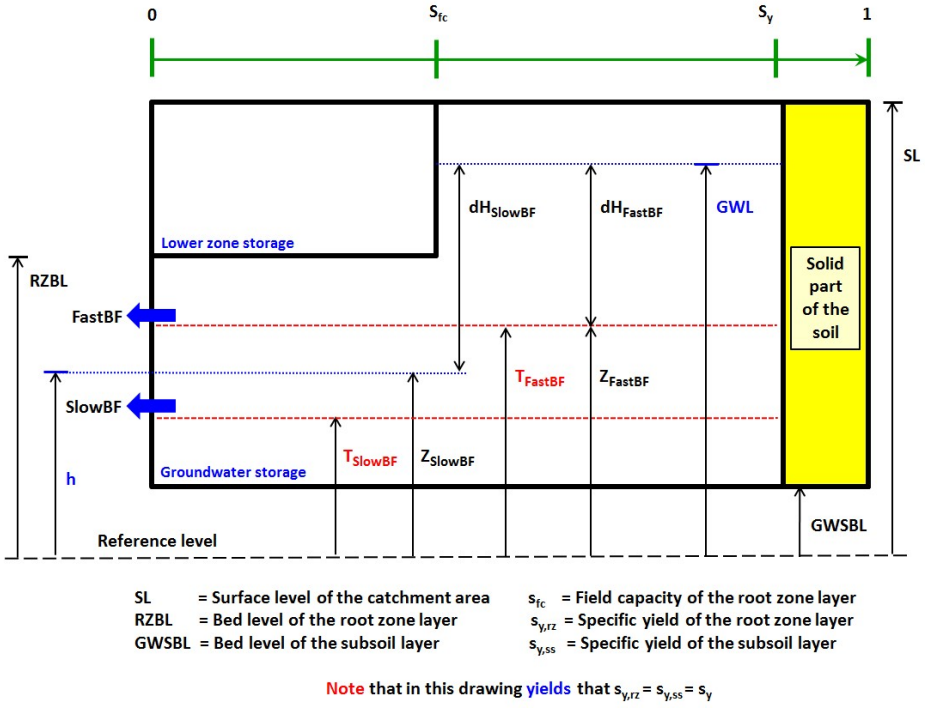


Figure 1.83: *Computing the Fast base flow component and the Slow base flow component. The solid part of the soil (coloured yellow) can not contain any groundwater.*

The fast base flow component ($FastBF$) and the slow base flow component ($SlowBF$) are

computed as follows (see Figure 1.83):

$$GWSD_{ss} = min(GWSD\ ,\ GWSD_{ss,max})$$

$$GWSD_{rz} = max(0\ ,\ GWSD - GWSD_{ss,max})$$

$$GWL = GWSBL + (GWSD_{ss}/1000\ s_{y,ss}) + ((GWSD_{rz}\ +\ L)/1000\ s_{y,rz})$$

$$Z_{FastBF} = max(h, T_{FastBF})$$

$$Z_{SlowBF} = max(h, T_{SlowBF})$$

$$dH_{FastBF} = max(0,\ GWL - Z_{FastBF})$$

$$dH_{SlowBF} = max(0,\ GWL - Z_{SlowBF})$$

$$GWSD_{Z_{FastBF}} = f(Z_{FastBF}\ ,\ SL\ ,\ RZBL\ ,\ GWSBL\ ,\ s_{y,rz}\ ,\ s_{fc}\ ,\ s_{y,ss})$$

$$GWSD_{Z_{SlowBF}} = f(Z_{SlowBF}\ ,\ SL\ ,\ RZBL\ ,\ GWSBL\ ,\ s_{y,rz}\ ,\ s_{fc}\ ,\ s_{y,ss})$$

$$V_{pot,FastBF} = max(0,\ GWSD - GWSD_{Z_{FastBF}})$$

$$V_{pot,SlowBF} = max(0,\ GWSD - GWSD_{Z_{SlowBF}})$$

$$CKFastBF_{\Delta t} = 1/(1 - (1 - 1/CKFastBF)^{\Delta t/86400})$$

$$CKSlowBF_{\Delta t} = 1/(1 - (1 - 1/CKSlowBF)^{\Delta t/86400})$$

$$FastBF_{pot} = min(V_{pot,FastBF}\ ,\ 1000(dH_{FastBF}/CKFastBF_{\Delta t})\ )$$

$$SlowBF_{pot} = min(V_{pot,SlowBF}\ ,\ 1000(dH_{SlowBF}/CKSlowBF_{\Delta t})\ )$$

$$GWOutflow_{pot} = FastBF_{pot} + SlowBF_{pot}$$

$$FastBF = FastBF_{pot} \quad \text{if} \quad GWOutflow_{pot} \leq V_{pot,FastBF}$$

$$FastBF = V_{pot,FastBF}\left(\frac{FastBF_{pot}}{GWOutflow_{pot}}\right) \quad \text{if} \quad GWOutflow_{pot} > V_{pot,FastBF}$$

$$SlowBF = SlowBF_{pot} \quad \text{if} \quad FastBF + SlowBF_{pot} \leq V_{pot,SlowBF}$$

$$SlowBF = V_{pot,SlowBF}\ -\ FastBF \quad \text{if} \quad FastBF + SlowBF_{pot} > V_{pot,SlowBF}$$

where:

$CKFastBF$	User-defined reservoir coefficient applied in routing the fast base component ($CKFastBF > 1$) [day].
$CKFastBF_{\Delta t}$	Reservoir coefficient used in each time-step Δt [s] for routing the fast base component [Δt].
$CKSlowBF$	User-defined reservoir coefficient applied in routing the slow base component ($CKSlowBF > 1$) [day].
$CKSlowBF_{\Delta t}$	Reservoir coefficient used in each time-step Δt [s] for routing the slow base component [Δt] .
dH_{FastBF}	Hydraulic head (or driving force) of the fast base flow component [m].
dH_{SlowBF}	Hydraulic head (or driving force) of the slow base flow component

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	[m].
$FastBF_{pot}$	Potential fast base flow component [mm].
$SlowBF_{pot}$	Potential slow base flow component [mm].
GWL	Groundwater level [m AD].
$GWOutflow_{pot}$	Potential groundwater outflow (=$FastBF_{pot}$ + $SlowBF_{pot}$) [mm].
$GWSBL$	Bed level of the groundwater storage [m AD].
$GWSD$	Water depth in the groundwater storage [mm].
$GWSD_{rz}$	Water depth in the root zone part of the groundwater storage [mm].
$GWSD_{ss}$	Water depth in the subsoil part of the groundwater storage [mm].
$GWSD_{ss,max}$	Maximum water depth in the subsoil part of the groundwater storage (see section 1.1.17.2 and Figure 1.78) [mm].
$GWSD_{Z_{FastBF}}$	Water depth in the groundwater storage if GWL equals Z_{FastBF} [mm].
$GWSD_{Z_{SlowBF}}$	Water depth in the groundwater storage if GWL equals Z_{SlowBF} [mm].
h	External water level [m AD].
L	Water depth in the lower zone storage [mm].
$RZBL$	Bed level of the root zone layer [m AD].
s_{fc}	Field capacity of the soil in the root zone layer [-].
SL	Surface level of the catchment area [m AD]
$s_{y,rz}$	Specific yield of the root zone layer (ratio of the volume of water in a fully saturated root zone layer to its total soil volume) [-].
$s_{y,ss}$	Specific yield of the subsoil layer (ratio of the volume of water in a fully saturated subsoil layer to its total soil volume) [-].
T_{FastBF}	Threshold for the fast base flow component. Only water in the groundwater storage above T_{FastBF} may flow out as fast base flow [m AD].
T_{SlowBF}	Threshold for the slow base flow component. Only water in the groundwater storage above T_{SlowBF} may flow out as slow base flow [m AD].
$V_{pot,FastBF}$	Volume of water in the groundwater storage, that is potentially available for drainage by the fast base flow component [mm].
$V_{pot,SlowBF}$	Volume of water in the groundwater storage, that is potentially available for drainage by the slow base flow component [mm].
Z_{FastBF}	Drainage level of the fast base flow component [m AD].
Z_{SlowBF}	Drainage level of the slow base flow component [m AD].
Δt	User-defined time-step [s].

1.1.17.14 External (ground)water flowing into the lower zone storage and groundwater storage

Inflow of external (ground)water into the soil ($GWInflow$ [mm]) occurs only if the external water level (h [m AD]) is above the groundwater level (GWL [m AD]). $GWInflow$ equals zero if $h \leq GWL$ (see Figure 1.84). The sequence of filling the lower zone storage and the groundwater storage by inflowing external (ground)water is as follows:

- ◇ Firstly, the subsoil part of the groundwater storage is completely filled-up
- ◇ Secondly, the lower zone storage is completely filled-up.
- ◇ Thirdly and finally, the root zone part of the groundwater storage is filed-up.

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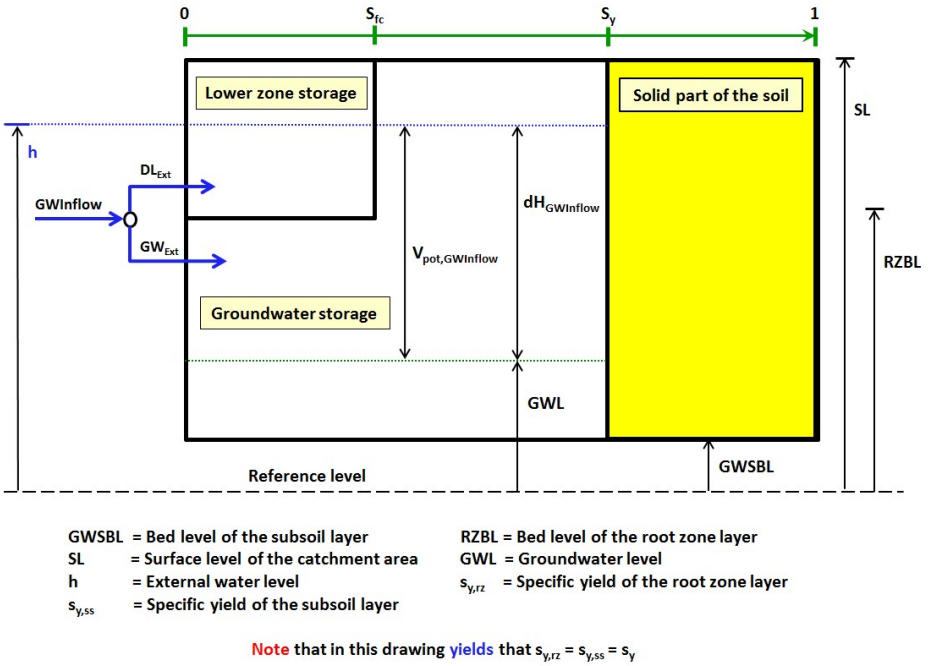


Figure 1.84: *Determining the inflow of external (ground)water ($GWInflow$). The solid part of the soil (coloured yellow) can not contain groundwater.*

The inflow of external (ground)water ($GWInflow$ [mm]), the part following into the lower zone storage (DL_{Ext} [mm]), and the part flowing into the groundwater storage (GW_{Ext} [mm]) are computed as follows (see Figure 1.84):

$$GWSD_{ss} = min(GWSD\ ,\ GWSD_{ss,max})$$

$$GWSD_{rz} = max(0\ ,\ GWSD - GWSD_{ss,max})$$

$$GWL = GWSBL + (GWSD_{ss}/1000\ s_{y,ss}) + ((GWSD_{rz}\ +\ L)/1000\ s_{y,rz})$$

$$dH_{GWInflow} = max(0, h - GWL)$$

$$V_{pot,GWInflow} = f(h\ ,\ GWL\ ,\ SL\ ,\ RZBL\ ,\ GWSBL\ ,\ s_{y,rz}\ ,\ s_{y,ss})$$

$$CKGWInflow_{\Delta t} = 1/(1 - (1 - 1/CKGWInflow)^{\Delta t/86400})$$

$$GWInflow = min(1000\ (dH_{GWInflow}/CKGWInflow_{\Delta t})\ ,\ V_{pot,GWInflow})$$

$$DL_{Ext} = max(0\ ,\ min(L_{max} - L\ ,\ GWSD + GWInflow - GWSD_{ss,max})$$

$$GW_{Ext} = GWInflow - DL_{Ext}$$

where:

$CKGWInflow$	User-defined reservoir coefficient, applied for the inflow of external (ground)water ($CKGWInflow > 1$) [day].
$CKGWInflow_{\Delta t}$	Reservoir coefficient used in each time-step Δt [s] for computing the inflow of external (ground)water [Δt].
$dH_{GWInflow}$	Hydraulic head (or driving force) for the inflow of external (ground)water [m].

GWL	Groundwater level [m AD].
$GWSBL$	Bed level of the groundwater storage [m AD].
$GWSD$	Water depth in the groundwater storage [mm].
$GWSD_{rz}$	Water depth in the root zone part of the groundwater storage [mm].
$GWSD_{ss}$	Water depth in the subsoil part of the groundwater storage [mm].
$GWSD_{ss,max}$	Maximum water depth in the subsoil part of the groundwater storage (see section 1.1.17.2 and Figure 1.78) [mm].
h	External water level [m AD].
L	Water depth in the lower zone storage [mm].
L_{max}	Maximum water depth in the lower zone storage (see section 1.1.17.2 and Figure 1.78) [mm].
$RZBL$	Bed level of the root zone layer [m AD].
s_{fc}	Field capacity of the soil in the root zone layer [-].
SL	Surface level of the catchment area [m AD].
$s_{y,rz}$	Specific yield of the root zone layer (ratio of the volume of water in a fully saturated root zone layer to its total soil volume) [-].
$s_{y,ss}$	Specific yield of the subsoil layer (ratio of the volume of water in a fully saturated subsoil layer to its total soil volume) [-].
$V_{pot,GWInflow}$	Volume of water in both the lower zone storage and the groundwater storage, that potentially can be filled-up by the inflow of external (ground)water [mm].
Δt	User-defined time-step [s].

1.1.17.15 Abstraction by the groundwater pump

Abstraction by the groundwater pump ($GW_{ABS,Act}$ [mm], see Figure 1.81) occurs only if the groundwater pump discharge ($GWPump$ [m^3/s]) is greater than zero. Water contained in the lower zone storage can not be abstracted by the groundwater pump. $GW_{ABS,Act}$ is computed as follows:

$$GW_{ABS,max} = GWSD$$

$$GW_{ABS,pot} = max(0\ ,(1000\Delta t\ /CatchmentArea)\ GWPump)$$

$$GW_{ABS,Act} = min(GW_{ABS,max}\ ,\ GW_{ABS,pot})$$

where:

$GW_{ABS,max}$	Maximum water depth in the groundwater storage that can be abstracted by the groundwater pump [mm].
$GW_{ABS,pot}$	Potential water depth that can be abstracted according to the user-defined groundwater pump discharge [mm].
$GWSD$	Water depth in the groundwater storage [mm].
Δt	User-defined time-step [s].

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1.1.17.16 Supply by the groundwater pump

Supply of water by the groundwater pump ($GW_{SUP,Act}$ [mm]), see Figure 1.81) occurs only if the groundwater pump discharge ($GWPump$ [m^3/s]) is less than zero. The sequence of filling the lower zone storage and the groundwater storage by the groundwater pump is as follows:

- ◇ Firstly, the subsoil part of the groundwater storage is completely filled-up
- ◇ Secondly, the lower zone storage is completely filled-up.
- ◇ Thirdly and finally, the root zone part of the groundwater storage is filed-up.

Supply of water by the groundwater pump ($GW_{SUP,Act}$) comprises of supply to the lower zone storage (DL_{GWPump} [mm]) and supply to the groundwater storage (GW_{GWPump} [mm]). DL_{GWPump} and GW_{GWPump} are computed as follows:

$$GW_{SUP,max} = (GWSD_{max} - GWSD) + (L_{max} - L)$$

$$GW_{SUP,pot} = max(0\ ,\ -(1000\Delta t\ /\ CatchmentArea)\ GWPump)$$

$$GW_{SUP,Act} = min(GW_{SUP,max}\ ,\ GW_{SUP,pot})$$

$$AD_{GWS,ss,GWPump} = max(0, GWSD_{ss,max} - GWSD))$$

$$DL_{GWPump} = min(L_{max} - L, GW_{SUP,Act} - min(AD_{GWS,ss,GWPump}\ , GW_{SUP,Act}))$$

$$GW_{GWPump} = GW_{SUP,Act} - DL_{GWPump}$$

where:

$AD_{GWS,ss,GWPump}$	Water-storage-depth available in the subsoil part of the groundwater storage for the supply of water by the groundwater pump [mm].
$GWSD$	Water depth in the groundwater storage [mm].
$GWSD_{max}$	Maximum water depth in the groundwater storage (see section 1.1.17.2 and Figure 1.78) [mm].
$GWSD_{ss,max}$	Maximum water depth in the subsoil part of the groundwater storage (see section 1.1.17.2 and Figure 1.78) [mm].
$GW_{SUP,max}$	Maximum water-storage-depth available for the supply of water by the groundwater pump. Sum of available water-storage-depth in the lower zone storage and groundwater storage. [mm].
$GW_{SUP,pot}$	Potential water depth that can be supplied according to the user-defined groundwater pump discharge [mm].
L	Water depth in the lower zone storage [mm].
L_{max}	Maximum water depth in the lower zone storage (see section 1.1.17.2 and Figure 1.78) [mm].
Δt	User-defined time-step [s].

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1.1.17.17 D-NAM output timeseries

The next table provides an overview of the output timeseries available for each D-NAM rainfall-runoff model. The second column contains for each output timeseries the symbols that are used in equations.

Please note that within the "Runoff node" output is to be switched on in the "Output options" tab in RR Settings input screen in SOBEK or in the Delft_3B.Ini file in Delft3D FM Suite 1D2D, in order to generate the output timeseries.

Table 1.16: *Overview of D-NAM output timeseries*

Name of Output Time Series	Symbols	Unit	Description
Rainfall	P	mm	Rainfall as defined in the (SOBEK2) "Meteo Task" block
PotEvap	EP	mm	Evapotranspiration as defined in the (SOBEK2) "Meteo Task" block.
ActEvap		mm	Sum of evaporation from surface storage ($E1$) and transpiration from the lower zone storage ($E2$).
Runoff		mm	Equals overland flow (OF) + interflow (IF) + fast base flow component ($FastBF$) + slow base flow component ($SlowBF$) - inflow of external (ground)water into the soil ($GWInflow$).
Outflow		m^3/s	Equals overland flow (OF) + interflow (IF) + fast base flow component ($FastBF$) + slow base flow component ($SlowBF$) - inflow of external (ground)water into the soil ($GWInflow$).
D-NAM ExternalWaterLevel	h	$m\,AD$	External water level.
D-NAM GWPumpDefinedDischarge	$GWPump$	m^3/s	Groundwater pump discharge.
D-NAM EvapSurfaceStorage	$E1$	mm	Evaporation from the surface storage.
D-NAM EvapRootZoneLayer	$E2$	mm	Transpiration from the root zone layer.
D-NAM EvapRootZonePartGWStorage	$E2_{GWS,rz}$	mm	Transpiration from the root zone part of the groundwater storage.
D-NAM EvapLowerZoneStorage	$E2_{LZS}$	mm	Transpiration from the lower zone storage.
D-NAM OverlandFlow	OF	mm	Overland flow (outflow from surface storage).
D-NAM Interflow	IF	mm	Interflow (outflow from surface storage).
D-NAM FastBaseFlow	$FastBF$	mm	Fast base flow component (outflow from groundwater storage).
D-NAM SlowBaseFlow	$SlowBF$	mm	Slow base flow component (outflow from groundwater storage).
D-NAM ExternalGWInflow	$GWInflow$	mm	Inflow of external (ground)water into the soil.
D-NAM ExternalGWInflowLowerZoneStorage	DL_{Ext}	mm	Inflow of external (ground)water into the lower zone storage.
D-NAM ExternalGWInflowGroundwaterStorage	GW_{Ext}	mm	Inflow of external (ground)water into the groundwater storage.
D-NAM GWPumpAbsGroundwaterStorage	$GW_{ABS,Act}$	mm	Abstraction of water from the groundwater storage by the groundwater pump.
D-NAM GWPumpSupLowerZoneStorage	DL_{GWPump}	mm	Supply of water into the lower zone storage by the groundwater pump.
D-NAM GWPumpSupGroundwaterStorage	GW_{GWPump}	mm	Supply of water into the groundwater storage by the groundwater pump.
D-NAM Infiltration	INF	mm	Water contained in the surface storage that infiltrates into the soil.

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Table 1.16: *Overview of D-NAM output timeseries*

Name of Output Time Series	Symbols	Unit	Description
D-NAM InfiltrationLowerZoneStorage	DL	mm	Water contained in the surface storage that infiltrates into the lower zone storage.
D-NAM Percolation	G	mm	Water contained in the surface storage that percolates into the groundwater storage.
D-NAM CapillaryRise	CR	mm	Capillary rise of water from the subsoil part of the groundwater storage into the lower zone storage.
D-NAM DepthSurfaceStorage	U	mm	Water depth in the surface storage.
D-NAM DepthLowerZoneStorage	L	mm	Water depth in the lower zone storage.
D-NAM DepthGroundwaterStorage	$GWSD$	mm	Water depth in the groundwater storage.
D-NAM VolumeSurfaceStorage	V_U	m^3	Volume of water contained in the surface storage.
D-NAM VolumeLowerZoneStorage	V_L	m^3	Volume of water contained in the lower zone storage.
D-NAM VolumeGroundwaterStorage	V_{GWS}	m^3	Volume of water contained in the groundwater storage.
D-NAM GroundWaterLevel	GWL	mAD	Groundwater level (measured in metres above datum).
D-NAM GroundWaterTableDepth	$GWTD$	mBS	Groundwater table depth (measured in metres below surface level).
D-NAM AvailableSoilStorage	AV_{soil}	m^3	Volume in the soil that is still available for the storage of water.
D-NAM BaseFlow	BF	mm	Equals fast base flow component ($FastBF$) + slow base flow component ($SlowBF$) - inflow of external (ground)water into the soil ($GWInflow$).
D-NAM GWPumpActualDischarge	$GWPump_{Act}$	m^3/s	Actual groundwater pump discharge. Positive if water is abstracted. Negative if water is supplied.
D-NAM GWPumpActual-DefinedDischarge	$GWPump_{Act-Defined}$	m^3/s	Actual groundwater pump discharge **minus** the user-defined groundwater pump discharge. If positive, the amount of discharge that could not be supplied. If negative, the amount of discharge that could not be abstracted.

1.1.17.18 Comparing the D-NAM model and the NAM model

Concepts of the Nielsen and Hansen (1973) NAM model implemented into the D-NAM model are:

1 Interflow occurs only if the water depth in the lower zone storage (L) is above the lower-zone-threshold for interflow (L_{TIF}). Interflow is a function of the relative moisture content in the lower zone storage above L_{TIF}. (see factor L_α in section 1.1.17.7).
2 Percolation may occur if the water depth in the lower zone storage is below field capacity (see section 1.1.17.10).
3 Transpiration from the root zone layer ($E2$) is a function of the relative moisture content in the root zone layer (see section 1.1.17.11).

Reasons for developing the D-NAM rainfall-runoff model are:

1 The NAM model stores water on the surface of a catchment area in three different stacked on each other storages, being the NAM surface storage, the NAM interflow storage and the NAM overland flow storage. As a result hereof, water contained in the latter two NAM

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storages is not available for evaporation and for infiltration into the soil.
In the D-NAM model all water on the surface of a catchment area is contained in one surface storage only and hence all water on the surface of a D-NAM model is at any point-in-time available for evaporation, interflow, infiltration into the soil and overland flow (see Figure 1.80).

2 The NAM surface storage, receiving precipitation has a watertight bed and a maximum water-storage-capacity. Hence, water contained in the NAM surface storage can not directly infiltrate into the soil. Only if the water depth in the NAM surface storage, after accounting for evaporation and inflow into the interflow storage, exceeds its maximum water-storage-capacity, the surplus of water in the NAM surface storage is divided into infiltration into the soil and inflow into the overland flow storage. This division of surplus water in the NAM surface storage is, however, made irrespective of the actual water depth in the NAM overland flow storage.
In the D-NAM model the surface storage has an unlimited water-storage-capacity and its bed is not watertight, allowing water to infiltrate into the soil (INF). Infiltrated water is limited by the actual water depth in the surface storage, the available water-storage-depth in the soil, and the user defined maximum infiltration and percolation capacity (see section 1.1.17.8).

Additional functionalities in the D-NAM model with respect to the NAM model are:

1 The D-NAM model makes a distinction between the specific yield of root zone layer and the specific yield of the subsoil layer (see section 1.1.17.2)
2 The NAM model is suited for free discharging catchment areas only. The D-NAM model allows for the inflow of external (ground)water (see section 1.1.17.14), that is required in modelling deltaic areas.
3 The D-NAM model determines overland flow using the Manning formula for open channel flow (see section 1.1.17.9).
4 The D-NAM model makes a distinction between a fast and a slow base flow component (see Figure 1.83).
5 The D-NAM model includes a groundwater pump, that can abstract water (see section 1.1.17.15 or supply water (see section 1.1.17.16).

1.1.18 RR-Urban Rainfall Runoff concept

1.1.18.1 Urban NWRW concept

The urban runoff model concept available in SOBEK and Delft3D FM 1D2D is based on the model described in the Dutch sewer system modelling guidelines C2100 (Clemens *et al.* (2004)), (DHV/Grontmij (1995)), which goes back to descriptions of the Dutch National Werkgroep Riolering and Waterkwaliteit as early as 1978 (NLingenieurs (1978)). The urban runoff model concept is a 0D rainfall-runoff model, applied for small areas connected to a sewer system with typical inflow points into the 1D system every 40m. The model thus describes the transformation of a rainfall timeseries into a runoff timeseries which is used as inflow into a 1D sewer model. Processes playing a role are:

- ◇ wetting and storage on surface
- ◇ infiltration
- ◇ evaporation
- ◇ delay of runoff
- ◇ dry weather flow (return flow from households/commercial/industrial water use)

The next figure illustrates the water balance: the runoff towards the sewer system is based on the rainfall, minus evaporation, minus infiltration and minus the change of storage including

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some delay in the runoff towards the sewer system.

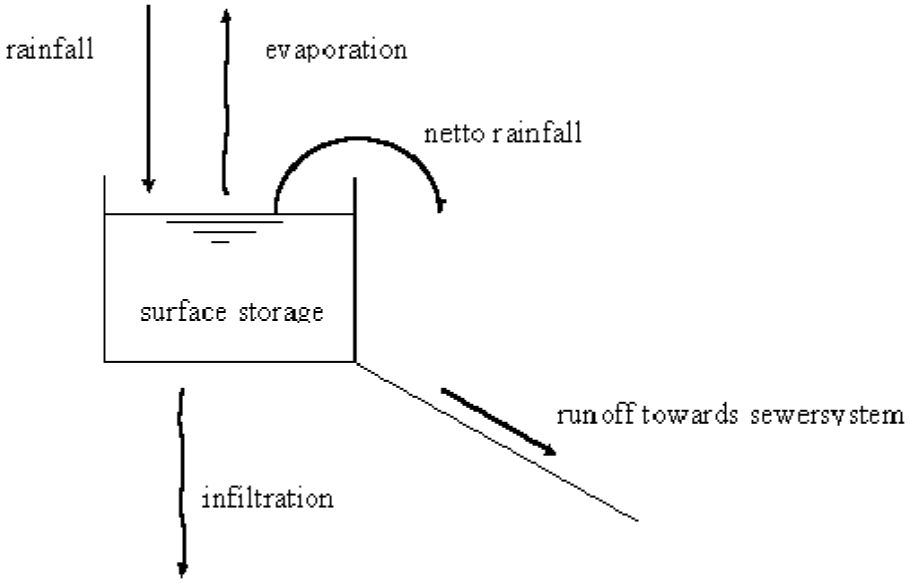


Figure 1.85: *Illustration of rainfall-runoff process*

The rainfall-runoff process
As a result of moistening and puddle forming, part of the rainfall will be stored temporarily on the surface. This storage is called surface storage. This storage is reduced by evaporation as well as infiltration. Different types of surfaces can be distinguished, depending on surface characteristics and slope. The model distinguishes four types of surfaces (closed paved, open paved, roof, unpaved) and three types of slopes (area with a slope, flat, stretched flat), thus twelve different area types. The slope of the surface and the infiltration capacity largely influence the rainfall-runoff process.

The infiltration of rainfall takes place in the open paved areas and unpaved areas. The infiltration capacity depends mostly on the type of surface and moisture condition. Other factors may also play an important role. For example, the infiltration capacity of brick paths depends on the condition of the openings between the bricks. The infiltration capacity of the unpaved areas depends on the vegetation, the kind of soil and the percentage of moisture in the subsurface. The description of the infiltration of the urban NWRW rainfall-runoff model is based on the Horton formula.

The delay of runoff depends on the average distance to the inflow location in sewer system, the slope and the geometry of the catchment. The formula which describes the runoff to the sewer system is the formula of the rational method, see also section 1.1.18.2.

Depending on the type of surface (closed paved, open paved, roofs or unpaved), different parameters can be defined for surface storage and infiltration. Infiltration is described using

the Horton time-varying formulation. For storage and runoff delay, besides the type of surface also the slope is relevant.

It is remarked that this model concept, as all RR modelling concepts, is considered a lumped conceptual model, although the spatial resolution in urban sewer applications is quite high, meaning that the RR catchments are rather small. This is related to the typical distance of 40m between sewer inflow points in the Netherlands and the computational possibilities when this model concept was developed (before the year 2000): it is designed for a combined RR-1D sewer system model. Nowadays, often 2D modelling is added. Sometimes only to model overflow from the 1D sewer system to the streets, but also by modelling the rainfall on the 2D grid covering the entire sewer system area. Delft3D FM Suite 1D2D also supports this 2D modelling with rainfall on the grid, with optional Horton infiltration formulation defined per 2D-gridcell depending on land use and including flow of runoff over the surface to the sewer pipes.

1.1.18.2 Delay of Runoff

The delay of runoff (in SOBEK-Urban Rainfall-Runoff) depends on the average distance to the inflow location in sewer system, the slope and the geometry of the catchment. The formula which describes the runoff to the sewer system is the formula of the Rational Method:

$$q = ch \tag{1.143}$$

where:

q	inflow into sewer [mm/min]
c	runoff factor [$1/min$]
h	rainfall, dynamic storage on catchment [mm]

The runoff factor, c, is a function of length, roughness and slope. Twelve different area types are described in the 'Dutch Guidelines for sewer systems computations, hydraulic functioning'. The types and default values are presented in Table 1.17.

Table 1.17: *Default parameters of Delay of Runoff (Rational Method)*

Number	Area type	Runoff type	Runoff factor, c (1/min)	Max. surface storage, h (mm)
1	closed paved	with a slope	0.5	0.0
2	closed paved	flat	0.2	0.5
3	closed paved	stretched flat	0.1	1.0
4	open paved	with a slope	0.5	0.0
5	open paved	flat	0.2	0.5
6	open paved	stretched flat	0.1	1.0
7	roof	with a slope	0.5	0.0
8	roof	flat	0.2	2.0
9	roof	stretched flat	0.1	4.0
10	unpaved	with a slope	0.5	2.0
11	unpaved	flat	0.2	4.0
12	unpaved	stretched flat	0.1	6.0

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An area 'with a slope' is an area with a slope more than 4 %. A stretched flat area is an area with the distance to the nearest inflow point in the sewer larger than 100 meters.

1.1.18.3 Horton equation

The description of the infiltration in the Urban version of the Rainfall Runoff module (NWRW model; Nationale Werkgroep Riolering en Waterkwaliteit (NLingenieurs, 1978)) is based on the formula of Horton:

Decreasing infiltration capacity:

$$f_t = f_e + (f_b - f_e)e^{-k_a t} \tag{1.144}$$

Recovering infiltration capacity:

$$f_t = f_b - (f_b - f_e)e^{-k_h t} \tag{1.145}$$

where:

f_t	infiltration capacity at moment of time t	$[mm/h]$
f_b	maximum infiltration capacity at t=0	$[mm/h]$
f_e	minimum infiltration capacity	$[mm/h]$
k_a	time factor decreasing infiltration capacity	$[1/h]$
k_h	time factor recovering infiltration capacity	$[1/h]$
t	time	$[h]$

The rate of decreasing and recovering infiltration capacity between the maximum value f_b and the minimum value f_e depends on the time factors k_a and k_h. Typical values used in the Netherlands are given in the table below. It is assumed that at the beginning of each rainfall event, the infiltration capacity is at its maximum f_b. Infiltration capacity is decreasing as long as there is water stored on the surface. If infiltration capacity equals the minimum value f_e, and there is still water on the surface, it will remain at the minimum value. Infiltration capacity will increase as soon as the surface is dry and it is not raining. If infiltration capacity equals the maximum value and there is no water on the surface, it will remain at the maximum value. If the infiltration is in between minimum and maximum infiltration capacity when the rainfall stops and there is no more water on the surface, recovery will start from the infiltration capacity at that moment and the infiltration capacity will rise slowly up to the maximum infiltration f_b.

Table 1.18: *Default parameters Horton equation*

nr	Area type	Runoff type	Infiltration capacity maximum, f_b (mm/hour)	Infiltration capacity minimum, f_e (mm/hour)	Time factor decreasing, k_a (1/hour)	Time factor recovering, k_h (1/hour)
1	closed paved	with a slope	0.0	0.0	0.0	0.0
2	closed paved	flat	0.0	0.0	0.0	0.0
3	closed paved	stretched flat	0.0	0.0	0.0	0.0
4	open paved	with a slope	2.0	0.5	3.0	0.1

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Table 1.18: *Default parameters Horton equation*

nr	Area type	Runoff type	Infiltration capacity maximum, f_b (mm/hour)	Infiltration capacity minimum, f_e (mm/hour)	Time factor decreasing, k_a (1/hour)	Time factor recovering, k_h (1/hour)
5	open paved	flat	2.0	0.5	3.0	0.1
6	open paved	stretched flat	2.0	0.5	3.0	0.1
7	roof	with a slope	0.0	0.0	0.0	0.0
8	roof	flat	0.0	0.0	0.0	0.0
9	roof	stretched flat	0.0	0.0	0.0	0.0
10	unpaved	with a slope	5.0	1.0	3.0	0.1
11	unpaved	flat	5.0	1.0	3.0	0.1
12	unpaved	stretched flat	5.0	1.0	3.0	0.1

1.1.18.4 Special areas

Besides the 12 standard area types (3 slopes combined with 4 types of surfaces), the user can define additional special areas. The first option is to add a special area for which similar parameters as the standard types are defined; this is called a user defined area. The second option is to add a green roof area. For a green roof area, different parameters are defined and used to compute the storage of water in the green roof and the runoff from the green roof. A third option is to add a swale filter (in Dutch: wadi) which is a trench able to store water and slowly release storm water. The first option is basically an option to specify an area which the same functional description as the 12 standard area types, but with different parameter values. The second and third option are described in the following subparagraphs.

1.1.18.5 Green roofs

A green roof consists of several layers, For the hydrology, relevant are only the description of the soil layer or growing medium (and the kind of vegetation), and the drainage layer. Precipitation falling on a green roof as rain, is first catched by the vegetation (=interception). When more precipitation falls, it falls through to the top of the growing medium where it forms a shallow water layer (in fact again an interception layer). With additional rainfall falling, infiltration into the soil or growing medium starts (infiltration). Next, the soil pores within the growing medium are filled and the water content increases. When the field capacity is reached, additional water infiltrating will lead to percolation of water from the growing medium to the drainage layer. Percolation starts when soil moisture is at field capacity (=water content which can be hold against gravity), which is before complete saturation is reached. The water from the drainage layer is discharged via gutters to the rain pipe. In case of an 'open' drainage layer (typically at light-weight green roofs, also called extensive green roofs) the drainage velocity is almost the same as the drainage velocity from a similar conventional roof. In case of a drainage layer consisting of sand or gravel (heavy-weight green roofs), the runoff is delayed when compared to a conventional roof.

After percolation has started, the runoff from a green roof is very similar to a conventional roof. There is only some delay due to the thickness of the growing medium, and and attenuation caused by the heterogeinity of the growing medium. When rainfall stops, percolation still continues while slowly reducing. Within one hour after the rainfall stops, percolation is typically

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negligable. The vegetation, via the roots, take water from the growing medium and evaporate this under the influence of sunlight (transpiration). Also water can directly evaporate from the growing medium to the atmosphere (evaporation). The processes of interception, transpiration and evaporation are together often called evapotranspiration. The actual evapotranspiration of green roofs depends on the potential evaporation and the available soil moisture in the growing medium. Not all soil moisture is available for evapotranspiration. And of course actual evapotranspiration also depends on the type of vegetation on the green roof. It is remarked that for small thicknesses of the growing medium, the vegetation only has a limited impact on evaporation, meaning that with vegatation the evapotranspiration of a green roof is only a little higher than without vegetation.

The growing medium is typically unsaturated. For a green roof on an open drainage layer, at the bottom side of the unsaturated zone there is not a groundwater level, but atmospheric conditions prevail. Usually the conductivity of the growing medium is much larger than the rainfall intensity, leading to unsaturated zone conditions. For instance, an extremely high rainfall intensity of 5 mm/minute corresponds to 7.2 m/day, which is comparable to the saturated conductivity of very fine sand. In most cases, rainfall intensity is lower and conductivity of the growing medium is larger due to the more coarse material and macro pores created by roots.

The water balance of a green roof is illustrated in Figure 1.86.

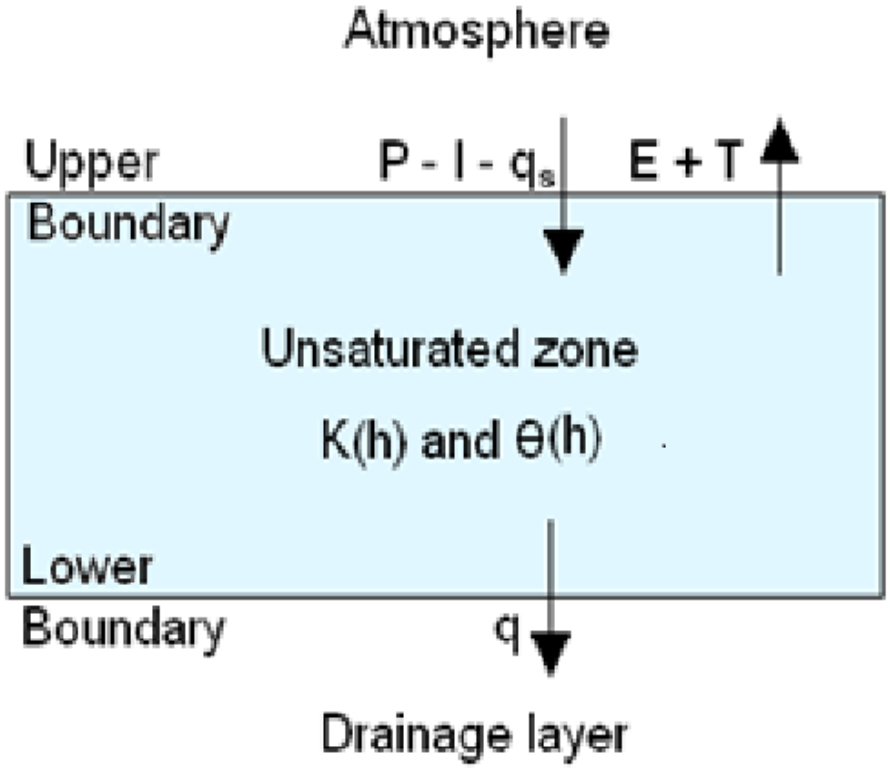


Figure 1.86: *Green roof water balance*

The water balance of the unsaturated zone of a green roof as shown in Figure 1.86 can be expressed as:

$$\frac{dS_t}{dt} = (P - I - q_s - E - T - q) * A \tag{1.146}$$

Where:

S_t	soil water storage in the unsaturated zone [m^3]
P	precipitation [$mm/timestep$]
I	interception [$mm/timestep$]
q_s	specific surface runoff [$mm/timestep$]
E	evaporation [$mm/timestep$]
T	transpiration [$mm/timestep$]
q	percolation out of the unsaturated zone bottom [$mm/timestep$]
A	surface area [m^2]
t	time [minutes]

The vertical flow through the unsaturated zone is described by the Richards equation:

$$\frac{d\theta}{dt} = \frac{d}{dz}[(K(\theta)(\frac{dh}{dz} + 1)] \tag{1.147}$$

Where:

K	unsatured hydraulic conductivity [$mm/minute$]
θ	soil moisture content (-)
h	pressure head (negative) (m)
z	vertical elevation above a reference level (m)

Given the often thin uniform layer of the growing medium we use a Van-Genuchten-approach for the soil with a uniforme porosity, and neglect the effect of hysteresis:

$$S_e = \frac{\theta(h) - \theta_r)}{(\theta_s - \theta_r)} = [1 + (\alpha|h|)^n]^{-m} \tag{1.148}$$

Where:

S_e	effective saturation [-]
θ_r	residual moisture content (moisture content at wilting point) [-]
θ_s	saturated moisture content [-]
α	inverse air entry value or bubbling pressure [$1/m$]
n	indicator for pore size distribution [-]
m	empirical shape parameter with $m = 1 - 1/n$ [-]

From this, a formula follows where the soil moisture content θ depends on the water pressure height h and a few empirical constants:

$$\theta(h) = \theta_r + \frac{\theta_s - \theta_r}{[1 + (\alpha|h|)^n]^m} \tag{1.149}$$

Next, the unsaturated conductivity can be expressed in the pressure height h and empirical constants:

$$K(h) = K_s S_e^I [1 - (I - S_e^{1/m})^m]^2 \tag{1.150}$$

Where:

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K_s	saturated hydraulic conductivity [mm/min]
Is	pore-connectivity parameter (-)

For the evapotranspiration, we use actual crop evapotranspiration. We use potential evaporation combined with a crop factor, as is also used in other RR concepts (e.g. at unpaved nodes). We assume evapotranspiration is maximal when the soil moisture content is equal to (or above) the soil moisture content at which percolation starts. While the soil moisture content reduces to the residual soil moisture contant (at wilting point), evapotranspiration gradually decreases to zero. We use a water balance for the unsaturated growing medium. The model concept is based on the Van-Genuchten-relations for flow through an unsaturated zone. Since we compute using a single soil moisture bucket, which represents the soil water retention characteristics for the whole thickness of the growing medium, another empirical variable is added in determing the percolation per timestep.

So, the following soil or growing medium parameters are used:

D	Thickness of growing medium (cm)
θ_r	Residual moisture content (cm^3/cm^3)
θ_s	Saturated moisture content (cm^3/cm^3)
K_s	Saturated hydraulic conductivity (cm/d)
α	Inverse air entry value or bubbling pressure (1/cm)
n	Indicator for pore size distribution (-)
I	Pore-connectivity parameter (-)
f_c	Crop factor (-)
θ_p	Moisture content above which percolation starts (cm^3/cm^3)

Boundary conditions are:

- ◇ the minimal soil moisture content is θ_r
- ◇ The evapotranspiration is maximum when soil moisture content is equal or larger than θ_p
- ◇ There is no percolation when the soil moisture content is lower than θ_p

The iteration scheme is as follows: Following other process descriptions in DRR, the state of the system at the start of the timestep determines the fluxes during the timestep. For each timestep, the following steps are followed:

- ◇ 1) Get the precipitation P_t and potential evaporation $E_{p,t}$ from the input
- ◇ 2) Determine the actual evapotranspiration $E_{a,t}$ from:

$$E_{a,t} = 0 \quad for \quad \theta_{t-1} <= \theta_r \tag{1.151}$$

$$E_{a,t} = f_c E_{p,t} \frac{\theta_{t-1} - \theta_r}{\theta_p - \theta_r} * e^{1 - \frac{\theta_{t-1} - \theta_r}{\theta_p - \theta_r}} \quad for \quad \theta_r < \theta_{t-1} < \theta_p \tag{1.152}$$

$$E_{a,t} = f_c E_{p,t} \quad for \quad \theta_{t-1} >= \theta_p \tag{1.153}$$

- ◇ 3) Determine the percolation q_t from

$$q_t = 0 \quad for \quad \theta_{t-1} <= \theta_p \tag{1.154}$$

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$$q_t = K(\theta)_{t-1}\frac{\theta_{t-1}-\theta_p}{\theta_s-\theta_p} \quad for \quad K(\theta)_{t-1}\frac{\theta_{t-1}-\theta_p}{\theta_s-\theta_p} <= \theta_{t-1}-\theta_p \tag{1.155}$$

$$q_t = (\theta)_{t-1}-\theta_p \quad for \quad K(\theta)_{t-1}(\theta)_{t-1}\frac{\theta_{t-1}-\theta_p}{\theta_s-\theta_p} > \theta_{t-1}-\theta_p \tag{1.156}$$

where $\frac{\theta_{t-1}-\theta_p}{\theta_s-\theta_p}$ is an empirical determine variable depending on the soil moisture content at the start of the timestep (θ_{t-1}), and the soil moisture content at which percolation starts (θ_p) and the saturated soil moisture content (θ_s).

◇ 4) Determine the soil moisture content at the end of the timestep (θ_t) from

$$\theta_t = \theta_r \quad for \quad \theta_{t-1}+P_t-E_{a,t}-q_t <= \theta_r \tag{1.157}$$

$$\theta_t = \theta_{t-1}+P_t-E_{a,t}-q_t \quad for \quad \theta_{t-1}+P_t-E_{a,t}-q_t > \theta_r \tag{1.158}$$

◇ 5) Determine the effective storage at the end of the timestep $S_{e,t}$ from

$$S_{e,t} = \frac{\theta_t-\theta_r}{\theta_s-\theta_r} \tag{1.159}$$

◇ 6) Determine the unsaturated conductivity at the end of the timestep ($K(\theta_t)$) from

$$K(\theta)_t = 0 \quad for \quad \theta_t <= \theta_p \tag{1.160}$$

$$K(\theta)_t = K_s S_{e,t}^I[1-(1-S_{e,t}^{\frac{1}{1-1/n}})^{1-1/n}]^2 \quad for \quad \theta_t > \theta_p \tag{1.161}$$

Remarks:

◇ In this way the parameter α, the air entry value, is superfluous. This variable is only interesting in case the has to be determined, but within this water balance method that is not needed.
◇ The empirical variable $\frac{\theta_{t-1}-\theta_p}{\theta_s-\theta_p}$ has been tested only for one specific thickness and soil growing medium.

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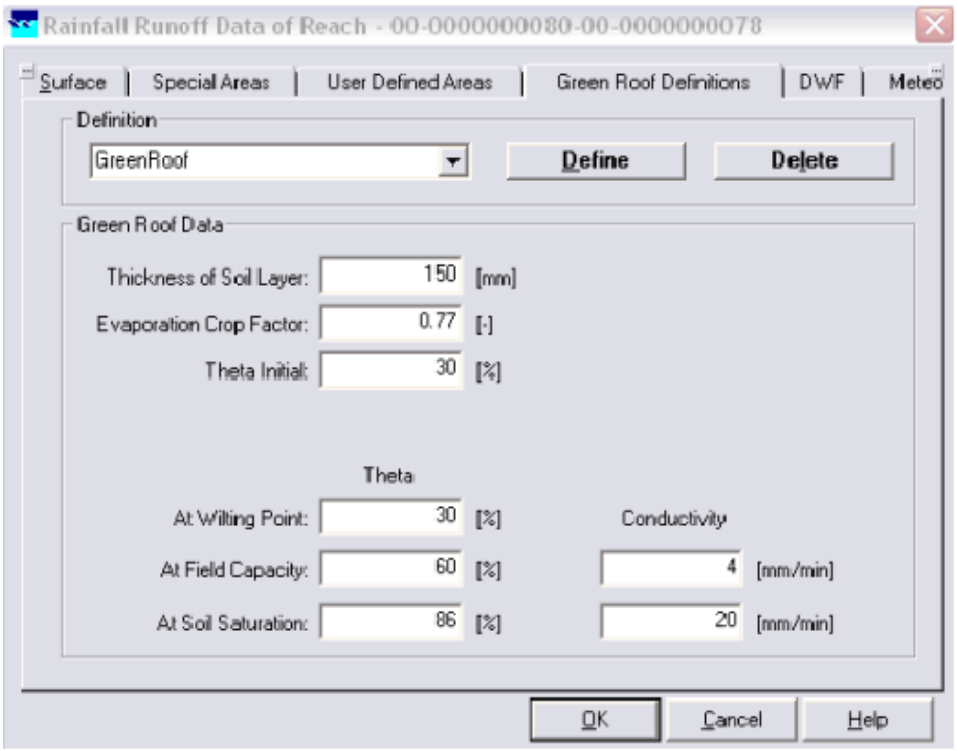


Figure 1.87: *Green roof parameters*

For every Green Roof definition, the user can specify the necessary parameters (see Figure 1.87). The following parameters have to be specified:

D	Thickness of soil layer in mm
f_c	Evaporation crop factor (-)
θ_i	Initial moisture content (%)
θ_r	Residual (wilting point) moisture content (%)
θ_p	Field capacity moisture content (%)
θ_s	Saturated moisture content (%)
K_s	Saturated hydraulic conductivity (mm/min)
K_p	Hydraulic conductivity at field capacity (mm/min)

For the green roofs, the following output parameters are generated:

- ◇ 1) Actual soil moisture content for each special area (Storage sp1..9) (m3)
- ◇ 2) Outflow towards sewer system for every area (Infl. Sewer sp1..9) (m3)

With these output parameters, the user can estimate the effects of green roofs on runoff and check inidividual output in detail.

1.1.18.6 Swale filter

The Swale drainage system is included as an extra option within the NWRW urban runoff model. The Swale filter is located in between the runoff from the different paved/unpaved/green roof areas and the inflow into the sewerage system. It consists of a long, rectangular shaped

trench which is usually filled with coarse sand or gravel. A swale filter offers water buffering and infiltration facilities. The idea is to catch runoff from roofs, roads and other paved area. Figure 1.88 below gives an example with two connected swale filters (or 'wadi' in Dutch).

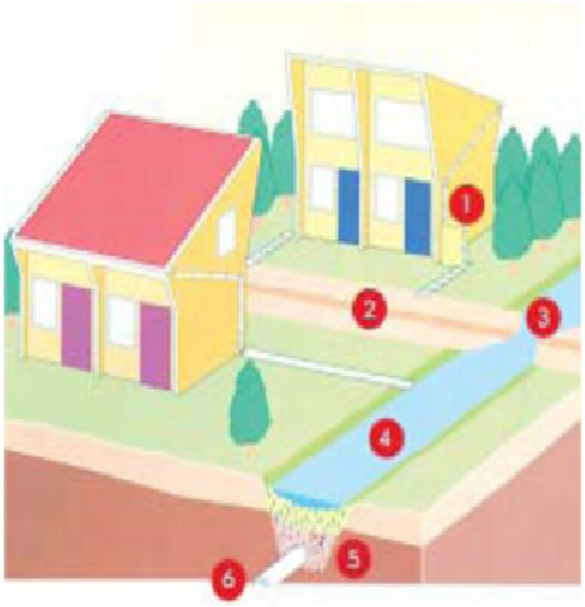


Figure 1.88: *Swale filter principle*

From an initial dry situation the rainfall on roofs (1) and roads and other paved areas (2) flows to the swale filter. The upstream swale filter (3) flows to the downstream swale filter (4). Part of the water can infiltrate into the ground (5) and the rest flows with some delay (6) to the downstream area.

Figure 1.89 shows how a swale filter functions.

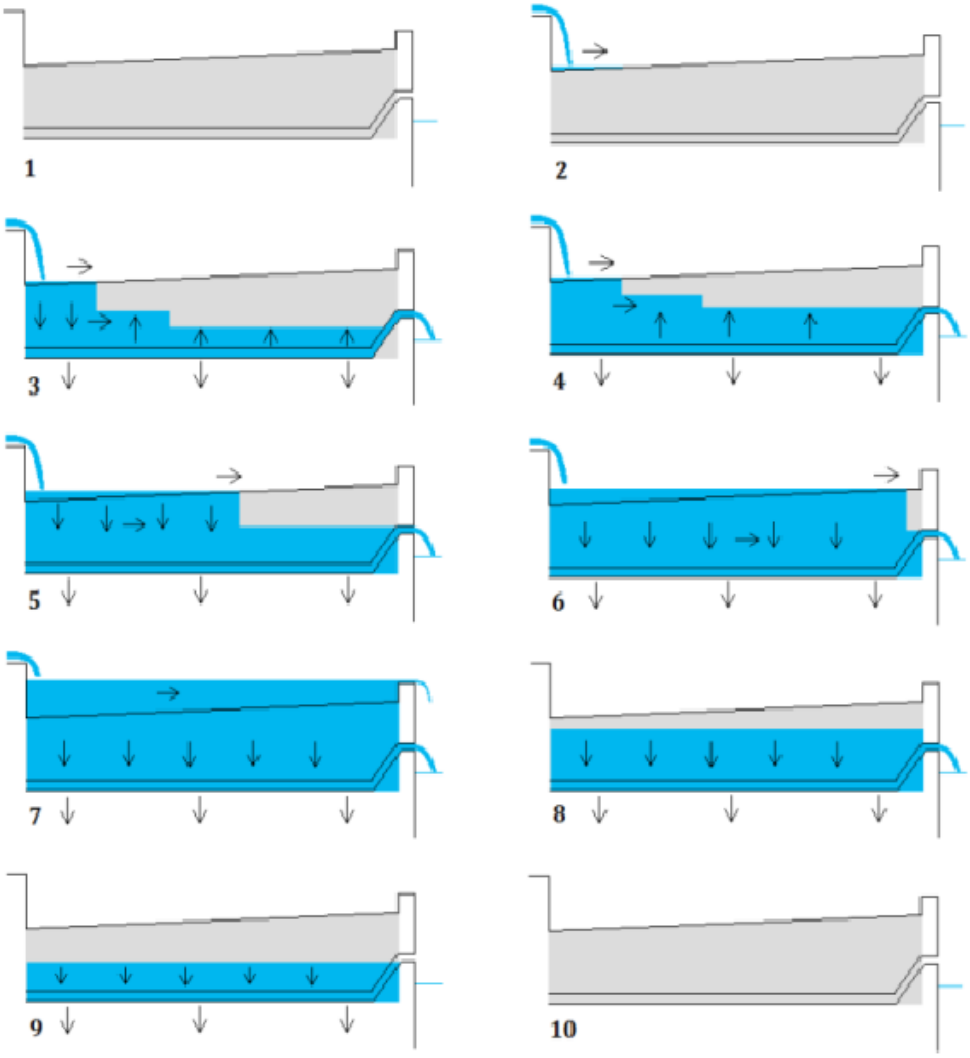


Figure 1.89: *Swale filter functioning after (Donkers (2010))*

From an initial dry situation (1), runoff water flows into the swale filter (2). At low groundwater levels, water can infiltrate (3). At higher rainfall rates, the runoff will exceed infiltration capacity and the water level in the swale filter will rise (4) and the drain outflow start functioning (3-6). When the infiltration trench is completely filled, also the overflow spillway starts discharging (7). After the rainfall and inflow into the swale filter stop, the trench will slowly empty (8-10).

Water balance computations of the swale filter are illustrated in Figure 1.90.

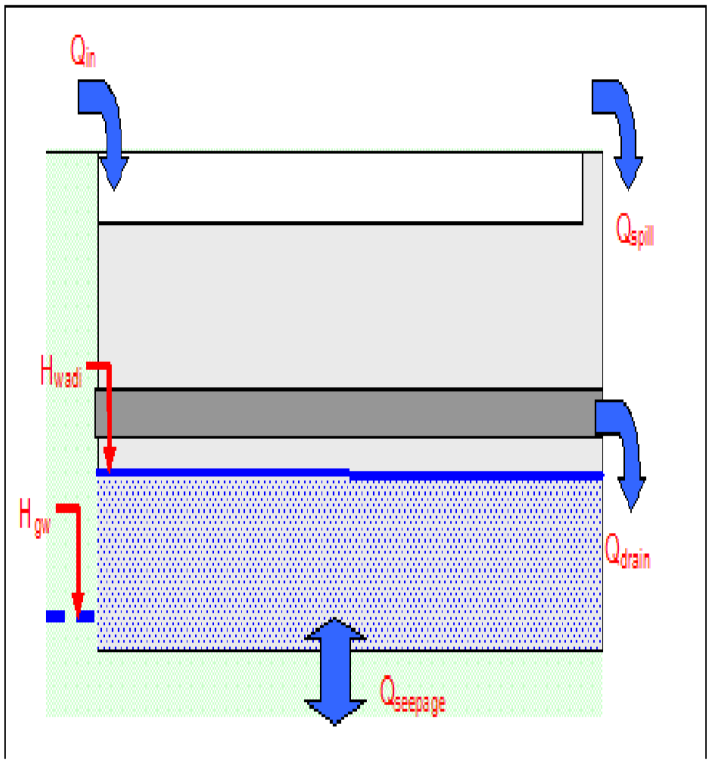


Figure 1.90: *Swale filter water balance*

The water balance is based on the following assumptions:

- ◇ The swale filter is filled with coarse material (gravel/sand) with high saturated conductivity. Together with the drainage capacity of the drain this makes sure that we can assume the groundwater level below the swale filter as horizontal.
- ◇ The infiltration capacity and unsaturated zone conductivity are high in relation to the conductivity of the lower soil;
- ◇ The grondwater level in the subsoil is assumed constant
- ◇ The outside water level is low, such that the swale filter outlet (both orifice and spillway) can discharge freely.

This simplifies the swale filter water balance to a bucket model with some incoming and outgoing fluxes as illustrated in Figure 1.90.

$$Q_{in} - \Delta S = Q_{spill} + Q_{drain} + Q_{seepage} \tag{1.162}$$

Where:

Q_{in}	runoff from roofs and paved area into the swale filter
ΔS	storage change per unit of time
Q_{spill}	discharge via the spillway
Q_{drain}	discharge via the drain outflow
$Q_{seepage}$	discharge from the swale filter to the underground

Rewriting equation Equation (1.162) using $\Delta S = \frac{h_{wadi,t+\delta t} - h_{wadi,t}) A n_p}{\Delta t}$ gives

$$h_{wadi,t+\Delta t} = h_{wadi,t} + (Q_{spill} + Q_{drain} + Q_{seepage}) \frac{\Delta t}{A n_p} \tag{1.163}$$

With the additional variables:

$h_{wadi,t}$	water level in the wadi (swale filter) at time t
Δt	timestep size
A	area of the wadi (swale filter), assumed constant independent of the water level
n_p	porosity of the wadi (swale filter) material

With known $h_{wadi,t}$ and Q_{in}, the other terms are related to the water level in the wadi:

$$Q_{spill} = c_{d,spill} * B_{spill} \frac{2}{3} \sqrt{(\frac{2}{3} g)} [h_{wadi,t} - Z_{spill}]^{1.5} \quad for \quad h_{wadi,t} > Z_{spill} \tag{1.164}$$

$$Q_{seepage} = \frac{(h_{wadi,t} - h_{gw,adj})}{W_{drain-underground}} \tag{1.165}$$

in which

$$h_{gw,adj} = max(h_{gw}, Z_{wadibedlevel}) \tag{1.166}$$

Where

$h_{gw,adj}$	groundwater level in the subsoil
Δt	timestep size
A	area of the wadi (swale filter), assumed constant independent of the water level
n_p	porosity of the wadi (swale filter) material
Z_spill	spillway crest level
$Z_wadibedlevel$	bed level of the swale filter (wadi)
$W_drain - underground$	drainage resistance for infiltration from the swale filter into the underground

Based on field experiments in the Netherlands (Utrecht, Leidse Rijn area), the drain outflow is default implemented as a free weir formula (note that an orifice formulation is also available):

$$Q_{drain} = c_{d,drain} * B_{drain} \frac{2}{3} \sqrt{(\frac{2}{3} g)} [h_{wadi,t} - Z_{drain}]^{1.5} \quad for \quad h_{wadi,t} > Z_{drain} \tag{1.167}$$

From calibration on field data it appeared that the weir coefficient $c_{d,drain}$ was very low: 0.0323. This could be related to the conductivity of the drain material.

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For every swale filter, the user can specify a number of parameters relating to the trench (swale) itself, the drain, and the spillway. These are shown in Figure 1.91 and consist of:

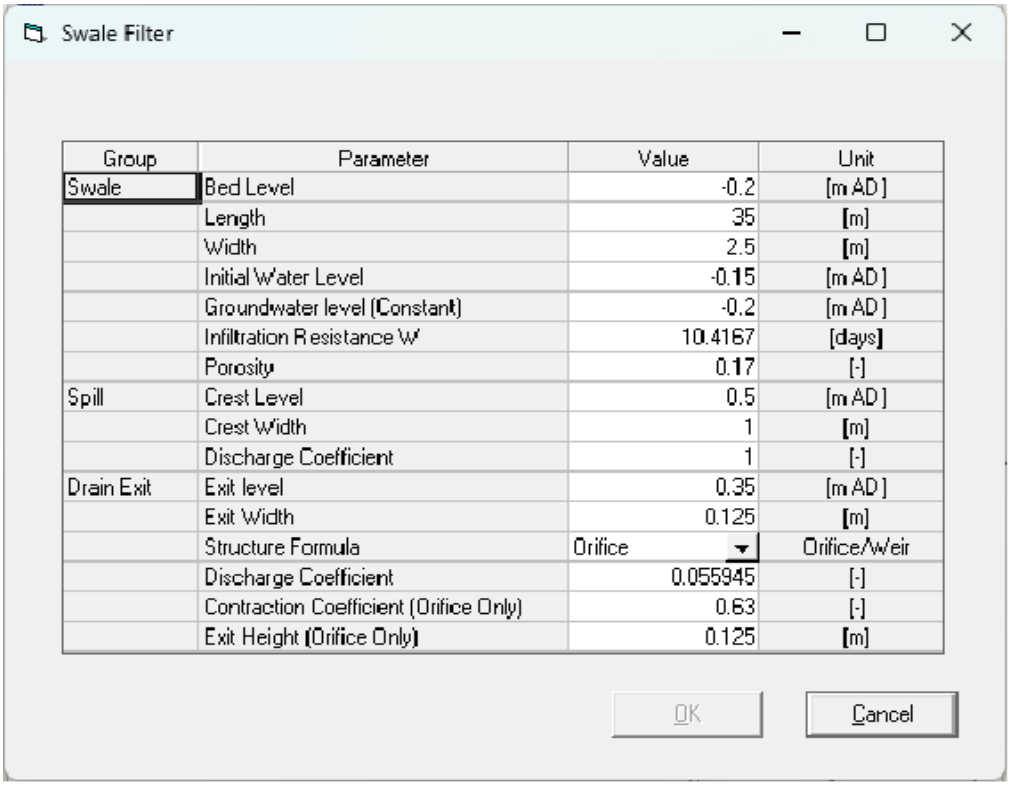


Figure 1.91: *Swale filter input data*

Bed level Bottom level of the trench

Length The length of the trench. Keep in mind that the swale is considered to be rectangular (oblong) in shape.

Width The (constant) width of the trench

Initial water level The initial water level inside the trench at the start of the simulation. This must not be lower than the bed level

Groundwater level The groundwater level in the surrounding soil. The difference between this level and the water level inside the trench determines whether the water exfiltrates or infiltrates from the trench. The groundwater level is considered to be constant duruing the simulation

Infiltration resistance W The infiltration resitance determines the magnitude of the flux through the interface between the trench and the surrounding soil by

Porosity The porosity of the soil inside the trench, in m3/m3

Spill level Spillway crest level, used in the spillway modelling using a standard weir formula, and assuming free flow.

Spill width Spillway crest width, used in the spillway modelling using a standard weir formula, and assuming free flow.

Discharge coefficient Spillway discharge coefficient, used in the spillway modelling using a standard weir formula, and assuming free flow.

Drain exit parameters The actual discharge through the drain exit depends on the design of the swale filter. To give users maximum flexibility, they can choose between an orifice type formula or a weir type formula to estimate the drain discharge and specify the necessary parameters.

Additional output for the swale filter are the inflow, infiltration, spill and drainage flows from the swale filter, and the water level and storage. This gives clear insight in the buffering effect of the swale filter.

1.2 DRR Input file formats: the Model Database 4.00

1.2.1 Philosophy

The DRR Model Database consists of a number of ASCII-files, which are logically separated into a number of layers; each layer covers a number of files. Each file contains the definition of one or more DRR-objects. In most cases the object has an attribute **id** which is the key to that particular object.

- ◇ For more information on the layers, see the chapter Subdivision into layers Database
- ◇ For more information on the design conventions, see the chapter Input conventions
- ◇ For exact description of certain input files, see the chapter of the module that has your interest:

1.2.2 RR (Rainfall Runoff)

Introduction

This document describes the model database (MDB) of DRR available in SOBEK version 2.16, and DRR implementation in D-HYDRO and Delft3D FM 1D2D Suite. **Note:** Not all described DRR functionality is available through the GUI.

The input formats for this module are in the process of being changed from '.tp' to an initialization file format (.ini). INI files are plain text (ASCII) and are used to set parameters for the RR program. The description of modifications to date include the replacement of topography layer files: 3b_runoff.tp, 3b_link.tp, and 3b_nod.tp with a single file, rr_topology.ini.

RR datafiles

Control layer

- ◇ Ini file (Delft_3B.Ini)

Topography layer Currently, the topology layer may be read in two ways: From three individual .tp files, or from a single rr_topology.ini file. The read protocol is specified in the control layer input file, DELFT_3B.Ini by setting USE_INI to either 0 (.tp protocol) or 1 (.ini protocol).

Files used for the .tp protocol:

- ◇ node file (3b_nod.tp)
- ◇ link file (3b_link.tp)
- ◇ runoff file (3b_runoff.tp)

Files used for the .ini protocol:

- ◇ node file (rr_topology.ini)

Paved area layer

- ◇ data file (paved.3b)

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- ◇ storage definition file (paved.sto)
- ◇ dry weather flow file (paved.dwa)
- ◇ table file (paved.tbl)

Unpaved area layer

- ◇ data file (unpaved.3b)
- ◇ storage definition file (unpaved.sto)
- ◇ alfa definition file (unpaved.alf)
- ◇ infiltration definition file (unpaved.inf)
- ◇ seepage definition file (unpaved.sep)
- ◇ table file (unpaved.tbl)

Greenhouse layer

- ◇ data file (greenhse.3b)
- ◇ roof storage file (greenhse.rf)
- ◇ silo definition file (greenhse.sil)

Open water layer

- ◇ data file (openwate.3b)
- ◇ seepage definition file (openwate.sep)
- ◇ table file (openwate.tbl)

Structure layer

- ◇ data file (struct3b.dat)
- ◇ definition file (struct3b.def)
- ◇ controller file (contr3b.def)
- ◇ table file (struct3b.tbl)

Boundary layer

- ◇ data file (bound3b.3b)
- ◇ table file (bound3b.tbl)
- ◇ boundaryconditions.bc file (D-HYDRO)

NWRW layer

- ◇ data file (pluvius.3b)
- ◇ dry weather flow file (pluvius.dwa)
- ◇ general data (pluvius.alg)
- ◇ table file (pluvius.tbl)

WWTP layer

- ◇ data file (wwtp.3b)
- ◇ table file (wwtp.tbl)

Industry layer

- ◇ data file (industry.3b)
- ◇ table file (industry.tbl)

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Sacramento layer

- ◇ data file (sacrmnto.3b)
- ◇ capacities and contents definition file (sacrmnto.cap)
- ◇ unit hydrograph definition file (sacrmnto.uh)
- ◇ other data file (sacrmnto.oth)

Meteo and fixed files

- ◇ storage coefficients soil types <..\..\fixed\bergcoef>
- ◇ crop factors agricultural crops <..\..\fixed\cropfact>
- ◇ Capsim soil types <..\..\fixed\bergcoef.cap>
- ◇ Capsim rootzone data <..\..\fixed\root_sim.inp>
- ◇ Capsim unsaturated zone data <..\..\fixed\unsa_sim.inp>
- ◇ greenhouse class file <..\..\fixed\kasklasse>
- ◇ greenhouse initialisation file <..\..\fixed\kasinit>
- ◇ greenhouse water use file <..\..\fixed\kasgebr>
- ◇ crop factors open water <..\..\fixed\crop_ow>
- ◇ rainfall file (1 or more events)
- ◇ evaporation file
- ◇ Default definition file <..\fixed\3bEdit.def>
- ◇ Optional: New format greenhouse class file <..\fixed\3B\NewKasKlasData.dat>
- ◇ Optional: New format greenhouse initialisation file <..\fixed\3B\NewKasInitData.dat>
- ◇ Optional: New format greenhouse water use file <..\fixed\3B\NewKasGebrData.dat>
- ◇ Optional: New format crop data <..\fixed\3B\NewCropData.dat>
- ◇ Optional: New format open water crop factor data <..\fixed\3B\NewCropOWData.dat>
- ◇ Optional: New format soil data <..\fixed\3B\NewSoilData.dat>

The files of the data layers are generated by either the SOBEK Network Editor (Netter) and associated programs (ModelEdit, Meteo, Settings) or the DeltaShell GUI of D-HYDRO and Delft3D FM 1D2D Suite. Also, model files can be generated using conversion tools. For instance, within SOBEK the old Dutch standard files for sewer system models can be imported (so-called SUF-HYD files), while within D-HYDOR and Delft3D FM 1D2D Suite the new csv format specified by RioNed is supported. SOBEK also supports a Diwa-HYDRA format and other old model formats. In the conversion of the SUF-HYD files or RioNed csv files, DRR input files for the NWRW layer are automatically generated.

1.2.3 Boundary layer

Data file: BoundaryCondition.bc

This file is the new boundary condition format used within D-HYDRO and Delft3D FM 1D2D Suite. The other files described in this subsection are only used within the SOBEK2 implementation of DRR. This is the main difference in input files between SOBEK2 and D-HYDRO / Delft3D FM 1D2D. Other differences between SOBEK2 and D-HYDOR implementations of DRR relate to the availability of functionality in the GUI. The computational DRR module is identical.

to be added

The new format for the RR-boundary conditions file is based on the BC-format for D-FlowFM. It can be used to combine multiple quantities for multiple locations in a single (ASCII) file. Each location is defined in a separate ‘BC block’, which consists of a column-wise table, (the data), and a header specifiying how this table should be interpreted. The option of providing bound-

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ary conditions realtime from D-Flow FM computational core is specified in the DIMR.XML file of the combined model. Typically the BoundaryCondition.bc file in that case is only filled with all locations and a constant suitable initial water level for each location.

Table 1.19: *Definition of boundary conditions.*

Keyword	Type	Default	Description
`[General]`			
`majorVersion`	Int	`1`	Major file version. Do not edit this.
`minorVersion`	Int	`0`	Minor file version. Do not edit this.
`fileType`		`boundConds`	File type. Do not edit this.
`[Boundary]`			
`name`	String		Name of the boundary location (node id)
`function`			Possible values: ◇ constant ◇ timeSeries
`time-interpolation`			Possible values: ◇ linear ◇ linear-extrapolate ◇ block-from (value holds from specified time step) ◇ block-to (value holds until next specified time step)
`quantity`			Possible values: ◇ time ◇ water_level
`unit`			Possible values (time): ◇ seconds since *begintime* ◇ minutes since *begintime* ◇ hours since *begintime* format: yyyy-MM-dd hh:mm:ss +00:00 (+00:00: time zone) Possible values (other than time): ◇ m (water level)
`{data}` `(a column per quant/unit)`			

Example RR-boundarycondition.bc file:

[General] fileVersion = 1.01 fileType = boundConds

[Boundary] name = 0-NOORD1 function = constant quantity = water_level unit = m 0

The next RR-boundary files are used within the SOBEK2 implementation.

Data file: Bound3B.3B

This file contains the data for the nodes of model type 6 (boundary)

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BOUN id ’1’ bl 0 -0.5 is 100. boun

With

id	node identification
bl	boundary level type bl 0 0.5 = fixed boundary level, of 0.5 m NAP bl 1 ’bound_1’ = variable boundary level with table identification ‘bound_1’ (data in Bound3B.Tbl file) bl 2 ’3’ = variable boundary level, with on-line coupling SOBEK. Data taken from id ‘3’ in the SOBEK-HIS-file.
is	initial salt concentration (mg/l)

Boundary table file: Bound3b.Tbl

Boundary water levels and salt concentrations as a function of time, in case of option 1. Data is known before the simulation starts.

BN_T id ’bnd_level’ nm ’IJsselmeer’ PDIN 1 0 ’ ’ pdin
TBLE
1997/01/01;00:00:00 -1.5 200<
1997/05/01;00;00;00 -1.45 220<
1997/10/01;00:00:00 -1.50 200<
tble
bn_t

With

id	table identification
nm	name of table PDIN ..pdin = period and interpolation method 0 0 ’ ’ = interpolation continuous, no period 1 0 ’ ’ = interpolation block, no period 0 1 ’365;00:00:00’ = interpolation continuous, period in DDD;HH:mm:ss 1 1 ’1;00:00:00’ = interpolation block, period in DDD;HH:mm:ss

TBLE .. tble contains the table, first column is the date (year month day hour minute second), the second column contains the boundary level in meters w.r.t. reference level (NAP), and the third column contains the salt concentration (only used if salt computation option is switched on).

1.2.4 Control layer

Ini file: DELFT_3B.Ini

This file is set-up with a Windows INI type of filestructure. If records are not present, defaults will be used.

```
[System]
CaseName= Test
DebugFile=0
DebugTime=1 1
DebugTime2=900 905
DebugTimeCapsim=0 0
Version=2.04
SkipBinfile=-1       ! Always -1
CaseSensitive=-1     ! -1=yes; 0=no. Default -1. (ARS 4988)
TimeMessage=-1       ! -1=yes.Default 0=no. Shows time on screen after reading input
            ! data and just before starting actual calculations, and immediately
            ! after finishing actual calculations

[OutputOptions]
```

```
OutputTimestep=1     *1=yes, 0=no;
OutputEvent=1     *1=yes,0=no
OutputOverall=1      * not used?
OutputDetail=47      *number event
OutputScreen=0       *1=yes, 0=no
OutputOpenwater=1       *1=output w.r.t. NAP, 0=w.r.t. referemce level
OutputGroundwater=1     *1=w.r.t. NAP, 0=w.r.t. surface level
OutputBoundary=2     *0=no, 1=yes, Rijnland format; 2=yes, Pluvius format
OutputAtTimestep=1      *Output every xx timestep
OutputAtTimestepOption=1  *1=Current value, 2=Average value, 3=Maximum value
Reduced Output=0     !-1=yes: only output at open water, structures.
                     ! Also balances and link flows.
                     ! 0=no (default)
ExtendedBalance=0       ! 0=no, -1=yes. Default=0
OutputRRPaved=0         ! 0=no, -1=yes. Default=-1. Output of RR-paved nodes
OutputRRUnpaved=1       ! 0=no, -1=yes. Default=-1. Output of RR-unpaved nodes
OutputRRGreenhouse=0    ! 0=no, -1=yes. Default=-1. Output of RR-greenhouse nodes
OutputRROpenWater=1     ! 0=no, -1=yes. Default=-1. Output of RR-open water nodes
OutputRRStructure=0     ! 0=no, -1=yes. Default=-1. Output of RR-structure nodes
OutputRRBoundary=0      ! 0=no, -1=yes. Default=-1. Output of RR-boundary nodes
OutputRRNWRW=0          ! 0=no, -1=yes. Default=-1. Output of RR-NWRW Urban runoff nodes
OutputRRWWTP=0          ! 0=no, -1=yes. Default=-1. Output of RR-Waste Water Treatment nodes
OutputRRIndustry=0      ! 0=no, -1=yes. Default=-1. Output of RR-Industry odes
OutputRRSacramento=0    ! 0=no, -1=yes. Default=-1. Output of RR-Sacramento nodes
OutputRRBalance=1       ! 0=no, -1=yes. Default=-1. Output of RR-water balance
OutputRRSalt=0          ! 0=no, -1=yes. Default=-1. Output of RR-salt computations
OutputRRRunoff=-1       ! 0=no, -1=yes. Default=-1. Output of RRRunoff nodes
                        ! (HBV, SCS, LGSI, Walrus, D-NAM)
OutputRRLinkFlows=1     ! 0=no, -1=yes. Default=-1. Output of RR-link flows

[Options]
GenerateNetCdfOutput=-1  !(-1=yes,0=no) Note that in D-HYDRO GUI NetCdf output is required.
GenerateHISOutput=-1     !(-1=yes,0=no) Note that in the SOEK2 GUI, HIS output is required,
                          ! while in the D-HYDRO / Delft3D FM 1D2D GUI it is not
MeteoNetCdfInput=0   (-1=yes,0=no) Default MeteoNetCdfInput is switched off.
PrecipitationNetCdfSeriesId ='precipitation'
   ! Precipitation series id in Netcdf file between single quotes, if MeteoNetCdfInput is true
EvaporationNetCdfSeriesId ='evaporation'
   ! Evaporation series id in Netcdf file between single quotes, if MeteoNetCdfInput is true
TemperatureNetCdfSeriesId ='temperature'
   ! Temperature series id in Netcdf file between single quotes, if MeteoNetCdfInput is true
Dirlisting=1      (1=yes,0=no; within CMT always 1; indicator use @ in filenames)
WeirSettingTargetLevel=0 (Weir setting yes/no adjusted to be <= upstream target level; 0=no)
UnsaturatedZone=0       (0=no,1=Capsim,)
InitGwlOption=-1     ! This option is relevant when the initial groundwater level is
            ! specified with respect to surface level
            ! If no Scurve is used, initialisation is done w.r.t. the specified
            ! constant surface level.
            ! If an Scurve is used, it depends on the switch InitGwlOption
            ! 0 =when using SCurve: initialisation of groundwaterlevel w.r.t
            ! specified constant surface level
            ! Default=-1= in case of using Scurve: initialisation of
            ! groundwaterlevel relative to lowest point of the Scurve
InitBcOption=0       !How to initialize storage coefficients unpaved area
            ! 0 = using open water target level (default)
            ! 1 = using initial groundwater level
InitCapsimOption=1      ! 0 = default old initialisatie,
            ! 1 = at equilibrium moisture,
            ! 2 = at moisture content for pF=2
            ! 3 = at moisture content for pF=3
CapsimPerCropArea=0     ! 0 = call Capsim with crop area averaged data once per
            ! unpaved area
            ! Default= -1=call Capsim for each crop area seperately
KvdlVariatieOpenwater=0 !0 = groundwater level follows changes open water level (default)
            !1 = treat changes open water level as rainfall/evaporation
KvdLInitOpenwater=0     !1=treat initial level difference open water - groundwater
```

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```
            !as rainfall/evaporation; see KvdLVariatieOpenwater  option.
            !default = 0
KvdLDimensionInDays=30  !Number of days history to be taken into account in Krayenhoff
            ! van de Leur calculations. If missing, backwards compatibility is
            ! accomplished by assuming 10 days, with a maximum of 999
            ! timesteps.  If the record is specified, there is no maximum
            ! anymore. Only a warning is given if the number of timesteps
            ! exceeds 1000.
UnpavedScurveAreas=100  ! Number of sub areas for SCurve; default=100.
UnpavedScurveAlfaOption=0 ! 0 = absolute, 1=relative w.r.t. surface level; Default=0.
DrainageDeltaH=-1       ! -1=parallell systems (default), 0=stacked system;
USE_INI =0              ! 0=read topology from three .tp files(default),
                        ! 1= read topology from single .ini file;
MinimumDepthCF=0.0      ! Minimum depth in Channel Flow boundaries
            ! Below this level RR-unpaved infiltration from CF are stopped
ControlModule=-1     (-1=with control module, 0=without)
MaalstopModule=-1       (-1=with RTC module, 0=without)
ModFlowModule=0      (-1=with Modflow module, 0=without) \textbf{not yet operational}
WLMModule=0       !( -1=with WLM, 0=without) \textbf{not yet operational}
WQModule=0        !( -1=with WQ module, 0=without) \textbf{not yet operational}
SaltComputation=1       ! -1=yes, 0=no.
SaltConcOutput=0     ! -1=yes, 0=no.
SaltConcRainfall=8      (8 mg/l)
MinFillingPercentage=10 (below 10\% no extractions for greenhouse from storage basins)
RestartIn=1       (Use binary restart input file; 1=yes, 0=no)
RestartOut=0      (write binary restart file)
LowestRestartGwl=       ! laagste toegestane restart grondwaterstand in m - maaiveld
SkipBoundaryLevelFromRestartFile=0 !0 or -1, default=0;
            !-1 = do not use boundary levels from restart file
MaxIterations=5      ! maximum aantal iteraties binnen 1 tijdstap (default=5)
DetailedGwlComputation=-1 ! --1 = yes (default)
StepGwlStorageCoefficient=0.01 !Default 0.01 m.
CheckRootzoneGwl=-1     !-1=yes, 0=no (default).Checks whether groundwater levels
            ! is in the rootzone and sets storage coefficient to 0.01
BinaryInput=0     ! -1=yes, 0=no
BinaryOutput=0       ! -1=yes, 0=no
Defaultdataset=0     !1=yes,0=no;  option from rainfall rainfall file
            (=1 default dataset, 0 for the 44-year series)
EvaporationYear=-1      !-1 = check on year = rainfall year (default),
            !xxxx=use data from year xxxx
GreenHouseYear=1951     !-1 = check on year = rainfall year (default),
            !xxxx=use data from year xxxx
VolumeCheckFactorToCF=1 ! A multiplier on the computed maximum volume to be exchanged.
       ! Default=1. A large value effectively means that the volume check is not active.
VolumeCheckFactorToOW=1.0 ! A multiplier on the computed maximum volume to be exchanged.
       ! Default=1.0. A large value effectively means that the volume check ! is not active.
DetailedVolumeCheckMessages=-1 ! Enables extra information in Sobek_3B log file
                               ! in order to show at which locations VolumeCheck is active.
CoefRz=0.1      ! 4 Interpolation coefficients for interpolation in Capsim tables
CoefGwl=100.    ! 2 coefficients, 2 power coefficients, for root zone and groundwater level
PowerRz=2.      ! Default power coefficients PowerRz=PowerGwl=2
PowerGwl=2.     ! weight coefficients CoefRz=0.1 and CoefGwl=100.
CheckBalance=0  ! 0 = no detailed output balance error per timestep, -1=yes
OpenWaterLevelComputations=Advanced
            ! Simple or Advanced. Default =Advanced
            ! Advanced = take into account storage on land of neighbouring
            ! nodes in computing open water levels
OpenWaterPrecipitation=0 ! 0 = rainfall at area at actual level, (Default)
            ! -1=at area at lowest surface level of connected nodes,
            ! if no paved/unpaved/greenhouse nodes connected, the area at
            ! maximum allowable level is used.
StructureOperation=     ! 0 = Operate structure depending on initial level. (=default)
            ! 1 = Use SetFractionTime to switch on/off.
            ! (Sept 1999, ARS xxxx) For internal use only.
DrownedWeirDepth=    ! Switch to set which depth is used in weir formula. Default=0.
                     ! 0 = lowest depth; 1 = maximum depth; 2=average depth;
```

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```
EmulateUnixOnPc=     ! default=false; for testing Sobek-Parallell functionality only

HeaderRunoffOutAlways=  ! default=false; for testing Sobek-Parallell functionality only.
```

For internal use only.

```
FixARS5176=-1       ! default=-1 (true), if =0 then false.
        !(ARS5176=surface runoff bug fixed in April/May 2000)
FixARS8842=-1       ! default=-1 (true), if =0 then false.
        !(ARS8842=added volume check to unpaved module drainage to
        ! open water, January 2002)
FixARS11610=-1    ! default=-1 (true), if =0 then false.
        ! (ARS11610=added volume check for drainage from unpaved area to boundary (CF node)
        ! (improved stability RR-CF on-line coupling)
FixARSControllerLvlCheck=   ! default=true, if =0 then false.
        !(Extra check that the computed weir controller flow satisfies the
        ! possible flow directions)
FixARS10084=-1  ! default=.true., if=-1 then true.
                ! if gw on surface, include water layer on surface in dh.
FixARS12253=-1  ! default=.false., if=-1 then true
                ! NWRW runoff delay correction (1s-1min)
CumulativeGroundwaterExceedance=0 ! 0=Sum, 1=Sum of squares above threshold; default=0
MessageInundation=0     ! 1 = detailed inundation message per node and timestep,
                        ! 0 = no message
MessageVolumeCheck=0     ! 1= yes, 0 = no message
MessagePerTimestep=0    ! 1= 1 message per inundation timestep, 0 = no message
DailyRainfallStartHour=0 ! Option to specify start hour of daily rainfall.
            ! Default=0. 8=daily rainfall starts at 8 o'clock in the morning
EstimateRemainingDuration=0 ! 0=no (default), -1=give estimate remaining runtime on screen
GenerateAanvoerAbr=0    ! 1= yes, 0=no; default=0
LargeBalanceErrorPercentage=1.0    ! Percentage ($>$0).
            ! An error message will be given if the total balance error for the
            ! RR simulation exceeds this percentage. Default value = 1.0
NewFormatCropFactors=0  ! 0=no, -1=yes. Option to use new format crop factors.
CropDefinition='Default'
            ! If NewFormatCropFactors=-1, CropDefinition= is used to specify
            ! the set of crop definitions used. Default set is named `Default'.
            ! In the crop definition, crop names and crop factors are specified.
OpenWaterCropDefinition='Default'
            ! Compare with CropDefinition, but now for open water crop factors.
NewFormatSoilData=0     ! 0=no (default), -1=yes. Option to use new format soil data.
SoilDefinition='Default'
            ! If NewFormatSoilData=-1, SoilDefinition= is used to specify
            ! the set of soil data used. Default set is named `Default'.
            ! In the soil definition, soil names and soil data like storage
            ! coefficients are defined
NewFormatKasdata = 0   ! 0=no (default), -1=yes. Option to use new format greenhouse data.
KasDefinition='Default'
            ! If NewFormatKasdata is used, then KasDefinition= is used to
            ! specify the set of fixed greenhouse data used. This set contains
            ! greenhouse initialisation data and greenhouse water use data.
ParseTokenMethod=0
            ! 0=no (default), -1=yes. Alternative method of parsing input files;
            ! at the moment only available for NWRW nodes.
HisConvergence=0
            ! 0=no (default), -1=yes. Option to generate an extra His file called
            ! Convergence.His containing convergence info
            ! (compare with FlowAnal.His file of Sobek-CF-SF)
StructComp=0      ! 0=no (default, backwards compatible), -1=yes.
            ! If Structcomp=.true., then the same internal iteration criteria are
            ! used for weir, orifice and friction nodes.
RunOffOutHis=0       ! -1 = Runoff.Out in HIS format, 0=no.
SeparateRWA\_DWA     ! -1=yes,separate DWA and RWA in Runoff.Out file, 0=no
DWAString='DWA'      ! String to add to NWRW location for DWA
```

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```
RWAString='RWA'      ! String to add to NWRW location for RWA
NWRWContinuous=-1       !-1=yes, 0=no. Simulate NWRW model continuously for series of
           ! events, using an additional continuous bui file covering all events.
ConstantHdeZBergC=-1    !0=(default) no; --1=constant Hellinga-deZeeuw coefficient used.
WriteRestartFileWithAdimC=0
           ! 0=no, -1=yes (default); includes AdimC Sacramento in restart file
VnotchWeirNrInterpolationpoints=14  ! number of interpolation points for Vnotch weir.
VnotchWeirh2h1Ratio=0.8 0.85 0.885 0.9 0.925 0.945 0.95 0.96 0.97 0.978  0.985 0.99 0.995 1.0
           ! Default values for H2H1 ratio Vnotch weir in RR.
           ! Should be given for specified number of interpolation points.
VnotchWeirDrownedFlowReductionfactor= 1.0 0.96 0.9 0.87 0.8 0.7 0.67 0.6 .5 0.4 0.3 0.2 0.1
           ! Default values for drowned weir flow reduction factor
RestartOrderStrict=-1   !-1=true, 0=false. Default =0.
ReduceRROpenWaterInfiltrationAtNegativeVolume=-1 !0=false, -1= true, default value = false

[TimeSettings]
TimestepSize=3600       (computation timestepsize in seconds)
Timeweightfactor=0.5    ! weight factor of t=t0 and t=t + delta t.
TimestepExchange=3600   ! exchange timestepsize in seconds,
           ! should be a multiple of the computation timestepsize
           ! Default = equal to computation timestep
           ! At the moment, only for Modflow coupling a multiple is allowed.
ExchangeOption=Detail   ! Detail, Average or Current;
! \textbf{At the moment only option Detail is impemented}!.
IrrigationStartDay=1    ! (start day of irrigation season,
                        ! only needed if unpaved irrigation is used)
IrrigationStartMonth=4  ! (start month of irrigation season,
                        ! only needed if unpaved irrigation is used)
IrrigationEndDay=1      ! (end day of irrigation season,
                        ! only needed if unpaved irrigation is used)
IrrigationEndMonth=10   ! (end month of irrigation season,
                        ! only needed if unpaved irrigation is used)
IrrigationFromHrs=0     ! (daily irrigation from hour,
                        ! only needed if unpaved irrigation is used)
IrrigationToHrs=24      ! (daily irrigation to hour,
                        ! only needed if unpaved irrigation is used)


NightFromHrs=23      ! night yours from  -  to
NightToHrs=7
WeekHrsasNight=1     ! Weekend hours (Saturday, Sunday) to be considered as night?
          ! 1=yes, 0=no
EvaporationFromHrs=7    ! daily evaporation from - to
EvaporationToHrs=19
PeriodFromEvent=0       ! period derived from rainfall event file. -1=yes (default);0=no.
StartTime=1995/01/01;00:30:00  ! Start date-time of simulation (subset of rainfall event)
EndTime=1995/01/01;01:55:00    ! End date-time of simulation period
```

1.2.5 General layer

Storage coefficients soil types <..\..\fixed\bergcoef>

The fixed input file with soil type names and storage coefficients as a function of the soil type and drainage basis. 12 soil types, 12 values for the drainage basis, followed by the table with 12 lines (1 line per drainage basis) with the storage coefficients for all soil types for that particular value of the drainage basis.

```
*Comment lines start with an * in column 1
*Bergingscoefficient, afh. ontw.diepte en grondsoort
*aantal grondsoorten (=12)
 1  'loamy, humous fine sand (mu  = 0.115 per m)'
 2  'peat (mu = 0.103 per m)'
 3  'heavy clay (mu = 0.089 per m)'
 4  'humous clay and peat (mu = 0.085 per m)'
 5  'light loamy, medium coarse sand (mu = 0.084 per m)'
```

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```
 6  'loamy silt (mu = 0.072 per m)'
 7  'humous clay and peat with silty top layer (mu = 0.069 per m)'
 8  'clay and light clay (mu = 0.062 per m)'
 9  'loamless, medium coarse and coarse sand (mu = 0.060 per m)'
10  'silt (mu = 0.058 per m)'
11  'very light clay (mu = 0.048 per m)'
12  'sand with a silty top layer (mu = 0.044 per m)'
*aantal ontw.diepten (=12) (tov maaiveld)
 -0.1 -0.2  -0.3  -0.4  -0.5  -0.6  -0.7  -0.8  -0.9  -1.0  -1.2 -1.5
*Grond1   2      3      4      5      6      7      8      9     10    11   12
0.0074 0.0077 0.0094 0.0063 0.0046 0.0049 0.0049 0.0048 0.0027 0.0029 0.0025 0.0013
0.0183 0.0153 0.0166 0.0134 0.0124 0.0103 0.0118 0.0092 0.0098 0.0066 0.0052 0.0032
0.0302 0.0226 0.0226 0.0206 0.0220 0.0158 0.0181 0.0132 0.0162 0.0105 0.0079 0.0052
0.0419 0.0305 0.0278 0.0279 0.0323 0.0211 0.0237 0.0171 0.0228 0.0145 0.0107 0.0074
0.0532 0.0387 0.0323 0.0350 0.0427 0.0262 0.0289 0.0206 0.0294 0.0186 0.0134 0.0096
0.0664 0.0467 0.0363 0.0420 0.0528 0.0310 0.0339 0.0240 0.0359 0.0226 0.0160 0.0120
0.0805 0.0545 0.0399 0.0486 0.0625 0.0364 0.0386 0.0271 0.0422 0.0265 0.0185 0.0143
0.0938 0.0621 0.0431 0.0551 0.0715 0.0416 0.0430 0.0300 0.0484 0.0303 0.0210 0.0167
0.1061 0.0702 0.0461 0.0613 0.0801 0.0466 0.0472 0.0328 0.0542 0.0341 0.0235 0.0191
0.1173 0.0784 0.0489 0.0673 0.0880 0.0514 0.0512 0.0355 0.0598 0.0377 0.0258 0.0214
0.1372 0.0939 0.0538 0.0786 0.1024 0.0605 0.0586 0.0404 0.0704 0.0446 0.0303 0.0261
0.1614 0.1158 0.0600 0.0941 0.1208 0.0729 0.0685 0.0470 0.0845 0.0541 0.0366 0.0329
```

Crop factors agricultural crops <**..\..\fixed\cropfact**>

The fixed input file with crop names and crop factors as a function of time.

1 year of data is enough, since it is assumed that the crop factors are constant over the years; the variation is taken into account in the reference evaporation data and not in the crop factors.

The header of the file contains the number of crops and crop names.

```
*Aantal gewassen
     16
*Namen
'grass           '
'corn            '
'potatoes        '
'sugarbeet       '
'grain           '
'miscellaneous       '
'non-arable land         '
'greenhouse area         '
'orchard         '
'bulbous plants      '
'foliage forest      '
'pine forest         '
'nature          '
'fallow          '
'vegetables          '
'flowers         '
*Gemiddelde 1    2    3    4    5    6    7    8    9    10   11   12   13   14   15   16
1996  1  1  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1  2  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1  3  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1  4  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1  5  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1  6  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1  7  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1  8  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1  9  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1 10  0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
1996  1 11  0.35 0.40 0.40 0.40 0.40 0.40 0.40 0.00 0.35 0.40 0.35 0.40 0.35 0.40 0.35 0.00
1996  1 12  0.35 0.40 0.40 0.40 0.40 0.40 0.40 0.00 0.35 0.40 0.35 0.40 0.35 0.40 0.35 0.00
```

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```
1996  1 13  0.35 0.40 0.40 0.40 0.40 0.40 0.40 0.00 0.35 0.40 0.35 0.40 0.35 0.40 0.35 0.00
etc.
```

Capsim soil types <**..\..\fixed\bergcoef.cap**>

File containing the names of the 21 soil types of the Staring reeks used in Capsim.

1 'Veengrond met veraarde bovengrond'
2 'Veengrond met veraarde bovengrond, zand'
3 'Veengrond met kleidek'
4 'Veengrond met kleidek op zand'
5 'Veengrond met zanddek op zand'
6 'Veengrond op ongerijpte klei'
7 'Stuifzand'
8 'Podzol (Leemarm, fijn zand)'
9 'Podzol (zwak lemig, fijn zand)'
10 'Podzol (zwak lemig, fijn zand op grof zand)'
11 'Podzol (lemig keileem)'
12 'Enkeerd (zwak lemig, fijn zand)'
13 'Beekeerd (lemig fijn zand)'
14 'Podzol (grof zand)'
15 'Zavel'
16 'Lichte klei'
17 'Zware klei'
18 'Klei op veen'
19 'Klei op zand'
20 'Klei op grof zand'
21 'Leem'

The list of 21 Capsim soil types in English:

Soil1	Decomposed clayey peat over eutrophic peat: peat soil with decomposed topsoil (hVb, hVc)
Soil2	Decomposed mesotrophic peat over a coarse textured, sandy subsoil: peat soil with decomposed topsoil (aVz, hVz)
Soil3	Humic very fine textured, clay topsoil over eutrophic peat: peat soil with a clay cover (pVb, kVb)
Soil4	Humic very fine textured, clay topsoil over coarse textured, sandy subsoil: peat soil with a clay cover (kVz)
Soil5	Humic, medium textured, sandy topsoil over coarse textured, sandy subsoil: peat soil with sand cover (iWz, iWp)
Soil6	Decomposed clayey peat over unripened clay: peat soil with decomposed topsoil (W0)
Soil7	Eolian, coarse textured sandy soil: sandy soil (Zd20, Zd21)
Soil8	Podzolic, coarse textured sandy soil: sandy soil (Hd21)
Soil9	Podzolic, medium textured sandy soil: sandy soil (Hn21)
Soil10	Podzolic, medium textured sandy soil over coarse textured sand:sandy soil (Hn21g)
Soil11	Podzolic, medium textured sandy soil over boulder clay: sandy soil (Hn23x)
Soil12	Plaggen, coarse textured sandy soil: sandy soil (zEZ21)
Soil13	Humic gleysol, coarse textured sandy soil: sandy soil (pZg23)
Soil14	Podzolic, coarse textured sandy soil: sandy soil (gHd30)
Soil15	Calcareous, medium textured, clay soil: alluvial soil (Mn25A)
Soil16	Medium textured clay soil: alluvial soil (Mn35A, Rd90A, Rd90C)
Soil17	Fine textured clay soil: alluvial soil (Rn44C, gMn83C, kMn48C, Rn47C)
Soil18	Fine textured clay over mesotrophic peat: alluvial soil (RvO1C, Mv41C)
Soil19	Medium textured clay over sand: alluvial soil (Mn22A)
Soil20	Medium textured clay over coarse textured sand: alluvial soil(n52A)
Soil21	Eolian, medium textured loam: lvss soil (BLd6)

Capsim rootzone data <**..\..\fixed\root_sim.inp**>

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This file contains the rootzone data. The lay-out of the file is column-wise, with:

column 1 = soil type
column 2 = crop type
column 3 = root zone thickness (mm)
column 4 = relative root zone storage for h1
column 5 = relative root zone storage for h2
column 6 = relative root zone storage for h3l
column 7 = relative root zone storage for h3h
column 8 = relative root zone storage for h4

The soil water pressure heads h1, h2, h3l, h3h, h4 are used to define evaporation reduction (see technical reference for more explanation).

```
     1      1    0.40   1.000   1.000   0.760   0.669   0.479
     1      2    0.60   1.000   1.000   0.757   0.695   0.479
     1      3    0.60   1.000   1.000   0.760   0.695   0.434
     1      4    0.60   1.000   1.000   0.760   0.695   0.434
     1      5    0.60   1.000   1.000   0.713   0.658   0.434
     1      6    0.60   1.000   1.000   0.713   0.658   0.434
     1      7    0.50   1.000   1.000   0.713   0.658   0.434
     1      8    0.50   1.000   1.000   0.757   0.695   0.479
     1      9    0.60   1.000   1.000   0.760   0.669   0.479
     1     10    0.50   1.000   1.000   0.760   0.669   0.479
     1     11    0.60   1.000   1.000   0.760   0.669   0.479
     1     12    0.60   1.000   1.000   0.760   0.669   0.479
     1     13    0.50   1.000   1.000   0.760   0.669   0.479
     1     14    0.20   1.000   1.000   0.760   0.669   0.479
     1     15    0.40   1.000   1.000   0.760   0.713   0.513
     1     16    0.60   1.000   1.000   0.760   0.713   0.548
     2      1    0.40   1.000   1.000   0.628   0.486   0.249
     2      2    0.50   1.000   1.000   0.673   0.580   0.302
     2      3    0.50   1.000   1.000   0.676   0.580   0.254
     2      4    0.50   1.000   1.000   0.676   0.580   0.254
     2      5    0.50   1.000   1.000   0.607   0.527   0.254
     2      6    0.50   1.000   1.000   0.607   0.527   0.254
     2      7    0.50   1.000   1.000   0.607   0.527   0.254
     2      8    0.50   1.000   1.000   0.673   0.580   0.302
     2      9    0.50   1.000   1.000   0.676   0.542   0.302
     2     10    0.50   1.000   1.000   0.676   0.542   0.302
etc.
```

Capsim unsaturated zone data <**..\..\fixed\unsa_sim.inp**>

This file contains the unsaturated zone data. The lay-out of the file is column-wise, with:

column 1 = soil type
column 2 = rootzone thickness (cm)
column 3 = groundwater level (meter below surface)
column 4 = root zone soil moisture storage in equilibrium conditions (mm)
column 5 = potential capillary rise (mm/day)
column 6 = storage coefficient (m/m)
1 10 0.00 61.6 5.000 0.000
1 10 0.10 61.0 5.000 0.010
1 10 0.20 59.8 5.000 0.018
1 10 0.30 58.7 5.000 0.029
1 10 0.40 57.7 3.546 0.038
1 10 0.50 56.7 2.392 0.049
1 10 0.60 55.9 1.780 0.062
1 10 0.70 55.1 1.397 0.070
1 10 0.80 54.4 1.134 0.081

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1 10 0.90 53.8 0.942 0.091
1 10 1.00 53.2 0.797 0.101
1 10 1.10 52.6 0.684 0.110
1 10 1.20 52.1 0.595 0.119
1 10 1.30 51.6 0.521 0.126
1 10 1.40 51.2 0.461 0.135
1 10 1.50 50.8 0.410 0.142
1 10 1.60 50.4 0.367 0.149
1 10 1.70 50.0 0.330 0.156
1 10 1.80 49.6 0.298 0.162
1 10 1.90 49.3 0.271 0.168
1 10 2.00 49.0 0.246 0.174
1 10 2.10 48.7 0.225 0.180
1 10 2.20 48.4 0.206 0.185
1 10 2.30 48.1 0.190 0.190
1 10 2.40 47.8 0.175 0.195
1 10 2.50 47.6 0.162 0.199
1 10 3.00 46.4 0.112 0.220
1 10 4.00 44.6 0.060 0.251
1 10 5.00 43.2 0.035 0.275
1 10 10.00 39.1 0.002 0.341
1 20 0.00 123.2 5.000 0.000
1 20 0.10 122.6 5.000 0.010
1 20 0.20 120.8 5.000 0.010
1 20 0.30 118.5 5.000 0.018
1 20 0.40 116.4 5.000 0.028
1 20 0.50 114.4 3.916 0.039
1 20 0.60 112.6 2.749 0.052
1 20 0.70 111.0 2.074 0.063 etc.

Greenhouse class file <**..\..\fixed\kasklasse**>

This file contains the definition of the greenhouse classes; the name, the maximum storage (in m3/ha), the maximum depth (m), and en evaporation indicator (0=no evaporation,1=yes)

```
*Kasklassen
*Code  Name  Max\_berging in m3/ha, Max.diepte, wel/geen verdamping uit bassins
1   '<500 m3/ha'      0       0.       0
2   ' 500-1000'     500       0.5      0
3   '1000-1500'    1000       1.       0
4   '1500-2000'    1500       1.5      1
5   '2000-2500'    2000       2.       1
6   '2500-3000'    2500       2.5      1
7   '3000-4000'    3000       3.       1
8   '4000-5000'    4000       4.       1
9   '5000-6000'    5000       5.       1
10 '>6000-m3/ha'   6000       5.       1
```

Greenhouse initialisation file <**..\..\fixed\kasinit**>

The greenhouse initialisation file defines the free space (available storage) at the start of the simulation for each greenhouse class. It contains data from 1951 up to 1999. This data is only used for defining the initial storage in the greenhouse storage basins.

```
*       Grootte bassin  (m3/ha)
*                 0   500  1000 1500 2000 2500 3000 4000 5000 6000
*Jaar Maand Dag   Ruimte voor waterberging (m3)
1951  1     1     0   0    0    0    31   100  215  519  829  1078
1951  1     2     0   0    0    0    8    77   191  495  803  1052
1951  1     3     0   5    5    5    13   82   196  500  808  1057
1951  1     4     0   0    0    0    0    67   181  484  792  1040
```

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```
1951  1     5     0   0    0    0    0    5    118  419  725  971
1951  1     6     0   0    0    0    0    0    108  409  715  960
1951  1     7     0   3    3    3    3    3    112  412  718  964
1951  1     8     0   0    0    0    0    0    85   384  689  934
1951  1     9     0   0    0    0    0    0    63   362  666  910
1951  1     10    0   0    0    0    0    0    32   330  633  876
1951  1     11    0   0    0    0    0    0    0    204  503  743
1951  1     12    0   0    0    0    0    0    0    181  479  718
1951  1     13    0   0    0    0    0    0    0    152  449  687
1951  1     14    0   0    0    0    0    0    0    127  424  661
```

Greenhouse water use file <..\..\fixed\kasgebr>

This file defines the actual water use (m3/ha) from the greenhouse storage basins by the greenhouse crops for each day. Values are depending on year and date, but are assumed independant of the size of the greenhouse storage basins (so independant of the greenhouse class).

```
*         Voor alle kasklassen hetzelfde
*Jaar Maand Dag    EACT
 1951   1     1       8.36
 1951   1     2       7.04
 1951   1     3       8.04
 1951   1     4       7.04
 1951   1     5       7.04
 1951   1     6       7.04
 1951   1     7       7.72
 1951   1     8       7.04
 1951   1     9       9.43
 1951   1    10       7.53
 1951   1    11       7.31
 1951   1    12      10.94
 1951   1    13       7.31
 1951   1    14       7.51
```

Crop factors open water <..\..\fixed\crop_ow>

The fixed input file with open water evaporation factors as a function of time. 1 year of data is enough; it is assumed that the evaporation factors are same for all years.

```
*Oppervlaktewater
     1
*Namen
'0.0 Oppervlaktewater        '
*Gemiddelde       1
    1994     1     1     0.00
    1994     1     2     0.00
    1994     1     3     0.00
    1994     1     4     0.00
    1994     1     5     0.00
    1994     1     6     0.00
    1994     1     7     0.00
    1994     1     8     0.00
    1994     1     9     0.00
    1994     1    10     0.00
    1994     1    11     0.50
    1994     1    12     0.50
    1994     1    13     0.50
etc.
```

Rainfall file (1 or more events)

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This file contains the rainfall data: the number of stations, the names (id's) of the rainfall stations, the number of rainfall events, the rainfall data timestep size (may be different from the computation timestep size), the start date and duration of each event and the rainfall data for each event.

```
*Name of this file: \SOB_LITE\FIXED\DEFAULT.BUI
*Date and time of construction: 17-04-1997   17:13:50
*Enige algemene wenken:
*Gebruik de default dataset voor overige invoer (altijd 1)
1
*Aantal stations
 1
*Namen van stations
'Station1'
*Aantal gebeurtenissen (omdat het 1 bui betreft is dit altijd 1)
*en het aantal seconden per waarnemingstijdstap (10800 = 3x3600)
1  10800
*Elke commentaarregel wordt begonnen met een * (asteriks).
*Eerste record bevat startdatum en -tijd, lengte van de gebeurtenis in dd hh mm ss
*Het format is: yyyymmdd:hhmmss:ddhhmmss
*Daarna voor elk station de neerslag in mm per tijdstap.
 1996 1 1 0 0 0 1 3 0 0
 0.2
 0.2
 0.2
 0.2
 0.2
 0.2
 0.2
 0.2
 0.2
```

Evaporation file

This file contains the evaporation data. It is assumed that meteo stations occur in the same order as in the rainfall file. If the evaporation file contains less stations, missing stations will use data from the first station in the file.

```
*Name of this file: \SOB\_LITE\FIXED\DEFAULT.EVP
*Date and time of construction: 10-03-2000   10:04:25
*Verdampingsfile
*Meteo data: evaporation intensity in mm/day
*First record: start date, data in mm/day
*Datum (year month day), verdamping (mm/dag) voor elk weerstation
*jaar maand dag verdamping[mm]
 1996    1    1    0
 1996    1    2    0
```

Default defition file <..\fixed\3bEdit.def>

This file contains some default values for records in the DRR MDB files for all types of DRR-nodes. The file is used in SOBEK2 by ModelEdit.

Optional: New format greenhouse class file file..\fixed\3B\NewKasKlasData.dat

This file contains the greenhouse class specification in a new format, which allows extensions by the user.

The file contains two type of records: the KASD records and KLAS records. In the KASD records the global definitions are placed, including references to the greenhouse initialisation and greenhouse water use definitions. The KLAS record contains the greenhouse class definition. The file can contain multiple KASD and KLAS records. Selection of the actual records used is done per case in the <Delft_3B.Ini> file (see description of the NewFormatKasData= and KasDefinition= records).

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An example input file:

KASD id 'Default' nm 'Default kas initialisation' nc 10 kk 'Default kasklassen' ki 'Default kas initialisation data' kg 'Default kasgebruik data' kasd

KASD id 'Low' nm 'Low kas initialisation' nc 10 kk 'Default kasklassen' ki 'Low kas initialisation data' kg 'Low kasgebruik data' kasd

KASD id 'Periodic test' nm 'Periodic test' nc 10 kk 'Default kasklassen' ki 'Periodic kas initialisation data' kg 'Periodic kasgebruik data' kasd

KASD id 'My own mix' nm 'My own mix' nc 10 kk 'Default kasklassen' ki 'Low kas initialisation data' kg 'Periodic kasgebruik data' kasd

with

id	greenhouse definition id
nm	greenhouse definition name
nc	number of greenhouse classes
kk	reference to greenhouse class definition
ki	reference to greenhouse initialisation definition
kg	reference to greenhouse water use definition

The KLAS record contains the greenhouse class definition.

KLAS id 'Default kasklassen'

nm '<500 m3/ha' ' 500-1000' '1000-1500' '1500-2000' '2000-2500' '2500-3000' '3000-4000' '4000-5000' '5000-6000' '>6000-m3/ha'

mxstorage 0 500 1000 1500 2000 2500 3000 4000 5000 6000

mxdepth 0. 0.5 1. 1.5 2. 2.5 3. 4. 5. 5.

evap 0 0 0 1 1 1 1 1 1 1

KLAS

with

id	greenhouse class definition id
nm	names of greenhouse classes
mxstorage	maximum storage per greenhouse class (m3/ha)
mxdepth	maximum depth per greenhouse class (m)
evap	0=no evaporation, 1=evaporation from greenhouse class storage basins (per class)

Optional: New format greenhouse initialisation file <..\fixed\3B\NewKasInitData.dat> This file contains the greenhouse initialisation data in a new format, which allows extensions by the user. The file contains INIT records. The INIT record contains a time table of greenhouse initial free storage. The table is a standard SOBEK time table with options for interpolation and periodicity. The file can contain multiple INIT records; the selection of the actual record used is based on the selected greenhouse definition from the greenhouse class file.

An example input file:

INIT id 'Default kas initialisation data' PDIN 1 0 pdin TBLE
'1951/01/01;00:00:00' 0 0 0 0 31 100 215 519 829 1078 <
'1951/01/02;00:00:00' 0 0 0 0 8 77 191 495 803 1052 <
'1951/01/03;00:00:00' 0 5 5 5 13 82 196 500 808 1057 <
'1951/01/04;00:00:00' 0 0 0 0 0 67 181 484 792 1040 <

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'1951/01/05;00:00:00' 0 0 0 0 0 5 118 419 725 971 <
'1951/01/06;00:00:00' 0 0 0 0 0 0 108 409 715 960 <
'1951/01/07;00:00:00' 0 3 3 3 3 3 112 412 718 964 <
'1951/01/08;00:00:00' 0 0 0 0 0 0 85 384 689 934 <
'1951/01/09;00:00:00' 0 0 0 0 0 0 63 362 666 910 <
'1951/01/10;00:00:00' 0 0 0 0 0 0 32 330 633 876 <
'1951/01/11;00:00:00' 0 0 0 0 0 0 0 204 503 743 <
'1951/01/12;00:00:00' 0 0 0 0 0 0 0 181 479 718 <
'1951/01/13;00:00:00' 0 0 0 0 0 0 0 152 449 687 <
'1951/01/14;00:00:00' 0 0 0 0 0 0 0 127 424 661 <
'1951/01/15;00:00:00' 0 0 0 0 0 0 0 86 382 617 <
'1951/01/16;00:00:00' 0 8 8 8 8 8 8 95 390 626 <
'1951/01/17;00:00:00' 0 0 0 0 0 0 0 0 212 442 <
'1951/01/18;00:00:00' 0 0 0 0 0 0 0 0 98 325 <
'1951/01/19;00:00:00' 0 0 0 0 0 0 0 0 60 286 <
'1951/01/20;00:00:00' 0 0 0 0 0 0 0 0 34 259 <
'1951/01/21;00:00:00' 0 0 0 0 0 0 0 0 0 219 <
'1951/01/22;00:00:00' 0 0 0 0 0 0 0 0 0 209 <
...
...
'1995/10/10;00:00:00' 0 125 361 799 1216 1668 2135 2994 3346 3419 <
'1995/10/11;00:00:00' 0 135 370 808 1226 1677 2145 3004 3356 3429 <
'1995/10/12;00:00:00' 0 150 385 823 1241 1693 2161 3020 3372 3446 <
'1995/10/13;00:00:00' 0 159 394 833 1250 1702 2170 3030 3382 3456 <
'1995/10/14;00:00:00' 0 176 412 851 1269 1721 2189 3049 3402 3476 <
'1995/10/15;00:00:00' 0 190 426 865 1283 1735 2204 3064 3417 3492 <
'1995/10/16;00:00:00' 0 203 439 878 1297 1749 2218 3079 3432 3507 <
'1995/10/17;00:00:00' 0 212 448 887 1305 1758 2227 3088 3441 3516 <
'1995/10/18;00:00:00' 0 224 460 899 1318 1770 2239 3100 3454 3530 <
'1995/10/19;00:00:00' 0 237 473 913 1331 1784 2253 3115 3469 3544 <
'1995/10/20;00:00:00' 0 240 477 916 1334 1787 2257 3118 3472 3548 <
'1995/10/21;00:00:00' 0 257 494 933 1352 1805 2274 3136 3491 3567 <
'1995/10/22;00:00:00' 0 274 511 951 1370 1823 2293 3155 3510 3586 <
'1995/10/23;00:00:00' 0 292 529 969 1388 1841 2311 3174 3529 3606 <
'1995/10/24;00:00:00' 0 309 546 987 1406 1860 2330 3193 3549 3626 <
'1995/10/25;00:00:00' 0 324 562 1002 1422 1876 2346 3209 3565 3643 <
'1995/10/26;00:00:00' 0 324 562 1002 1421 1875 2345 3209 3565 3642 <
'1995/10/27;00:00:00' 0 310 547 987 1406 1860 2330 3192 3548 3625 <
'1995/10/28;00:00:00' 0 325 562 1002 1422 1876 2346 3209 3564 3642 <
'1995/10/29;00:00:00' 0 342 579 1019 1439 1893 2363 3226 3583 3660 <
'1995/10/30;00:00:00' 0 357 594 1035 1455 1909 2379 3243 3599 3677 <
'1995/10/31;00:00:00' 0 371 609 1050 1470 1924 2395 3259 3616 3694 <
'1995/11/01;00:00:00' 0 275 511 950 1368 1821 2290 3150 3504 3579 <
'1995/11/02;00:00:00' 0 254 490 928 1346 1798 2267 3127 3480 3554 <
'1995/11/03;00:00:00' 0 262 498 936 1354 1806 2275 3135 3488 3562 <
'1995/11/04;00:00:00' 0 274 510 949 1367 1819 2288 3148 3502 3576 <
'1995/11/05;00:00:00' 0 289 525 964 1382 1834 2303 3164 3517 3592 <
'1995/11/06;00:00:00' 0 299 535 974 1392 1845 2314 3175 3528 3603 <
tble
init
INIT id 'Low kas initialisation data' PDIN 1 0 pdin TBLE
'1951/01/01;00:00:00' 0. 0. 0. 0. 0. 24. 74. 223. 322. 372. <
'1951/01/02;00:00:00' 0. 0. 0. 0. 0. 0. 45. 193. 292. 341. <
'1951/01/03;00:00:00' 0. 0. 0. 0. 0. 0. 44. 192. 291. 340. <
'1951/01/04;00:00:00' 0. 0. 0. 0. 0. 0. 24. 172. 270. 318. <
'1951/01/05;00:00:00' 0. 0. 0. 0. 0. 0. 0. 101. 197. 243. <
'1951/01/06;00:00:00' 0. 0. 0. 0. 0. 0. 0. 86. 182. 227. <
'1951/01/07;00:00:00' 0. 0. 0. 0. 0. 0. 0. 84. 179. 225. <
'1951/01/08;00:00:00' 0. 0. 0. 0. 0. 0. 0. 50. 145. 190. <
'1951/01/09;00:00:00' 0. 0. 0. 0. 0. 0. 0. 21. 115. 159. <
'1951/01/10;00:00:00' 0. 0. 0. 0. 0. 0. 0. 0. 77. 120. <

PROOF

'1951/01/11;00:00:00' 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. <
'1951/01/12;00:00:00' 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. <
'1951/01/13;00:00:00' 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. <
...
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'1995/10/10;00:00:00' 0. 72. 73. 408. 842. 1334. 1629. 2348. 2772. 2893. <
'1995/10/11;00:00:00' 0. 76. 78. 412. 847. 1339. 1634. 2353. 2778. 2898. <
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'1995/10/14;00:00:00' 0. 96. 98. 433. 868. 1360. 1656. 2376. 2802. 2923. <
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'1995/10/16;00:00:00' 0. 109. 111. 446. 882. 1375. 1670. 2391. 2817. 2940. <
'1995/10/17;00:00:00' 0. 111. 113. 448. 884. 1377. 1673. 2394. 2820. 2943. <
'1995/10/18;00:00:00' 0. 116. 119. 454. 890. 1383. 1679. 2400. 2827. 2950. <
'1995/10/19;00:00:00' 0. 122. 125. 461. 897. 1390. 1686. 2407. 2834. 2957. <
'1995/10/20;00:00:00' 0. 122. 124. 460. 896. 1389. 1685. 2406. 2834. 2957. <
'1995/10/21;00:00:00' 0. 126. 129. 464. 900. 1394. 1690. 2412. 2839. 2963. <
'1995/10/22;00:00:00' 0. 130. 133. 469. 905. 1399. 1695. 2418. 2845. 2969. <
'1995/10/23;00:00:00' 0. 134. 138. 474. 910. 1404. 1701. 2423. 2852. 2976. <
'1995/10/24;00:00:00' 0. 139. 143. 479. 916. 1410. 1707. 2430. 2859. 2983. <
'1995/10/25;00:00:00' 0. 143. 147. 483. 920. 1415. 1712. 2435. 2864. 2989. <
'1995/10/26;00:00:00' 0. 137. 140. 477. 914. 1408. 1705. 2428. 2857. 2982. <
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'1995/10/28;00:00:00' 0. 120. 123. 459. 896. 1390. 1687. 2410. 2839. 2964. <
'1995/10/29;00:00:00' 0. 124. 127. 464. 901. 1395. 1692. 2416. 2845. 2970. <
'1995/10/30;00:00:00' 0. 127. 131. 468. 905. 1399. 1697. 2420. 2850. 2975. <
'1995/10/31;00:00:00' 0. 131. 135. 472. 909. 1404. 1701. 2425. 2855. 2981. <
'1995/11/01;00:00:00' 0. 29. 32. 367. 802. 1295. 1591. 2312. 2739. 2861. <
'1995/11/02;00:00:00' 0. 0. 2. 337. 772. 1265. 1560. 2280. 2706. 2827. <
'1995/11/03;00:00:00' 0. 0. 1. 336. 772. 1264. 1560. 2280. 2706. 2827. <
'1995/11/04;00:00:00' 0. 1. 3. 338. 773. 1266. 1562. 2282. 2708. 2830. <
'1995/11/05;00:00:00' 0. 3. 5. 340. 776. 1269. 1564. 2285. 2711. 2834. <
'1995/11/06;00:00:00' 0. 4. 6. 341. 777. 1270. 1566. 2287. 2713. 2836. <
tble
init
INIT id 'Periodic kas initialisation data' PDIN 1 1 '31536000' pdin TBLE
'1994/01/01;00:00:00' 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. <
'1994/02/01;00:00:00' 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. <
'1994/03/01;00:00:00' 0. 10. 10. 10. 10. 10. 10. 10. 10. 10. <
'1994/04/01;00:00:00' 0. 20. 20. 20. 20. 20. 20. 20. 20. 20. <
'1994/05/01;00:00:00' 0. 30. 30. 30. 30. 30. 30. 30. 30. 30. <
'1994/06/01;00:00:00' 0. 40. 40. 40. 40. 40. 40. 40. 40. 40. <
'1994/07/01;00:00:00' 0. 50. 50. 50. 50. 50. 50. 50. 50. 50. <
'1994/08/01;00:00:00' 0. 50. 50. 50. 50. 50. 50. 50. 50. 50. <
'1994/09/01;00:00:00' 0. 40. 40. 40. 40. 40. 40. 40. 40. 40. <
'1994/10/01;00:00:00' 0. 30. 30. 30. 30. 30. 30. 30. 30. 30. <
'1994/11/01;00:00:00' 0. 20. 20. 20. 20. 20. 20. 20. 20. 20. <
'1994/12/01;00:00:00' 0. 10. 10. 10. 10. 10. 10. 10. 10. 10. <
'1994/12/31;23:59:00' 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. <
tble
init

Optional: New format greenhouse water use file <..\fixed\3B\NewKasGebrData.dat> This file contains the greenhouse water use data in a new format, which allows extensions by the user. The file can contain several GEBR records, in which typical water uses of greenhouses are specified in a standard SOBEK time table. Selection of the GEBR record to be used is done by the selection of the greenhouse definition from the greenhouse class file. An example input file is given below.

GEBR id 'Default kasgebruik data' PDIN 1 0 pdin TBLE
'1951/01/01;00:00:00' 8.36 <

'1951/01/02;00:00:00' 7.04 <
'1951/01/03;00:00:00' 8.04 <
'1951/01/04;00:00:00' 7.04 <
'1951/01/05;00:00:00' 7.04 <
'1951/01/06;00:00:00' 7.04 <
'1951/01/07;00:00:00' 7.72 <
'1951/01/08;00:00:00' 7.04 <
'1951/01/09;00:00:00' 9.43 <
'1951/01/10;00:00:00' 7.53 <
'1951/01/11;00:00:00' 7.31 <
'1951/01/12;00:00:00' 10.94 <
'1951/01/13;00:00:00' 7.31 <
'1951/01/14;00:00:00' 7.51 <
'1951/01/15;00:00:00' 9.81 <
'1951/01/16;00:00:00' 10.61 <
'1951/01/17;00:00:00' 7.31 <
'1951/01/18;00:00:00' 9.17 <
'1951/01/19;00:00:00' 7.31 <
'1951/01/20;00:00:00' 7.31 <
...
...
'1995/11/10;00:00:00' 9.88 <
'1995/11/11;00:00:00' 12.61 <
'1995/11/12;00:00:00' 12.88 <
'1995/11/13;00:00:00' 12.18 <
'1995/11/14;00:00:00' 9.65 <
'1995/11/15;00:00:00' 7.41 <
'1995/11/16;00:00:00' 7.81 <
'1995/11/17;00:00:00' 6.70 <
'1995/11/18;00:00:00' 8.21 <
'1995/11/19;00:00:00' 7.15 <
'1995/11/20;00:00:00' 13.20 <
tble
gebr
GEBR id 'Low kasgebruik data' PDIN 1 0 pdin TBLE
'1951/01/01;00:00:00' 2.17 <
'1951/01/02;00:00:00' 1.84 <
'1951/01/03;00:00:00' 2.09 <
'1951/01/04;00:00:00' 1.84 <
'1951/01/05;00:00:00' 1.84 <
'1951/01/06;00:00:00' 1.84 <
'1951/01/07;00:00:00' 2.01 <
'1951/01/08;00:00:00' 1.84 <
'1951/01/09;00:00:00' 2.45 <
'1951/01/10;00:00:00' 1.96 <
'1951/01/11;00:00:00' 2.67 <
'1951/01/12;00:00:00' 3.97 <
...
...
'1995/12/07;00:00:00' 2.11 <
'1995/12/08;00:00:00' 2.86 <
'1995/12/09;00:00:00' 2.24 <
'1995/12/10;00:00:00' 1.82 <
'1995/12/11;00:00:00' 1.65 <
'1995/12/12;00:00:00' 1.94 <
'1995/12/13;00:00:00' 1.73 <
'1995/12/14;00:00:00' 2.23 <
'1995/12/15;00:00:00' 1.93 <
'1995/12/16;00:00:00' 2.38 <
'1995/12/17;00:00:00' 2.11 <
'1995/12/18;00:00:00' 1.56 <

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'1995/12/19;00:00:00' 1.48 <
'1995/12/20;00:00:00' 2.18 <
'1995/12/21;00:00:00' 1.69 <
'1995/12/22;00:00:00' 1.43 <
'1995/12/23;00:00:00' 1.60 <
'1995/12/24;00:00:00' 2.75 <
'1995/12/25;00:00:00' 2.42 <
'1995/12/26;00:00:00' 1.83 <
'1995/12/27;00:00:00' 3.30 <
'1995/12/28;00:00:00' 3.79 <
'1995/12/29;00:00:00' 3.16 <
'1995/12/30;00:00:00' 2.56 <
'1995/12/31;00:00:00' 1.61 <
tble
gebr
GEBR id 'Periodic kasgebruik data' PDIN 1 1 '31536000' pdin TBLE
'1995/01/01;00:00:00' 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 <
'1995/02/01;00:00:00' 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 <
'1995/03/01;00:00:00' 5.00 5.00 5.00 5.00 5.00 5.00 5.00 5.00 5.00 5.00 <
'1995/05/01;00:00:00' 6.00 6.00 6.00 6.00 6.00 6.00 6.00 6.00 6.00 6.00 <
'1995/07/01;00:00:00' 7.00 7.00 7.00 7.00 7.00 7.00 7.00 7.00 7.00 7.00 <
'1995/09/01;00:00:00' 6.00 6.00 6.00 6.00 6.00 6.00 6.00 6.00 6.00 6.00 <
'1995/10/01;00:00:00' 5.00 5.00 5.00 5.00 5.00 5.00 5.00 5.00 5.00 5.00 <
'1995/11/15;00:00:00' 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 <
'1995/12/31;23:59:00' 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 <
tble
gebr

Optional: New format crop data <**..\fixed\3B\NewCropData.dat**> This file contains the crop data (names and crop factors) in a new format, which allows extensions by the user. The file contains 3 type of records:

- ◇ **CRPD records,** containing the general crop definitions (id, name, number of crops, and references to the crop name definition and crop factor definition)
- ◇ **NAME records,** containing the crop name definitions.
- ◇ **CRF records**, containing a time table with crop factors**.**

An example file is given below,

with

id	id
nm	name
nc	number of crops
cn	crop name definition
cf	crop factor definition

PDIN pdin = as in other SOBEK time tables: interpolation and periodicity options

TBLE tble = as in other SOBEK time tables

BBB2.2

CRPD id 'Default' nm 'Nederlandse gewasdefinitie' nc 16 cn 'Dutch Crops' cf 'Dutch Crop factors' crpd
CRPD id 'Example' nm 'Example crop definition' nc 1 cn 'Example Crops' cf 'Example Crop factors' crpd
CRPD id 'Taiwan' nm 'Taiwan crop definition' nc 7 cn 'Taiwan Crops' cf 'Taiwan Crop factors' crpd
NAME id 'Dutch Crops' nm
'grass '
'corn '

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'potatoes '
'sugarbeet'
'grain '
'miscellaneous '
'non-arable land'
'greenhouse area'
'orchard '
'bulbous plants'
'foliage forest'
'pine forest '
'nature '
'fallow '
'vegetables '
'flowers '
name
NAME id 'Taiwan Crops' nm 'paddy1' 'paddy2' 'fishpond' 'upland1' 'upland2' 'upland3' 'fallow' name
NAME id 'Example Crops' nm 'all crops' name
CRF id 'Dutch Crop factors' nm 'Dutch Crop factors' PDIN 1 1 '31536000' pdin TBLE
'1994/01/01;00:00:00' 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.0
<
'1994/01/11;00:00:00' 0.35 0.40 0.40 0.40 0.40 0.40 0.40 0.00 0.35 0.40 0.35 0.40 0.35 0.40 0.35 0.0
<
'1994/01/21;00:00:00' 0.49 0.56 0.56 0.56 0.56 0.56 0.56 0.00 0.49 0.56 0.49 0.56 0.49 0.56 0.49 0.0
<
'1994/02/01;00:00:00' 0.56 0.64 0.64 0.64 0.64 0.64 0.64 0.00 0.56 0.64 0.56 0.64 0.56 0.64 0.56 0.0
<
'1994/02/11;00:00:00' 0.70 0.80 0.80 0.80 0.80 0.80 0.80 0.00 0.70 0.80 0.70 0.80 0.70 0.80 0.70 0.0
<
'1994/02/21;00:00:00' 0.70 0.70 0.70 0.70 0.70 0.80 0.70 0.00 0.70 0.70 0.70 0.80 0.70 0.80 0.70 0.0
<
'1994/03/01;00:00:00' 0.84 0.72 0.72 0.72 0.72 0.96 0.72 0.00 0.84 0.48 0.84 0.96 0.84 0.96 0.84 0.0
<
'1994/03/11;00:00:00' 0.91 0.65 0.65 0.65 0.65 1.04 0.65 0.00 0.91 0.65 0.91 1.04 0.91 1.04 0.91 0.0
<
'1994/03/21;00:00:00' 0.91 0.52 0.52 0.52 0.52 1.04 0.52 0.00 0.91 0.78 0.91 1.04 0.91 1.04 0.91 0.0
<
'1994/04/01;00:00:00' 1.04 0.52 0.52 0.52 0.52 1.04 0.52 0.00 1.04 0.91 1.04 1.04 1.04 1.04 1.04 0.0
<
'1994/04/11;00:00:00' 1.04 0.39 0.39 0.39 0.65 1.04 0.39 0.00 1.04 0.91 1.04 1.04 1.04 1.04 1.04 0.0
<
'1994/04/21;00:00:00' 1.04 0.26 0.26 0.26 0.78 1.04 0.26 0.00 1.04 0.91 1.04 1.04 1.04 1.04 1.04 0.0
<
'1994/05/01;00:00:00' 1.04 0.52 0.52 0.52 0.91 1.04 0.13 0.00 1.04 1.04 1.04 1.04 1.04 1.04 1.04 0.0
<
'1994/05/11;00:00:00' 1.04 0.65 0.65 0.65 1.04 1.04 0.13 0.00 1.04 1.04 1.04 1.04 1.04 1.04 1.04 0.0
<
'1994/05/21;00:00:00' 1.04 0.78 0.78 0.78 1.17 1.04 0.13 0.00 1.04 1.04 1.04 1.04 1.04 1.04 1.04 0.0
<
'1994/06/01;00:00:00' 1.05 0.92 0.92 0.92 1.31 1.05 0.13 0.00 1.05 1.05 1.05 1.05 1.05 1.05 1.05 0.0
<
'1994/06/11;00:00:00' 1.05 1.05 1.05 1.05 1.31 1.05 0.13 0.00 1.05 1.05 1.05 1.05 1.05 1.05 1.05 0.0
<
'1994/06/21;00:00:00' 1.05 1.18 1.18 1.18 1.18 1.05 0.13 0.00 1.05 0.92 1.05 1.05 1.05 1.05 1.05 0.0
<
'1994/07/01;00:00:00' 1.03 1.16 1.16 1.16 1.16 1.03 0.13 0.00 1.03 0.77 1.03 1.03 1.03 1.03 1.03 0.0
<
'1994/07/11;00:00:00' 1.02 1.14 1.14 1.14 1.14 1.02 0.13 0.00 1.02 0.64 1.02 1.02 1.02 1.02 1.02 0.0
<
'1994/07/21;00:00:00' 0.99 1.12 1.12 1.12 0.99 0.99 0.12 0.00 0.99 0.50 0.99 0.99 0.99 0.99 0.99 0.0
<
'1994/08/01;00:00:00' 0.97 1.09 1.09 1.09 0.73 0.97 0.12 0.00 0.97 0.12 0.97 0.97 0.97 0.97 0.97 0.0

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<
'1994/08/11;00:00:00' 0.95 1.07 0.95 1.07 0.36 0.95 0.12 0.00 0.95 0.12 0.95 0.95 0.95 0.95 0.95 0.0
<
'1994/08/21;00:00:00' 0.94 1.06 0.83 1.06 0.24 0.94 0.12 0.00 0.94 0.12 0.94 0.94 0.94 0.94 0.94 0.0
<
'1994/09/01;00:00:00' 0.94 1.05 0.70 1.05 0.23 0.94 0.23 0.00 0.94 0.23 0.94 0.94 0.94 0.94 0.94 0.0
<
'1994/09/11;00:00:00' 0.94 1.05 0.59 1.05 0.35 0.94 0.35 0.00 0.94 0.35 0.94 0.94 0.94 0.94 0.94 0.0
<
'1994/09/21;00:00:00' 0.94 1.05 0.47 1.05 0.47 0.94 0.47 0.00 0.94 0.47 0.94 0.94 0.94 0.94 0.94 0.0
<
'1994/10/01;00:00:00' 0.70 0.40 0.40 0.40 0.40 0.80 0.40 0.00 0.70 0.40 0.70 0.80 0.70 0.80 0.70 0.0
<
'1994/10/11;00:00:00' 0.63 0.45 0.45 0.45 0.45 0.72 0.45 0.00 0.63 0.45 0.63 0.72 0.63 0.72 0.63 0.0
<
'1994/10/21;00:00:00' 0.56 0.48 0.48 0.48 0.48 0.64 0.48 0.00 0.56 0.48 0.56 0.64 0.56 0.64 0.56 0.0
<
'1994/11/01;00:00:00' 0.56 0.56 0.56 0.56 0.56 0.64 0.56 0.00 0.56 0.56 0.56 0.64 0.56 0.64 0.56 0.0
<
'1994/11/11;00:00:00' 0.49 0.56 0.56 0.56 0.56 0.56 0.56 0.00 0.49 0.56 0.49 0.56 0.49 0.56 0.49 0.0
<
'1994/11/21;00:00:00' 0.42 0.48 0.48 0.48 0.48 0.48 0.48 0.00 0.42 0.48 0.42 0.48 0.42 0.48 0.42 0.0
<
'1994/12/01;00:00:00' 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.0
<
'1994/12/31;23:59:00' 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.0
<
tble
crf
CRF id 'Taiwan Crop factors' nm 'Taiwan Crop factors' PDIN 1 1 '31536000' pdin TBLE
'1994/01/01;00:00:00' 0.60 0.00 1.20 0.20 0.00 0.80 0.40 <
'1994/01/11;00:00:00' 0.80 0.60 1.20 0.20 0.40 1.00 0.40 <
'1994/01/21;00:00:00' 1.00 0.80 1.20 0.20 0.40 1.10 0.40 <
'1994/02/01;00:00:00' 1.10 1.00 1.20 0.50 0.60 0.00 0.40 <
'1994/02/11;00:00:00' 1.00 1.10 1.20 1.10 0.80 0.00 0.40 <
'1994/02/21;00:00:00' 1.10 1.10 1.20 1.10 1.00 0.00 0.40 <
'1994/03/01;00:00:00' 1.15 1.10 1.20 1.10 1.10 0.00 0.40 <
'1994/03/11;00:00:00' 1.15 1.15 1.20 0.90 1.10 0.00 0.40 <
'1994/03/21;00:00:00' 1.15 1.15 1.20 0.90 0.80 0.00 0.40 <
'1994/04/01;00:00:00' 1.00 1.00 1.20 0.70 0.80 0.00 0.40 <
'1994/04/11;00:00:00' 1.00 1.00 1.20 0.60 0.60 0.00 0.40 <
'1994/04/21;00:00:00' 1.00 0.80 1.20 0.60 0.60 0.00 0.40 <
'1994/05/01;00:00:00' 1.00 0.80 1.20 0.40 0.60 0.00 0.40 <
'1994/05/11;00:00:00' 0.80 0.60 1.20 0.40 0.40 0.00 0.40 <
'1994/05/21;00:00:00' 0.80 .04 1.20 0.40 0.00 0.00 0.40 <
'1994/06/01;00:00:00' 0.80 .05 1.20 0.40 0.00 0.00 0.40 <
'1994/06/11;00:00:00' 0.80 .00 1.20 0.40 0.00 0.00 0.40 <
'1994/06/21;00:00:00' 0.80 .00 1.20 0.40 0.00 0.00 0.40 <
'1994/07/01;00:00:00' .00 .00 1.20 0.00 0.00 0.00 0.40 <
'1994/12/31;23:59:00' .00 .00 1.20 0.00 0.00 0.60 0.40 <
tble
crf
CRF id 'Example Crop factors' nm 'Example Crop factors' PDIN 1 1 '31536000' pdin TBLE
'1994/01/01;00:00:00' 0.90 <
'1994/06/11;00:00:00' 0.90 <
'1994/12/31;23:59:00' 0.90 <
tble
crf

Optional: New format open water crop factor data <..\fixed\3B\NewCropOWData.dat> This file contains the open water crop factors in a new format, which allows extensions by the user. This file

is similar to the new format crop data file, but somewhat simpler because there is only 1 ‘crop’, i.e. open water. There are two type of records, the CROW and the CRFO records. The file can contain several CROW and CRFO records. Selection of the actual records used is done by using the options NewFormatCropFactors=-1 and OpenWaterCropDefinition= in the Delft_3B.Ini file. An example input file:

CROW ’Default’ nm ’Nederlandse openwaterverdamping’ nc 1 cn ’Open water Netherlands’ cf ’Dutch Open water crop factors’ crow
CROW id ’Example’ nm ’Example open water evaporation’ nc 1 cn ’Open water Example’ cf ’Example Open water crop factors’ crow
CROW id ’Taiwan’ nm ’Taiwan open water evaporation’ nc 1 cn ’Open water Taiwan’ cf ’Taiwan Open water crop factors’ crow
CRFO id ’Dutch Open water crop factors’ nm ’Dutch Open water Crop factors’ PDIN 1 1 ’31536000’ pdin TBLE
’1994/01/01;00:00:00’ 0.00 <
’1994/01/11;00:00:00’ 0.50 <
’1994/01/21;00:00:00’ 0.70 <
’1994/02/01;00:00:00’ 0.80 <
’1994/02/11;00:00:00’ 1.00 <
’1994/02/21;00:00:00’ 1.00 <
’1994/03/01;00:00:00’ 1.20 <
’1994/03/11;00:00:00’ 1.30 <
’1994/05/21;00:00:00’ 1.30 <
’1994/06/01;00:00:00’ 1.31 <
’1994/06/21;00:00:00’ 1.31 <
’1994/07/01;00:00:00’ 1.29 <
’1994/07/11;00:00:00’ 1.27 <
’1994/07/21;00:00:00’ 1.24 <
’1994/08/01;00:00:00’ 1.21 <
’1994/08/11;00:00:00’ 1.19 <
’1994/08/21;00:00:00’ 1.18 <
’1994/09/01;00:00:00’ 1.17 <
’1994/09/21;00:00:00’ 1.17 <
’1994/10/01;00:00:00’ 1.00 <
’1994/10/11;00:00:00’ 0.90 <
’1994/10/21;00:00:00’ 0.80 <
’1994/11/01;00:00:00’ 0.80 <
’1994/11/11;00:00:00’ 0.70 <
’1994/11/21;00:00:00’ 0.60 <
’1994/12/01;00:00:00’ 0.00 <
’1994/12/31;23:59:00’ 0.00 <
tble
crfo
CRFO id ’Taiwan Open water crop factors’ nm ’Taiwan Open water Crop factors’ PDIN 1 1 ’31536000’ pdin TBLE
’1994/01/01;00:00:00’ 1.00 <
’1994/02/01;00:00:00’ 1.05 <
’1994/03/01;00:00:00’ 1.10 <
’1994/04/01;00:00:00’ 1.15 <
’1994/05/01;00:00:00’ 1.10 <
’1994/06/01;00:00:00’ 1.05 <
’1994/07/01;00:00:00’ 1.00 <
’1994/08/01;00:00:00’ 0.95 <
’1994/09/01;00:00:00’ 0.95 <
’1994/10/01;00:00:00’ 1.00 <
’1994/12/31;23:59:00’ 1.00 <
tble
crfo
CRFO id ’Example Open water crop factors’ nm ’Example Open water Crop factors’ PDIN 1 1 ’31536000’ pdin TBLE
’1994/01/01;00:00:00’ 1.2 <

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'1994/12/31;23:59:00' 1.2 <
tble
crfo

Optional: New format soil data <..\fixed\3B\NewSoilData.dat>

This file contains the soil data in a new format, which allows extensions by the user. The file contains several type of records, and is used to specify data on the soil types (both with or without Capsim).

- ◇ **SLDF records:** containing general definitions and references to other records.
- ◇ **NAME records**, containing the names of the soil types (both with or without Capsim) (the names are actually only relevant for the user interface)
- ◇ **SDEF records**, containing the drainage depths (for use without Capsim)
- ◇ **STAB records**, containing the tables with storage coefficients for use without Capsim; values are specified for the number of specified soil types (from the SLDF record) and the number of drainage depths (from the STAB record)
- ◇ **SCNV records**, containing conversion tables from soil types to use without Capsim to Capsim soil types.

An example is given below.

SLDF id 'Default' nm 'Dutch default soils no Capsim' ns 12 sn 'Default soil names' nd 12 sd 'Default soil depths' st 'Default table soil storage coefficients' nsc 21 snc 'Capsim soil names' cv 'Default conversion' sldf

SLDF id 'BergcoefARS4633' nm 'Dutch default soils no Capsim' ns 12 sn 'Default soil names' nd 12 sd 'Default soil depths' st 'Soil storage coefficients ARS4633' nsc 21 snc 'Capsim soil names' cv 'Default conversion' sldf

with

id	id of soil definition
nm	name of soil definition
ns	number of soil types, no Capsim
nsc	aantal bodemtypen, Capsim
sn	soil names definition, no Capsim
snc	soil names definition Capsim
nd	number of drainage depths in table SDEF record
st	soil storage coefficient table for ns soil types and sd drainage depths (no Capsim)
cv	id conversion tables

NAME id 'Default soil names' nm 'loamy, humous fine sand (μ=0.115 per m)'
'peat (μ=0.103 per m)'
'heavy clay (μ=0.089 per m)'
'humous clay and peat (μ=0.085 per m)'
'light loamy, medium coarse sand (μ=0.084 per m)'
'loamy silt (μ=0.072 per m)'
'humous clay and peat with silty top layer (μ=0.069 per m)'
'clay and light clay (μ=0.062 per m)'
'loamless, medium coarse and coarse sand (μ=0.060 per m)'
'silt (μ=0.058 per m)'
'very light clay (μ=0.048 per m)'
'sand with a silty top layer (μ=0.044 per m)'
name

NAME id 'Capsim soil names' nm
'Veengrond met veraarde bovengrond'
'Veengrond met veraarde bovengrond, zand'
'Veengrond met kleidek'
'Veengrond met kleidek op zand'

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'Veengrond met zanddek op zand'
'Veengrond op ongerijpte klei'
'Stuifzand'
'Podzol (Leemarm, fijn zand)'
'Podzol (zwak lemig, fijn zand)'
'Podzol (zwak lemig, fijn zand op grof zand)'
'Podzol (lemig keileem)'
'Enkeerd (zwak lemig, fijn zand)'
'Beekeerd (lemig fijn zand)'
'Podzol (grof zand)'
'Zavel'
'Lichte klei'
'Zware klei'
'Klei op veen'
'Klei op zand'
'Klei op grof zand'
'Leem'
name

with

id	name definition id
nm	crop names for all crops (the number of names should match with the number specified in the SLDF record for this name definition id)

SDEF id 'Default soil depths' dp -0.1 -0.2 -0.3 -0.4 -0.5 -0.6 -0.7 -0.8 -0.9 -1.0 -1.2 -1.5 SDEF

with

id	soil drainage depth definition id
dp	drainage depths (corresponding with nd drainage depths as specified in the SLDF record)

STAB id 'Default table soil storage coefficients' nm 'Default ns*nd storage coefficient values' sc
0.0050 0.0080 0.0230 0.0070 0.0030 0.0070 0.0060 0.0130 0.0020 0.0050 0.0070 0.0030
0.0145 0.0195 0.0385 0.0175 0.0100 0.0155 0.0150 0.0235 0.0085 0.0125 0.0140 0.0080
0.0243 0.0313 0.0497 0.0273 0.0190 0.0243 0.0243 0.0310 0.0157 0.0200 0.0200 0.0130
0.0342 0.0433 0.0585 0.0370 0.0290 0.0325 0.0338 0.0375 0.0225 0.0268 0.0252 0.0178
0.0486 0.0544 0.0654 0.0462 0.0390 0.0400 0.0414 0.0430 0.0292 0.0330 0.0300 0.0226
0.0635 0.0652 0.0713 0.0548 0.0490 0.0472 0.0480 0.0477 0.0360 0.0390 0.0342 0.0273
0.0777 0.0754 0.0766 0.0630 0.0580 0.0540 0.0540 0.0519 0.0424 0.0444 0.0380 0.0317
0.0910 0.0851 0.0810 0.0709 0.0678 0.0604 0.0596 0.0556 0.0486 0.0495 0.0415 0.0360
0.1034 0.0944 0.0850 0.0783 0.0763 0.0663 0.0649 0.0590 0.0547 0.0543 0.0447 0.0401
0.1148 0.1031 0.0887 0.0852 0.0845 0.0719 0.0697 0.0621 0.0605 0.0588 0.0476 0.0440
0.1350 0.1193 0.0950 0.0983 0.0992 0.0820 0.0786 0.0676 0.0713 0.0670 0.0529 0.0513
0.1595 0.1409 0.1027 0.1155 0.1182 0.0951 0.0902 0.0745 0.0858 0.0777 0.0597 0.0611 stab

STAB id 'Soil storage coefficients ARS4633' nm 'Default ns*nd storage coefficient values' sc
0.0074 0.0077 0.0094 0.0063 0.0046 0.0049 0.0049 0.0048 0.0027 0.0029 0.0025 0.0013
0.0183 0.0153 0.0166 0.0134 0.0124 0.0103 0.0118 0.0092 0.0098 0.0066 0.0052 0.0032
0.0302 0.0226 0.0226 0.0206 0.0220 0.0158 0.0181 0.0132 0.0162 0.0105 0.0079 0.0052
0.0419 0.0305 0.0278 0.0279 0.0323 0.0211 0.0237 0.0171 0.0228 0.0145 0.0107 0.0074
0.0532 0.0387 0.0323 0.0350 0.0427 0.0262 0.0289 0.0206 0.0294 0.0186 0.0134 0.0096
0.0664 0.0467 0.0363 0.0420 0.0528 0.0310 0.0339 0.0240 0.0359 0.0226 0.0160 0.0120
0.0805 0.0545 0.0399 0.0486 0.0625 0.0364 0.0386 0.0271 0.0422 0.0265 0.0185 0.0143
0.0938 0.0621 0.0431 0.0551 0.0715 0.0416 0.0430 0.0300 0.0484 0.0303 0.0210 0.0167
0.1061 0.0702 0.0461 0.0613 0.0801 0.0466 0.0472 0.0328 0.0542 0.0341 0.0235 0.0191
0.1173 0.0784 0.0489 0.0673 0.0880 0.0514 0.0512 0.0355 0.0598 0.0377 0.0258 0.0214
0.1372 0.0939 0.0538 0.0786 0.1024 0.0605 0.0586 0.0404 0.0704 0.0446 0.0303 0.0261
0.1614 0.1158 0.0600 0.0941 0.1208 0.0729 0.0685 0.0470 0.0845 0.0541 0.0366 0.0329 stab

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with

id	soil storage coefficient table id
nm	name
sc	storage coefficient vales (for nd drainage depths and ns soil types)

SCNV id 'Default conversion' nocap 'Default soil names' cap 'Capsim soil names'

c1 10 1 17 18 9 11 6 16 8 21 15 12 c2 2 2 7 7 7 7 9 9 1 1 1 12 1 5 11 8 3 4 12 12 6 scnv

with

id	soil conversion tabel id
nocap	soil names definitie zonder Capsim
cap	soil names definitie met Capsim
c1	conversion table of soil type (no Capsim) to soil type Capsim
c2	conversion table of soil type Capsim to soil type (no Capsim)

1.2.6 Greenhouse layer

Data file: Greenhse.3b

This file contains the data for the nodes of model type 3 (greenhouse or glasshouse area)

GRHS id '1' na 10 ar 1000. 0. 0. 3000. 0. 0. 0. 0. 0. 0. sl 1.0 as 0. sd 'roofstor_1mm' si 'silo_typ1' ms 'meteostat1' is 50.0 grhs

With

id	node identification
na	number or areas (default=10)
ar	area (in m2) as a table with areas for all greenhouse classes (na values)
as	greenhouse area connected to silo storage [m^2]
sl	surface level in m NAP
sd	storage definition on roofs
si	silo definition
ms	identification of the meteostation
is	initial salt concentration ([mg/l])

Roof storage file: Greenhse.rf

This file contains the roof storage definitions for the nodes of model type 3 (greenhouse area)

STDF id 'roofstor_1mm' nm 'roof_1mm' mk 1. ik 0. stdf

Where:

id	storage identification
nm	name (optional)
mk	maximum storage on roofs (in mm). Default 1 mm. Initial value is zero by default.
ik	initial storage on roofs (in mm). Default 0. **NOT READ. Default value zero used.**

Silo definition file: Greenhse.sil

This file contains the silo definitions for the nodes of model type 3 (greenhouse or glasshouse area)

SILO id 'silo_typ1' nm 'silo_200m3' sc 200.0 pc 0.2 silo

With:

id	silo identification

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nm	name (optional)
sc	silo capacity (m3/ha). Default 0.
pc	silo pump capacity (m3/s). Default 0.

1.2.7 Industry layer

dat=industry.dat

This file contains the data for the nodes of model type 15: industrial demands and/or discharges.

INDU id '1' dm 'Qdemand' ds 'Qdis' rf 90. sc. 100. co 1 indu

With

id	node identification
dm	name of demand table (refer to tbl file)
ds	name of discharge table (refer to tbl file)
rf	return flow percentage
sc	salt concentration of discharge
co	computation option: discharge as a table, of a percentage of allocation 1 = discharge and salt concentration as a table (ds field) 2 = fixed percentage and fixed salt concentration (rf and sc fields)

Optional fields:

redow	reduction level open water specified as absolute level (0) or relative to target level (1)
redlv	level at open water node below which industrial abstraction is reduced
redbnlv	level at boundary node below which industrial abstraction is reduced
redrf	return flow reduction option (0=same ratio as demand reduction; otherwise no reduction of return flow)

tab=industry.tbl

Contains demand flow and discharge flow and salt concentration tables

DISC id 'Qdis' PDIN 1 1 '1;00:00:00' pdin
TBLE
1997/01/01;00:00:00 0.50 1000.0 <
1997/12/31;00:00:00 0.50 1000.0 <
tble
disc

DEMD id 'Qdemand' PDIN 1 1 '1;00:00:00' pdin
TBLE
1997/01/01;00:00:00 0.60 <
1997/12/31;00:00:00 0.60 <
tble
demd

With

id	table identification
nm	table name

PDIN ..pdin = period and interpolation method

0 0 ' ' = interpolation continuous, no period 1 0 ' ' = interpolation block, no period 0 1 '365;00:00:00' = interpolation continuous, period in DDD;HH:mm:ss 1 1 '1;00:00:00' = interpolation block, period in DDD;HH:mm:ss

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After the date and time fields in the table of the DISC record, the first value is the discharge (m3/s), the second value is the salt concentration (g/m3). The table of the DEMD records only contains a demand (m3/s).

1.2.8 NWRW layer

Data file: Pluvius.3B

This file contains the data for the nodes of model type 7 (NWRW rainfall-runoff nodes, Urban runoff model)

NWRW id '1' sl 2.0 ar 1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. np 3 dw '125_lcd' ms 'meteostat1' nwrw

or

NWRW id '1' sl 2.0 na 2 aa 123 456 at 1 0 nw 'special1' 'special2' ar 1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. np 3 dw '125_lcd' ms 'meteostat1' nwrw

or

NWRW id '4' sl 1 ar 0 30000 0 0 0 0 0 0 0 0 0 0 np 0 dw '1' ms '' aaf 1 uw 1 wspl 0.50 wdl 0.35 wbl -0.2 wl 35 ww 2.5 wsw 1.0 wsdc 1.0 wdeh 0.125 wdew 0.125 wdf 0 wddc 0.055945 wdcc 0.63 wdiwl -0.15 wgwl -0.2 wir 10.4167 wp 0.17 nwrw

or

NWRW id '3' sl 1 ar 0 30000 0 0 0 0 0 0 0 0 0 0 na 2 aa 100 222 at 1 0 nw 'GreenRoofTest1' 'UserDefined1' np 0 dw '1' ms '' aaf 1 uw 1 wbl -0.2 wl 35 ww 2.5 wdiwl -0.15 wgwl -0.2 wir 10.4167 wp 0.17 wspl 0.5 wsw 1 wsdc 1 wdl 0.35 wdew 0.125 wdf 0 wddc 0.055945 wdcc 0.63 wdeh 0.125 nwrw

With

id	node identification
sl	surface level (in m) (optional input data)
ar	12 area types as combination of 3 kind of slopes (with a slope, flat, flat stretched) and 4 types of surfaces (closed paved, open paved, roofs, unpaved) a1=closed paved, with a slope a2=closed paved, flat a3=closed paved, flat stretched a4=open paved, with a slope .. a7=roofs, with a slope .. a10=unpaved, with a slope
np	number of people
dw	dry weather flow identification
ms	identification of the meteostation
na	number of special areas with special inflow characteristics
aa	special area in m2 (for number of areas as specified after the na keyword)
at	special area type (for number of areas as specified after the na keyword), with type at 1 = green roof, and type at 0 = user defined
nw	reference to definition of special inflow characteristics (for na special areas)
uw	use wadi (swale filter) 1=yes, 0=no
wbl	wadi bed level [m AD]
wl	wadi length [m]
ww	wadi width [m]
wdiwl	wadi initial water level [m AD]
wgwl	wadi groundwater level [m AD]
wir	wadi infiltration resistance W [days]
wp	wadi porosity [-]
wspl	wadi spillway crest level [m AD]
wsw	wadi spillway crest width [m]
wsdc	wadi spillway discharge coefficient [-]
wdl	wadi drain exit level [m AD]
wdew	wadi drain exit width [m]
wdf	wadi drain structure formula (0=orifice, 1=weir)

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wddc	wadi drain discharge coefficient [-]
wdcc	wadi drain contraction coefficient (orifice only) [-]
wdeh	wadi drain exit height (orifice only) [m]

Remarks:

- The number of standard areas (3 type of slopes, 4 types of surfaces, 12 possible combinations) is hard-coded.

- Salt concentrations are not implemented for this type of node.

DWA file: Pluvius.Dwa

DWA id '125_lcd' nm '125_liters per day' do 1 wc 12. wd 125. wh 1.5 1.5 1.5 1.5 1.5 3.0 4.0 5.0 6.0 6.5 7.5 8.5 7.5 6.5 6.0 5.0 5.0 5.0 4.0 3.5 3.0 2.5 2.0 2.0 dwa

id	dwa identification
nm	name
do	dry weather flow computation option 1 = nr. of people * constant DWA per capita per hour 2 = nr. of people * variable DWA per capita per hour 3 = 1 * constant DWA per hour 4 = 1 * variable DWA per hour 5 = using a table
wc	water use per capita as a constant value per hour (l/hour)
wd	water use per capita per day (l/day)
wh	water use per capita per hour (24 percentages, total should be 100%)
dt	dwa table id

NWRW table file: Pluvius.Tbl

This file contains the DWA discharges as a function of time. The lay-out is similar to other time tables.

DW_T id 'DWA_table' nm 'DWA_table' PDIN 1 0 ' ' pdin
TBLE
1997/01/01;00:00:00 0.5 <
1997/05/01;00;00;00 0.55 <
1997/10/01;00:00:00 0.5 <
1997/12/31;23:59:00 0.50 <
tble
dw_t

With

id	table identification
nm	name of table

PDIN ..pdin = period and interpolation method

0 0 ' ' = interpolation continuous, no period 1 0 ' ' = interpolation block, no period 0 1 '365;00:00:00' = interpolation continuous, period in DDD;HH:mm:ss 1 1 '1;00:00:00' = interpolation block, period in DDD;HH:mm:ss

TBLE .. tble contains the table, first column is the date (year month day hour minute second), second column the DWA discharge in m3/s.

General data: Pluvius.Alg

This file contains the general data for the nodes of model type 7 (NWRW rainfall-runoff nodes)

PLVG id '-1' nm ' ' ru 0.5 0.2 0.1 ms 0.0.5 1.0 0 0.5 1.0 0. 2. 4. 2. 4. 6. ix 0. 2.0 0. 5.0 im 0. 0.5 0. 1.0 ic 0.0 0.1 0.0 0.1 dc 0.0 3.0 0.0 3.0 od 0 or 0 plvg

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With:

id	identification (default id –1; this record is the default general definition)
nm	name
ru	runoff-delay factor for 3 types of slopes (with a slope, flat, flat stretched)
ms	maximum storage (in mm per m2) for 12 types of area (see plv file)
ix	maximum infiltration capacities in mm/hour for 4 types of surface (closed paved, open paved, roofs, unpaved)
im	minimum infiltration capacities in mm/hour for 4 types of surface (closed paved, open paved, roofs, unpaved)
ic	decrease in infiltration capacity (1/hour) for 4 types of surface
dc	increase in infitration capacity (1/hour) for 4 types of surface
od	option for infiltration from depressions (1=yes, 0=no)
or	option for infiltration from runoff (1=yes, 0=no)

or for a user-defined special area (with similar parameters as the standard types)

PLVA id 'UserDefined1' nm 'UserDef1' ru 0.4 ms 2 ef 1 ix 5 im 1 ic 0.4 dc 0.1 od 1 or 0 plva

With:

id	identification definition for user defined special inflow area
nm	name
ru	runoff-delay factor
ms	maximum storage (in mm per m2)
ix	maximum infiltration capacities in mm/hour
im	minimum infiltration capacities in mm/hour
ic	decrease in infiltration capacity (1/hour)
dc	increase in infitration capacity (1/hour)
ef	evaporation crop factor (default=1.0)
od	option for infiltration from depressions (1=yes, 0=no)
or	option for infiltration from runoff (1=yes, 0=no)

or for a green-roof area:

PLGR id 'GreenRoofTest2' nm 'Test2' sl 50 cf 1 thi 99 thwp 10 thfc 30 thsat 50 kp 0.02 ksat 0.06 plgr

With:

id	identification definition for green roof special inflow area
nm	name
sl	soil layer thickness [m]
cf	crop factor evaporation [-]
thi	$\theta_{initial}$, initial soil moisture content [$\%$]
thwp	$\theta_{wiltingpoint}$, soil moisture content at wilting point [$\%$]
thfc	$\theta_{fieldcapacity}$, soil moisture content at field capacity [$\%$]
thsat	$\theta_{saturation}$, soil moisture content at saturation [$\%$]
kp	conductivity at field capacity [mm/min]
ksat	conductivity at saturation [mm/min]

Remark: The conductivity k_p at field capacity should be <= the saturated conductivity k_{sat}.
Remark: The soil moisture content at wilting point is <= the soil moisture content at field capacity, and the soil moisture content at field capacity is <= the soil moisture content at saturation.

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1.2.9 Open water layer

Data file: Openwate.3b

This file contains the data for the nodes of model type 4 (open water area)

OPWA id '1' ml 0.0 rl 0.0 al 2 na 6 ar 10000. 110000. 120000. 130000 140000. 150000.

lv -1. -0.8 -0.6 -0.4 -0.2 0. bl -2.0 tl 0 -0.9 sp 'seep_1' ms 'meteostat1' is 75.0 opwa

With

id	node identification
al	area-level relation (only used by ModelEdit) 1 = constant area, 2=interpolation, 3=lineair
na	number of area/level combinations. Default 6. NOT READ.
ar	6 values of area (in m2)
lv	6 corresponding values of level (m NAP) in increasing order
ml	maximum allowable level (m NAP)
rl	reference level (m NAP)
bl	bottom level (m NAP) Default 1 meter below lowest value from area-level relation.
sp	seepage definition.
ms	identification of the meteostation
is	initial salt concentration (mg/l)
tl	target level; constant or reference to a table. tl 0 -0.9 = initial groundwater level as a constant, with value -0.9 m NAP. tl 1 'Tlv-Table' = target open water levels as a table, with table id Tlv-Table.

Seepage definition file: Openwate.sep

This file contains the seepage definitions for the nodes of model type 4 (open water)

SEEP id 'seep_1' nm 'constant seepage_1mm' co 1 sp 1. ss 500. seep
SEEP id 'seep_2' nm 'simple variable seepage' co 2 cv 3.0 h0 'H0Table' ss 500. seep
SEEP id 'seep_3' nm 'variable seepage using coupling with Modflow' co 3 cv 3.0 ss 500. seep

With:

id	seepage identification
nm	name
co	computation option seepage 1 = constant seepage (default) 2 = variable seepage, using C and a table for H0 3 = variable seepage, using C and H0 from Modflow If the co field is missing, co 1 will be assumed.
sp	Seepage or percolation (mm/day) Positive numbers represent seepage, negative numbers represent percolation. Default 0.
ss	salt concentration seepage (mg/l). Default 500 mg/l. This value is only important for positive seepage values.
cv	Resistance value C for aquitard
h0	reference to a table with H0 values

Remark: File could be combined with similar file from unpaved layer.

Open water table file: Openwate.tbl

This file contains the tables for target levels and variable H0 values for the open water nodes.

Table input of the target open water level as function of time.

OW_T id 'targetlevel table1' PDIN 1 0 '1;00:00:00' pdin
TBLE
1995/01/01;00:00:00 0.3 <

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1995/04/15;00:00:00 0.4 <
1995/09/14;23:00:00 0.4 <
1995/09/15;00:00:00 0.3 <
1995/12/31;23:59:00 0.3 <
tble
ow_t

With

id	table identification
nm	name of table

PDIN ..pdin = period and interpolation method 0 0 ' ' = interpolation continuous, no period 1 0 ' ' = interpolation block, no period 0 1 '365;00:00:00' = interpolation continuous, period in DDD;HH:mm:ss 1 1 '1;00:00:00' = interpolation block, period in DDD;HH:mm:ss

TBLE .. tble contains the table, first column is the date (year month day hour minute second), second column the target level in m NAP. The < sign is a separator.

H0 table definitions:

H0_T id 'H0table' PDIN 1 1 '1;00:00:00' pdin
TBLE
1995/01/01;00:00:00 0.4 <
1995/02/01;00:00:00 0.5 <
1995/03/01;00:00:00 0.5 <
1995/04/01;00:00:00 0.4 <
1995/05/01;00:00:00 0.3 <
1995/06/01;00:00:00 0.2 <
1995/07/01;00:00:00 0.1 <
1995/08/01;00:00:00 0.0 <
1995/09/01;00:00:00 0.0 <
1995/10/01;00:00:00 0.1 <
1995/11/01;00:00:00 0.2 <
1995/12/01;00:00:00 0.3 <
tble
h0_t

With:

id	table identification

PDIN .. pdin = period and interpolation method (for description see above)

1.2.10 Paved area layer

Data file: Paved.3B

This file contains the data for the nodes of model type 1 (paved or impervious area)

The keyword PAVE is used to mark the beginning of a record, and in lower case characters it marks the end of a record. The keywords within the record may appear in any order (same for other files). Some keywords within the records may be missing; in that case defaults will be used. If essential keywords (e.g. the id) are missing, error message will follow.

PAVE id '1' ar 100000. lv 1.0 sd 'stor_1mm' ss 0 qc 0 0.2 0 qo 1 1 so 0.5 0.5 si 0 0 ms 'meteostat1' is 100. np 5000 dw '1' ro 1 ru 0.5 qh 'Qhrelation' pave

With

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id	node identification
ar	area (in m2)
lv	street level (m NAP)
sd	storage identification
ss	sewer system type (0=mixed, 1=separated, 2=improved separated)
qc	capacity of sewer pump (m3/s) qc 0 0.2 0.0 = capacity as a constant, with value 0.2 (mixed/rainfall sewer) and 0.0 (DWA in separated or improved separated systems). **So, first value is for mixed/rainfall sewer, second value for the dry weather flow (DWA) sewer.** qc 1 'qctable1' = capacity as a table, with table identification qctable1.
qo	1 1 = both sewer pumps discharge to open water (=default) 0 0 = both sewer pumps discharge to boundary 0 1 = rainfall or mixed part of the sewer pumps to open water, DWA-part (if separated) to boundary 1 0 = rainfall or mixed part of the sewer discharges to boundary, DWA-part (if separated) to open water 2 2 = both sewer pumps discharge to WWTP 2 1 = rainfall or mixed part of the sewer pumps to open water, DWA-part (if separated) to WWTP etc. **Note: first position of record is allocated to DWA sewer, second position is allocated to mixed/rainfall sewer; 0=to boundary, 1= to openwater, 2=to WWTP. In all other keywords the order is just the other way around!!!!**
so	sewer overflow level (first value for RWA/Mixed sewer, second value for DWA sewer). If missing, the surface level will be used. The level is used to verify whether sewer overflows can occur (no overflows can occur if the related boundary or open water level is higher)
si	sewer inflow from open water/boundary possible yes/no (1=yes,0=no); first value for RWA/Mixed sewer, second value for DWA sewer). Default value is 0, meaning that no external inflow is possible.
s	identification of the meteostation by a character id If this id is missing in the rainfall file, data from the first station in the rainfall file will be used.
is	initial salt concentration (g/m3). Default 0.
np	number of people
dw	dry weather flow identification
ro	runoff option 0 = default, no delay (=previous situation) 1 = using runoff delay factor (as in NWRW model) 2 = using Qh relation
ru	runoff delay factor in (1/min) (as in NWRW model) (only needed and used if option ro 1 is specified)
qh	reference to Qh-relation (only needed and used if option ro 2 is specified)

Spilling of paved area sewers occurs to the downstream open water node (if existing), otherwise to the downstream boundary. In case no downstream open water node and no downstream boundary exists, an error message will be given.

Storage Definition file: Paved.Sto

This file contains the storage definitions for the nodes of model type 1 (paved or impervious area)

STDF id 'stor_1mm' nm 'storage 1mm' ms 1. is 0. mr 7.0 0.0 ir 0.0 0.0 stdf

With

id	storage identification
nm	name (optional)
ms	maximum storage on streets (mm). Default 1 mm.
is	initial storage on streets (mm). Default 0.
mr	maximum storage sewer (mm). Default 7 mm.
ir	initial storage in sewer (mm). Default 0.

For **mr** and **ir** different sewer systems are distinghuished (mixed systems, separated systems, improved separated system). The first value is for mixed and rainfall sewer, the second value for DWA sewer.

DWF definition file: Paved.dwa

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This file defined the dry weather flow for paved area. It is similar to the DWA file for the NWRW rainfall-runoff model.

DWA id '125_lcd' nm '125 liter per capita per day' do 1 wc 12. wd 125. wh 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 sc 50. dwa

id	dwa identification
nm	dwa name
do	dwa computation option see NWRW dwa description.
wc	water use per capita as a constant value per hour (l/hour)
wd	water use per capita per day (l/day)
wh	water use per capita per hour (24 percentages, total should be 100%)
sc	salt concentration (mg/l) of DWA. Default 400 mg/l. The sc keyword is **NOT READ. Default value always used.**

Table definition file: Paved.tbl

For table input of the sewer pump capacity as function of time.

QC_T id 'qctable1' nm 'Test' PDIN 1 0 ' ' pdin
TBLE
1997/01/01;00:00:00 1.0 0.0 <
1997/12/31;00:00:00 1.0 0.0 <
tble
qc_t

With

id	table identification
nm	table name

PDIN ..pdin = period and interpolation method 0 0 ' ' = interpolation continuous, no period 1 0 ' ' = interpolation block, no period 0 1 '365;00:00:00' = interpolation continuous, period in DDD;HH:mm:ss 1 1 '1;00:00:00' = interpolation block, period in DDD;HH:mm:ss

TBLE .. tble contains the table, first column is the date (year month day hour minute second), second and third column the capacity in m3/s. Third column is default zero, can only be used for separated sewer systems. The < sign is used to separate the rows.

Period can be indicated in DDD;HH:mm:ss indicates a yearly period, for example 365;00:00:00. This option is especially useful for open water target levels as a table.

The period is indicated within quotes; this is also done in the SOBEK2 1D modeldatabase.

For a Qh-table the following lay-out is used:

QHTB id 'Qhrelation' nm 'Qhrelation table' TBLE
0.0 0.0 <
0.01 0.5 <
0.03 1.0 <
0.05 1.5 <
0.07 2.0 <
tble qhtb

With the keywords:

id	reference to Qh-relation
nm	name of Qh-relation

TBLE . . . tble = table, with 1st column = Q in mm/s,2nd column = h in mm.

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1.2.11 Runoff layer

Data file (sacrmnto.3b)

This file contains the data for the nodes Sacramento, HBV, SCS, and external timeseries, and optionally for bifurcation nodes. Also the data records for WALRUS, LGSI and NAM are added in this file, but they are described seperately

SACR id ’5’ ar 319000000 ms ’1’ ca ’Capacities&Contents’ uh ’UnitHydrograph’ op ’OtherParameters’ sacr

EXTR id ‘nodeid’ ar 7010324 rs ‘rs7001’ ms ‘ms7001’ Extr

HBV id ‘nodeid’ ar 7010324 sl 300 snow ‘1’ soil ‘1’ flow ‘2’ hini’3’ ts ‘Moezeltemperature’ ms ‘Moezel’ aaf 0.9 hbv

SCS id ‘nodeid’ ar 7010324 slp 0.530683 le 2650.826 cn 50 ms ‘7001’ uh 1 amc 1 scs

SCS id ‘nodeid2’ ar 7010324 slp 0.530683 le 2650.826 cn 50 ms ‘7001’ uh 0 tl 1.0 scs

SCS id ‘nodeid3withbaseflow’ ar 7010324 slp 0.530683 le 2650.826 cn 50 ms ‘7001’ uh 0 tl 1.0 amc 1 bu 1 gsmax 100 gsinit 50 kr 0.01 scs

SCS id ‘nodeid4GreenAmptInfiltration’ ar 7010324 slp 0.530683 le 2650.826 cn 50 ms ‘7001’ uh 0 tl 1.0 ga 1 ksat 2.54 psi 50.8 theta_dmax 0.2 scs

SCS id ’RR_194’ ar 1561388 uhdr 1 tl 3.0 uhdf 0.5 bfdr 0.5 bfsq 1 bfim 1 cn 1 69 uh 2 cp 1.0 amc 3 ms ’Lelydorp’ aaf 1 scs

BIFU id ’lnk_4’ nl 4 li ‘763’ ‘760’ ‘761’ ‘762’ fr 0.10 0.30 0.30 0.30 bifu

With

id	node identification
ar	area in m2
ca	name of capacity and content definition (refer to sacrmnto.cap file)
uh	name of unit hydrograph definition (refer to sacrmnto.uh file)
op	name of definition of other parameters (refer to sacrmnto.oth file)
rs	runoff station
ms	meteo station
ts	temperature station
sl	surface level (altitude)
snow	reference to snow definition in SNOW record
soil	reference to soil definition in SOIL record
flow	reference to flow definition in FLOW record
hini	reference to initial water contents definition in HINI record
slp	average basin slope (m/m)
le	flow path length to outlet (m)
cn	SCS curve number
uh	SCS unit hydrograph option (optional) default value uh 0 = HEC-HMS unit hydrograph; other option uh 1 = SCS dimensionless unit hydrograph 2 = Snyder unit hydrograph.
tl	SCS time lag (hours), optional input, unless uh=2 (Snyder) (if not specified, computed using SCS time lag formula from SCS curve number, average basin slope and flow path length)
amc	antecedent moisture condition. 1=dry, 2=average, 3=wet. The keyword is optional, if missing a default value of 2 is assumed.
aaf	area adjustment factor. Optional keyword, default value 1. Value should be $>$=0.
bu	baseflow used; 0=no, 1=yes. Default 0
gsmax	groundwater storage, maximum value (mm)
gsinit	groundwater storage, initial value (mm)

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kr	recession coefficient Kr (1/day)

The following additional options can be used if a HEC-HMS unit hydrograph (uh=0) is specified for SCS nodes:

tl	optional time lag (in hours); if not specified, the SCS formula will be used to estimate the time lag
R	reservoir coefficient R (hour) of the HEC-HMS linear reservoir; if not specified or zero, no linear reservoir is assumed and outflow equals the inflow according to the time-area diagram HEC-HMS formula.

The following additional options can be used if a Snyder unit hydrograph (uh=2) is specified for SCS nodes:

uhdf	unit hydrograph decay rate, the fraction of peak flow at which exponential decay will start, max 0.5.
uhdr	exponential decay rate (1/hour) of the Unit Hydrograph
cp	peaking constant that determines the slope of the Unit Hydrograph. Region dependent factor.
bfsq	Base Flow starting Q, starting base flow (in mm/hour) at the start of the simulation
bfdr	Base flow decay rate, either linear (mm/hour*hour) or exponential (1/hour) depending on interpolation method chosen.
bfim	Base flow interpolation method. 0 = no base flow, 1 = constant, 2 = exponential, 3 = linear

The following additional options can be used if a Green-Ampt infiltration option is specified for SCS nodes.

ga	Green-Ampt infiltration used (1) or not (0 or keyword not specified)
ksat	saturated hydraulic conductivity [mm/hour], only used if Green-Ampt option ga=1 is specified
psi	suction head at wetting front [mm], , only used if Green-Ampt option ga=1 is specified
theta_dmax	maximum soil moisture deficit [-], only used if GreenAmpt option ga=1 is specified

Note that ksat and psi input is in mm and mm/hour (not in inches and inches/hour). The computations are based on the EPA-SWMM manual (which describes everything in imperial units using inches and inches/hour). Relevant non-dimensionless constants are adjusted in RR using 1 inch=25.4 mm.

For bifurcation nodes, optional a distribution over the downstream links can be specified (if omitted, an equal distribution is assumed) For this purpose, the following keywords are available in the BIFU record;

id	Bifurcation node id
nl	Number of downstream links
li	Downstream link ids (between single quotes)
fr	Fraction of total discharge going over each downstream link, in the order of the specifiek links via keyword li The sum of all fractions should be equal to 1.0

In the SACR, HBV and SCS record also the aaf keyword is optionally present to specify area adjustment factor. There are no other changes in the SACR, HBV, EXTR, SNOW, SOIL, FLOW, HINI records in the Sacrmnto.3B file.

Capacities and contents definition file (sacrmnto.cap)

This file contains the definitions of the capacities, initial contents, and reservoir coefficients of the Sacramento soil storage reservoirs.

CAPC id 'Capacities&Contents' uztwm 50 uztwc 50 uzfwm 150 uzfwc 100 lztwm 150 lztwc 150 lzfsm 200 lzfsc 100 lzfpm 150 lzfpc 150 uzk .08 lzsk .05 lzpk .003 CAPC

With

id	identification
uztwm	upper zone tension water storage capacity (maximum content)
uztwc	upper zone tension water initial content
uzfwm	upper zone free water storage capacity (maximum content)
uzfwc	upper zone free water initial content
lztwm	lower zone tension water storage capacity
lztwc	lower zone tension water initial content
lzfsm	lower zone supplemental free water storage capacity
lzfsc	lower zone supplemental free water initial content
lzfpm	lower zone primary free water storage capacity
lzfpc	lower zone primary free water initial content
uzk	upper zone drainage rate
lzsk	lower zone secondary drainage rate
lzpk	lower zone primary drainage rate

Unit hydrograph definition file (sacrmnto.uh)

UNIH id 'UnitHydrograph' uh 1 0 0 0

HBV database records in the <sacrmnto.3b> file

The graphical user interfaces of SOBEK2 and D-HYDRO write the input parameters of a HBV catchment into six different type of HBV database records, that are stored in the <sacrmnto.3b> database file. These HBV database records are:

1 The **HBV** database record (main record):

HBV id 'nodeid' ar 7010324 sl 300 snow '1' soil '1' flow '2' hini'3' ts 'Moezeltemperature' ms 'Moezel' aaf 0.9 hbv

where:

id	HBV node id
nm	optional node name
ar	Catchment surface area [m^2]
sl	Surface level of the catchment area (optional) [m AD]
snow	reference to definition of SNOW record, containing HBV snowmelt parameters
soil	reference to definition of SOIL record, containing HBV soil parameters
flow	reference to definition of FLOW record, containing HBV flow reservoir coefficients and runoff parameters
hini	reference to definition of HINI record, condinting HBV definition of initial reservoir contents
ts	temperature station, referring to temperature time series to be used
ms	meteorological station, referring to rainfall and evaporation time series to be used
aaf	Area adjustment factor for rainfall (default 1.0)

2 The **SNOW** database record:

SNOW id '1' nm 'DefaultDinkel' mc 4 sft 0 smt 0 tac 6 fe 0.005 fwf 0.1 snow

where:

id	Snow definition id
nm	Optional definition name
mc	Snow melding constant [mm/day/degree Celsius], usually between 3.0 and 6.0, typical value 4.0
sft	Snowfall temperature [degree Celsius], between -1.0 and 2.0. Typical value 0.0
smt	Snowmelt temperature [degree Celsius], between -1.0 and 2.0. Typical value 0.0
tac	Temperature altitude constant [degree Celsius per km] , usually between 4 and 10, typical value 6.0
fe	Freezing efficiency [-], usually between 0 and 0.01, typical value 0.005
fwf	Free water fraction [-] typically 0.1

3 The **SOIL** database record:

SOIL id 'Soil1' nm 'DefaultSoil' be 3 fc 100 ef 0.1 soil

where:

id	Soil definition id
nm	Optional definition name
be	Beta, dimensionless exponent in equation relating runoff volume SP to soil moisture content SM [-], usually between 1.0 and 4.0, typcal value 3.0
fc	Field capacity [mm]
ef	Fraction of field capacity above which actual evapotranspiration equals potential evapotranspiration [-], between 0 and 1, typically 0.7

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4 The **FLOW** database record:

FLOW id '1' nm 'DefaultFlow' kb 0.001 ki 0.25 kq 0.45 qt 20 mp 5 flow

where:

id	FLOW definition id
nm	Optional definition name
kb	Base flow reservoir coefficient [1/day], typical value 0.01
ki	Interflow reservoir coefficient [1/day], typical value 0.1
kq	Quickflow reservoir coefficient [1/day], typical value 0.3 $0 < k_b < k_i < k_q < 1$
qt	Threshold for upper zone reservoir content [mm], below the threshold there is no quick runoff, Typical values between 10 and 40 mm.
mp	Maximum percolation [mm/day], usually between 0.5 and 1.0 mm/day

5 The **HINI** database record:

HINI id '1' nm 'DefaultHINI' ds 0 fw 0 sm 0.95 uz 6.6 lz 50 hini

where:

id	FLOW definition id
nm	Optional definition name
ds	Initial dry snow content [mm]
fw	Initial free water content [mm]
sm	Initial soil moisture content as a fraction of field capacity [-], between 0 and 1.
uz	Initial upper zone content [mm]
lz	Iniital lower zone content [mm], typical initial value is initial runoff divided by the base flow constant

1.2.13 NAM rainfall runoff model

The SOBEK2 graphical user interface writes all D-NAM input parameters into six different type of NAM database records, that are stored in the <SACRMNTO.3B> database file. The Deltashell GUI does not support the D-NAM model concept.

1 **NAM** database record (Main record)

NAM id '1' nm '1' ar 5000000 sp '##1' lwd '##1' sr '##1' bf '##1' gw '##1' ms 'Station1' aaf 1 nam

where:

id	ID of D-NAM node
nm	NAME of D-NAM node (optional)
ar	Catchment area [m^2]
sp	Reference to NAMS database record with id=##1, containing the soil parameters definition
lwd	Reference to NAML database record with id=##1, containing the levels and initial water depth definition
sr	Reference to NAMR database record with id=##1, containing the surface runoff parameters definition
bf	Reference to NAMB database record with id=##1, containing the base flow parameters definition
gw	Reference to NAMG database record with id=##1, containing groundwater pump definition
ms	Meteo-station
aaf	Area adjustment factor [-]

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2 **NAMS** database record (Soil parameters definition)

NAMS id '##1' nm 'SoilDef01' infcap 11 sfc 0.8 syrz 0.9 syss 0.4 capo 0 capc 12 capt PDIN 0 0 " pdin
TBLE
.2 15 <
.5 12 <
1 10 <
2 10 <
tble
pero 0 perc 8 pert PDIN 0 0 " pdin
TBLE
0 1 <
1 5 <
2 8 <
4 16 <
tble
nams

where:

id	ID of this NAMS database record, containing a particular soil parameter definition
nm	Name of this NAMS database record, containing a particular soil parameter definition (shown in GUI)
infcap	Maximum infiltration capacity ($INFCap$) [mm/hr]
sfc	Field capacity of root zone layer (s_{fc}) [-]
syrz	Specific yield of root zone layer ($s_{y,rz}$) [-]
syss	Specific yield of subsoil layer ($s_{y,ss}$) [-]
capo	Option for capillary rise (CR), 0 = Constant, 1 = Table
capc	Capillary rise (CR) [mm/day] as a Constant
capt	Capillary rise (CR) [mm/day] as Function of groundwater table depth in default SOBEK format
pero	Option for Percolation capacity (G), 0 = Constant, 1 = Table
perc	Percolation Capacity (G) as a Constant [mm/day]
pert	Percolation Capacity (G) as Function of groundwater table depth [mm/day] in default SOBEK format

3 **NAML** database record (Levels and initial water depth definition)

NAML id '##1' nm 'IniValDef01' sl 6.66 rzbl 4 gwsbl 3 ui 25 li 400 gwsdi 45 naml

where:

id	ID of this NAML database record, containing a particular level and initial water depth definition
nm	Name of this NAML database record, containing a particular level and initial water depth definition (shown in GUI)
sl	Surface level (SL) [m AD]
rzbl	Bed level of the root zone layer ($RZBL$) [m AD].
gwsbl	Bed level of the subsoil layer or groundwater storage ($GWSBL$) [m AD]
ui	Initial water depth in the surface storage (U_i) [mm]
li	Initial water depth in lower zone storage (L_i) [mm]
gwsdi	Initial water depth in groundwater storage ($GWSD_i$) [mm]

4 **NAMR** database record (Runoff parameters definition)

NAMR id '##1' nm 'Runoff' cl 5000 ss 0.001 man 0.02 utof 5 ckif 1000 utif 1 ltif 5 namr

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where:

id	ID of this NAMR database record, containing a particular runoff parameters definition
nm	Name of this NAMR database record, containing a particular runoff parameter definition (shown in GUI)
cl	Catchment length (CL) [m]
ss	Averaged slope of the catchment surface area towards its overland outflow point (S) [-]
man	Manning value (n) for the roughness of the surface catchment area [$s/m^{1/3}$]
utof	Threshold, used in determining if water in the surface storage may flow out as overland flow [mm]
ckif	Reservoir coefficient used in routing interflow [days]
utif	Threshold, used in determining if water in the surface storage may flow out as interflow [mm]
ltif	Lower-zone-storage-threshold, used in determining if water in the surface storage may flow out as interflow [mm]

5 **NAMB** database record (Base flow parameters definition)

NAMB id '##1' nm 'BaseFlow' ckfastbf 1500. ckslowbf 2500. ckgwinf 3000. ltp 10. tfastbf -1.0 tslowbf -5.0 namb
where:

id	ID of this NAMB database record, containing a particular base flow parameter definition
nm	Name of this NAMB database record, containing a particular base flow parameter definition (shown in GUI)
ckfastbf	Reservoir coefficient used in routing fast base flow ($CKFastBF$) [days]
ckslowbf	Reservoir coefficient used in routing slow base flow ($CKSlowBF$) [days]
ckgwinf	Reservoir coefficient used for the inflow of external (ground)water ($CKGWInflow$) [days]
ltp	Threshold, used in determining if water in the lower zone storage may percolate to the groundwater storage (L_{TP}) [mm]
tfastbf	Threshold for fast base flow (T_{FastBF}) [m AD]. Only water in the groundwater storage above T_{FastBF} can flow out as fast base flow]
tslowbf	Threshold for slow base flow (T_{SlowBF}) [m AD]. Only water in the groundwater storage above T_{SlowBF} can flow out as slow base flow

Note: $CKGWInflow \geq CKSlowBF \geq CKFastBF \geq 1$.



6 **NAMG** database record (Groundwater pump definition)

NAMG id '##1' nm 'GWPump' po 0 pu 0.1 pt PDIN 0 1 '365;00:00:00' pdin
TBLE
'2001/01/01;00:00:00' 0.0 <
'2001/03/31;23:59:00' 0.0 <
'2001/04/01;00:00:00' 0.1 <
'2001/09/30;23:59:00' 0.1 <
'2001/10/01;00:00:00' 0.0 <
'2001/12/31;00:00:00' 0.0 <
tble
namg

where:

id	ID of this NAMG database record, containing a particular groundwater pump definition
nm	Name of this NAMG database record, containing a particular groundwater pump

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	definition (used in GUI)
po	Pumping option, 0 = Constant, 1 = Time Series
pu	Discharge of the groundwater pump ($GWPump$) as a Constant [m^3/s]
pt	Discharge of the groundwater pump ($GWPump$) as a Time series [m^3/s] in default SOBEK format

1.2.14 Walrus rainfall runoff model

The <delft_3b.ini> file:

The <delft_3b.ini> file contains switch (UseWalrusorWagmod=ModelName):

- ◇ SOBEK 2: Switch set as UseWalrusorWagmod=WALRUS, means that the Walrus model is run. Switch set as UseWalrusorWagmod=WAGMOD, means that the WAGMOD model is run. Limitation, the Walrus and Wagmod model cannot jointly be used during a simulation.
- ◇ D-HYDRO: Switch is default set as UseWalrusorWagmod=WALRUS, since the old Wagmod model is not supported in the SOBEK3, D-HYDRO and Delft3D FM 1D2D GUI.

Following five numerical parameters can be specified in the <delft_3b.ini> file, which control the Walrus internal timestep (i.e. possible reductions of the user-defined timestep):

1 Walrus_min_deltime: minimum timestep in seconds (default=60.0)
2 Walrus_maxhchange: maximum change in the water depth of the surface water reservoir and/or quickflow reservoir in 1 internal timestep in mm (default =10.0).
3 Walrus_minh: minimum water depth in the surface water reservoir and/or quickflow reservoir in mm (default =0.001).
4 Walrus_max_pstep: maximum precipitation in 1 internal Walrus timestep in mm (default =10.0).
5 Walrus_max_substeps: maximum number of internal timesteps in 1 computation timestep (default=288)

Note: For more information on the Walrus time step reduction procedure, reference is made to section "Possible time step reductions and applied threshold values" in the technical reference manual.

Walrus database records in the <sacrmnto.3b> file

The SOBEK graphical user interface writes the input parameters of a Walrus catchment into six different type of Walrus database records, that are stored in the <sacrmnto.3b> database file. The D-HYDRO GUI does not yet support the editing of Walrus data, the Walrus records have to be edited by the user.

1 The **WALR** database record (main record):

WALR id 'node id' nm 'node name' ar 1000 sl 20.1 wa 0 cw 200 wit 'wetness index table' cv 4 cg 5000000 cq 10 va 0 vit 'Veq table' ba 0 bit 'evaporation reduction table' qa 0 qit 'Qh interpolation table' cs 4 xs 1.5 as 0.01 cd 1500 hst 0 hsmin 0 hstable '1' st 21 b 4.05 psi_ae 121 theta_s 0.395 hs0 0 hq0 0 dg0 0 dv0 0 q0 1 ms 'meteostation' msevap 'evap' fxs 'sw supply/extraction' fxg 'seepage' aaf 1.0 walr

where:

id	Walrus node id
nm	Walrus node name
ar	Catchment surface area [m^2]
sl	Surface level of the catchment area (optional) [m AD]
wa	-1 (wetness index is determined, applying the Walrus analytical formulation for wetness index and value for parameter cw), or

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	0 (wetness index is determined, applying the user-defined interpolation table for wetness index).
cw	Wetness index parameter [mm] (default=200). Value for cw is only used if wa=-1.
wit	id of a user-defined interpolation table for the wetness index, which is to be specified on a WETN database record. This interpolation table is only used if wa=0.
cv	Vadose zone relaxation time constant [hours] (default=4).
cg	Constant parameter applied in computing groundwater drainage or infiltration [mm hour] (default=5000000).
cq	Quickflow constant reservoir coefficient [hours] (default=10).
va	-1 (equilibrium storage deficit is determined, applying the Walrus analytical formulation for equilibrium storage deficit), or 0 (equilibrium storage deficit is determined, applying the user-defined interpolation table for equilibrium storage deficit).
vit	id of a user-defined interpolation table for equilibrium storage deficit, which is to be specified on a VEQD database record. This interpolation table is only used if va=0.
ba	-1 (transpiration reduction factor β is determined, applying the Walrus analytical formulation for β), or 0 (transpiration reduction factor β is determined, applying the user-defined interpolation table for β).
bit	id of a user-defined interpolation table for the transpiration reduction factor β, which is to be specified on a BETA database record. This interpolation table is only used if ba=0.
qa	-1 (discharge Q flowing out of the surface water reservoir is determined, applying the Walrus analytical formulation for Q), or 0 (discharge Q flowing out of the surface water reservoir is determined, applying the user-defined interpolation table for Q).
qit	id of a user-defined interpolation table for discharge Q flowing out of the surface water reservoir, which is to be specified on a WQH database record. This interpolation table is only used if qa=0.
cs	Bankfull discharge of the surface water reservoir [mm/hour] (default=4). Value for cs is only used if qa=-1.
xs	Exponent in Walrus analytical formulation for computing discharge (Q) flowing out of the surface water reservoir (default=1.5). Value for xs is only used if qa=-1.
as	Surface water area fraction, ratio of the area of the surface water reservoir and the catchment surface area [-] (default=0.1)
cd	Channel depth or bed level of the surface water reservoir [mm below SL] (default=1500).
hst	-1 (value of $h_{S,min}$ in the Walrus analytical formulation for discharge (Q) flowing out of the surface water reservoir is defined as a time table), or 0 (a constant value is defined for $h_{S,min}$ in the Walrus analytical formulation for discharge (Q) flowing out of the surface water reservoir).
hsmin	For a water depth in the surface water reservoir equal to $h_{S,min}$, the outflow of the surface water reservoir Q equals zero. Value for hsmin is only used if qa=-1.
hstable	id of the user-defined time table for hsmin, which is to be specified on a HSTT database record. This time table is only used if qa=-1 and hst=-1.
st	Soil type (default; st=22 loamy sand). For soil types st=21 up to st=33 yields that values for b, ψ_{ae} and θ_S are read from Table 1.20. For soil type st=34 (custom) yields that the values for b, ψ_{ae} (or psi_ae) and θ_S (or theta_s) are applied, that are specified hereunder.
b	Pore size distribution parameter [-]. This value for b is only used if soil type st=34 (custom) is selected.
psi_ae	Air entry pressure [mm]. This value for psi_ae is only used if soil type st=34 (custom) is selected.
theta_s	Soil moisture content at saturation [-]. This value for theta_s is only used if soil type st=34 (custom) is selected.
hs0	Initial water depth in surface water reservoir [mm]
hq0	Initial water level in quickflow reservoir [mm]
dg0	Initial groundwater table depth [mm]
dv0	Initial water-storage deficit (or availabe water storage capacity in vadose zone of soil water reservoir [mm]
q0	Initial discharge flowing out of the surface water reservoir [mm/hour]

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ms	Reference (or id) to meteo-station with rainfall time-series in [mm/timestep]
msevap	Reference (or id) to meteo-station with potential evapotranspiration time-series in [mm/timestep] (to give the option to specify evaporation not only as daily values, but in detail)
fxs	Reference (or id) to surface water supply or extraction time-series in [mm/timestep] (from rainfall file)
fxg	Reference (or id) to groundwater seepage or extraction time-series in [mm/timestep] (from rainfall file)
aaf	Area adjustment factor for rainfall (default 1.0)

Please note that:

- ◇ Input parameters ar, cw, cv, cg, cq, cs, cd, xs, hsmin, as, hs0, hq0, dg0, dv0 and q0 should be larger than or equal to zero.
- ◇ For input parameters hsmin and cd should yield that hsmin < cd. Since, the $q = 0$ level (coinciding with hsmin) should be smaller than the channel depth (cd).
- ◇ The groundwater reservoir area fraction (a_G) is not defined in the WALR database record, since a_G= 1-as.
- ◇ If a restart file is used, the values for hs0, hq0, dg0, dv0 and q0 are read from the restart file. So the values specified in the WALR database record are neglected.
- ◇ Time-series for precipitation (P), potential evapotranspiration (ET_{pot}), groundwater seepage and extraction (fxg) and external surface water supply or extraction (fxs) are assigned to a particular meteo-station (ms). The id of this meteo-station as well as the time-series data is read from a particular rainfall file.

Table 1.20: *Soil types and related Walrus parameter values.*

Walrus soil type **Nr.**	**Soil type** **Description**	**Pore-size distribution parameter** **b [-]**	**Air entry pressure** Ψ_{ae} **[-]**	**Soil moisture content at saturation** θ_s **[-]**
21	sand	4.05	121	0.395
22	loamy_sand	4.38	90	0.410
23	sandy_loam	4.9	218	0.435
24	silt_loam	5.3	786	0.485
25	loam	5.39	478	0.451
26	sandy_clay_loam	7.12	299	0.42
27	silt_clay_loam	7.75	356	0.477
28	clay_loam	8.52	630	0.476
29	sandy_clay	10.4	153	0.426
30	silty_clay	10.4	490	0.492
31	clay	11.4	405	0.482
32	cal_Hupsel	2.63	90	0.418
33	cal_Cabauw	16.77	9	0.639
34	custom	User defined	User defined	User defined

Note: Soil type descriptions "cal_Hupsel" and "cal_Cabauw" are respectively the calibrated dataset for the Hupselse Beek catchment and the Cabauw catchment in the Netherlands.

2 The **WETN** database record (Wetness index (W) interpolation table)

WETN id ‘wetness index table’ nv 4 dv 0 100 200 300 wi 1.0 0.7 0.1 0.0 wetn

where:

id	Table id (matching with id given after keyword wit in the WALR database record)
nv	Number of values (length of table)
dv	Water-storage deficit (d_V) values [mm]. For dv should yield dv ≥ 0; consecutive values for dv should be increasing.
wi	Wetness index (W) [-] For wi should yield $0 \leq$ wi ≤ 1; consecutive values for wi should be non-increasing.

3 The **BETA** database record (Transpiration reduction (β) interpolation table)

BETA id 'evaporation reduction table' nv 5 dv 0 100 200 300 400 er 1.0 0.99 0.9 0.75 0.0 beta

where:

id	Table id (matching with id given after keyword bit in the WALR record)
nv	Number of values (length of table)
dv	Water-storage deficit (d_V)values [mm]. For dv should yield dv ≥ 0; consecutive values for dv should be increasing.
er	Transpiration reduction factor (β) [-]. For er should yield $0 \leq$ er ≤ 1; consecutive values for er should be non-increasing.

4 The **VEQD** database record (Equilibrium water-storage deficit ($d_{V,eq}$) interpolation table)

VEQD id 'Veq table' nv 6 dg 0 250 500 1000 1500 2000 veq 0.0 0.0 100 250 500 900 veqd

where:

id	Table id (matching with id given after keyword vit in the WALR record)
nv	Number of values (length of table)
dg	Groundwater table depth (d_G) [mm]. For dg should yield dg ≥ 0; consecutive values for dg should be increasing.
veq	Equilibrium water-storage deficit ($d_{V,eq}$) [mm]. For veq should yield veq ≥ 0; consecutive values for veq should be non-decreasing.

5 The **WQH** database record (Discharge relationship ($Q = f(h_S)$ interpolation table)

WQH id 'Qh interpolation table' nv 4 hs 0 1000 1500 2000 q 0.0 1.0 2.0 3.0 wqh

where:

id	Table id (matching with id given after keyword qit in the WALR record)
nv	Number of values (length of table)
hs	Surface water depth (h_S) [mm]. For hs should yield hs ≥ 0; consecutive values for hs should be increasing.
q	Discharge (Q) flowing out of the surface water reservoir [mm/hour]. For q should yield q ≥ 0; consecutive values for q should be non-decreasing.

6 The **HSTT** database record (Time-series for $h_{S,min}$)

HSTT id '1' nm 'table hsmin'
PDIN 1 0 '365;00:00:00' pdin
TBLE
'2016/01/01;00:00:00' 0.0 <

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'2016/04/01;00:00:00' 20.0 <
'2016/10/01;00:00:00' 0.0 <
'2016/12/31;23:59:00' 0.0 <
tble
hstt

where:

id	Table id (matching with id given after keyword hstable in the WALR database record)
nm	Name of table
PDIN 1 0 '365;00:00:00' pdin	First value: 0 (linear interpolation) / 1(block interpolation) Second value: 0 (no periodicity) / 1 (periodic) Third value: period in 'days;hh:mm:ss' (only used if second value equals 1)
TBLE . . . tble	Free number of sub-records allowed. Each sub-record consecutively contains: date 'YYYY/MM/DD;hh:mm:ss', value and <, denoting the end of a sub-record.

1.2.15 LGSI rainfall runoff model

LGSI nodes in the <3b_nod.tp> file

In the network file describing the DRR network, the node type for LGSI is 67 and the model type is 22 (also see Topography layer)

Example 3b_nod.tp file with an LGSI node:

NODE id '1' ri '-1' mt 1 '22' nt 67 ObID '3B_LGSI' px 229701.199308936 py 480620.1730032 node
NODE id '2' ri '-1' mt 1 '6' nt 47 ObID '3B_BOUNDARY' px 231897.586877857 py 482816.1863686 node

LGSI database records in the <sacrmnto.3b> file

The D-HYDRO and Delft3D FM 1D2D do not support the LSGI editing in the user interface. The LGSI records have to be edited by the use and stored in the file <sacrmnto.3b>. These LGSI database records are:

1 The **LGSI** database record (main record):

LGSI id '1' na 2 ar 9220091 10718300 st 'low' 'high' erd 1.5 5 ersd 0.3 1 rdr 10 10 ddr 0.62 0.85 fas 0.8 0.93 tets 0.65 0.65 alp 1 1 n 4 4 qout 0.0 0.35 hdif 7.5 c 130 ms 'Rijs-uur' 'Rijs-uur' init 'initlow' 'inithigh' dd 'delay' lgsi

where:

id	LGSI node id
nm	LGSI node name
na	number of subareas, default 1 but in LGSI tyhpical use is with 2 subareas
ar	Catchment surface area for each subarea [m^2]
st	statistical type for each subarea (1=normal, 2=gamma distribution)
erd	evapotranspiration reduction depth for each subarea [m]
ersd	evapotranspiration standard deviation reduction depth for each subarea [m]
rdr	drainage resistance for each subarea [$days$]
ddr	drainage depth for each subarea [m]

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fas	fraction non-drainaged for each subarea [$-$]
tets	soil porosity for each subarea [$-$]
alp	Van Genuchten parameter 1 [$1/m$]
n	Van Genuchten parameter 2 [$-$]
qout	outflow or percolation to deep gw layers [mm/day]
hdif	head difference [m]
c	flow resistance from high area to low area [$days$]
ms	meteostation code for both subareas
dd	delay definition (only for draining subarea)

2 The **LGST** database record (LGSI input depending on the *type* of distribution)

LGST id 'low' ty 1 maxsd 0.9 minsd 0.55 nb 0.6 nusdmax 1.3 rex 12 rov 11 fp 0.15 fow 0.01 lgst
LGST id 'high' ty 2 ag 0.1 bg 12.2 lgst

where:

id	LGST definition id (matching with id given after keyword st in the LGSI record)
maxsd	maximum standard deviation of normal distribution (m)
minsd	minimum standard deviation of normal distribution (m)
nb	parameter nb, shape parameter of normal distribution (-)
nusdmax	parameter nusdmax, average of normal distribution (m)
rex	flow resistance water courses (days)
rov	flow resistance overland flow (days)
fp	fraction ponding
fow	fraction open water
ag	parameter a (shape) of gamma distribution (-)
bg	parameter b (scale) of gamma distribution (m)

3 The **LGSC** database record (LGSI initial *conditions*)

LGSC id 'inithigh' ig 16.777 lgsc LGSC id 'initlow' ig 1.185 lgsc

where:

id	LGSC definition id (matching with id given after keyword xxx in the LGSI record)
ig	initial groundwater level [m below surface]

4 The **LGSD** database record (LGSI *delay* definitions)

LGSD id 'delay' dt 3600 nt 120 dc 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125
0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125 0.0125
0.0125 0.0125 0.0125 0.014583333 0.014583333 0.014583333 0.014583333 0.014583333 0.014583333
0.014583333 0.014583333 0.014583333 0.014583333 0.014583333 0.014583333 0.014583333
0.014583333 0.014583333 0.014583333 0.014583333 0.014583333 0.014583333 0.014583333
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0.008333333 0.008333333 0.008333333 0.008333333 0.008333333 0.008333333 0.008333333
0.004166667 0.004166667 0.004166667 0.004166667 0.004166667 0.004166667 0.004166667
0.004166667 0.004166667 0.004166667 0.004166667 0.004166667 0.004166667 0.004166667
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0.004166667 0.004166667 0.004166667 0.002083333 0.002083333 0.002083333 0.002083333
0.002083333 0.002083333 0.002083333 0.002083333 0.002083333 0.002083333 0.002083333
0.002083333 0.002083333 0.002083333 0.002083333 0.002083333 0.002083333 0.002083333
0.002083333 0.002083333 0.002083333 0.002083333 0.002083333 0.002083333 lgsd

where:

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id	LGSI delay definition id (matching with id given after keyword xxx in the LGSI record)
dt	specified timestep size of the dc values [seconds]
nt	number of timesteps
dc	delay coefficients (in total nt values)

Note: if the computation timestep is different from the specified delay timestep size, the dc coefficients are rescaled to fit the actual time step. For instance, the record

LGSD id 'delay' dt 86400 nt 5 dc 0.3 0.35 0.2 0.1 0.05 lgsd

where delay coefficients are specified in daily timesteps, will be converted to hourly dc values when using an hourly computation timestep, resulting in 120 hourly dc values of respectively 24*0.0125, 24*0.014583, 24*0.00833, 24*0.004167, and 24*0.002083.

1.2.16 Topography layer

[Topography layer with .tp protocol]

The topography layer consists of three files which describe the topography of the network. The first file contains the list of all the nodes in the schematisation, the second file the list of links (connections) between these nodes. The third file is only used in case of Sobek-Urban models using the NWRW node type. The Topography layer in fact defines all the network objects of which the attribute data are described in the other layers.

Node file: 3b_nod.tp

This file contains the node definitions for Sobek-Rural applications. The header line of this file contains in the first positions the string 'BBB2.2'. The keyword NODE is used to mark the beginning of a record, and in lower case characters it marks the end of a record.

NODE id '3B72' nm 'Node1' ri '2' mt 1 '6' nt 23 ObID 'SBK_SBK-3B-REACH' px 11404.2 py 123768.5 node

With

id	node identification
nm	name of the node
ri	branch identification
mt	model nodetype
nt	netter nodetype
ObID	Object id
px	position X (X coordinate)
py	position Y (Y coordinate)

Remarks :

Remark 1: no weather station in this file!

Remark 2: For aggregation purposes it is very well possible that this file should be extended with fields such as

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au = administrative unit, or rb = river basin

so that e.g. total areas or total waste loads can be determined later on by administrative unit and/or river basins. Also a ‘multi-select’ option of changing some input data for all nodes within one administrative unit, or setting default values per river basin, could be nice options for a user.

Remark 3: Another suggestion is to include a second name in the file, to be able to have the node names in two languages.

Remark 4: the following model types are available:

1	Paved area
2	Unpaved area
3	Greenhouse
4	Open water (only in SOBEK2)
5	(Internally reserved for all structures)
6	Boundary
7	NWRW
8	pump (only in SOBEK2)
9	weir (only in SOBEK2)
10	orifice (only in SOBEK2)
11	Manning resistance (only in SOBEK2)
12	Q-h relation (only in SOBEK2)
14	WWTP
15	Industry
16	Sacramento
18	External Runoff Timeseries (only in SOBEK2)
19	HBV
20	SCS
21	Ow Rain
22	LGSI
23	Wagmod/Walrus (in D-HYDRO only Walrus)
30	RR-Connection
31	D-NAM
32	RR-Bifurcation

Remark 5: the Object id is used by RR to distinguish RR-boundaries from RR-CF connection nodes. The RR model type of these different objects is the same (mt 1 ‘6’) , but there Netter types and Object id’s are different.

Link file: 3B_Link.Tp

This file contains the link definitions for Sobek-Rural applications. The header line of this file contains in the first positions the string ‘BBB2.2’.
BRCH id ’1’ nm ’Tak1’ ri ’-1’ mt 1 ‘0’ bt 11 ObID ‘3B-LINK’ bn ’1’ en ’2’ brch

With:

id	link identification
nm	name of the link
ri	branch identification
mt	model type
bt	branch type
ObID	Object identification
bn	identification of begin node (‘from’ node)
en	identification of end node (‘to’ node)

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The branch identification is only used for special (aggregation) purposes of the NWRW rainfall-runoff model which is used in Sobek-Urban applications.

The branch type and Object Id are not used by the RR-computational core, but are used by user-interface programs (in SOBEK2: Netter, in D-HYDRO the Deltashell). The model type of the links is used to check whether RR-routing links are used or not.

Runoff file: 3B_Runoff.Tp

This file contains the NWRW node definitions for Sobek-Urban applications.

NODE id '0-28' ri '1' mt 1 '7' nt 3 ObID 'SBK_CONN&RUNOFF' px 198002.5 py 457582.5 node

With

id	node identification
nm	name of the node
ri	branch identification
mt	model nodetype
nt	netter nodetype
ObID	Object id
px	position X (X coordinate)
py	position Y (Y coordinate)

The branch identification is only used for special (aggregation) purposes of the NWRW rainfall-runoff model which is used in Sobek-Urban applications.

1.2.17 RR-Routing link layer

Data file (3B_Rout.3b)

This file contains the data for the links of model type 30: RR-Routing links. The file contains two type of records, ROUT and RDEF records. The ROUT records contain for each routing link a reference to the routing definition, while the RDEF records contain the routing definition.

ROUT id 'l_8' di '1' rout

ROUT id 'link_id' di 'def-id' rout

With

id	link identification
di	definition of routing id to be used

RDEF id '1' nm 'Donau' nl 2 x 0.3 0.12 k 2.7 4.06 qm 4000 qi 2400 qo 2750 rdef

With

id	routing definition id
nm	routing definition name
rt	routing type, type 0 = Muskingum
nl	number of layers

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x	x coefficient per layer; there are as many coefficients as layers
k	k coefficient per layer; there are as many coefficients as layers
qm	maximum discharge Q per layer, except for the last layer
qi	initial link inflow
qo	initial link outflow

1.2.18 Unpaved area layer

Data file: Unpaved.3B

This file contains the data for the nodes of model type 2 (pervious or unpaved area)
UNPV id '1' na 16 ar 1. 0. 3. 0. 0. 6. 0. 0. 0. 10. 11. 12. 13. 14. 15. 16. lv 1.0 ga 110. co 1 su 0 sd 'ovhstor_1mm' ad 'alfa_1' sp 'seepage_1' ic 'infcap_5' bt 1 0 ig 0 0.5 mg 0.8 gl 1.5 ms 'meteostat1' is 100. unpv
or, when the unpaved irrigation option is used,
UNPV id 'unpaved' na 16 ar 1000000 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 lv 1.5 co 3 rc 1 sd '1' ad '-1' ed '1' sp '1' ic '1' bt 101 ig 0 -999.99 su 0 ' ' gl 5 mg 1.5 ms 'Station1' aaf 1 is 100 irr 2 irreff 1.2 pFcrit 2.7 pFtarget 2.4 initsup 6.0 initdur 5 wait 0 maxsup 12. unpv

With

id	node identification
na	number of areas (at the moment fixed at 16)
ar	area (in m2) for all crops. In the user interface either the total area can be specified, or the different areas per crop. In case the total area is specified, it is put at the first crop (grass).
ga	area for groundwater computations. Default = sum of crop areas.
lv	surface level (=ground level) in m NAP
co	computation option (1=Hellinga de Zeeuw (default), 2=Krayenhoff van de Leur, 3=Ernst)
rc	reservoir coefficient (for Krayenhoff van de Leur only);
su	Indicator Scurve used su 0 = No Scurve used (Default) su 1 'Scurve-id' = Scurve used; Unpaved.Tbl contains defniition of table with id 'Scurve-id'.
sd	storage identification
ad	alfa-level identification (for Hellinga de Zeeuw drainage formula only)
ed	Ernst definition (for Ernst drainage formula only)
sp	seepage identification.
ic	infiltration identification
bt	soil type (from file BERGCOEF or BergCoef.Cap) Indices $>$100 are from Bergcoef.Cap.
ig	initial groundwater level; constant, or as a table ig 0 0.2 = initial groundwater level as a constant, with value 0.2 m below the surface. ig 1 'igtable1' = initial groundwater level as a table, with table identification igtable1.
mg	maximum allowed groundwater level (in m NAP)
gl	initial depth of groundwater layer in meters (for salt computations)
ms	identification of the meteostation
is	initial salt concentration (mg/l) Default 100 mg/l
irr	irrigation option; 0 = no irrigation (default), 1 = from connected RR open water or RR-boundary, 2 = from local groundwater, 3 = from external system
irreff	irrigation efficiency factor (only when irrigation option is switched on)
pFcrit	critical pF, typical 2.7 (only when irrigation option is switched on)
pFtarget	target pF, typical 2.4 (only when irrigation option is switched on)
initsup	initial gift, typical 2.5 mm/day (only when irrigation option is switched on)
initdur	initial duration in days (only when irrigation option is switched on)
wait	minimum waiting time in days between end of previous irrigation supply and start of new irrigation supply. If not specified, 0 days is used.
maxsup	maximum gift in mm/day (only when irrigation option is switched on)

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Note: when irrigation is used, the Delft_3b.ini file should contain specification of the irrigation season.

Alfa definition file: Unpaved.alf

This file contains the definitions of alfa-factors and related levels for the nodes of model type 2 (unpaved or pervious area). Also the Ernst definitions are included in this file.

ALFA id 'alfa_1' nm 'set1 alfa factors' af 5.0 0.9 0.7 0.6 0.3 0.03 lv 0. 1.0 2.0 alfa

ERNS id 'Ernst_1' nm 'Ernst definition set1' cvi 300 cvo 30 30 30 cvs 1 lv 0. 1.0 2.0 erns

With

id	alfa-factors identification
nm	name
af	alfa factors (say a1 to a6) for Hellinga-de Zeeuw formula (1/day). a1 = alfa factor surface runoff a2 = alfa factor drainage to open water, top soil layer a3 = alfa factor drainage to open water, second layer a4 = alfa factor drainage to open water, third layer a5 = alfa factor drainage to open water, last layer a6 = alfa factor infiltration
cvi	Resistance value (in days) for infiltration from open water into unpaved area
cvo	Resistance value (in days) for drainage from unpaved area to open water, for 3 layers
cvs	Resistance value (in days) for surface runoff
lv	three levels below surface (say lv1, lv2, lv3), separating the zones with various alfa-factors (or Ernst resistance values) for drainage. a2 is used between surface level and lv1 m below the surface. a3 is used between lv1 and lv2 m below the surface. a4 is used between lv2 and lv3 m below the surface a5 is used below lv3 m below surface.

Storage definition file: Unpaved.sto

This file contains the storage definitions for the nodes of model type 2 (unpaved or pervious area)

STDF id 'ovhstor_1mm' nm '1 mm storage' ml 1. il 0. STDF
With:

id	storage identification
nm	name
ml	maximum storage on land (mm). Default 1 mm.
il	initial storage on land (mm). Default 0.

Seepage definition file: Unpaved.sep

This file contains the seepage definitions for the nodes of model type 2 (unpaved or pervious area)

SEEP id 'seep_1' nm 'constant seepage_1mm' co 1 sp 1. ss 500. seep

SEEP id 'seep_2' nm 'simple variable seepage' co 2 cv 3.0 h0 'H0Table' ss 500. seep

SEEP id 'seep_3' nm 'variable seepage using coupling with Modflow' co 3 cv 3.0 ss 500. seep

With:

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id	seepage identification
nm	name
co	computation option seepage 1 = constant seepage (Default) 2 = variable seepage, using C and a table for H0 3 = variable seepage, using C and H0 from Modflow If the co field is missing, co 1 will be assumed.
sp	Seepage or percolation (mm/day) Positive numbers represent seepage, negative numbers represent percolation. Default 0.
ss	salt concentration seepage (mg/l). Default 500 mg/l. This value is only important for positive seepage values.
cv	Resistance value C for aquitard
h0	reference to a table with H0 values Remark 1: This file has similar structure as a similar file of the open water layer.

Infiltration file: Unpaved.inf

This file contains the infiltration definitions for the nodes of model type 2 (unpaved or pervious area)

INFC id 'infcap_5' nm 'inf_5mm' ic 5. INFC

With:

id	infiltration identification
nm	name
ic	infiltration capacity of the soil, constant. (mm/hour) Remark: no variable infiltration capacity implemented yet.

Table file: Unpaved.tbl

This file can contain three kinds of tables:

- ◇ initial groundwater level as a function of time
- ◇ S curve definitions
- ◇ H0 definitions for variable seepage

Table input of the initial groundwater level as function of time.
IG_T id 'igtable1' PDIN 1 1 '0;10:00:00' pdin
TBLE
1995/01/01;00:00:00 0.3 <
1995/03/01;00:00:00 0.2 <
1995/05/01;00:00:00 0.3 <
1995/07/01;00:00:00 0.4 <
1995/09/01;00:00:00 0.5 <
1995/11/01;00:00:00 0.4 <
tble
ig_t

With

id	table identification

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PDIN ..pdin = period and interpolation method 0 0 ’ ’ = interpolation continuous, no period 1 0 ’ ’ = interpolation block, no period 0 1 ’365;00:00:00’ = interpolation continuous, period in DDD;HH:mm:ss 1 1 ’1;00:00:00’ = interpolation block, period in DDD;HH:mm:ss

TBLE .. tble contains the table, first column is the date (year month day hour minute second), second column the initial groundwater level in m below surface. The < sign is a separator.

S-curve definitions:

This table is not time depending, so different from the other types of tables in this file.

SC_T id ’ScurveTable’ nm ’ScurveTable1’ PDIN 1 1 ’0’ pdin
TBLE
0 -5.1 <
10 -5.0 <
80 -4.9 <
90 -4.2 <
100 -4.0 <
tble
sc_t

H0 table definitions:

H0_T id ’H0table’ PDIN 1 1 ’1;00:00:00’ pdin
TBLE
1995/01/01;00:00:00 0.4 <
1995/02/01;00:00:00 0.5 <
1995/03/01;00:00:00 0.5 <
1995/04/01;00:00:00 0.4 <
1995/05/01;00:00:00 0.3 <
1995/06/01;00:00:00 0.2 <
1995/07/01;00:00:00 0.1 <
1995/08/01;00:00:00 0.0 <
1995/09/01;00:00:00 0.0 <
1995/10/01;00:00:00 0.1 <
1995/11/01;00:00:00 0.2 <
1995/12/01;00:00:00 0.3 <
tble
h0_t

With:

id	table identification

PDIN ..pdin = period and interpolation method (for description see above at IG_T record)

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1.2.19 WWTP layer

The WWTP layer represents nodes of model type 14 (WWTP) or waste water treatment plants (in Dutch: RWZI RioolWaterZuiveringsInstallaties).

Table file (WWTP.Tbl)

The tbl file may contain measured WWTP discharge flow (m3/s)

MEAS id 'WWTPTable' nm 'Test' PDIN 0 0 ' ' pdin
TBLE
1997/01/01;00:00:00 -1.0 <
1997/01/01;00:00:01 0.5 <
1997/01/02;00:00:00 0.55 <
1997/01/06;00:00:00 0.55 <
1997/01/06;00:00:01 0.5 <
1997/01/08;00:00:00 0.6 <
1997/01/10;00:00:00 0.6 <
1997/01/11;00:00:00 -1.0 <
1997/01/20;00:00:00 -1.0 <
1997/01/20;00:00:01 0.5 <
1997/01/30;00:00:00 0.54 <
1997/02/01;00:00:00 -1.0 <
tble
meas

With

id	table identification
nm	table name

PDIN ..pdin = period and interpolation method 0 0 ' ' = interpolation continuous, no period 1 0 ' ' = interpolation block, no period 0 1 '365;00:00:00' = interpolation continuous, period in DDD;HH:mm:ss 1 1 '1;00:00:00' = interpolation block, period in DDD;HH:mm:ss

Convention: Missing data is indiciated with -1. In that case the discharge of the WWTP is equal to the computed inflow of the WWTP.

No salt concentration included in input table, is derived from inflow.

Data file (WWTP.3b)

This file contains the data of the WWTP node (RR model type 14).

WWTP id '1' tb 0 wwtp

With

id	node identification
tb	table used yes/no tb 0 = no table of measured data; the WWTP outflow is equal to the sum of the inflows. tb 1 'WWTPTable' = table of measured data with id 'WWTPTable'

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